

MAGNETOM

Replacement of Parts System

Magnet System

Avanto
Espree

Document Version

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This chapter provides an overview of the field replaceable units (FRUs) for the magnet system.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. compressor) and the protective earth wire connected to the "main component" (e.g. SEP).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.1 FRU Overview

Tab. 1 Magnet

FRU (Magnet / Page 12)	Time	Instructions	Final Checks	Tools
Burst disk for quench valve	120 min Additional ramping time	(→ Replacing the Quench Valve Burst Disk / Page 12)	Leakage check	- Torque wrench 0.8 Nm - Allen key 4 mm, 3 mm sockets - Socket wrench - Helium detector - Heat gun
OVC burst disk	45 min	(→ Replacing the OVC Burst Disk / Page 36)	n.a.	- Standard non-magnetic tool kit - Torque wrench 5–20 Nm (material number 08396116) 13 mm socket - Heat gun and paper towelling
Auxiliary Vent Burst Disc	30 min	(→ Replacing the Auxiliary Vent Burst Disk / Page 42)	Leakage check	Non-magnetic tool kit
Pressure switch transducer	60 min	(→ Replacing the Pressure Switch Transducer Assembly / Page 52)	Leakage check	Non-magnetic tool kit Helium detector
Vent assembly	60 min	(→ Replacing the Vent assembly / Page 64)	Leakage check	Heat gun Helium detector

Tab. 2 Refrigerator

FRU (Refrigerator / Page 74)	Time	Instructions	Final checks	Tools
Cold head	400 min (Ramping time is included)	(→ Cold Head Replacement / Page 74)	■ Leakage check ■ Efficiency test	■ Standard tool set ■ Aeroquip open-end wrenches (3 off) (10125650) ■ Cold head release screws (0113021) ■ Cold head balance line (10129376) ■ Torque wrench capable of measuring 4 Nm to 8 Nm. ■ Lint-free paper towel ■ Degreasing cleaner (e.g. petroleum ether or alcohol-based cleaner- sourced locally) ■ Protective clothing (gloves, protective goggles or face protection shield, protective clothing, protective shoes). ■ Hot air blower ■ Apiezon 'N'-grease ■ Magnet power supply 3600 ■ NW16 blanking flange ■ O₂ monitor ■ Hand-held helium leak detector

FRU (Refrigerator / Page 74)	Time	Instructions	Final checks	Tools
Compressor CSW-71	60 min	(→ Replacing the compressor CSW-71 / Page 99)	■ Check helium pressure ■ Leakage check	Screwdriver Open-ended wrench: 13 mm Open-ended wrench: 1"; 1 3/16" Helium leak detector Lifting bars order number 513790 Ground test meter (Fluke 1503, or equivalent)
Compressor F70	60 min	(→ Replacing the Compressor F70 / Page 109)	■ Check helium pressure ■ Leakage check	Screwdriver Open-ended wrench: 13 mm Open-ended wrench: 1"; 1 3/16" Helium leak detector Lifting bars order number 513790 Ground test meter (Fluke 1503, or equivalent)
Replacing the F70 Compressor with Tooling Kit / 11060700	120 min	(→ Replacing the F70 Compressor with Tooling kit / Page 145)	■ Leakage check ■ Check the shield temperature ■ Check helium pressure	■ Transportation lifting kit (material number 11060700, F70 compressor kit). ■ 14 mm and 17 mm open-ended wrench. ■ 17 mm socket. ■ Torque wrench (0–20 Nm) ■ 8 mm Allen (Hex) key. ■ 7 mm socket and ratchet.

Tab. 3 Magnet Supervisory

FRU (Magnet Supervisory Unit / Page 190)	Time	Instructions	Final checks	Tools
MSUP (Avanto/ Espree /Avanto Dot upgrade)	60 min	(→ MSUP for Avanto / Avanto Dot Upgrade / Espree / Page 190)	■ ERDU test	Non-magnetic tool kit ERDU test load
MSUP battery (Avanto/ Espree /Avanto Dot upgrade)	40 min	(→ ERDU battery / Page 232)	n.a.	Non-magnetic tool kit Screwdriver, 10 mm nut, 5 mm Allen key.
MSUP (Avanto/ Espree (VB19)/ Avanto Dot Ex-factory)	60 min	(→ Replacing the Magnet Supervisory Unit (MSUP 2306) / Page 235)	■ ERDU test	Non-magnetic tool kit ERDU test load
MSUP controller (Avanto (VB19)/ Espree /Avanto Dot Ex-factory)	60 min	(→ Replacing the Magnet Supervisory Unit Controller Assembly (MSUP 2306) / Page 256)	■ ERDU test	Non-magnetic tool kit
ERDU battery (Avanto / Espree (VB19)/Avanto Dot Ex-factory)	45 min	(→ Replacing the ERDU battery box / Page 273)	n.a.	Non-magnetic tool kit

Tab. 4 Alarm Box

FRU (Alarm Box / Page 276)	Time	Instructions	Final checks	Tools
Alarm box (Avanto/Avanto Dot upgrade / Espree)	60 min	(→ Alarm box for Avanto/Avanto Dot Upgrade / Espree / Page 276)	- Check LEDs - ERDU test	Non-magnetic tool kit ERDU test load
Alarm box (Avanto / Espree (VB19)/Dot upgrade)	60 min	(→ Alarm Box for Avanto, Espree (VB19)/ Avanto Dot Ex-factory / Page 280)	- Check LEDs - ERDU test	

Tab. 5 Shim power supply

FRU (Shim power supply / Page 294)	Time	Instructions	Final checks	Tools
Shim power supply	60 min	(→ Replacing the Shim Power Supply (SPSU) / Page 294)	- Phantom Shim - Shim Check	Standard tool set

Tab. 6 Shim Filter Unit

FRU (Replacing the Shim Filter Unit / Page 296)	Time	Instructions	Final checks	Tools
Shim Filter Unit	60 min. additional ramping time	(→ Replacing the Shim Filter Unit / Page 296)	- Phantom Shim - Shim Check	Standard tool set

2.1 Replacing the Quench Valve Burst Disk



CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » The magnet must be RAMPED DOWN before attempting this procedure.



CAUTION

Pressurized helium vessel.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » The magnet must be depressurized before attempting this procedure.
- » Personal safety equipment must be worn.



CAUTION

Cold helium gas.

Personal injury may occur.

- » Protective clothing, gloves and eye protection must be worn.



CAUTION

Air ingress.

Air ingress must be avoided. Block open ports with paper towelling to minimize air ingress, but maintain a flow of helium gas.

- » Do not use cloth towels.
- » Blank the NW40 quench valve port, whilst the quench valve is removed.
- » The blank is supplied in the spares kit.

**CAUTION**

If the quench valve or auxiliary vent burst disk breaks:

Failure to observe the following will result in ICE FORMING within the magnet.

- » Replace the graphite, or auxiliary vent bursting disk IMMEDIATELY.
- » If an auxiliary vent burst disk is not available, fit the red blanking plate immediately.
- » Inform Magnet HSC.
- » **DO NOT RAMP THE MAGNET** until the burst disks have been replaced.



This procedure must be completed as a single continuous operation. The system must not be left unattended.

2.1.1 Required time and tools

2.1.1.1 Time

180 min (excluding ramping time)

2.1.1.2 Tools

- Heat gun
- Helium leak detector (part no **10614850**)
- 13mm wrench
- 3mm, 4mm, 6mm allen keys
- Torque wrench
- 3mm and 4mm torque wrench allen keys
- Paper towelling
- Thread lock compound e.g. PermaBond A113 (part no **05834788**)
- **MSUP-Display Unit**

Spare parts

- The **26 psi burst disk** can be:
 - » Part number **10129362 (or)**
 - » Part number **08396470 (is still suitable)***
- NW40 blank red AL anodized (part no 10105764)
- NW40 High pressure seal. (part no 10100171)
- Silicon O-ring (part no 10100140)

- **Mk1**
 - Fibre gasket (4 x 9mm)- quench valve to turret flange (part no 08396538)
 - Fibre gasket (12 x 9mm)- quench valve to quench valve manifold (part no 08396504)
- **Mk2**
 - O-ring - nitrile BS4518 (part no 10100147)

Torque Settings

Tab. 7 Torque Settings

	Nm	ft. lbs
M4 Burst disk cap screws	0.8	0.6
M5 Burst disk/Spring Plate cap screws	5	3.8
M8 Quench Valve Backplate (Mk1 only)	10	7.2
NW25 High pressure Clamp	7	5.3
NW40 High pressure Clamp	7	5.3

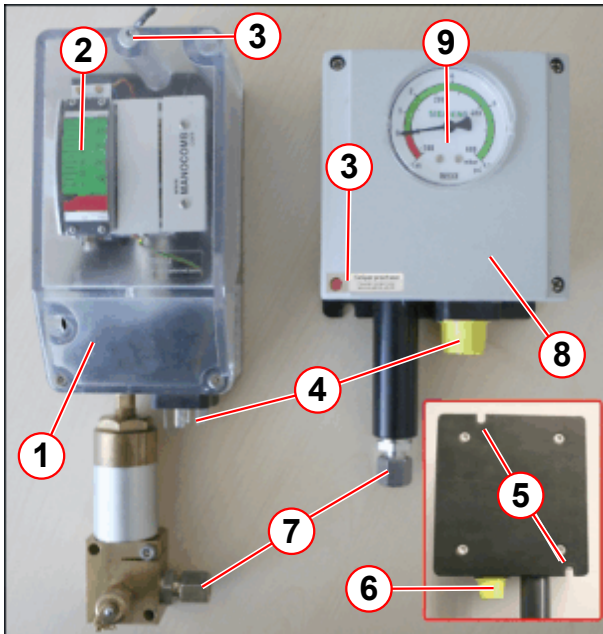
2.1.2 Replacement Procedure

Pressure switch transducer assembly

- The **WIKA** Pressure switch transducer assembly , **replaces** the **Manocomb** pressure switch, when stocks are exhausted.
- The **new** WIKA switch (part number **10615712**) will **replace** the **current unit** (part number **10126452**), when stocks are exhausted.

- The **functionality** of the new switch is **identical**, and the same fixings are used for installation.

Fig. 1: Pressure switch transducers Manocomb and WIKA



- (1) Manocomb Transducer assembly
- (2) Linear gauge
- (3) Tamperproof seal
- (4) Front/rear electrical connections
- (5) Mounting slots (rear view)
- (6) Protection caps
- (7) Pressure line connection
- (8) Wika Transducer assembly (part no. 10126452)
- (9) Dial gauge

Fig. 2: WIKA Pressure switch transducer (part no 10615712)



The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1 Manocomb switch.

2.1.2.1 Pressure heater control

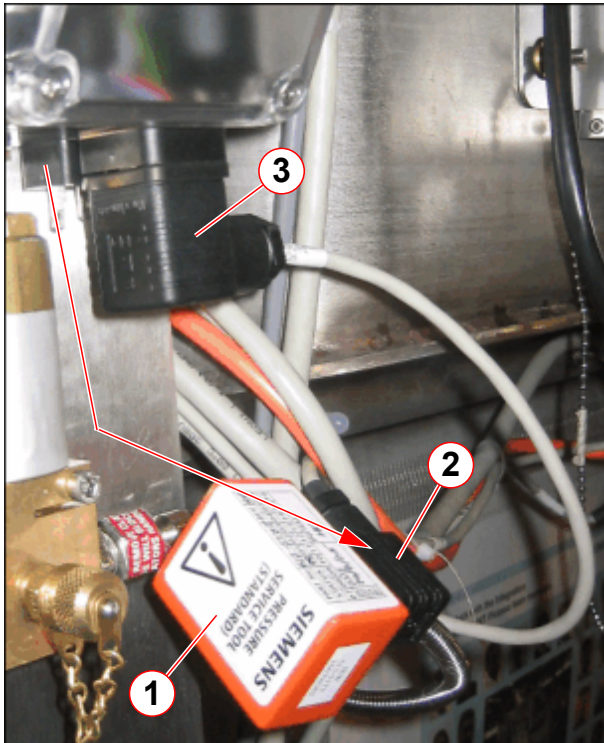
The **Pressure Service Tool** is used to **deactivate** the pressure heater control circuit. The **electrical connectors** are labeled **pressure sensor** (rear), and **pressure switch** (front).

- **Fit the pressure service tool:**
 - **Disconnect** the rear electrical connector on the **pressure switch transducer**.
 - Fit the **pressure service tool** to the connector lead. (→ 2/ Fig. 4 Page 16)
 - The **Alarm Box** will sound **PRESSURE ERROR**.
 - Press the **ACKNOWLEDGE** button to cancel the alarm.
- The **pressure** reading will be set to **~18 psi (A)**, which will **disable the pressure heater**.

To **reactivate** the pressure heater control circuit, re-connect the cable to the rear connector on the pressure switch transducer. Cancel any alarms.

Mk2 Manocomb Pressure switch

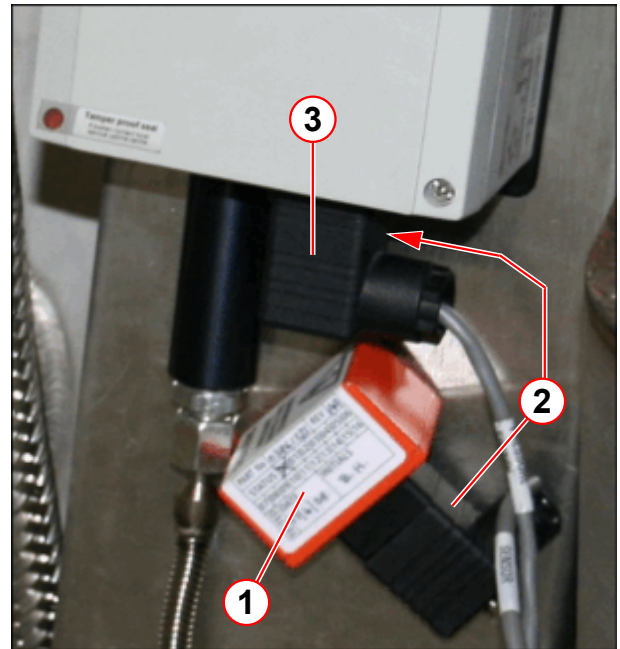
Fig. 3: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

WIKA Pressure switch

Fig. 4: Fit the pressure service tool

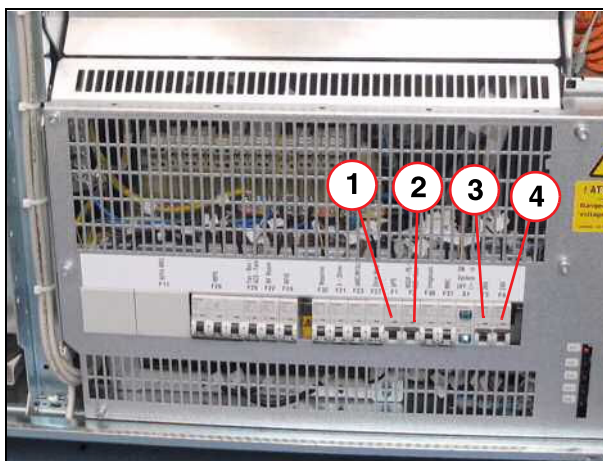


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 5: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service **pressure tool** is **not available**,

- **Switch OFF** the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » **Pressure heater is powered down.**
- The **ERDU circuit** is still functional.
- **Helium level and system temperatures** can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, **only 2 minutes** of power is available.

2.1.2.2 Mk1 Venting

Quench valve removal

! WARNING

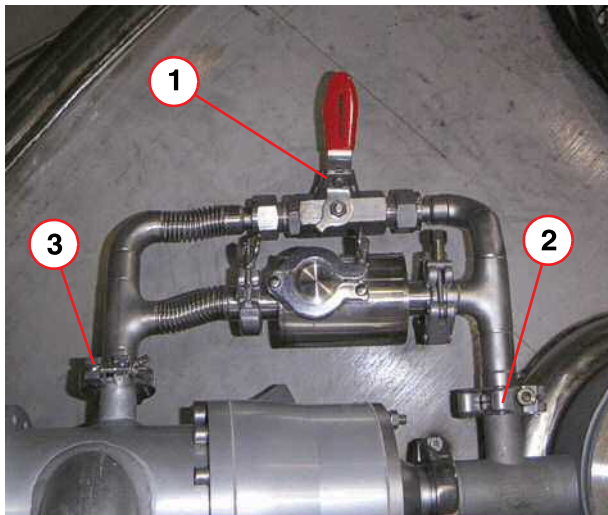
Risk of asphyxiation!!!

The magnet MUST NOT BE AT FIELD during this procedure.

» **Do not remove the quench line when the magnet is at field.**

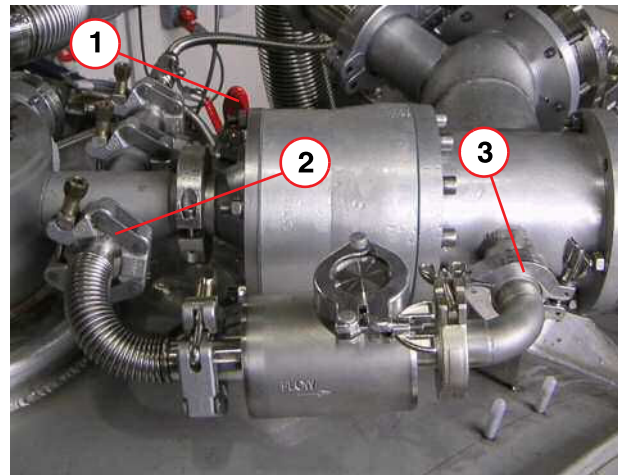
- **Ramp** the magnet to **zero field**.
- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 15).
- **Open** the **magnet bypass valve** to **depressurize** the helium vessel.

Fig. 6: OR105 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 7: OR122 Mk1 Vent valve assembly

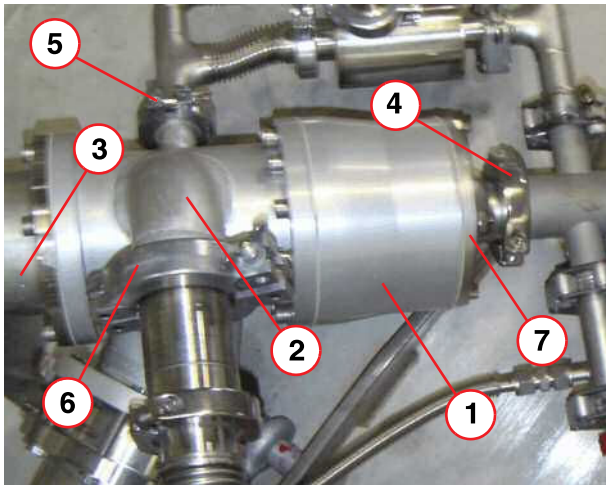


- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

- When the magnet has reached **zero pressure**, **CLOSE** the **magnet bypass valve** to prevent **air ingress**.
- Warm up the **quench elbow** and **turret venting** to room temperature, using a heat gun.

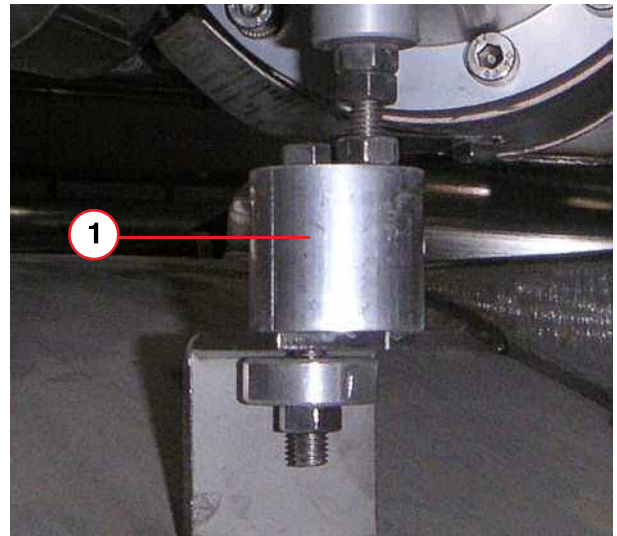
- To access the **quench valve**, first remove the **OVC burst disk**.

Fig. 8: OR105 Mk1 Manifold/Quench Valve



- (1) Quench valve
- (2) Mainfold
- (3) Quench elbow
- (4) NW40 high pressure clamp
- (5) NW25 low pressure clamp
- (6) NW50 low pressure clamp
- (7) Turret flange

Fig. 9: Mk1 Support Bracket



- (1) Support Bracket

- **Disconnect** the **quench elbow**, or straight coupling if fitted. (→ 3/ Fig. 8 Page 18)
- **Slacken** the support **bracket** underneath the **quench valve manifold**. (→ 1/ Fig. 9 Page 18)
- **Remove** the NW25 (low pressure) **clamp** to release the **16psi (A) valve assembly**. (→ 5/ Fig. 8 Page 18)
- **Remove** the NW50 (low pressure) **clamp** to release the **aux vent line**. (→ 6/ Fig. 8 Page 18)
- **Remove** the (12 off) M8 cap **screws** securing the **quench valve manifold** to the quench valve .
- Place the manifold **to one side**, allowing access to the **quench valve**.
- **Re-open** the magnet **bypass valve** (to ensure zero pressure), and **close** again.
- **Mark the orientation** of the quench valve. Then **remove** the 4 off M8 nuts and Nordlock washers that secure the quench valve to the turret flange. (→ 7/ Fig. 8 Page 18)



Remove the 4 off M8 nuts and nordlock washers in a diametric sequence. Undo the nuts slowly and evenly - the valve is spring loaded.

- **Lift the quench valve** from the turret flange.

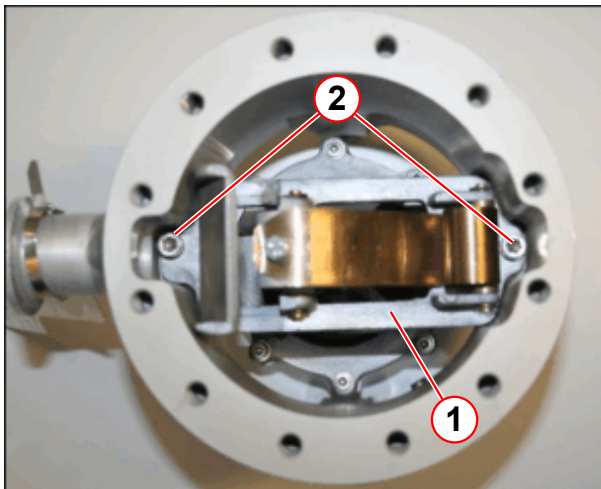


To minimize air ingress, immediately cover the open turret port temporarily, using the NW40 blank.

Use well compacted paper towelling if the blank is not available.

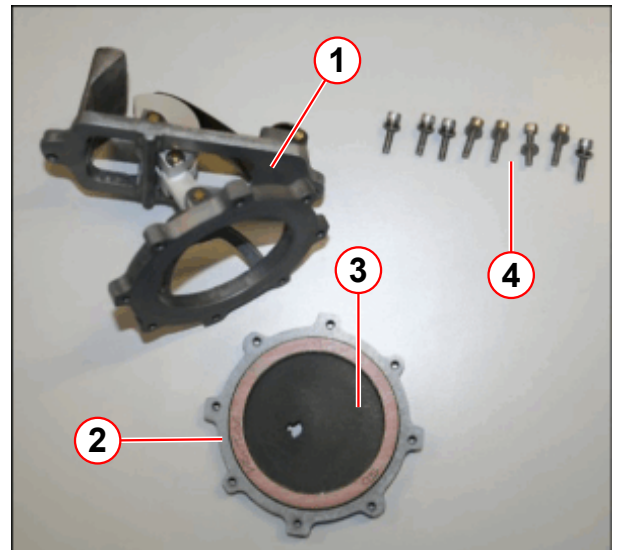
Quench Valve Inspection

Fig. 10: Mk1 Quench valve assembly (low pressure side)



- (1) Burst disc assembly
- (2) M5 cap head screws (2 off)

Fig. 11: Burst disk assembly



- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

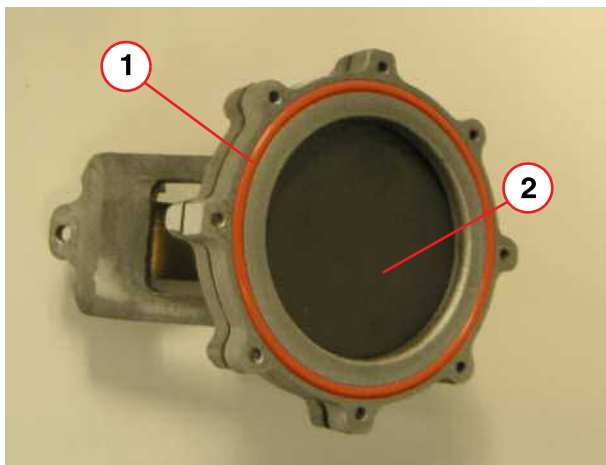
- Unscrew and remove the (2 off) M5 cap head screws and lift out the **burst disk assembly** from the quench valve body.



There is no requirement to remove the spring plate on the Mk1 quench valve.

Burst Disk Inspection

Fig. 12: Quench Valve Bursting Disk Assembly



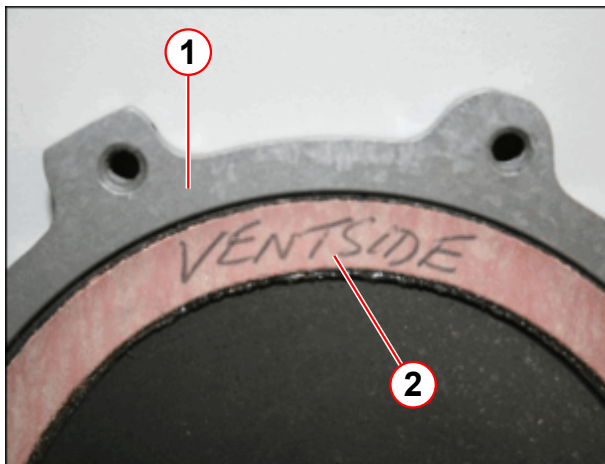
- (1) O-ring
- (2) Burst disk

- **Inspect** the burst disk.
 - » If the burst disk has **ruptured** or is **cracked**, it must be replaced.
- **Inspect** the burst disk **O-ring** (→ Inspecting the burst disk O-Ring / Page 22).

To replace the burst disk:-

- Unscrew and remove the 8 x M4 cap screws (→ 4/ Fig. 11 Page 19).
- Remove the broken burst disk and discard. **Ensure any small fragments are removed from the quench line.**
- Ensure that the new burst **disk** and **clamp faces** are **clean**.

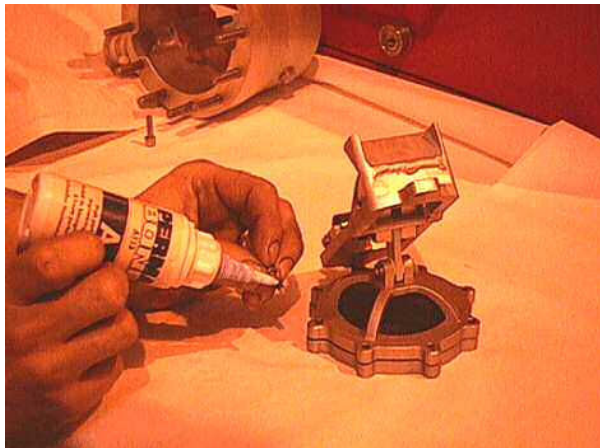
Fig. 13: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

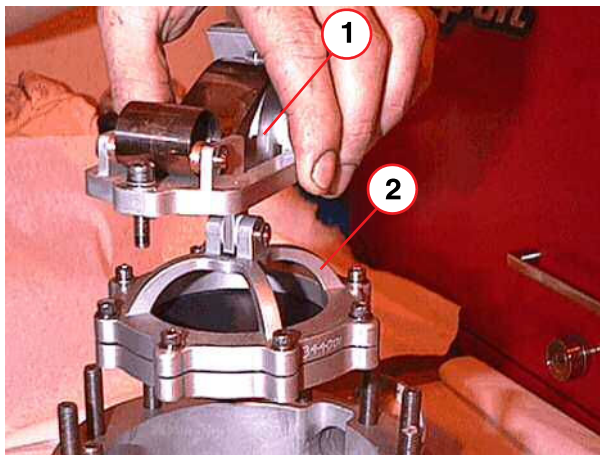
- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 14: Thread locking compound



- **Re-assemble** the burst disk assembly. **DO NOT APPLY GREASE.**
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound and torqued down evenly to 0.8 Nm (0.6 ft. lbs).

Fig. 15: Mk1 Refitting the spring plate

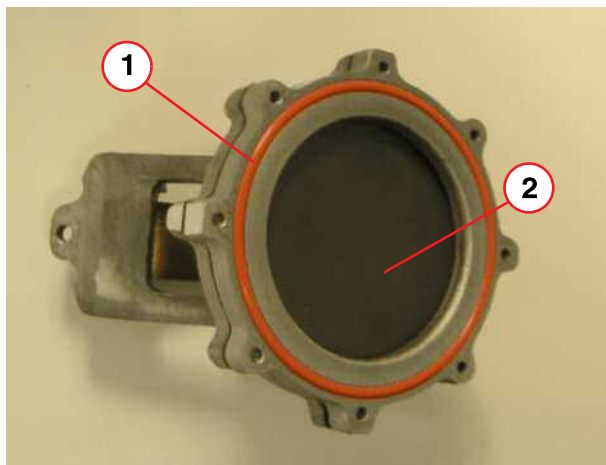


- (1) Spring plate
(2) Burst disc assembly

- **Re-fit** the spring plate/burst disk assembly into the quench valve body with the (2 off) M5 cap screw and 'Nordlock' washers.
- **Torque to 5 Nm (3.8 ft. lbs).**

Inspecting the burst disk O-Ring

Fig. 16: Quench Valve Bursting Disk Assembly



- (1) O-ring
(2) Burst disk

- **Inspect the O-ring** visually for damage.
- If the O-ring needs to be replaced, be very **careful not to scratch** the O-ring **groove**.
- Use a **non-metallic** lever to remove the O-ring.



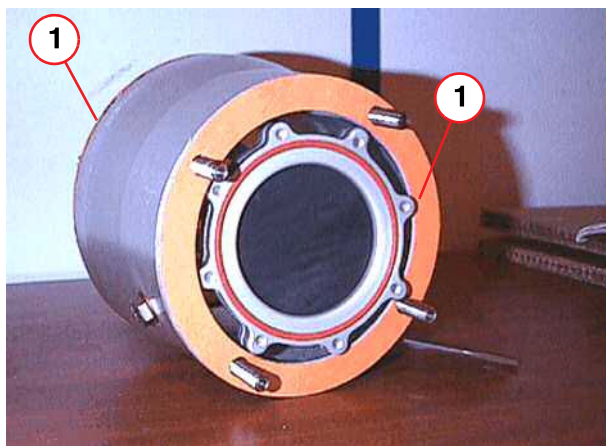
It is recommended that the O-ring is replaced each time the quench valve is removed. **DO NOT** apply grease to the O-ring or groove.

- **Clean** the O-ring **groove** and O-ring before **re-fitting**.
- **Ensure** the O-ring is **seated evenly** all around the **groove**.



If there is any damage to the O-ring (even the smallest of cracks or blemishes) it must be replaced.

Fig. 17: Mk1 Inspecting the fiber gasket



- (1) Fiber gasket

- Inspect the **turret flange fibre gasket**. Replace if damaged.
 - » Ensure the **drain screw** is **oriented correctly**.

Re-fitting the Quench Valve

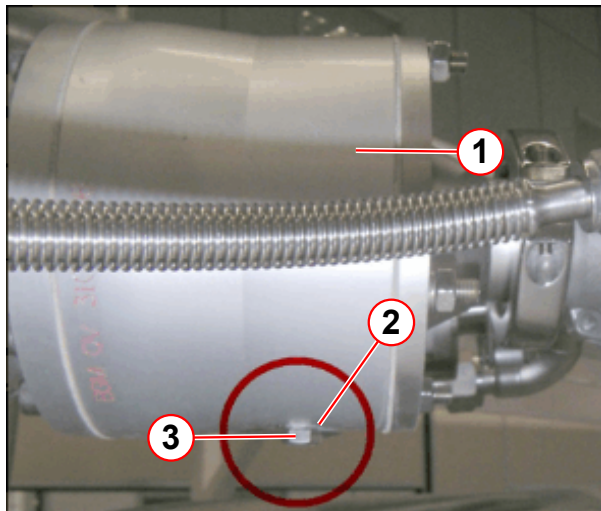
- Check the **gasket surfaces** on both the quench valve and the quench manifold are clean and dry.



If there is any damage to the fibre gaskets (even the smallest of cracks or blemishes), they **MUST** be replaced.

- Remove the **paper towelling** from the turret port. Ensure the surfaces are **clean** and **dry**.
- **Ensure correct orientation.**

Fig. 18: Orientate the drain screw correctly



- (1) Quench valve
- (2) Washer
- (3) Screw

Mk1

- **Immediately** refit the **quench valve assembly** to the turret flange.
- **Check the quench valve orientation.**
- **DO NOT** remove the drain screw and washer.



CAUTION

Water in quench valve!

Risk of quench valve failure and ice.

- » **Check quench valve orientation with drain screw at the bottom.**

- Tighten the 4 off M8 nuts and Nordlock washers evenly, in a diametric pattern, (i.e. tighten the nuts diagonally opposite). **Torque** the nuts to **10 Nm**.



By tightening the 4 off M8 nuts, the internal spring mechanism will be pulled down against the flange face.

- **Re-fit** the **quench manifold** to the **quench valve** with the (12 off) M8 cap screws.
- Use **new** gasket seals if necessary.
- **Re-fit** the **aux vent** line using the **NW50** (low pressure) clamp. (→ 6/Fig. 8 Page 18)
- **Re-fit** the **16psi (A) valve assembly** using the **NW25** (low pressure) clamp and seal. (→ 5/Fig. 8 Page 18)

- **Re-fit** the **OVC burst disk**.
- **Reactivate** the pressure heater (→ Pressure heater control / Page 15).

Final checks



CAUTION

Pressurized helium vessel!

As the cold head/compressor is switched off automatically during the replacement, the system pressure may build to **ABOVE 15.3 psi (A)**.

- » Allow the system pressure to stabilize at **15.3 psi (A)**.

- **Re-pressurize** the helium vessel to **15.3 psi (A)** by setting the pressure heater to **100% duty cycle**, using the MSUP-Display Unit.
- **Check** that the **compressor** re-starts.
- **Re-set** the **pressure heater** to normal operation when the helium vessel is at **15.3 psi (A)**.

Leak check

- **Leak test** the burst disk and O-ring **assembly**.



If the leak persists, the quench valve will need to be disassembled again, and the source of the leak rectified.

Final Tasks

- **Re-connect** the quench elbow. Use **new** seals if necessary.
- **Tighten** the support **bracket** underneath the quench valve **manifold**.
(→ 1/ Fig. 20 Page 25)



IMPORTANT: A burst disk identification tag is fitted to the turret flange. Replace the old identification tag of the broken burst disk with the new burst disk tag.

- **Ramp** the magnet back **to field**.
- **Replace** the **cover panels**.

2.1.2.3 Mk2 Venting



The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1 Man-
o-omb switch.

**WARNING**

Risk of asphyxiation!!!

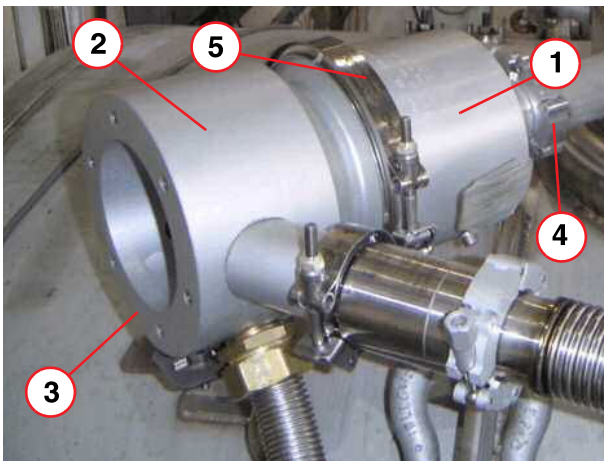
The magnet MUST NOT BE AT FIELD during this procedure.

- » **Do not remove the quench line when the magnet is at field.**

Quench valve removal

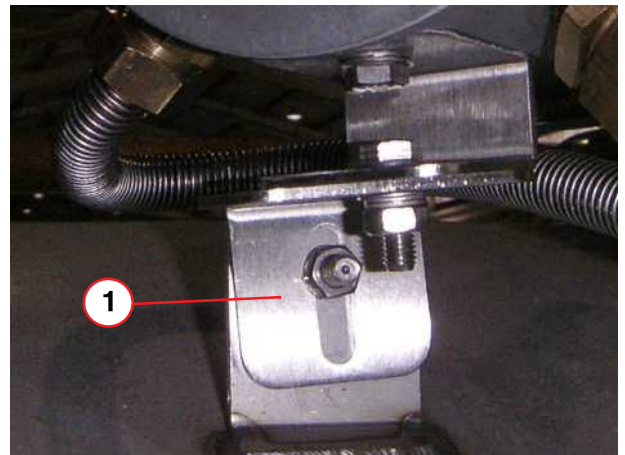
- **Ramp** the magnet to **zero field**.
- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 15).
- **Open** the **magnet bypass valve** to **depressurize** the helium vessel.
- When the magnet has reached **zero pressure**, **CLOSE** the **magnet bypass valve** to prevent air ingress.
- Warm up the **quench elbow** and **turret venting** to room temperature, using a heat gun.

Fig. 19: Mk2 Manifold/Quench Valve



- (1) Quench valve
- (2) Manifold
- (3) Elbow mounting
- (4) NW40 clamp (high pressure)
- (5) V-clamp

Fig. 20: Mk2 Support bracket



- (1) Support bracket

- Disconnect the **quench elbow**, or straight coupling if fitted. (→ 3/Fig. 19 Page 25)
- Slacken the **support bracket** underneath the quench valve **manifold**. (→ 1/Fig. 20 Page 25)
- **Remove** the **V-clamp** securing the quench valve manifold to the quench valve. (→ 5/Fig. 19 Page 25)
- Place the manifold to one side, allowing access to the quench valve.
- **Re-open** the magnet **bypass valve** (to ensure zero pressure), and **close** again.
- **Mark the orientation** of the quench valve, then remove the **NW40 (high pressure) clamp and seal**, securing the quench valve to the turret. (→ 4/Fig. 19 Page 25)

- Lift the quench valve from the turret flange.

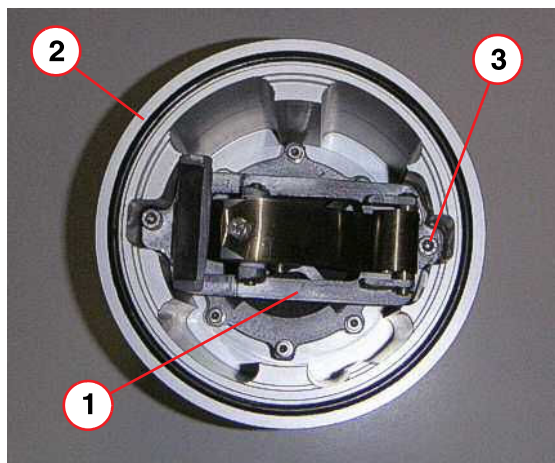


To minimize air ingress, immediately cover the open turret port temporarily, using the NW40 blank.

Use well compacted paper towelling if the blank is not available.

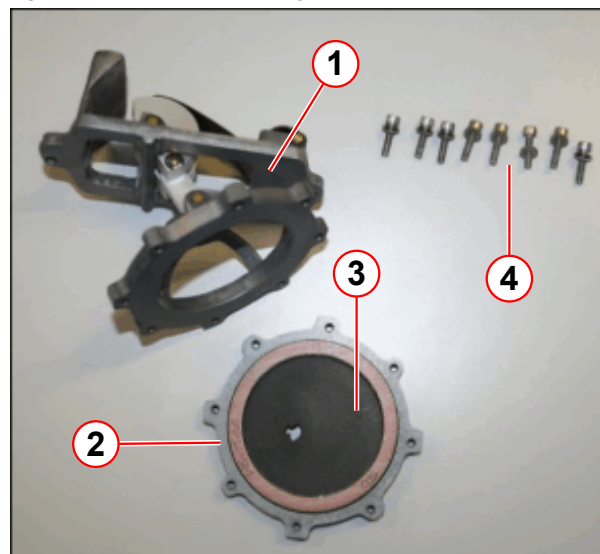
Quench Valve Inspection

Fig. 21: Mk2 Quench valve



- (1) Burst disk assembly
- (2) O-ring
- (3) M5 Cap head screws (2 off)

Fig. 22: Burst disk assembly



- (1) Spring plate assembly
- (2) Disk housing
- (3) Broken disk
- (4) M4 screws (8 off)

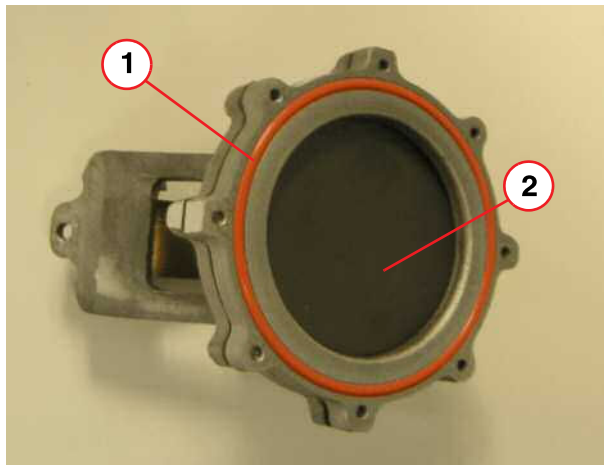
- **Evenly unscrew** and remove the (2 off) M5 cap head screws and **extract the spring plate assembly** from the quench valve body.



The spring exerts significant force on these screws.

Burst disk inspection

Fig. 23: Quench Valve Bursting Disk Assembly



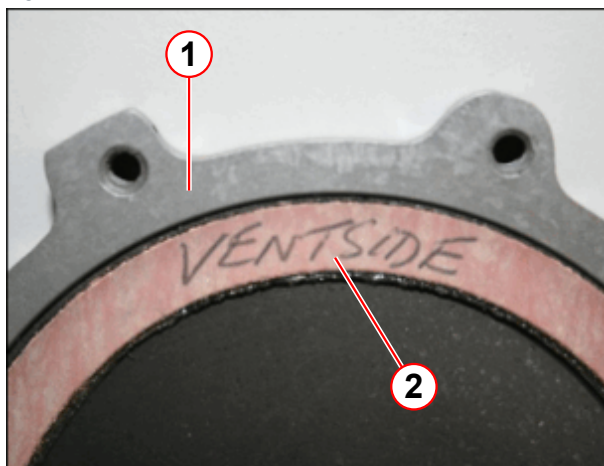
- (1) O-ring
- (2) Burst disk

- **Inspect** the burst disk.
 - » If the burst disk has **ruptured** or is **cracked**, it must be replaced.
- If the burst disk is **intact**, inspect the burst disk **O-ring** (→ Inspecting the O-Ring / Page 29).

To replace the burst disk:-

- Unscrew and remove the 8 x M4 cap screws.
- Remove the broken burst disk and discard. **Ensure any small fragments are removed from the quench line.**
- Ensure that the new burst **disk** and **clamp faces** are **clean**.

Fig. 24: Burst disk fitment



- (1) Burst disk housing
- (2) Burst disk (Vent side uppermost)

- Fit the replacement disk.
- The words '**Vent Side**' (written on the burst disk gasket) must be facing the spring plate assembly

Fig. 25: Thread locking compound



- **Re-assemble** the burst disk assembly. **DO NOT APPLY GREASE.**
- The (8 off) M4 cap screws and plain washers should be locked with threadlock compound and torqued down evenly to **0.8 Nm (0.6 ft. lbs).**

Fig. 26: Mk2 Refitting the spring plate



- (1) Spring plate
(2) Burst disc assembly

- **Re-fit the spring plate/burst disk assembly** into the quench valve body with the (2 off) M5 cap screws and 'Nordlock' washers.

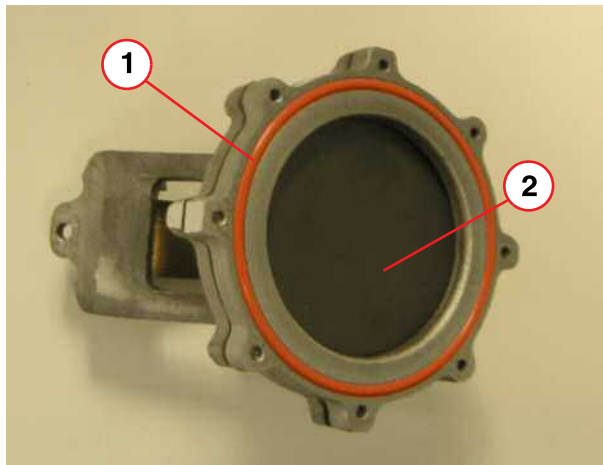


The spring will need to be tensioned before the screws can be fitted. This operation will require two people.

- **Torque** evenly to **5Nm (3.8 ft. lbs).**

Inspecting the O-Ring

Fig. 27: Quench Valve Bursting Disk Assembly



- (1) O-ring
(2) Burst disk

- **Inspect the O-ring** visually for damage.
- If the O-ring needs to be replaced, be very **careful not to scratch** the O-ring **groove**.
- Use a **non-metallic** lever to remove the O-ring.



It is recommended that the O-ring is replaced each time the quench valve is removed. **DO NOT** apply grease to the O-ring or groove.

- **Clean** the O-ring groove and O-ring **before re-fitting**.
- **Ensure** the O-ring is **seated evenly** all around the groove.



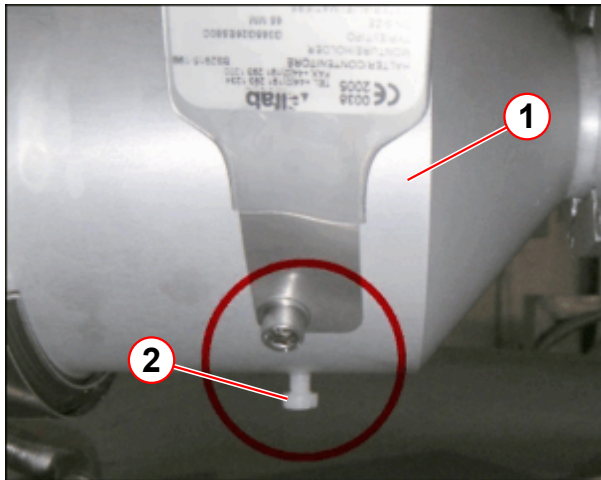
If there is any damage to the O-ring (even the smallest of cracks or blemishes) it must be replaced.

Refitting the Quench Valve

- **Check** the quench valve manifold **O-ring** is in good condition. **Replace** if damaged.
- **Remove** the **NW40** or **paper towelling** from the turret **port**. Ensure the surfaces are **clean** and **dry**.
- **Ensure correct orientation**.

- **Immediately** re-fit the **quench valve assembly** to the turret flange, using the NW40 (high pressure) clamp and a **new** metal seal. (→ 4/ Fig. 19 Page 25)

Fig. 28: Orientate the drain screw correctly



- (1) Quench valve
(2) Drain screw

Mk2

- Check the quench valve orientation.
- Check the drain screw is **REMOVED**.

! CAUTION

Water in quench valve!

Risk of quench valve failure and ice.

- » Check quench valve orientation with drain screw at the bottom.
- » Check the drain screw is removed.

- **Reactivate** the pressure heater (→ Pressure heater control / Page 15).

2.1.2.4 Final checks

! CAUTION

Pressurized helium vessel!

As the cold head/compressor is switched off automatically during the replacement, the system pressure may build to ABOVE 15.3 psi (A).

- » Allow the system pressure to stabilize at 15.3 psi (A).

- **Re-pressurize** the helium vessel to **15.3 psi (A)** by setting the pressure heater to **100% duty cycle**, using the MSUP-Display Unit.
- Check that the **compressor** re-starts.
- **Re-set** the **pressure heater** to normal operation when the helium vessel is at **15.3 psi (A)**.

Leak check

- **Leak test** the burst disk and O-ring **assembly**.



If the leak persists, the quench valve will need to be disassembled again, and the source of the leak rectified.

Final Tasks

- **Re-fit** the quench valve **manifold** to the quench **valve** using the **V-clamp**.
(→ 5/Fig. 19 Page 25)
- **Re-connect** the quench elbow. Use **new** seals if necessary.
- **Tighten** the support **bracket** underneath the quench valve **manifold**.
(→ 1/Fig. 20 Page 25)



IMPORTANT: A burst disk identification tag is fitted to the turret flange. Replace the old identification tag of the broken burst disk with the new burst disk tag.

- **Ramp** the magnet back **to field**.
- **Replace** the **cover panels**.

2.1.3 Replacing the High Altitude Bursting Disk (Mobile System)



WARNING

Risk of asphyxiation!!!

The magnet MUST NOT BE AT FIELD during this procedure.

- » **Do not remove the quench line when the magnet is at field.**



It is not necessary to de-pressurize the system to fit the High Altitude bursting disk kit.

2.1.3.1 Required time and tools

Time

40 min (excluding ramping time)

Tools

- 13mm spanner/wrench

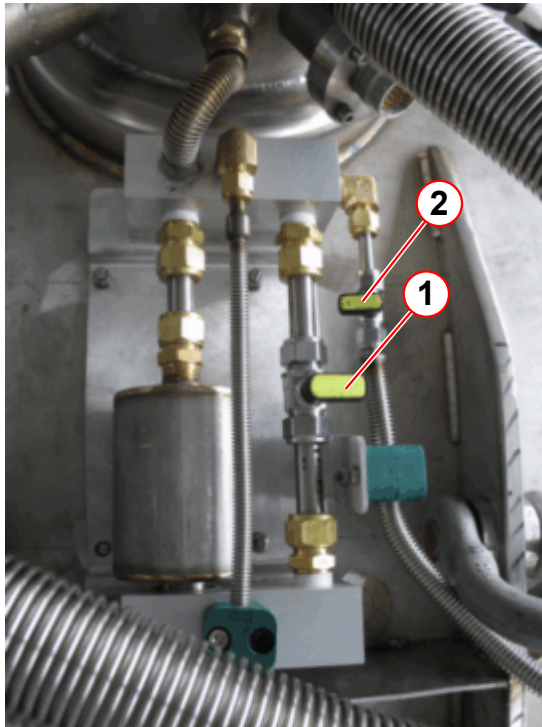
Spare parts

- High Altitude Burst Disk Kit (part no 10109300)

2.1.3.2 Replacement Procedure

High Altitude Bursting Disk Removal

Fig. 29: MK2 Venting Configuration



- (1) Magnet Bypass Valve
(2) Cold Head Bypass Valve

- Ensure that the magnet bypass valve is closed. (→ 1/Fig. 29 Page 32)
- The cold head bypass valve must be closed when the cold head is operational. (→ 2/Fig. 29 Page 32)

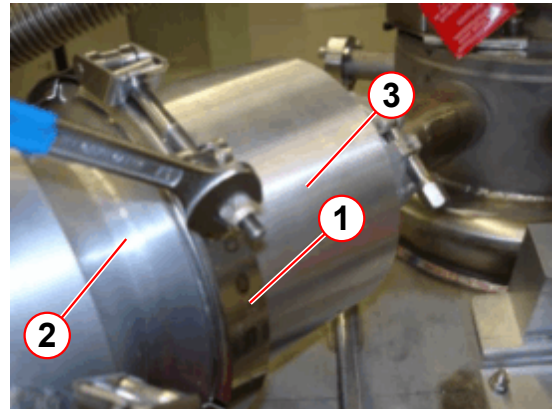
- Slacken the clamp on the manifold support bracket. (→ 1/Fig. 36 Page 35)
- Slacken the nut and release the bolt on the V clamp securing the joint between the quench valve outlet and the manifold. (→ 1/Fig. 31 Page 33)

Fig. 30: Manifold Support Bracket



(1) Support Clamp

Fig. 31: Manifold V Clamp



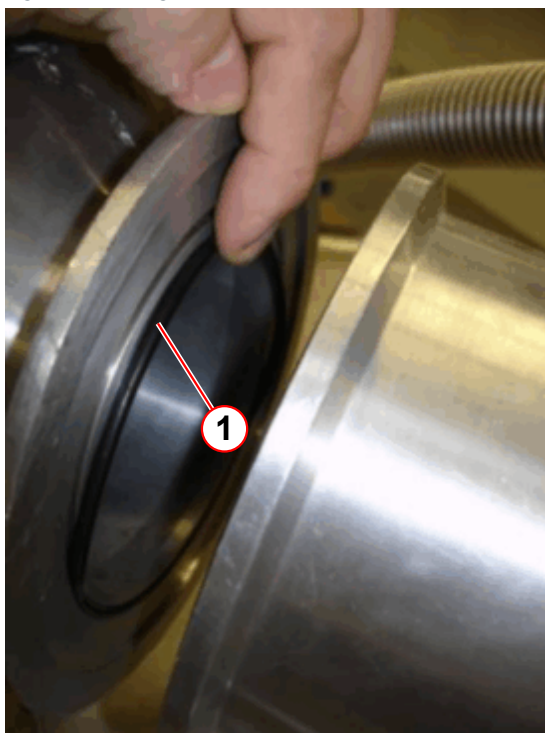
(1) V Clamp
(2) Manifold
(3) Quench Valve

- Remove the O-ring in the manifold flange.
- Remove the broken burst disk.
- Clean the flange mating surfaces.

2.1.3.3 Refitting the High Altitude Bursting Disk

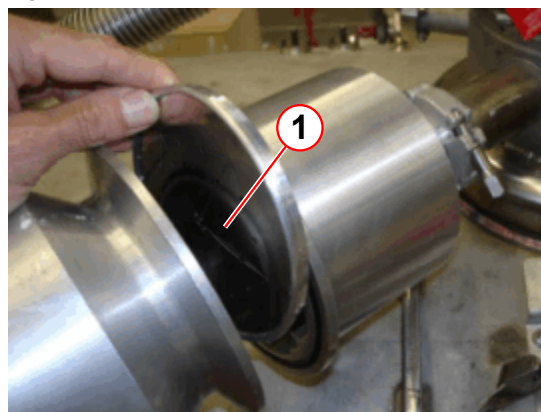
- Fit the new O-ring seal provided into the O-ring groove in the manifold flange. (→ 1/Fig. 32 Page 34)
Ensure that the O-ring and groove are clean.
- Ensure that the flange mating surfaces are clean then locate the bursting disk in the recess in the quench valve flange so that the words VENT SIDE on the face of the disk face outwards. (→ 1/Fig. 33 Page 34)
- Take care not to touch the central portion of the disk as it is very delicate and can be damaged easily. (→ 1/Fig. 34 Page 34)

Fig. 32: O Ring Fitment



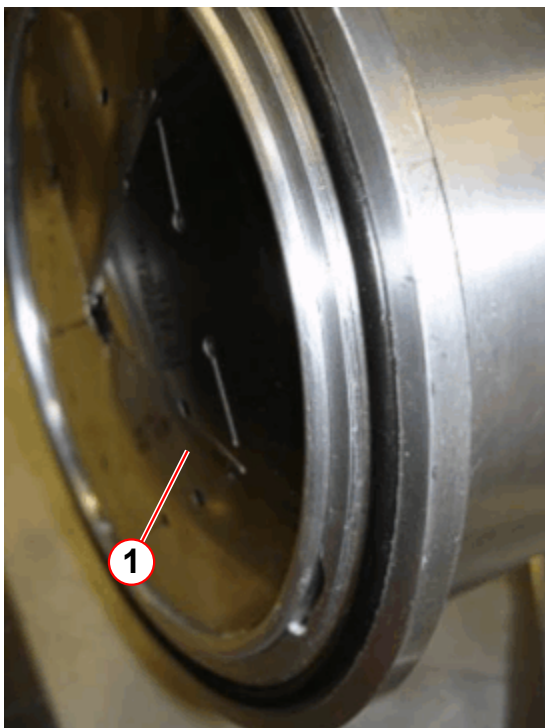
(1) O Ring and Groove

Fig. 33: Burst Disc Fitment



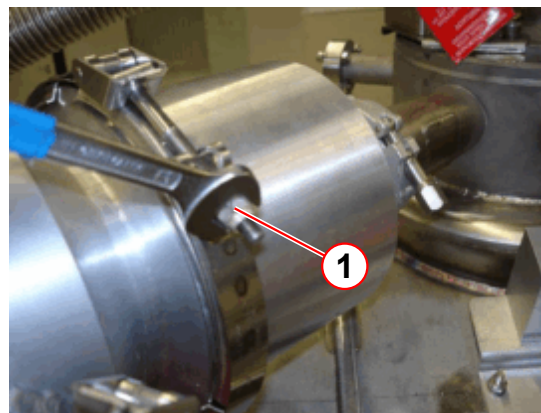
(1) Burst Disc Words "VENT SIDE"

Fig. 34: Burst Disc



(1) "VENT SIDE"

Fig. 35: V Clamp Fitment



(1) Self Locking Nut

- Close the joint and ensure that the disk is still centered in the quench valve flange and refit the V clamp.

- Tighten the nut until there is a sharp increase in resistance indicating that the self locking nut is fully home. (→ 1/Fig. 35 Page 34)

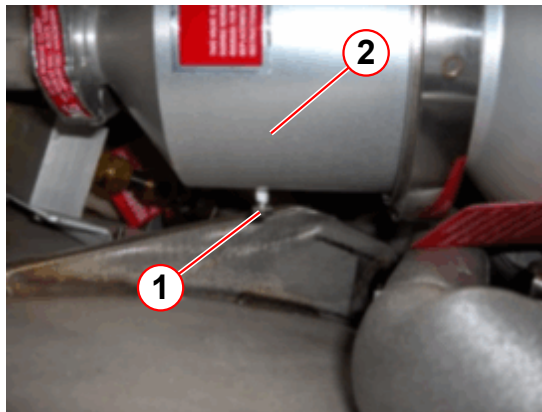
Fig. 36: Manifold Support Bracket



(1) M8 Nuts & Bolts

- Retighten the bolt and M6 nut on the manifold support bracket. (→ 1/Fig. 37 Page 35)

Fig. 37: Drain Screw



(1) Drain Screw
(2) Quench Valve

- Check that the drain screw is fitted in the quench valve body. (→ 1/Fig. 37 Page 35)
- If not, fit the drain screw supplied in the high altitude kit.



For the high altitude venting to be effective the M6 drain screw and washer must be fitted into the quench valve body.

2.1.3.4 Final Work Checks

- Re-check that all the newly made joints are secure.
- Check that the magnet has remained at its normal operating pressure.

2.2 Replacing the OVC Burst Disk



This operation can be performed with the magnet at field.



CAUTION

Risk of asphyxiation

Cold helium gas

If the magnet quenches, the RF-room will be filled with cold helium gas.

Failure to adhere to safety precautions may result in serious bodily injury.

- » Protective clothing, gloves and eye protection have to be worn.
- » Unauthorized personnel must be excluded from the imaging room.
- » Ensure there is a clear path to exit the imaging suite.



CAUTION

Cold helium gas.

Personal injury may occur.

- » Protective clothing, gloves and eye protection must be worn.

2.2.1 Required time and tools

Time required

- 120 min (excluding ramping time)

Tools required

- Standard non-magnetic tool kit.
- Torque wrench 5–20 Nm (material number 08396116).
- 13 mm socket.
- Heat gun and paper towelling.

2.2.2 Spare parts

- Mk1 OVC (0.4) bar burst disk (material no 08396207)
- Mk2 OVC (0.4) bar burst disk (material no 10103214 or 10121914¹)

1. Change of manufacturer

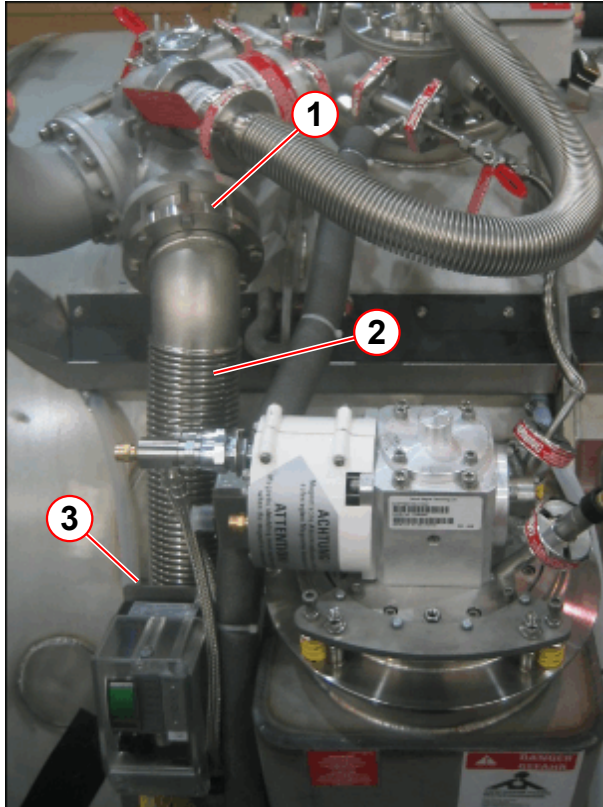
2.2.3 Replacement Procedure

The OVC burst disk is designed to remain intact after a magnet quench. It is fitted to stop helium gas reaching the seals on the DOP (drop off port) port during a quench or helium filling.

2.2.3.1 Mk1 Venting

- Remove the appropriate system covers to gain access to the magnet venting.

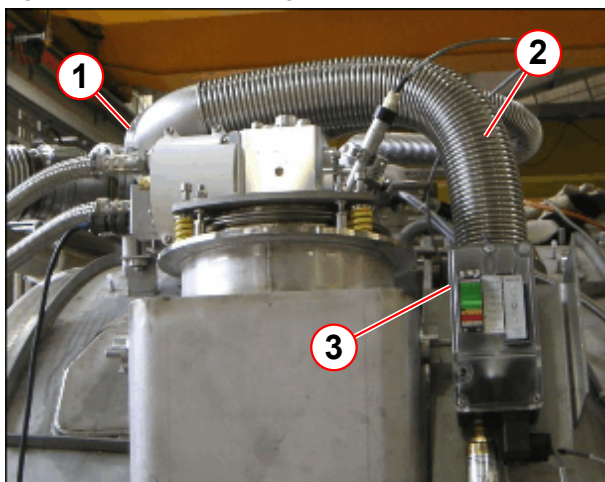
Fig. 38: OR105 Mk1 Venting



- (1) OVC burst disk
- (2) Flexible vent pipe
- (3) Connection to OVC port (behind gauge)

- Take care when removing the OR105 flexible vent pipe. It may be under tension.
- To assist with the burst disk removal the lower flexible vent connection to the OVC port can be taken off.

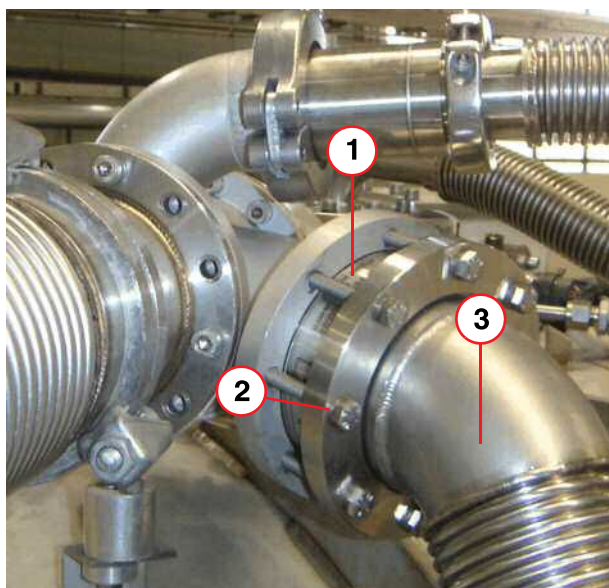
Fig. 39: OR122 Mk venting



- (1) OVC burst disk
- (2) Flexible vent pipe
- (3) OVC port location (behind gauge)

- The OR122 venting is longer than the OR105. There is less tension in the flexible pipe.
- When screws have been removed the flexible vent pipe can be moved upwards and away from the burst disk.

Fig. 40: OR105 Mk1 OVC Venting

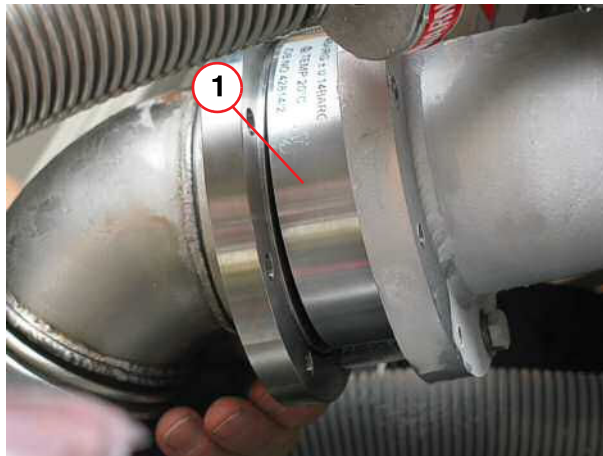


- (1) OVC Burst Disk
- (2) Screw
- (3) OVC Vent tube

To remove the OVC burst disk:

- Remove the 8 x M8 screws in a diametric pattern.
- Keep all screws and washers for later fitting the new burst disk.

Fig. 41: OVC burst disk



(1) OVC burst disk

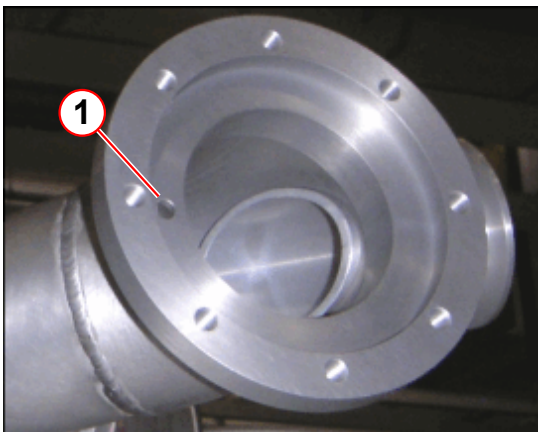
- Move the flexible vent pipe away from the burst disk.
- Lift out the broken OVC burst disk and the 2 x gasket seals.



Do not touch the metal membrane; it is very fragile

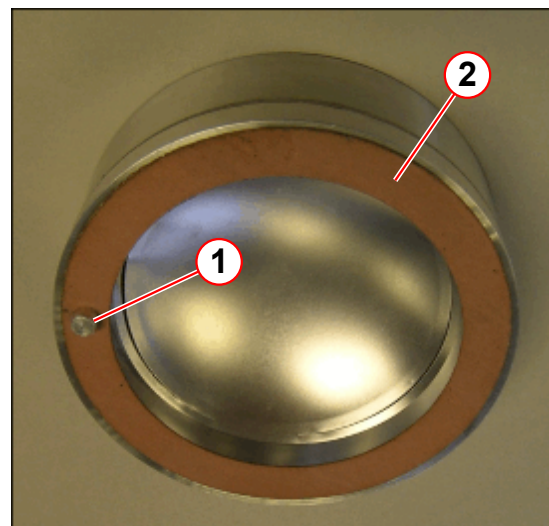
- Check that the mating faces of the flanges to the burst disk are clean and dry.
- Check the 2 x gasket seal faces are clean and undamaged. Replace if necessary.
- Check the position and alignment for the locating pin on the new burst disk.
 - » The locating pin must fit into the mating hole in the turret flange.

Fig. 42: Turret assembly flange



(1) Locating hole

Fig. 43: Burst disk and gasket



(1) Locating pin
(2) Gasket seal x 2

- The turret flange should be warm and dry before fitting the new burst disk.
- Carefully fit the new OVC burst disk. The raised pin must be aligned with the hole in the mating flange.

- Fit the 8 x M8 screws (bottom screw first). Continue fitting the screws in a diametric pattern.
- Before fully tightening the screws check that the burst disk is fitting flush between the 2 flanges.
- Torque the 8 x M8 screws to 15 Nm.
- Tighten the flexible pipe connection at the lower OVC port if removed at the start, (OR105)
- Replace the appropriate system cover.

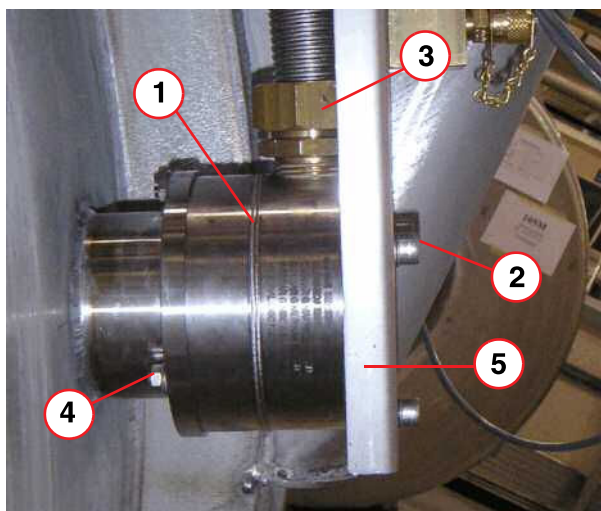
2.2.3.2 Mk2 Venting



The Mk2 OVC burst disk replacement procedure only applies to those magnets fitted with the OVC port and venting, (→ Fig. 44 Page 40).

- **Remove** the appropriate **cover panels**.

Fig. 44: Mk2 OVC Burst Disk Assy



- (1) OVC burst disk
- (2) Cap head screw (3 off)
- (3) OVC vent sealing nut
- (4) Nut (3 off)
- (5) Pressure gauge plate

- **Remove** the (3 off) cap head **screws** from the pressure gauge **plate**.

- **Lift** the pressure gauge and plate clear
- **Disconnect** the **vent tube** by undoing the brass sealing nut.
- **Remove** the (3 off) **nuts** from the rear of the burst disk OVC retaining flange. (→ 4/ Fig. 44 Page 40)
- **CAREFULLY lift** out the bursting disk.



Do not touch the metal membrane; it is very fragile.

- **Fit** the new **bursting disk** to the OVC retaining flange with the (3 off) nuts.
- **Re-connect** the **vent tube** to the burst disk housing.

- **Re-fit** the pressure gauge **plate** with the (3 off) screws. (→ 2/Fig. 44 Page 40)
- **Refit** the appropriate **cover panels**.

2.3 Replacing the Auxiliary Vent Burst Disk

CAUTION

Risk of asphyxiation.

Failure to adhere to safety precautions may result in serious bodily injury.

- » The magnet must be RAMPED DOWN before attempting this procedure.

WARNING

Risk of asphyxiation.

Failure to observe the following may result in dizziness and loss of consciousness.

- » Do not vent helium gas directly into the magnet room.
- » Personal oxygen monitors must be worn if a fixed room monitor has not been installed.

CAUTION

Pressurized helium vessel.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » The magnet must be depressurized before attempting this procedure.
- » Personal safety equipment must be worn.

CAUTION

If the quench valve or auxiliary vent burst disk breaks:

Failure to observe the following will result in ICE FORMING within the magnet.

- » Replace the graphite, or auxiliary vent bursting disk IMMEDIATELY.
- » If an auxiliary vent burst disk is not available, fit the red blanking plate immediately.
- » Inform Magnet HSC.
- » DO NOT RAMP THE MAGNET until the burst disks have been replaced.

**CAUTION****RISK OF ICING!!**

Air ingress will cause INTERNAL ICING OF THE TURRET.

- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk/O-rings IMMEDIATELY.
- » Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.
- » If ice is suspected contact Magnet HSC immediately.

**CAUTION**

Cold helium gas.

Personal injury may occur.

- » Protective clothing, gloves and eye protection must be worn.

2.3.1 Required time and tools

2.3.1.1 Time

60 min (not including ramping)

2.3.1.2 Tools

- Helium leak detector (part no **10614850**)
- Standard (non magnetic) tool kit
- **MSUP-Display Unit**

2.3.1.3 Spare Parts

Mk1 & Mk2

- Aux vent (1.8) bar metal burst disk (part no **10103133** or **10121913**¹)
- NW40 (high pressure) metal seal (part no **10100171**)

Mk1

- NW50 (low pressure) O-ring seal (part no **08396512**)

Mk2

- O-ring 0 BS4518 (part no **10100151**)

1. Change of manufacturer

2.3.2 Replacement procedure

**CAUTION****RISK OF ICING!!**

Air ingress will cause INTERNAL ICING OF THE TURRET.

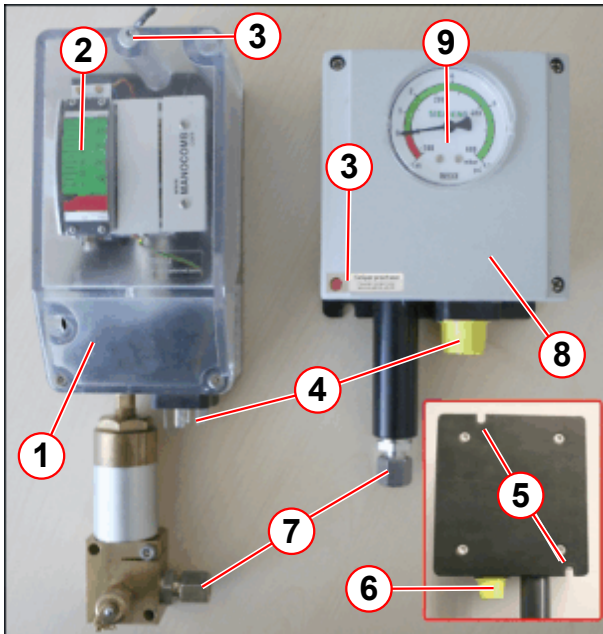
- » Replacement of burst disk(s) must be performed by a qualified CSE.
- » Replace broken or damaged burst disk/O-rings IMMEDIATELY.
- » Open the bypass valve to check that there is a flow of gas. If there is no gas flow, there may be an ice blockage. Close the bypass valve.
- » If ice is suspected contact Magnet HSC immediately.

Pressure switch transducer assembly

- The **WIKA** Pressure switch transducer assembly , **replaces** the **Manocomb** pressure switch, when stocks are exhausted.
- The **new** WIKA switch (part number **10615712**) will **replace** the **current unit** (part number **10126452**), when stocks are exhausted.

- The **functionality** of the new switch is **identical**, and the same fixings are used for installation.

Fig. 45: Pressure switch transducers Manocomb and WIKA



- (1) Manocomb Transducer assembly
- (2) Linear gauge
- (3) Tamperproof seal
- (4) Front/rear electrical connections
- (5) Mounting slots (rear view)
- (6) Protection caps
- (7) Pressure line connection
- (8) WIKA Transducer assy (part no. 10126452)
- (9) Dial gauge

Fig. 46: WIKA Pressure switch transducer (part no 10615712)



2.3.2.1 Pressure heater control

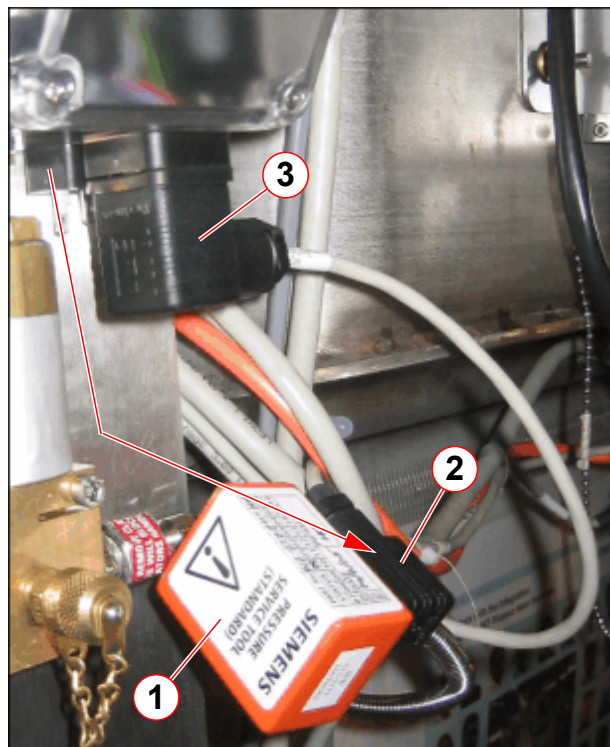
The **Pressure Service Tool** is used to **deactivate** the pressure heater control circuit. The **electrical connectors** are labeled **pressure sensor** (rear), and **pressure switch** (front).

- **Fit the pressure service tool:**
 - **Disconnect** the rear electrical connector on the **pressure switch transducer**.
 - Fit the **pressure service tool** to the connector lead. (→ 2/ Fig. 48 Page 46)
 - The **Alarm Box** will sound **PRESSURE ERROR**.
 - Press the **ACKNOWLEDGE** button to cancel the alarm.
- The **pressure** reading will be set to **~18 psi (A)**, which will **disable the pressure heater**.

To **reactivate** the pressure heater control circuit, re-connect the cable to the rear connector on the pressure switch transducer. Cancel any alarms.

Mk2 Manocomb Pressure switch

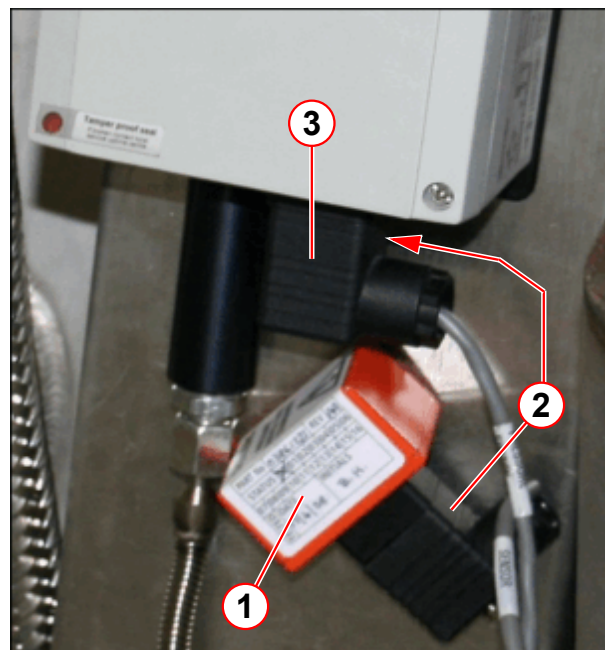
Fig. 47: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

WIKA Pressure switch

Fig. 48: Fit the pressure service tool

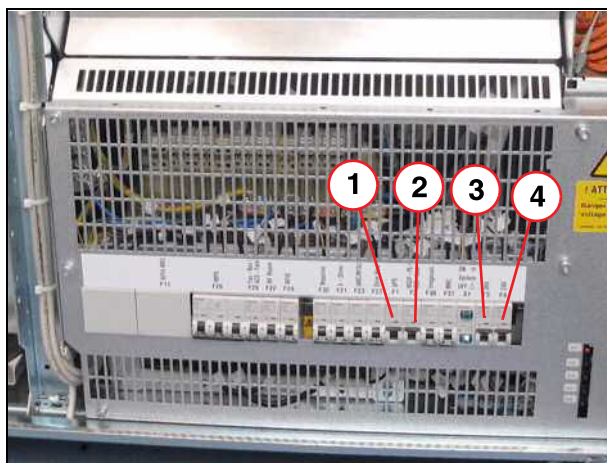


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 49: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service **pressure tool** is **not available**,

- **Switch OFF** the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » **Pressure heater is powered down.**
- The **ERDU circuit** is still functional.
- **Helium level** and **system temperatures** can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, **only 2 minutes** of power is available.

2.3.2.2 Auxiliary burst disk replacement

WARNING

Risk of asphyxiation!!!

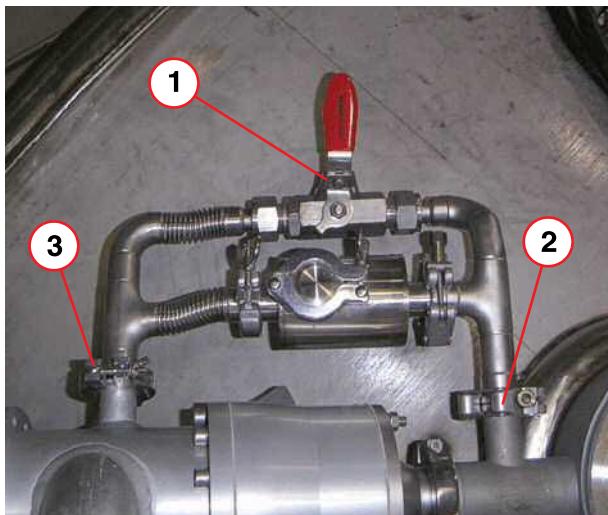
The magnet MUST NOT BE AT FIELD during this procedure.

» **Do not remove the quench line when the magnet is at field.**

Mk1 and Mk2 Venting

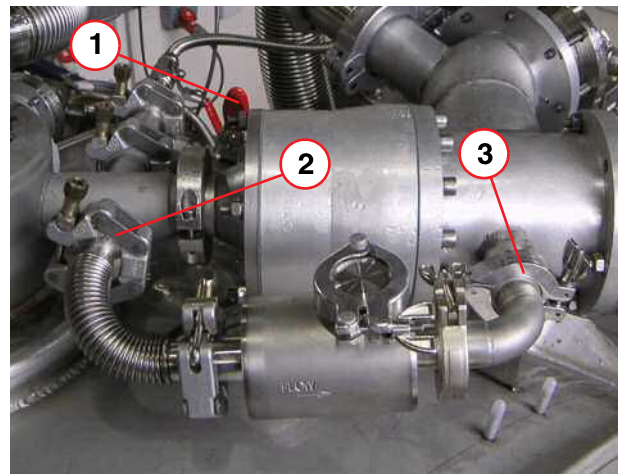
- **Ramp** the magnet to **zero field**.
- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 45).
- **Open** the **magnet bypass valve** to **depressurize** the helium vessel.
- When the magnet has reached **zero pressure**, **CLOSE** the bypass valve.
- Warm the **venting** to room temperature, using a heat gun.

Fig. 50: OR105 Mk1 Vent valve assembly



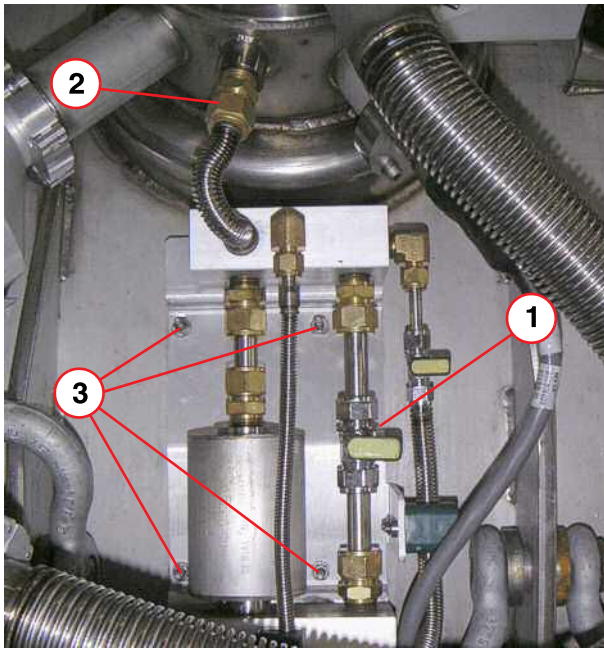
- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 51: OR122 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 52: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

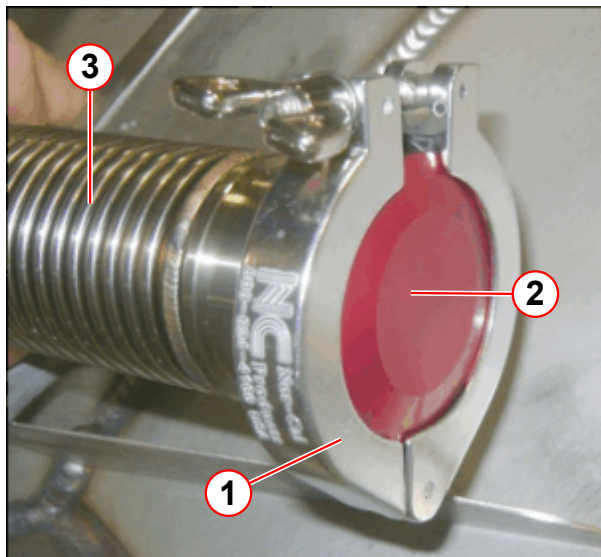
- **Visually inspect** the new aux vent burst disk **metal membrane** for damage.



Do not touch the metal membrane; it is very fragile.

- Ensure the **sealing surfaces** are **clean** and free of damage.
- Ensure the **NW40** (high pressure) seal surfaces are **clean**.

Fig. 53: Aux vent - fitting the NW40 blanking plate



- (1) NW40 high-load clamp
- (2) NW40 red anodized blanking plate
- (3) Aux vent line

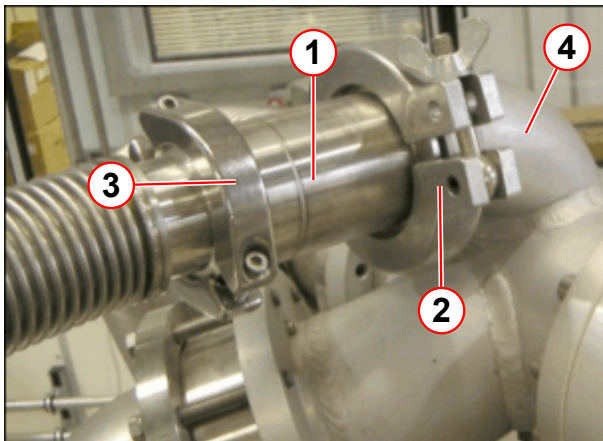
If damaged in any way:

- Undo and remove the **NW40 high pressure clamp**.
- **Remove** the broken burst disk and metal seal.
- Fit the **red NW40 blanking plate** in place of the burst disk.
- **Refit** the NW40 high pressure clamp.
 - » Contact **Magnet HSC** immediately.
 - » Order a replacement aux bursting disk.



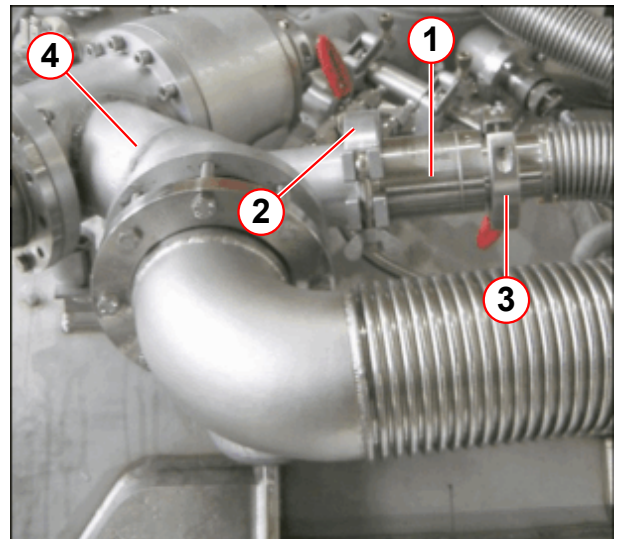
Do NOT ramp the magnet with the NW40 blanking plate fitted!
The magnet can only be ramped when a new auxiliary burst disk has been fitted.

Fig. 54: OR105 Mk1 Auxiliary burst disk assembly



- (1) Aux burst disk assy
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

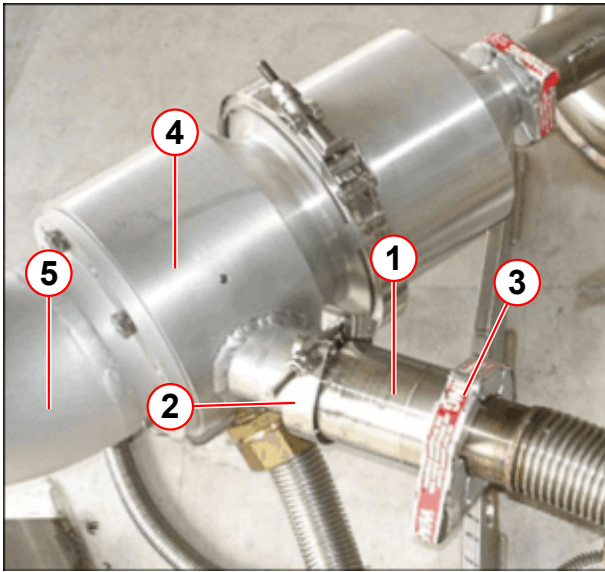
Fig. 55: OR122 Mk1 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) NW50 low load clamp (low pressure)
- (3) NW40 high load clamp (high pressure)
- (4) Low pressure manifold

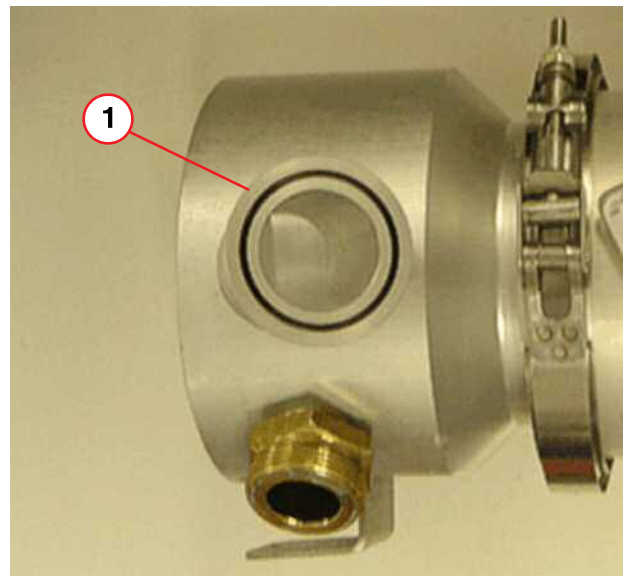
- **For Mk1 Venting** - Undo and remove the **NW50 clamp** (→ 2/ Fig. 54 Page 49) on the **low pressure** side of the aux burst disk.

Fig. 56: Mk2 Auxiliary burst disk assembly



- (1) Aux burst disk
- (2) V-clamp (low pressure)
- (3) NW40 clamp (high pressure)
- (4) Manifold
- (5) Quench elbow

Fig. 57: Mk2 Auxiliary vent interface



- (1) O-ring seal for aux vent

- **For Mk2 Venting** - Undo and remove the **hinged V-clamp** (→ 2/ Fig. 56 Page 50) from the **low pressure** side of the aux burst disk.
- **Undo** and **remove** the **NW40** clamp on the **high pressure** side of the aux burst disk.
- **Remove** the old **burst disk** and metal **seal** (or the **red anodized blanking plate**).
 - **Note:** The **blanking plate** will be present if the aux burst disk has previously ruptured.



Do not touch the metal membrane; it is very fragile.

- **Fit** the new burst disk and **new** high pressure seal **immediately** (on the **high pressure** side), and **re-fit the clamp**. Torque to **7Nm**. (→ Tab. 7 Page 14)
- **Do NOT** leave the magnet venting through the flexible tube.



DO NOT overtighten the NW40 high pressure clamp (→ Tab. 7 Page 14).

Check the O-ring

- **Mk1** - Check that the **NW50 O-ring seal** is undamaged. **Replace if required**.
- **Mk2** - Ensure the **O-ring seal** is undamaged and that the surface is clean. **Replace if required**.

Mk1 and Mk2 Venting:

- **Reactivate** the pressure heater. (→ Pressure heater control / Page 45).

- **OR switch ON** circuit breaker **F3** in the **ACC**.

2.3.2.3 Leak Check

- Allow approx 20 minutes for the helium vessel to **re-pressurize** to **15.3 psi (A)**.
- **Check for leaks** through the metal **membrane**.
- **Re-fit** the burst disk on the **low pressure** side:-
 - **Mk1 Venting** - using the **NW40** clamp
 - **Mk2 Venting** - using the hinged **V-clamp** and seal
- **Check for leaks** around the **clamps**.

2.3.3 Final checks



CAUTION

Pressurized helium vessel!

As the cold head/compressor is switched off automatically during the replacement, the system pressure may build to **ABOVE 15.3 psi (A)**.

» Allow the system pressure to stabilize at **15.3 psi (A)**.

- **Re-pressurize** the helium vessel to **15.3 psi (A)** by setting the pressure heater to **100% duty cycle**, using the MSUP-Display Unit.
- **Check** that the **compressor** re-starts.
- **Re-set** the **pressure heater** to normal operation when the helium vessel is at **15.3 psi (A)**.
- **Ramp** the magnet back to **field**.
- **Replace** the **cover panels**.

2.4 Replacing the Pressure Switch Transducer Assembly



This operation can be performed with the magnet at field.



WARNING

Risk of asphyxiation.

Failure to observe the following may result in dizziness and loss of consciousness.

- » Do not vent helium gas directly into the magnet room.
- » Personal oxygen monitors must be worn if a fixed room monitor has not been installed.



CAUTION

Pressurized helium vessel.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » The magnet must be depressurized before attempting this procedure.
- » Personal safety equipment must be worn.



CAUTION

Cold helium gas.

Personal injury may occur.

- » Protective clothing, gloves and eye protection must be worn.

2.4.1 Required time and tools

2.4.1.1 Time

60 min

2.4.1.2 Tools

- Helium leak detector (part no **10614850**)
- Standard non-magnetic tool kit
- Pressure service tool (part no **03861221**)
- **MSUP-Display Unit**

Spare parts

- Pressure Switch Transducer assembly:-
 - **Mk1** (part no **08396199**)
 - **Mk2** (part no **10100411**)
 - **WIKA** (part no **10126452**)
 - **WIKA (new)** (part no **10615712**)



The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1. Contact Magnet HSC for assistance.

- NW25 (high pressure) metal seal - **Mk1** (part no **01564025**)
- Compression Ferrule - **Mk2 and WIKA** (part no **10113289**)

2.4.2 Replacement procedure

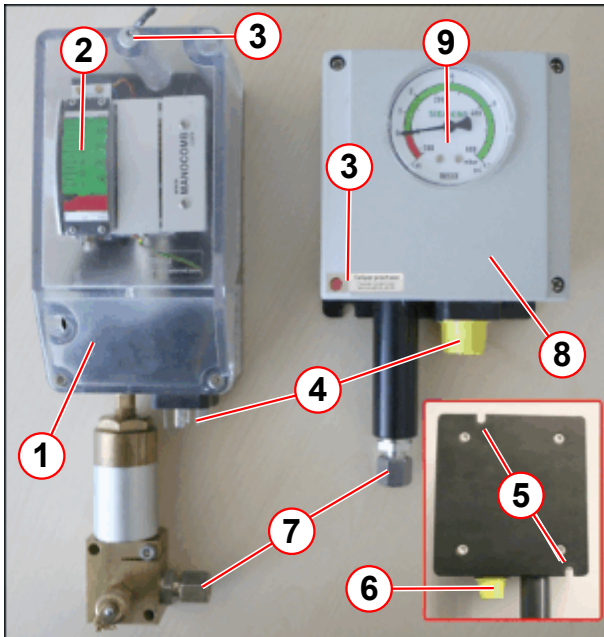
For the **Avanto**, the Pressure Switch Transducer assembly is located to the **left** of the Summitomo side sock. For the **Espre**, it is located to the **right**.

Pressure switch transducer assembly

- The **WIKA** Pressure switch transducer assembly , **replaces** the **Manocomb** pressure switch, when stocks are exhausted.
- The **new** WIKA switch (part number **10615712**) will **replace** the **current unit** (part number **10126452**), when stocks are exhausted.

- The **functionality** of the new switch is **identical**, and the same fixings are used for installation.

Fig. 58: Pressure switch transducers Manocomb and WIKA



- (1) Manocomb Transducer assembly
- (2) Linear gauge
- (3) Tamperproof seal
- (4) Front/rear electrical connections
- (5) Mounting slots (rear view)
- (6) Protection caps
- (7) Pressure line connection
- (8) WIKA Transducer assy (part no. 10126452)
- (9) Dial gauge

Fig. 59: WIKA Pressure switch transducer (part no 10615712)



The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1 Manocomb switch.

2.4.2.1 Pressure heater control

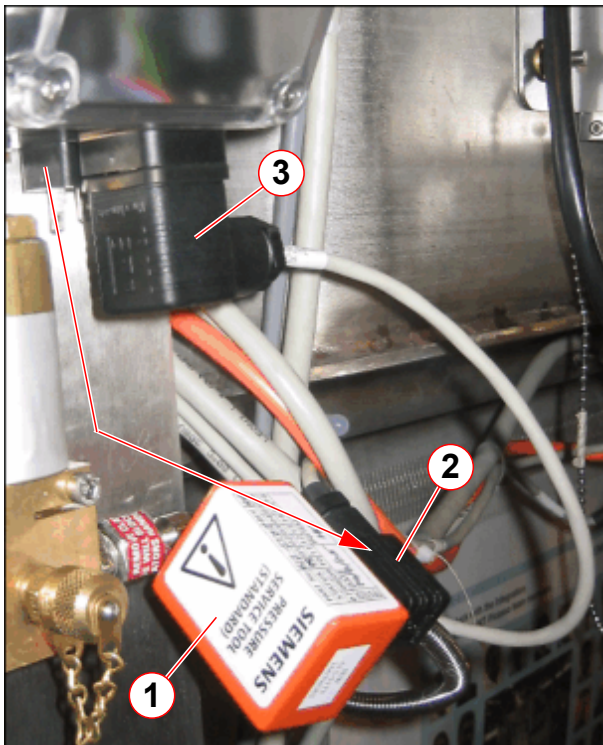
The **Pressure Service Tool** is used to **deactivate** the pressure heater control circuit. The **electrical connectors** are labeled **pressure sensor** (rear), and **pressure switch** (front).

- **Fit the pressure service tool:**
 - **Disconnect** the rear electrical connector on the **pressure switch transducer**.
 - Fit the **pressure service tool** to the connector lead. (→ 2/ Fig. 61 Page 55)
 - The **Alarm Box** will sound **PRESSURE ERROR**.
 - Press the **ACKNOWLEDGE** button to cancel the alarm.
- The **pressure** reading will be set to **~18 psi (A)**, which will **disable the pressure heater**.

To **reactivate** the pressure heater control circuit, re-connect the cable to the rear connector on the pressure switch transducer. Cancel any alarms.

Mk2 Manocomb Pressure switch

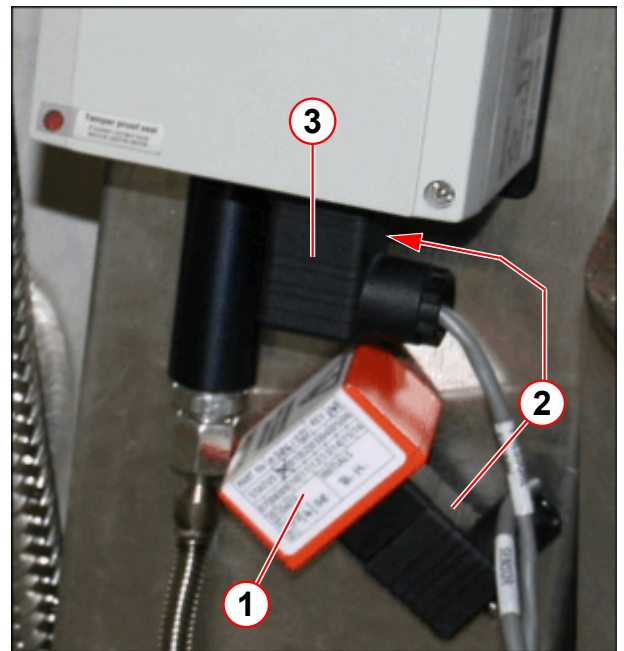
Fig. 60: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

WIKA Pressure switch

Fig. 61: Fit the pressure service tool

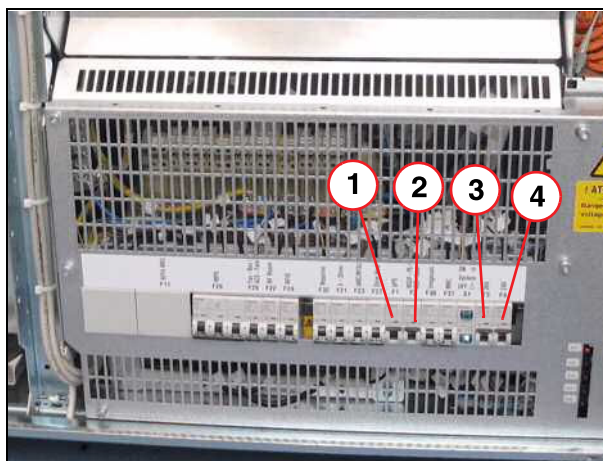


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 62: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service **pressure tool** is **not available**,

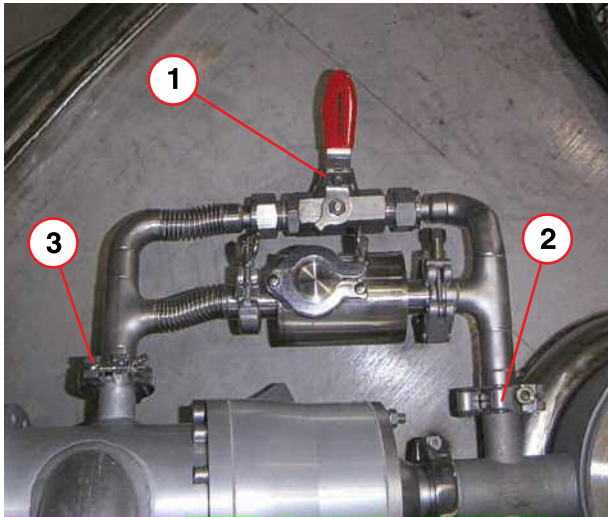
- **Switch OFF** the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » **Pressure heater is powered down.**
- The **ERDU circuit** is still functional.
- **Helium level** and **system temperatures** can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, **only 2 minutes** of power is available.

2.4.2.2 Mk1 Venting

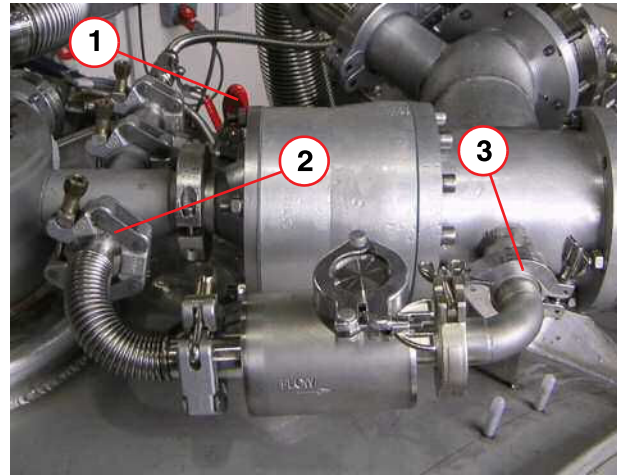
- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 54).

Fig. 63: OR105 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 64: OR122 Mk1 Vent valve assembly



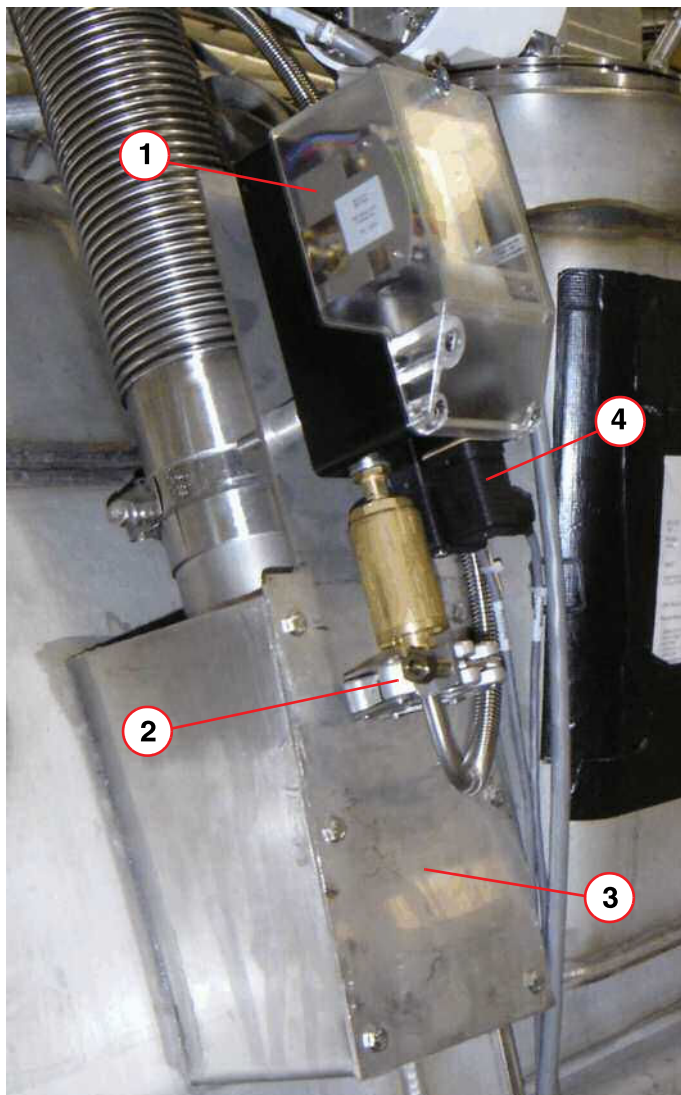
- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

- **Open** the **magnet bypass valve** to **depressurize** the helium vessel.
- When the magnet has reached **zero** pressure, **CLOSE** the **magnet bypass valve**.
- Warm the **venting** to room temperature, using a heat gun.

2.4.2.3 Remove the Pressure switch assembly - Mk1 Venting

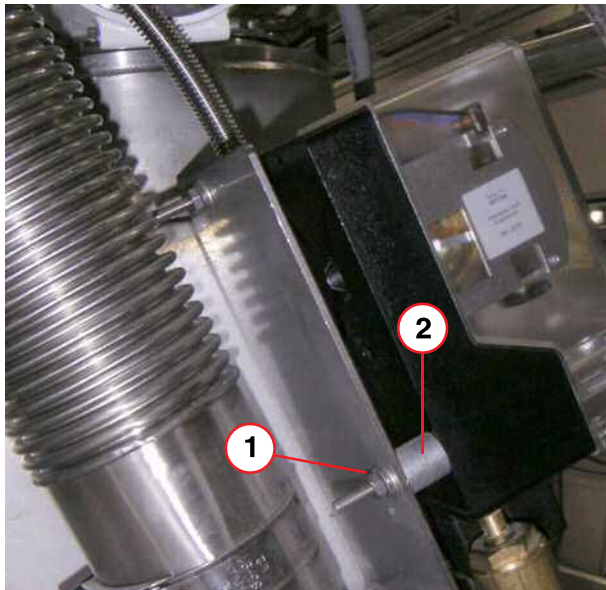
- **Disconnect** the remaining **electrical connector**.
- Remove the **NW25 (high pressure)** clamp and seal.(→ 2/ Fig. 65 Page 57)
- **Disconnect** the pressure line.
 - » **Helium gas** will vent from the **pressure line**.
 - » **Plug** the **end** of the **pressure line** with **paper towelling**.
 - » **Fit** the new unit **without delay**.

Fig. 65: Mk1 Pressure Switch Transducer



- (1) Transducer assembly
- (2) NW25 high load clamp and seal
- (3) Mounting plate
- (4) Electrical connections (front and rear)

Fig. 66: Mk1 Backplate



- (1) M6 nut and washer (2 off)
(2) Spacer (2 off)

- **Remove** the (2 off) cap hd screws securing the pressure switch to the **backplate**. (Each screw has a nut, washer and spacer).
- **Remove** the old pressure switch assembly.
- **Fit** the new pressure **switch** to the **backplate**, re-using the (2 off) cap hd screws, nuts, washers and spacers.
(Note: Earlier systems were not fitted with spacers).

Fig. 67: High load clamp



- **Remove** the **paper plug** from the end of the line.
- Check that all **sealing surfaces** are **clean**.
- **Re-connect** the pressure line using the NW25 (high pressure) clamp and **new seal**. (→ 2/ Fig. 65 Page 57)



DO NOT overtighten the high pressure clamp (→ Tab. 7 Page 14).

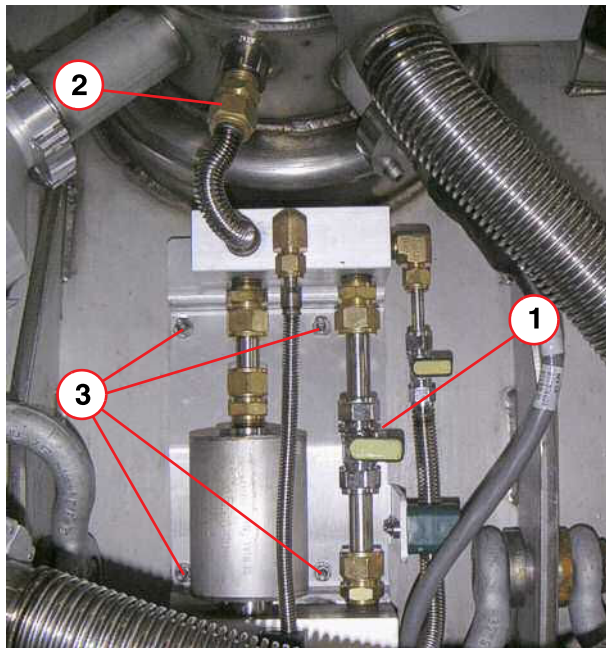
- **Remove** the **pressure service tool** from the electrical connector.
 - **Reconnect** the electrical connector(s) to the switch.
 - **Activate** the pressure heater (→ Pressure heater control / Page 54)

- **OR** switch **ON** circuit breaker **F3** in the **ACC**.

2.4.2.4 Mk2 Venting

- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 54).

Fig. 68: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

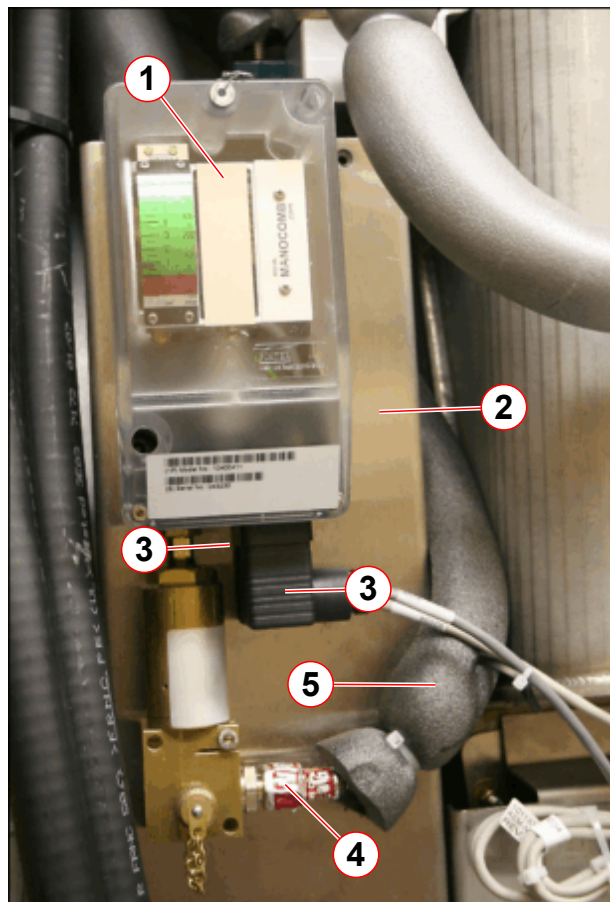
- **Open** the **magnet bypass valve** to **depressurize** the helium vessel
- When the magnet has reached **zero pressure**, **CLOSE** the magnet **bypass valve** to prevent air ingress.
- Warm the **venting** to room temperature, using a heat gun.

Remove the Pressure switch assembly

- **Disconnect** the remaining electrical connector.
- **Remove** the compression nut and **disconnect** the **pressure line**.
 - » **Helium gas** will vent from the **pressure line**.
 - » **Plug** the end of the **pressure line** with **paper towelling**.
 - » **Fit** the new unit **without delay**.

Mk2 Manocomb Pressure switch

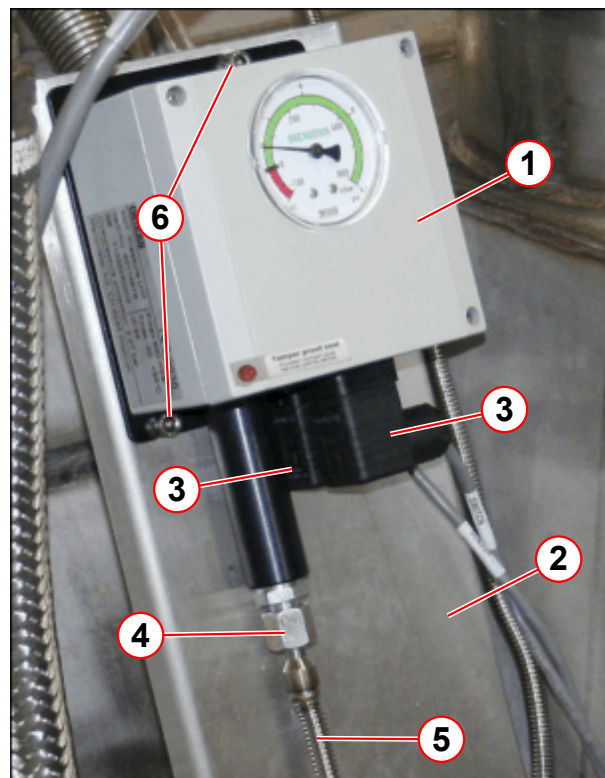
Fig. 69: Pressure switch assembly



- (1) Pressure switch assembly
- (2) Mounting plate
- (3) Magnet pressure sensor cables (front and rear)
- (4) Brass compression nut
- (5) Pressure line

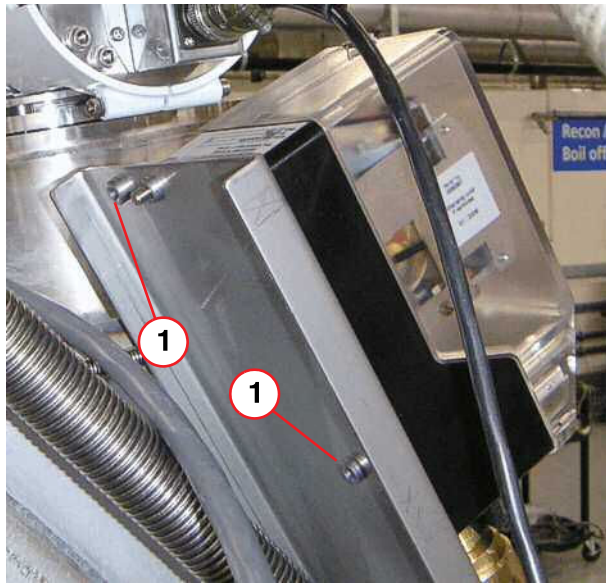
WIKA Pressure switch

Fig. 70: WIKA Pressure switch assembly



- (1) Pressure switch assembly
- (2) Mounting plate
- (3) Magnet pressure sensor cables (front and rear)
- (4) Compression nut
- (5) Pressure line
- (6) M6 x 20mm cap head bolts (2 off)

Fig. 71: Mk2 Backplate



(1) Captive nut (2 off)

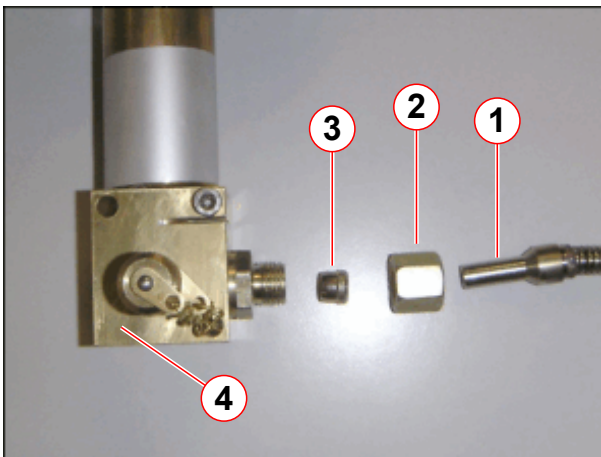
- **Remove** the 2 off M6 bolts securing the pressure switch to the **mountingplate**.
 - The bolts screw into 2 captive nuts on the rear of the **mounting plate** (pos 1).
- **Remove** the old pressure switch assembly.

Fit the new assembly

- **Fit** the **new** pressure switch to the backplate, using the 2 off M6 bolts
- **Fit** a **new** compression **ferrule**.
- Remove the **paper plug** from the vent line
- **Re-connect** the **pressure line**. **DO NOT** overtighten the nut.

Manacomb compression joint

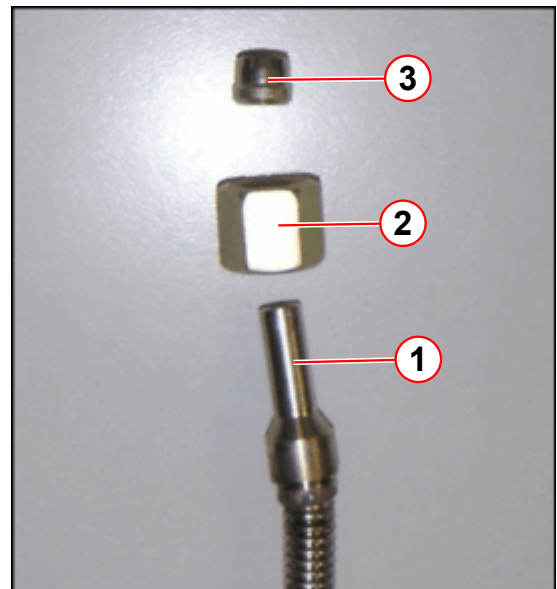
Fig. 72: Pressure switch compression joint



- (1) Pressure line
- (2) Compression nut
- (3) Compression ferrule
- (4) Transducer body

WIKA compression joint

Fig. 73: WIKA Pressure switch compression joint



- (1) Compression line
- (2) Compression nut
- (3) Ferrule

- Remove the **pressure service tool** from the electrical connector.
 - **Reconnect** the electrical connectors(s) to the switch.
 - **Activate** the pressure heater (→ Pressure heater control / Page 54).
- **OR switch ON** circuit breaker **F3** in the **ACC**.

2.4.3 Final checks

⚠ CAUTION

Pressurized helium vessel!

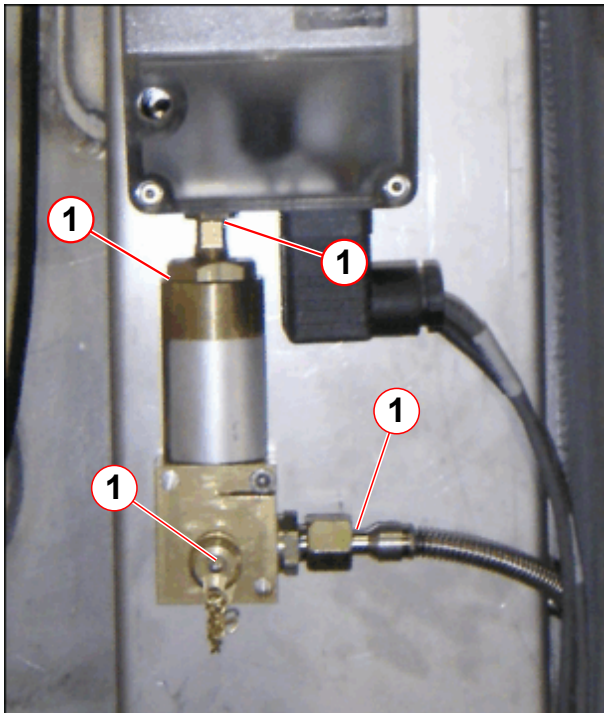
As the cold head/compressor is switched off automatically during the replacement, the system pressure may build to **ABOVE 15.3 psi (A)**.

» Allow the system pressure to stabilize at **15.3 psi (A)**.

- **Re-pressurize** the helium vessel to **15.3 psi (A)** by setting the pressure heater to **100% duty cycle**, using the MSUP-Display Unit.
- **Check** that the **compressor** re-starts.
- **Re-set** the **pressure heater** to normal operation when the helium vessel is at **15.3 psi (A)**.

Manocomb leak check points

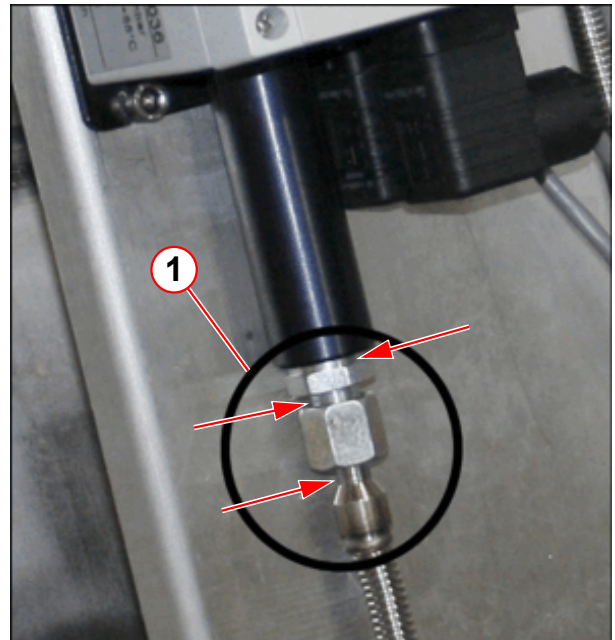
Fig. 74: Pressure switch assembly - leak checks



(1) Leak check points

WIKA leak check points

Fig. 75: WIKA Pressure switch assembly - leak checks



(1) Leak check points

Leak check the new **joint**:

- If there is a **small** leak, first try tightening the joint (carefully!) and **re-test**.
- If the **leak persists**, de-pressurize the magnet and **re-make** the joint using a **new compression ferrule**.
- **Replace** the appropriate **cover panels**.

2.5 Replacing the Vent assembly



This operation can be performed with the magnet at field.



CAUTION

Pressurized helium vessel.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » The magnet must be depressurized before attempting this procedure.
- » Personal safety equipment must be worn.



WARNING

Risk of asphyxiation.

Failure to observe the following may result in dizziness and loss of consciousness.

- » Do not vent helium gas directly into the magnet room.
- » Personal oxygen monitors must be worn if a fixed room monitor has not been installed.



CAUTION

Cold helium gas.

Personal injury may occur.

- » Protective clothing, gloves and eye protection must be worn.

2.5.1 Required time and tools

2.5.1.1 Time

120 min

2.5.1.2 Tools

- Helium leak detector (part no **10614850**)
- Standard (non magnetic) tool kit
- NW16 blank, O-ring and low pressure clamp
- Heat gun
- Pressure service tool (**03861221**)

- MSUP-Display Unit

Spare parts

- Mk1
 - 16 psi (A) vent assembly (part no **08396223**)
 - NW25 (high pressure) metal seal (part no **01564025**)
- Mk2
 - 16 psi (A) valve assembly (with seal kit) (part no **10117836**)

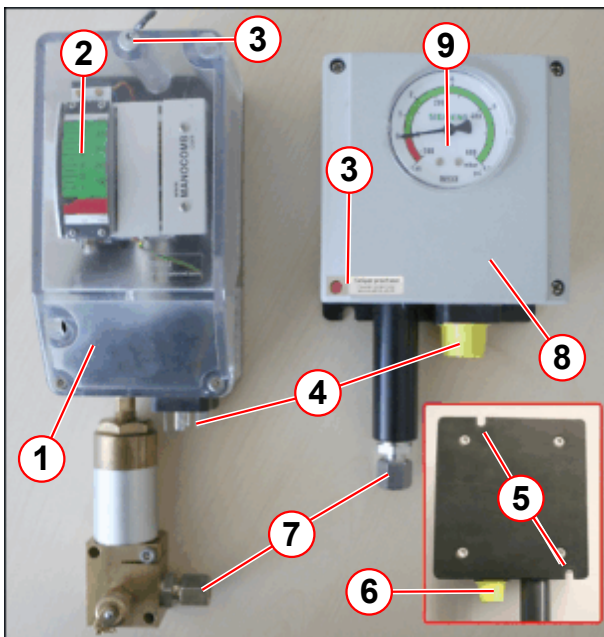
2.5.2 Replacement procedure

Pressure switch transducer assembly

- The **WIKA** Pressure switch transducer assembly , **replaces** the **Manocomb** pressure switch, when stocks are exhausted.
- The **new** WIKA switch (part number **10615712**) will **replace** the **current unit** (part number **10126452**), when stocks are exhausted.

- The **functionality** of the new switch is **identical**, and the same fixings are used for installation.

Fig. 76: Pressure switch transducers Manocomb and WIKA



- (1) Manocomb Transducer assembly
- (2) Linear gauge
- (3) Tamperproof seal
- (4) Front/rear electrical connections
- (5) Mounting slots (rear view)
- (6) Protection caps
- (7) Pressure line connection
- (8) WIKA Transducer assy (part no. 10126452)
- (9) Dial gauge

Fig. 77: WIKA Pressure switch transducer (part no 10615712)



The WIKA Pressure switch transducer is NOT directly interchangeable with the MK1 Manocomb switch.

2.5.2.1 Pressure heater control

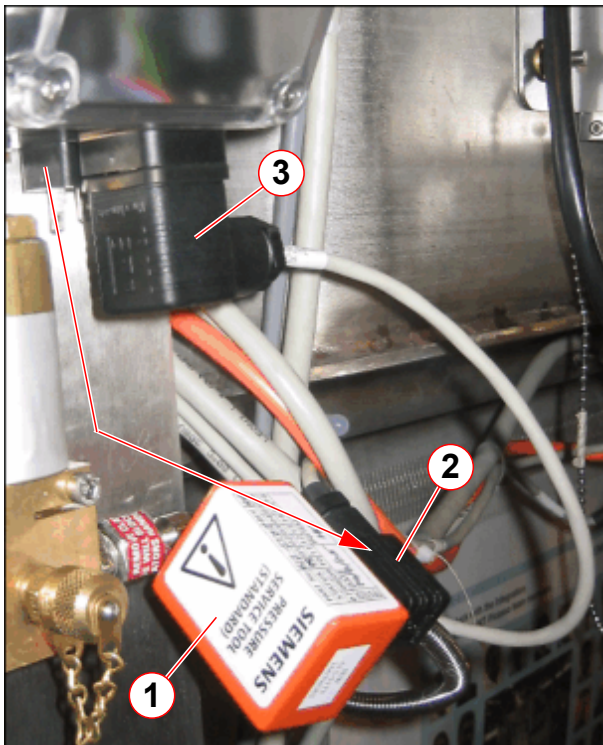
The **Pressure Service Tool** is used to **deactivate** the pressure heater control circuit. The **electrical connectors** are labeled **pressure sensor** (rear), and **pressure switch** (front).

- **Fit the pressure service tool:**
 - **Disconnect** the rear electrical connector on the **pressure switch transducer**.
 - Fit the **pressure service tool** to the connector lead. (→ 2/ Fig. 79 Page 67)
 - The **Alarm Box** will sound **PRESSURE ERROR**.
 - Press the **ACKNOWLEDGE** button to cancel the alarm.
- The **pressure** reading will be set to **~18 psi (A)**, which will **disable the pressure heater**.

To **reactivate** the pressure heater control circuit, re-connect the cable to the rear connector on the pressure switch transducer. Cancel any alarms.

Mk2 Manocomb Pressure switch

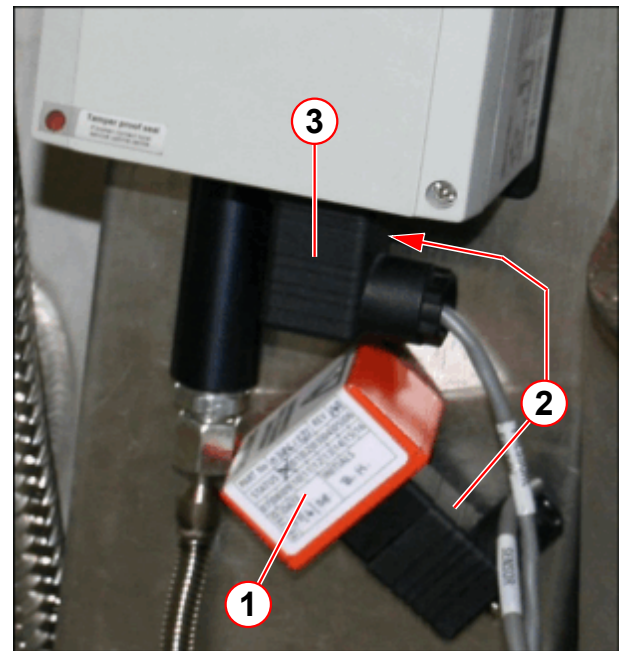
Fig. 78: Fit the pressure service tool



- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch

WIKA Pressure switch

Fig. 79: Fit the pressure service tool

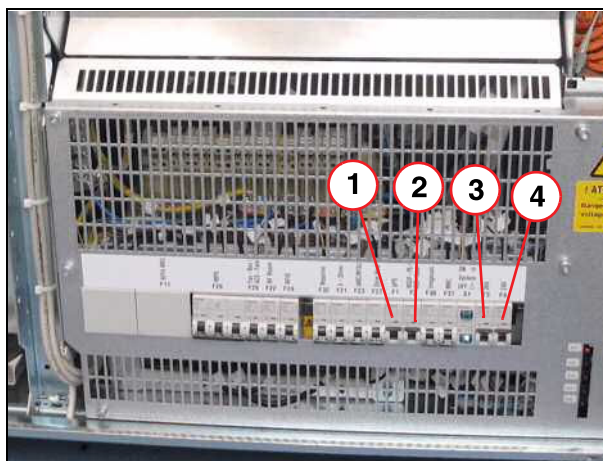


- (1) Pressure service tool
- (2) Rear pressure sensor
- (3) Front pressure switch



Do not leave the system unattended with the pressure service tool attached.

Fig. 80: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

If the service **pressure tool** is **not available**,

- **Switch OFF** the MSUP at circuit breaker F3 (fuse) in the ACC.
 - » **Pressure heater is powered down.**
- The ERDU circuit is still functional.
- **Helium level and system temperatures** can still be obtained via the MSUP-Display Unit (via battery power only).

NOTE: On battery power, **only 2 minutes** of power is available.

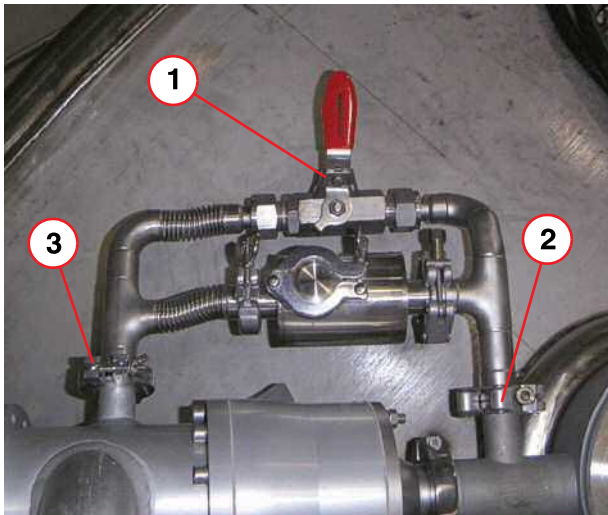
2.5.2.2 Mk1 Venting



Mk1 and Mk2 parts are NOT interchangeable.

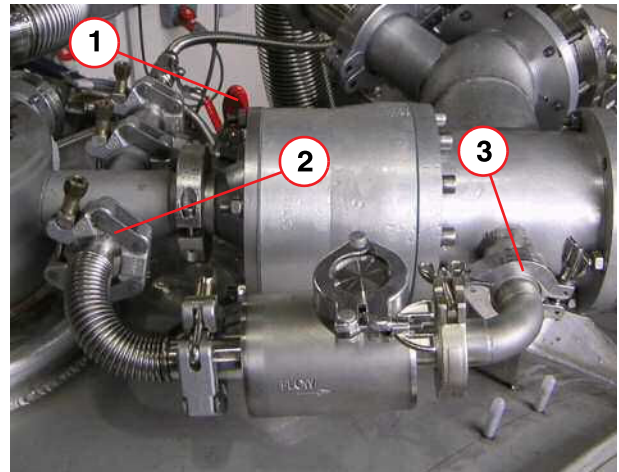
- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 66).

Fig. 81: OR105 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

Fig. 82: OR122 Mk1 Vent valve assembly



- (1) Bypass valve
- (2) NW25 High pressure clamp (turret)
- (3) NW25 Low pressure clamp (manifold)

- **Open** the **magnet bypass valve** to **depressurize** the helium vessel.
- When the magnet has reached **zero** pressure, **CLOSE** the **magnet bypass valve**.
- Use a heat gun to **de-ice** the high and low pressure **clamps**.
- **Undo** both **clamps**, (the **low pressure** side **first**), and remove the valve assembly.



When removing the venting assembly, the port must NOT be left open.
Ensure new components are readily available.

- **Immediately** fit the replacement valve assembly, firstly at the **high pressure** side, using the clamp and a **new** NW25 metal **seal**.
- **Re-fit** the NW25 low pressure clamp. (The low pressure seal may be re-used).



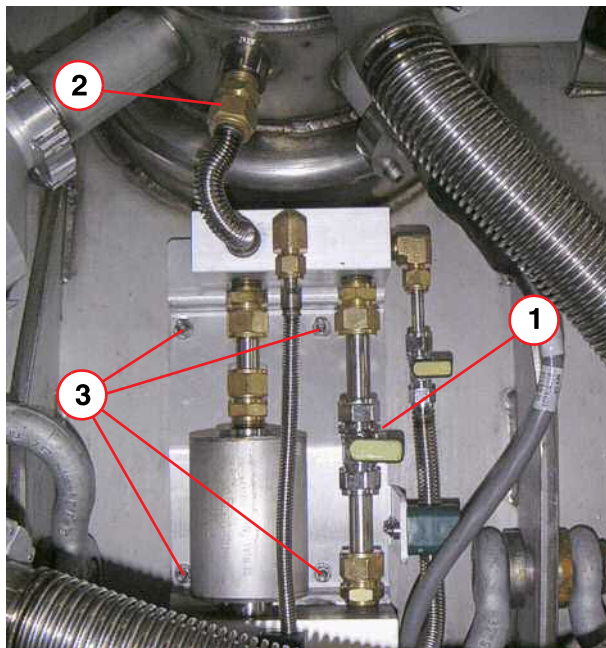
Make sure that the turret flange is dry before connecting the venting kit.

- Ensure that the **cold head bypass valve** is **closed**.
- **Reactivate** the pressure heater (→ Pressure heater control / Page 66).
- **Replace** the appropriate **cover panels**.

2.5.2.3 Mk2 Venting

- **Remove** the appropriate **cover panels**.
- **Deactivate** the pressure heater (→ Pressure heater control / Page 66).
- **Depressurize** the helium vessel by opening the **magnet bypass valve**.
(→ 1/ Fig. 83 Page 69)
- When the magnet has reached **zero pressure**, **CLOSE** the **magnet bypass valve** to prevent air ingress.
- Use a heat gun to **de-ice** the turret and manifold connections.

Fig. 83: Mk2 Vent valve assembly



- (1) Bypass valve
- (2) Turret connection
- (3) Mounting screws (4 positions)

- **Remove** the (4 off) **nuts** and 'Nord-lock **washers** securing the valve assembly plate to the cryostat. (Pos 3)

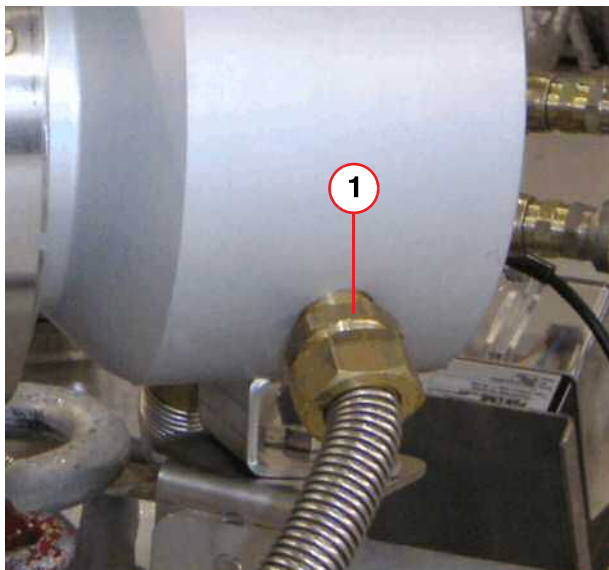


Early Mk2 vent assemblies may be connected to the quench valve exhaust tube and not directly connected to the turret (as shown in (→ 2/ Fig. 83 Page 69)).

- **Check** that the new vent valve assembly is in **good condition**, and that its sealing surfaces are **clean**.
- **Plug** the end of the new **pressure transducer line** with **paper towelling**.

- Close the **magnet bypass** and **cold head bypass** valves on the **new assembly**.

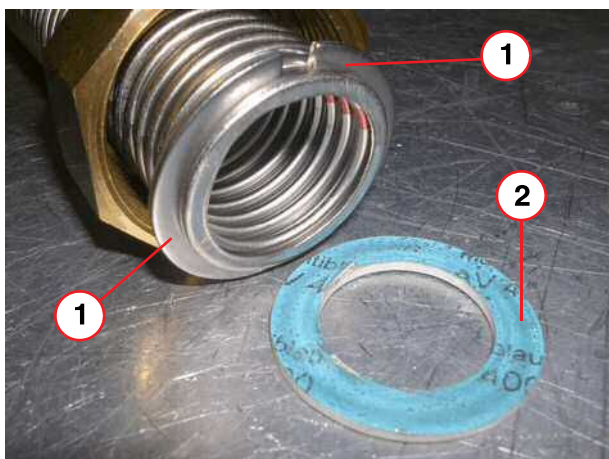
Fig. 84: Mk2 Manifold connection



(1) Manifold connection

- **Disconnect** the **manifold** connection.

Fig. 85: Mk2 16 psi (A) vent manifold connection

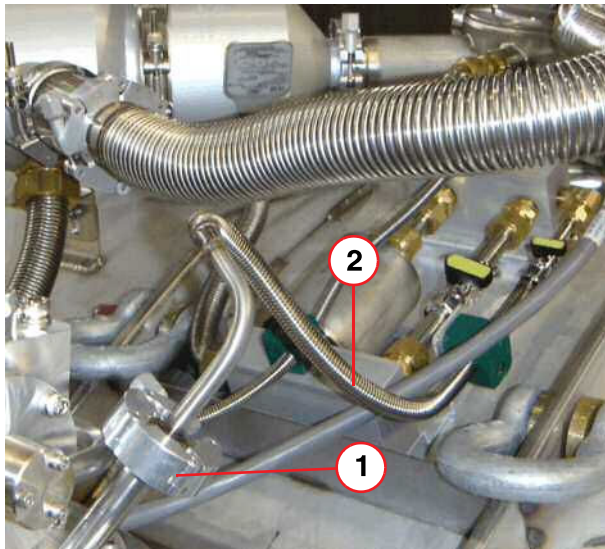


(1) Split rings
(2) Gasket

Take care not to lose the **split rings** during this process!

- Ensure the **split rings** do not fall out of position.
- Be careful to retain the gasket when disconnecting the vent line.
 - The **gasket** is not retained by the split rings and nut.

Fig. 86: Mk2 Coldhead vent line



- (1) NW16 high load clamp and seal
(2) Coldhead bypass line

- **Remove** the NW16 clamp (high pressure) from the **cold head bypass line**.
- **Temporarily** plug the open cold head **port** with the paper towelling to prevent **air ingress**.

- **Unscrew** the **turret connection** and **temporarily** block the open port with a plug of **paper towelling** to prevent **air ingress**. (→ 2/ Fig. 83 Page 69)
- **Unscrew** the vent line connection to the **pressure switch**. (→ Fig. 88 Page 72)
- **Remove** the valve assembly.
- **Fit** the **new valve assembly** to the cryostat with the (4 off) **nuts** and 'Nordlock **washers** (→ 3/ Fig. 83 Page 69)
- **Remove** the **paper towelling** plug from the **turret port**. Ensure **seal surfaces** are **clean**.



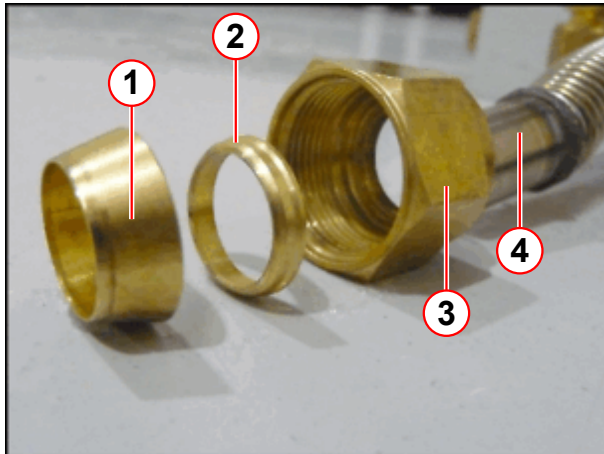
CAUTION

Cold helium gas.

Personal injury may occur.

- » Protective clothing, gloves and eye protection must be worn.

Fig. 87: Fitment of compression seals

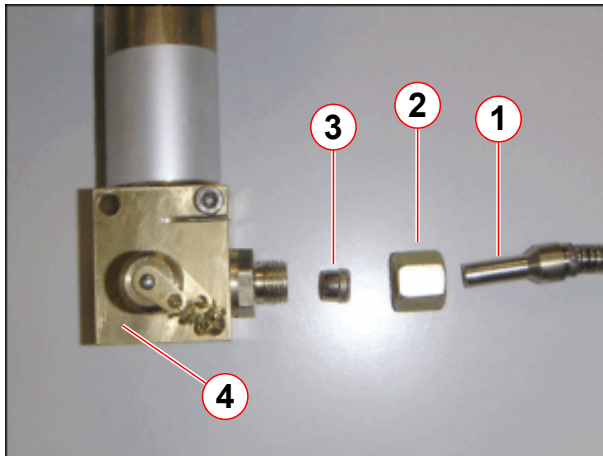


- (1) Small tapered ferrule
- (2) Compression ferrule
- (3) Nut
- (4) Turret vent pipe

- **IMMEDIATELY re-fit the turret connection** (high pressure).
 - Use a **new** compression ferrule.
 - **Tighten** the turret connection.
- **Remove the plug** of paper towelling from the **cold head bypass line port**.
- **Open the cold head bypass line:**
 - allow the **line to vent** for a few seconds to clear it of **air**.
 - **re-connect** the bypass line using a **new NW16** clamp and metal seal.
- **CLOSE** the cold head **bypass valve**.

Manacomb compression joint

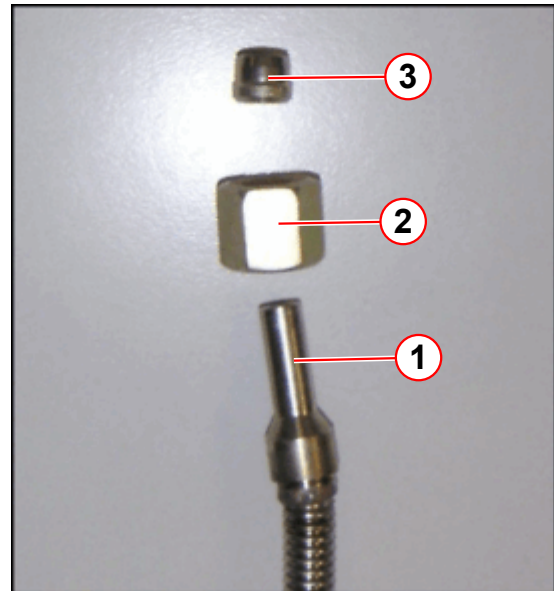
Fig. 88: Pressure switch compression joint



- (1) Pressure line
- (2) Compression nut
- (3) Compression ferrule
- (4) Transducer body

WIKA compression joint

Fig. 89: WIKA Pressure switch compression joint



- (1) Compression line
- (2) Compression nut
- (3) Ferrule

- **Remove the plug** of paper towelling in the **transducer line**.
- **Check the sealing surfaces are clean**, and then re-fit.

- Fit a **new** compression ferrule and **re-connect** the pressure line **DO NOT** overtighten the nut.
- Ensure the **cold head** bypass valve is **closed**.
- **Re-connect** the pressure line - **DO NOT** overtighten the nut
- **Reactivate** the pressure heater (→ Pressure heater control / Page 66).
- **OR switch ON** circuit breaker **F3** in the **ACC**.

2.5.3 Final checks



CAUTION

Pressurized helium vessel!

As the cold head/compressor is switched off automatically during the replacement, the system pressure may build to **ABOVE 15.3 psi (A)**.

» Allow the system pressure to stabilize at **15.3 psi (A)**.

- **Re-pressurize** the helium vessel to **15.3 psi (A)** by setting the pressure heater to **100% duty cycle**, using the MSUP-Display Unit.
- Check that the **compressor** re-starts.
- **Re-set** the **pressure heater** to normal operation when the helium vessel is at **15.3 psi (A)**.
- **Leak test** the entire **venting system**.
- **Replace** the appropriate **cover panels**.

3.1 Cold Head Replacement

3.1.1 Cold Head Replacement

This procedure describes the replacement of the **Sumitomo cold head** on all 4K magnet systems.

There are several operational differences between the 4K magnets during the cold-head replacement. Before, commencing the replacement procedure read the relevant magnet procedures from CB-DOC for:

- Depressurization of the magnet - **Installation - Depressurizing the magnet.**
- Operating pressures, SESO monitoring with 2306 MSUP - **Startup - Filling the magnet - Verifying the electronic low pressure switch.**
- Operating pressures, manual operation with 2657B MSUP - **Startup - Filling the magnet - Verifying the electronic low pressure switch.**
- Turret leak check - **Startup - Turret leak check.**



On all 4K recondensing magnets the complete cold head must be replaced if all other troubleshooting checks have been completed, but have made no improvement to the cooling efficiency.



Only Siemens HSC personnel, or CSEs who are trained and accredited in cold head exchange are permitted to perform the cold head exchange procedure.
Service personnel training on the Sumitomo cold head is offered by Siemens Magnet Technology.



Verifying the electronic low pressure switch in conjunction with the cold head replacement.
Refer to Startup > Filling the magnet > Verifying the electronic low pressure switch.
Document the test in the Service Software. Install & Maintenance > Maintenance Tests > Preventive Maintenance.

3.1.1.1 Safety



DANGER

Risk of asphyxiation

Failure to observe the following may result in the loss of consciousness leading to death.

- » Oxygen monitors must be active during the cold head replacement.
- » Be prepared. Complete the replacement of the cold head quickly to minimize gas loss into the room.

**WARNING**

Magnetic field

Risk of injury and equipment damage.

- » The magnet must be ramped down to zero field before replacing the cold head.

**CAUTION**

Pressurized helium vessel.

Failure to observe safety measures may result in serious injury and cold burns.

- » The magnet must be depressurized before attempting this procedure.

**CAUTION**

Risk of injury

The cold head weighs 25-30 kg. Take care when lifting.

- » Lift only by the motor end and support at the 50 K stage.
- » A 2nd person must be available when lifting and replacing the cold head.
- » Do not carry the cold head up or down the platform steps.
- » Always wear personal safety clothing and equipment when replacing the cold head.

3.1.1.2 Time

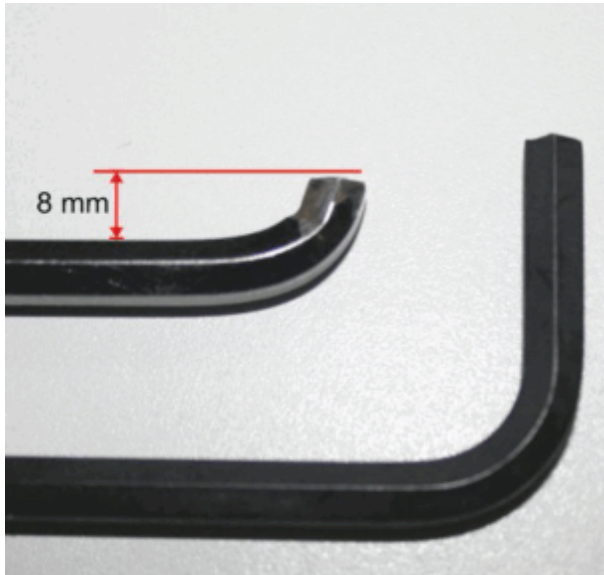
8 hrs is required. This includes:

- Setup time and ramping the magnet down.
- Replacing the cold head.
- Ramping the magnet to field.
- Final shim checks.

3.1.1.3 Tools

- Standard tool set.

Fig. 90: M5 Allen key - (cut down)



For Mk1 cold heads a shortened 5 mm Allen (Hex) key is required.:

- M5 Allen (Hex) key cut down to 8 mm (source locally).

Fig. 91: 4K Coldhead special tools



- (1) Aeroquip spanners
(2) Cold head release screws
(3) Cold-head balance line

- Aeroquip open-end wrenches (3 off) (**10125650**).
 - 11/8" - 5/8" open ended wrench.
 - 1" - 3/4" open ended wrench.
 - 13/16" open ended wrench.
- Cold head, Fridge release screws (**10113021**).
 - 2 x release screws are required.
- Cold-head balance line (**10129376**).

- Pressure tool (**03861221**). (2657B MSUP only)
- Torque wrench capable of **4 Nm** to **8 Nm**.
- Lint free paper towelling.
- De-greasing cleaner (e.g. Petroleum Ether or Alcohol based cleaner- sourced locally).

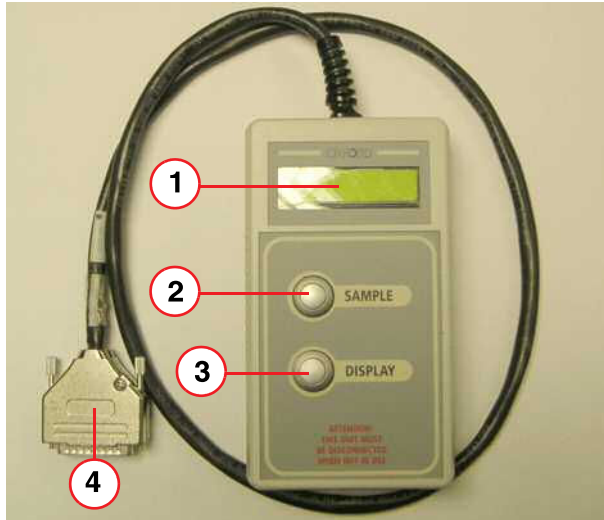


The de-greasing cleaner must not leave a residue on the surface of the side sock.

- Protective clothing (gloves, protective goggles or face protection shield, protective clothing, protective shoes).
- Hot air blower (**09702937**, 240v or **09702945**, 110v).
- Apiezon 'N'-grease (**10099740**).

- 3600 magnet power supply kit (**8395613**).
- NW16 blanking flange (**08396249**).
- **O₂ monitor** (room or personal). (Personal monitor material no **10142116**).

Fig. 92: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

- Helium remote display unit (**07758522**)
 - Option for systems fitted with 2657B MSUP.



Systems fitted with a 2306 MSUP. Use the host computer for activating and de-activating the pressure heater and compressor unit.

Fig. 93: Edwards Leak detector



- Suitable leak detector, capable of measuring leaks in the range 1×10^{-5} .
- **Equipment:**
 - Held held helium leak detector (material number: **10614850**).
 - or:
 - Helium mass spectrometer (material number: **05137224**).

3.1.1.4 Spare Parts

- 4K Sumitomo cold head assembly (material no: **08395860**).
- NW16 Helicoflex high load seal (material no: **08396496**).

- Universal cold head shield (material number **10620364**).
- Aeroquip 1/2" gasket seals, (material **03228017**).

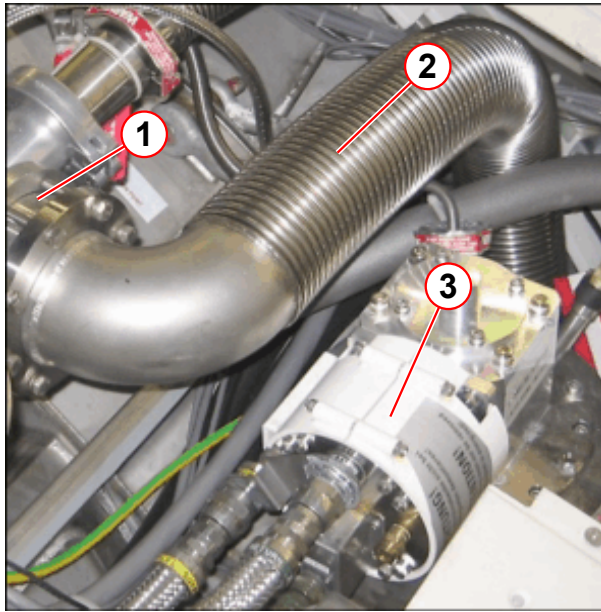
3.1.1.5 Prerequisites



This procedure **MUST NOT** be performed with the magnet **AT FIELD**.

- Complete the **Cold Head Exchange Report 4K units** with the requested information.
 - » Send a copy of the form to magnet HSC, (→ Healthcare Magnet WS CS, magnet.wscshsc.healthcare@siemens.com)
- Inspect the new cold head:
 - » Check that the material number is correct.
 - » Check for visible damage.
 - » Check the 10-pin temperature connector.
 - » If any faults found, return and order a replacement.
 - » If correct and no faults found, continue with the procedure.
- Remove the appropriate magnet covers.
- Check that there is a **sufficient service height** above the cold head before removal.
 - The **minimum height** above the cold head flange must be **50 cm** to enable removal.
- Download MSUP logs.
- Check the magnet site field current.
- Ramp the magnet to zero field.
- Check that an oxygen monitor is activated in the work area.
- Secure the service platform to the system frame using the straps.

Fig. 94: OR122 Cold head position



- (1) OVC burst disk assembly
- (2) OVC vent tube
- (3) Cold head

Espre (OR122) Mk1 Venting

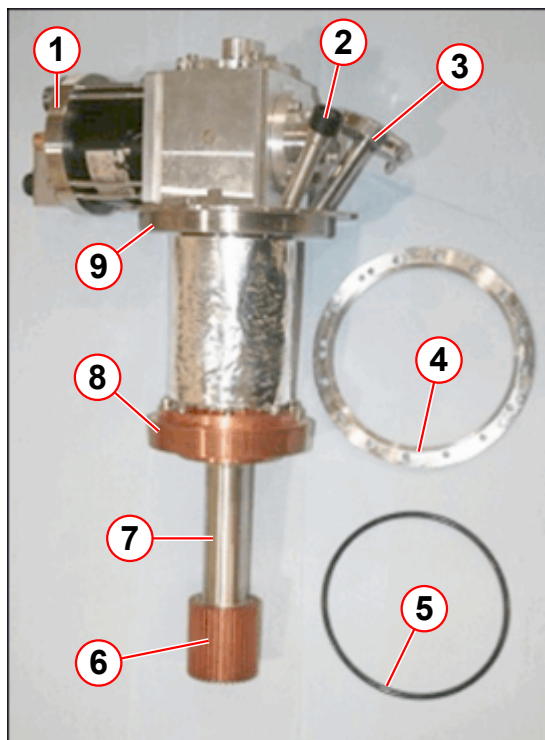
For the Mk1 system only, to allow replacement of the cold head:

- Remove the OVC burst disk assembly.

3.1.2 Removing the Cold Head

3.1.2.1 Preparing the New Cold Head

Fig. 95: Sumitomo 4K Cold head assembly



- (1) Motor end plate
- (2) 10 pin temperature sensor connector
- (3) NW16 vent port
- (4) O-ring clamp plate
- (5) O-ring seal
- (6) 4K copper stage
- (7) Thin wall Tube
- (8) 50K copper stage
- (9) Cold head flange

The preparation of the new cold head can be completed during the ramp down and depressurization of the magnet.



Never lift or support the cold head on the thin walled tube. (→ 7/ Fig. 95 Page 80). Damage to the tube could prevent the cold head from working.



To avoid contamination do not touch any of the copper parts or the super-insulation with bare hands. Contamination reduces the efficiency. Clean all surfaces with de-greasing fluid if contaminated.

Fig. 96: Unpacking the cold head



- Remove all protective materials from the new cold head.

- Clean and dry the first stage **50K** copper mating face.
- Only use the specified de-greasing cleaner on the cold head components.
- **DO NOT USE ANY GREASE** on the 50K copper mating face.

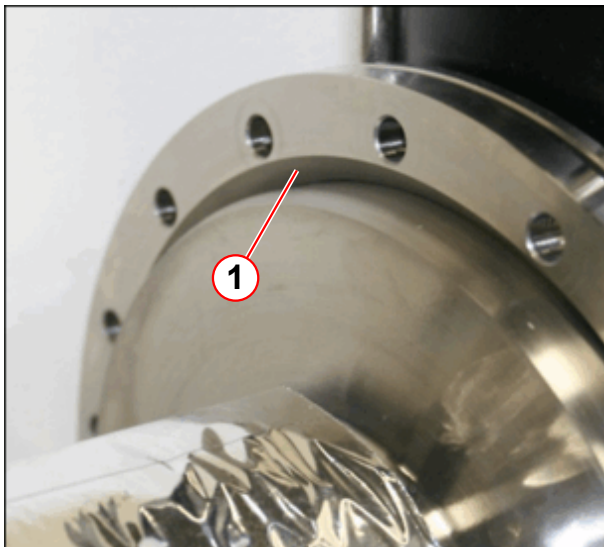
Fitting the O-ring

- Separate the cold head flange, O-ring and clamp ring.
- Clean the clamp ring and cold head flange using de-greasing cleaner.
- Place the O-ring clamp ring over the cold head on the flange.
 - The tapered edge facing towards the **50K** stage.
- **Lightly** coat the O-ring with N-grease and fit into position over the cold head flange.



The smallest amount of N-grease must be used to provide a shiny surface to the O ring.

Fig. 97: O-ring clamp plate tapered face



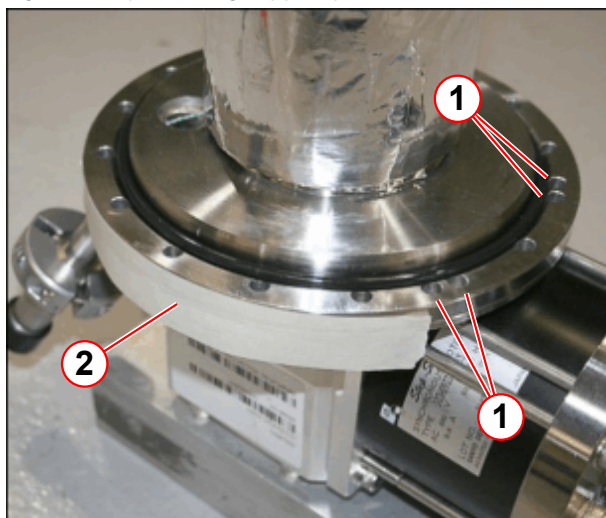
(1) Tapered face

Fig. 98: 4K Cold head clamping ring fitment



- (1) O-ring fitted correctly for installation
- (2) O-ring clamp plate
- (3) Cold head flange

Fig. 99: Taped O-ring support plate



- (1) Double holes aligned
(2) Tape

- Align the double holes on the clamp plate with the holes on the cold head flange (pos1).
- Secure together using an adhesive tape (pos 2).



Take care not to contaminate the cold head with the greased O-ring.



Remove the tape when the cold head has been installed in the side sock. Remove **BEFORE** the screws are torqued down.

- Keep the new cold head clean.
 - Cover the cold head with the plastic bag supplied with the original packaging. Remove when until ready to fit.

3.1.2.2 Depressurize the Magnet

- Deactivate the pressure heater using the suitable method for the system type:
 - fit the pressure control tool (2657B MSUP)
 - **Or** isolate the 36-V power to the MSUP (F2 /F3) in the **ACC**, or **ECA**.
 - **Or** de-activate from the **MRC host computer**. SeSo/Magnet&Cooling/Magnet Status/MSUP Control - Sets the pressure heater mode, **select Off button**.
- Cancel the alarms activated.
- **Open** the magnet bypass valve to depressurize the helium vessel.
 - Wait for the magnet to reach **zero pressure**.
- Check that the compressor unit has automatically switched **off** during depressurization.
- Warm up and **close** the magnet bypass valve to prevent air ingress.

3.1.2.3 Removing the Old Cold Head

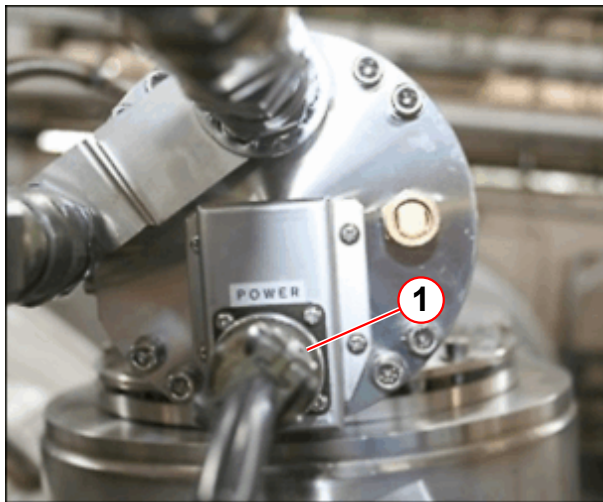
CAUTION

Risk of cold burns.

The end of the cold head is very cold and drips liquid oxygen. This causes injury and damage to components.

- » Personal protective equipment must be used.
- » Allow to warm up naturally. Do not use a hot air blower.
- » Store carefully in a safe place without causing damage to the floor or any work surface.

Fig. 100: Cold head motor electrical power cable



(1) Motor power connection

- Disconnect the cold head motor electrical power cable.

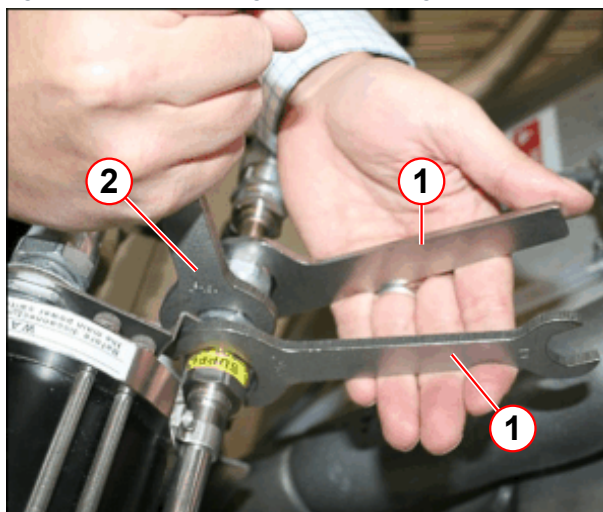
CAUTION

Contamination of Refrigeration circuit!

Failure to remove the Aeroquip connectors correctly leads to loss of pressure and air ingress into the refrigeration circuit. This can result in contamination, causing damage to the cold head.

- » Always use 3 wrenches when disconnecting the gas lines.

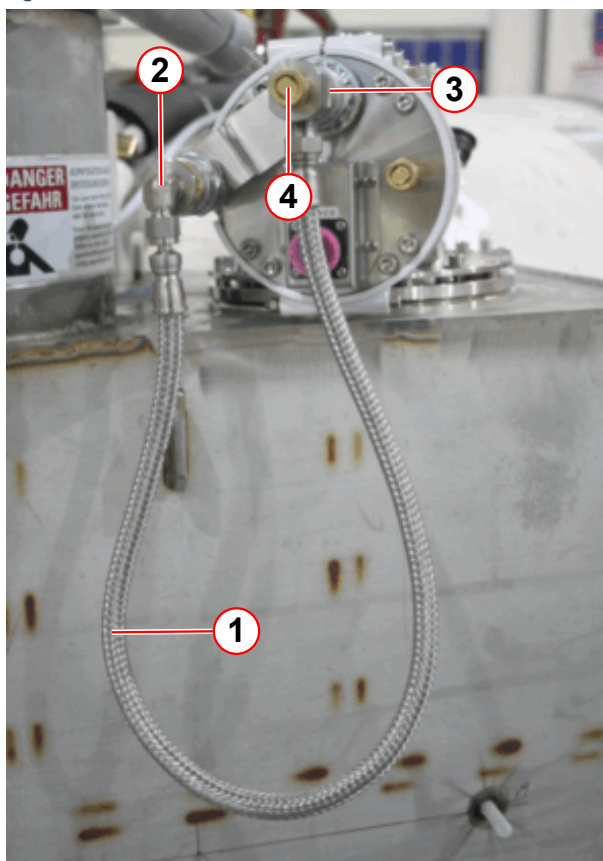
Fig. 101: Disconnecting the cold head gas lines



- (1) Hold
(2) Turn

- Remove the supply and return cold head gas lines.
- Move the gas lines away from the working area.

Fig. 102: Cold head balance line connection



- (1) Balance line
(2) Supply
(3) Return
(4) Pressure relief valve

A pressure relief valve (pos 4), is fitted to the return end of the **cold head balance line (10129376)**. This protects the cold head from over-pressurization and contamination during removal.

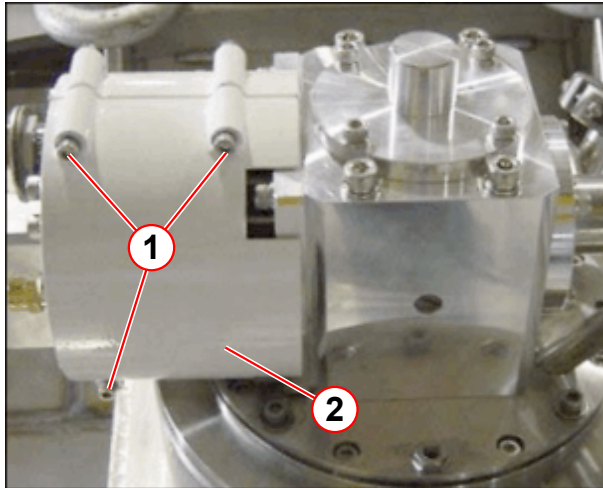
- Connect the cold head balance line to the **return** and **supply** couplings on the cold head.



There are 2 types of cold head shield with different screw fixing methods.

- **Type 1:** 3 x M5 x 70 mm Skt hd screws with washers, Nordlock washers and nuts.
 - Shield painted white.
- **Type 2:** 3 x M5 x 35 mm Skt hd screws with Nordlock washers.
 - Shield plated, (Non corrosive metal finish).

Fig. 103: Removing the iron shield

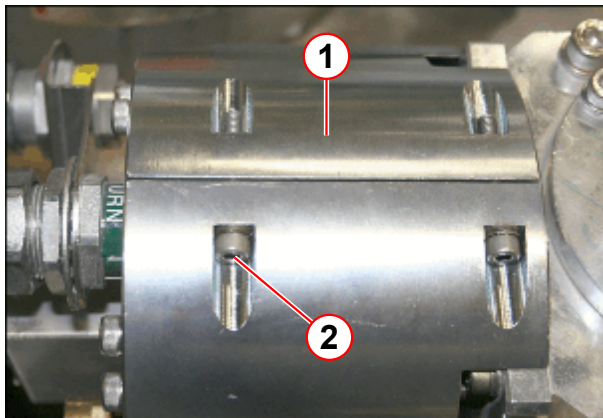


- (1) M5 x 70 mm skt head screws (3 off)
- (2) Iron shield

Type 1 shield removal

- To remove the iron shield:
 - Remove the (3 off) **M5 x 70 mm** socket head screws, Nordlock washers, washers and nuts.
 - **Carefully** lift clear the 2 sections of the iron shield from the cold head.

Fig. 104: Universal Cold head shield



- (1) Shielding component
- (2) M5 x 35 mm screw (x3)

Type 2 shield removal

- To remove the iron shield:
 - Remove the (3 off) **M5 x 35 mm** socket head screws and Nordlock washers.
 - **Carefully** lift clear the 2 sections of the iron shield from the cold head.

**WARNING**

Risk of cold burns. High helium gas pressure.

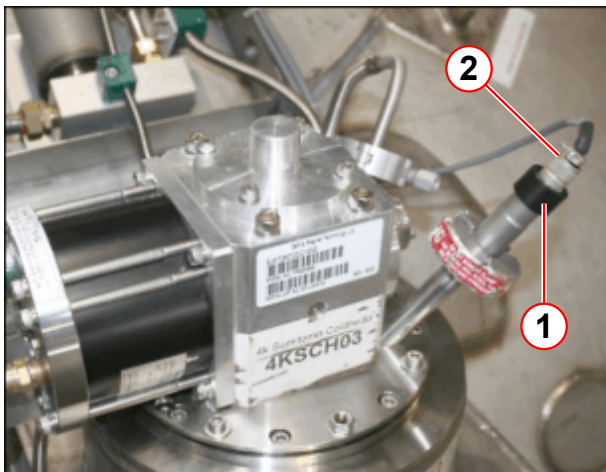
Risk of personal injury may occur.

- » Do not unscrew or remove the black temperature sensor sealing cap.
- » Remove the temperature sensor connector with a straight pull away from the sealing cap.

Temperature sensor

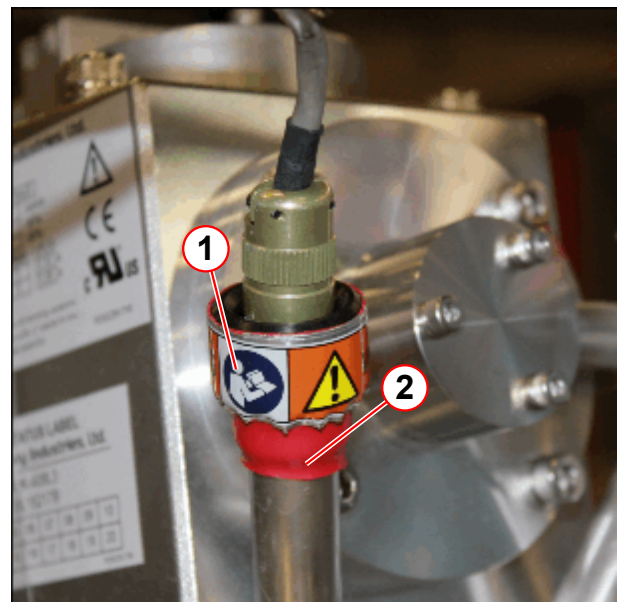
- **DO NOT** turn or try to remove the 'black' sealing cap (pos.1).
- Remove the temperature sensor connector (pos. 2), with a straight pull away from the black sealing cap.
- Move the cable to a safe position away from the cold head during the replacement.

Fig. 105: Cold head connections



- (1) Temperature sensor 'Black' sealing cap
(2) Temperature sensor connector and cable

Fig. 106: Protective heat shrink & warning label



- (1) Warning label
(2) Heat shrink



New cold heads are now supplied (from July 2015) with protective heat shrink and a warning label fitted around the temperature sensor connector.

- Warm and dry the cold head flange using a hot air blower if cold.
- Check that the cold head bypass valve is **CLOSED**.



Before, removing the cold head check the original orientation to ensure correct fitting of the new cold head.

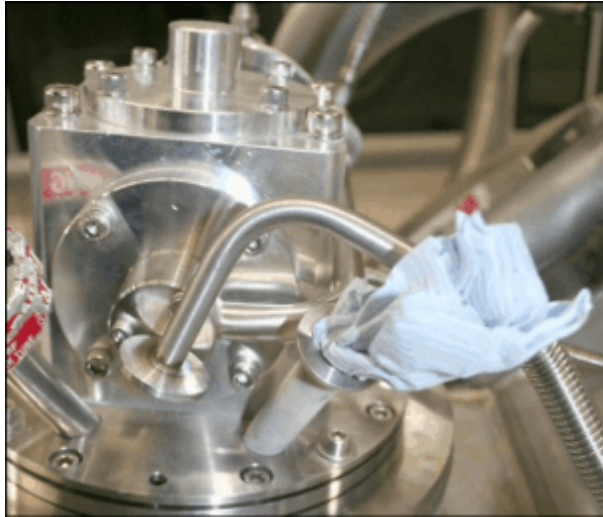
Make a reference mark if necessary.

- When the helium vessel is at atmospheric pressure:
 - Disconnect the high load NW16 clamp from the cold head port.
 - Remove the cold head vent line away from the cold head.



Air ingress must be avoided by blocking open ports with lint-free paper towelling

Fig. 107: Cold head interface



(1) Paper towelling

- Block the open NW16 port with lint free paper towelling.
- If available a NW16 blanking flange and seal can also be used.



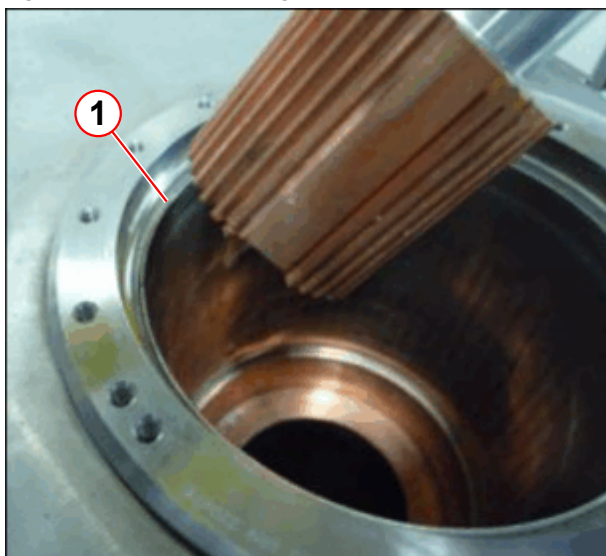
CAUTION

Loss of vacuum

Take care when removing or fitting a cold head.

- » Damage to the sidesock weld could lead to the loss of the system vacuum.

Fig. 108: Cold head fitting



(1) Sidesock weld

- Take care when removing the cold head.
 - Avoid hitting the sidesock weld with the cold head.

3.1.2.4 Mk1 Sidesock

The alignment of the O-ring clamp plate with the cold head flange would not allow the removal of the **M6 x 16 mm** screws under the cold head motor. The **lower** screws of the motor clamp plate must be **removed first**.



CAUTION

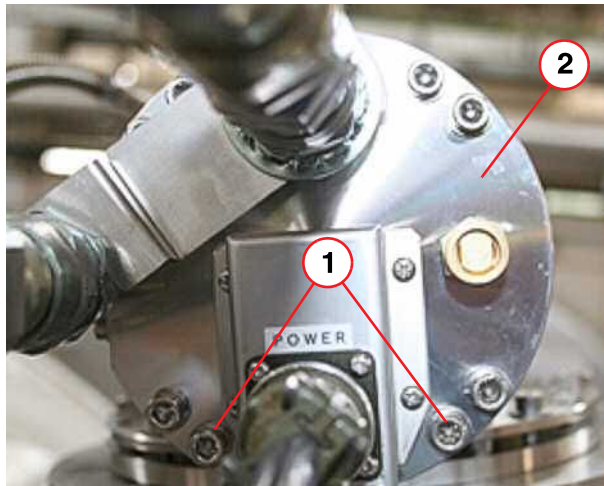
Risk of injury.

High pressure contained within the cold head housing.

» **Remove and replace the M6 x 80 mm motor clamp plate bolts one at a time.**

- Remove the lower (left) M6 x 80 mm skt hd screws from the motor clamp plate.
- Remove the M6 x 16 mm Skt hd screw directly under the O-ring clamp plate.
 - It is only possible to remove these screws using a **cut down 5 mm Allen (Hex) key**.

Fig. 109: 4K Cold head - motor end plate lower screws

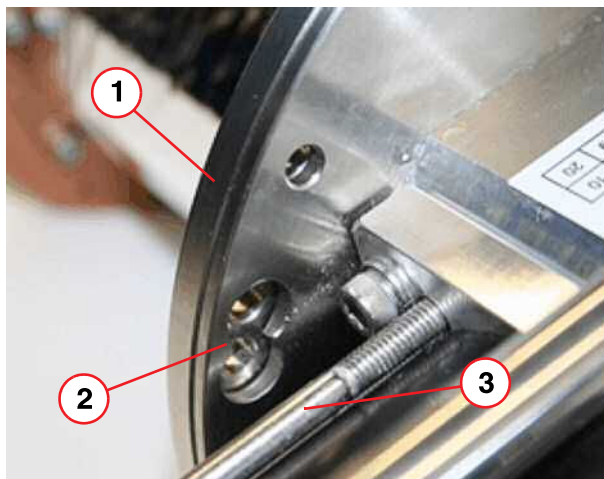


- (1) M6 x 80 mm (lower) skt hd screws
- (2) Motor end plate

- Re-fit the M6 x 80 mm skt hd screw immediately and torque to **8 Nm** (6ft/lbs).
- Repeat the process.
- Remove the lower (right) M6 x 80 mm screw to enable the removal of the M6 x 16 mm skt hd screw.
- Remove the 4 remaining M6 x 16 mm screws securing the O ring clamp plate.
- Remove the 4 off M6 x 25 mm cold head flange screws.

3.1.2.5 Mk2 Sidesock

Fig. 110: 4K Cold head flange/O-ring clamp plate alignment

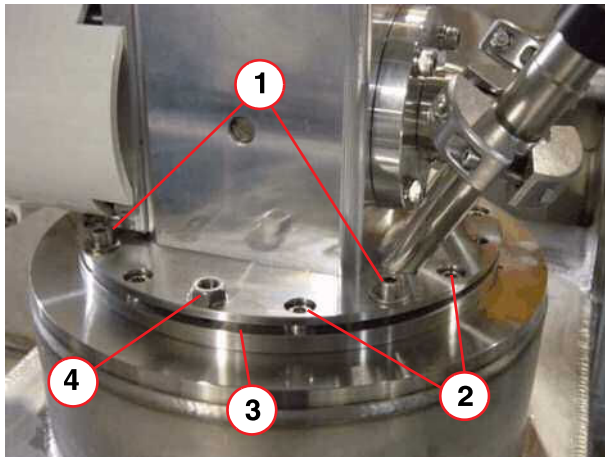


- (1) O-ring clamp plate
- (2) Cold head flange (showing universal screw holes)
- (3) M6 x 80 mm skt hd screw

The cold head flange has a **double screw** location hole (pos 2).

This allows installation of the O-ring clamp plate without the need to remove the M6 x 80 mm motor head screws.

Fig. 111: 4K Cold head screw positions



- (1) M6 x 25mm skt hd screws & washers (4 off equi-distant)
- (2) M6 x 16mm skt hd screws (8 off)
- (3) Gap of 2-4mm
- (4) Cold head release screw position (2 off - parallel)

- Remove the 4 off M6 x 25 mm cold head flange screws.
- Remove the 8 off M6 x 16 mm skt hd screws and washers from the O-ring clamp plate.



Tape can be placed around the cold head top plate and flange. This stops the O-ring and clamp plate falling during removal.

Fig. 112: Fitment of cold head release screws



- Screw the 2 cold head fridge release screws to the M8 nuts on the cold head flange.
- Confirm there is a small positive pressure inside the helium vessel (0.1 psi (G)).
- Check that the magnet bypass valve is **closed**.
- Tighten the cold head fridge screws evenly to disengage the cold head from the side sock.



CAUTION

Risk of cold burns.

The internal components of the cold head are very cold and will drip liquid oxygen.

» Safety gloves and eye protection must be worn.

Fig. 113: Removing the cold head



- Lift out the old cold head from the magnet side sock in one smooth vertical motion:
- Do not damage the lower section of the cold head by contact with any hard or sharp edges of the sidesock.

Fig. 114: Liquid oxygen drips



- Avoid contamination of capillary and 50k stage of the magnet
 - » Move the cold head clear of the sidesock within **3 sec.**
- The capillary tube could become blocked if liquid oxygen drips inside the side sock.
- Place the old cold head on a clean surface. Avoid any damage of the thin walled tubing and 4K stage.

3.1.3 Fitting the Cold Head

- Clean around the top plate of the sidesock using lint free paper towelling and degreasing cleaner:
 - the **O-ring** sealing surface must be **completely clean** to avoid leaks.

- **Do not** clean **inside** the side sock!

! CAUTION

Risk of air ingress.

Internal icing of side sock.

- » Fit the replacement cold head into the side sock immediately.



Care must be taken to ensure that the O-ring does not slip while the new cold head is being inserted into the side sock.



Do not lift or support the cold head on the thin walled tube and 4K stage. Damage to the tube prevents the cold head from working.

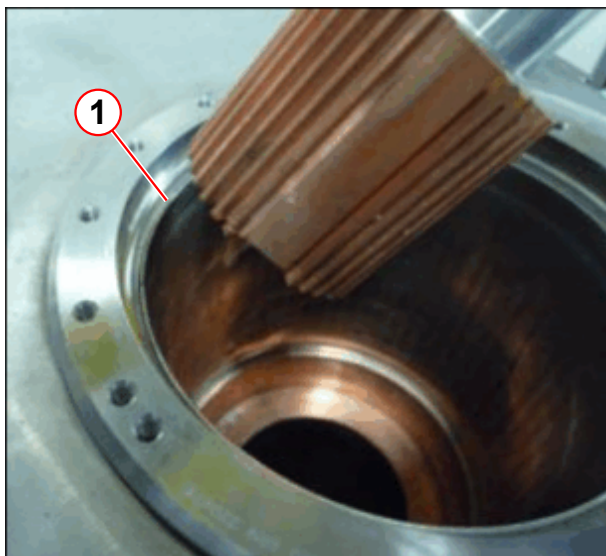
! CAUTION

Loss of vacuum

Take care when removing or fitting a cold head.

- » Damage to the sidesock weld could lead to the loss of the system vacuum.

Fig. 115: Cold head fitting



(1) Sidesock weld

- Take care when fitting the replacement cold head.
 - Avoid hitting the sidesock weld with the cold head.

**CAUTION**

Risk of injury

Trapped or pinched fingers.

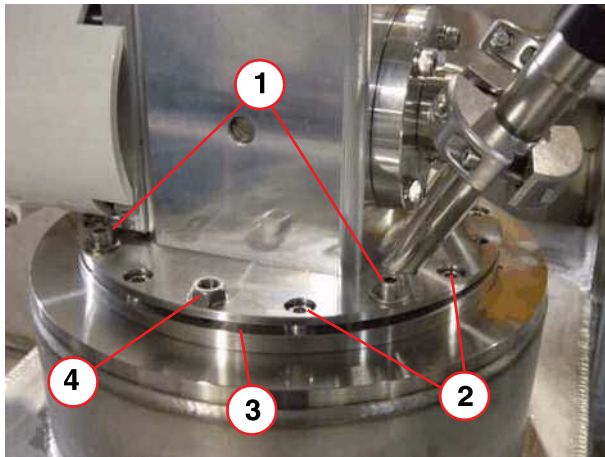
- » Carefully lower the cold head into the sidesock.

Fig. 116: Fitting the cold head



- Lift the new cold head in the **correct orientation** that it locates in the side sock.
 - Carefully lower vertically into the side sock.
 - Keep the cold head in the **upright position** as much as possible when lowering into the side sock.
 - A flow of gas is expelled from the side sock until the cold head is seated onto the O-ring seal.
-
- Check that the cold head is **fully seated** and located squarely in the side sock, with **NO movement**.
 - Remove the tape used to hold the O-ring clamp plate in position.
 - **Open** the magnet bypass valve:
 - » this prevents excess pressure build-up behind the O-ring seal, before it is clamped in place.

Fig. 117: 4K Cold head screw positions



- (1) M6 x 25mm skt hd screws & washers (4 off equi-distant)
- (2) M6 x 16mm skt hd screws (8 off)
- (3) Gap of 2-4mm
- (4) Cold head release screw position (2 off - parallel)

- Loosely fit the 8 off M6 x 16 mm screws through the cold head flange to secure the O-ring clamp plate to the side sock.
- Fit the 4 off M6 x 25 mm screws and plain M6 washers in 4 positions, equi-distant around the flange.
 - **Torque** each screw down in a diametric pattern, to 4 Nm (3 ft./lbs).
- **Tighten** the 8 off M6 x 16 mm screws in a **diametric pattern**, to 8 Nm (6 ft./lbs).



Mk1 cold heads - fit the 4 x M6 x 16 screws first. Then fit the 2 x screws under the cold head motor on at a time.



With the cold head fitted correctly, there is a clearance of 2-4 mm between the cold head flange and the O-ring clamp plate

- **Close** the magnet bypass valve.
- **Open** the cold head vent line to purge any air before connecting to the cold head.
- Connect the cold head vent line between the cold head and turret, using a new NW16 high load seal.
- Connect the temperature sensor cable.

- Connect the cold head motor power supply line.

Fig. 118: Aeroquip seal lodged in the flex line



- (1) Aeroquip seal
(2) Cold head compressor line

- Before, connecting the compressor unit gas lines check the condition of the Aeroquip flat gasket seals.
 - » **Check that** the Aeroquip seal is not **damaged** or **missing** from the compressor unit gas lines.
 - » **Check that** cold head Aeroquip seals are fitted after removing the balance line.



Clean surface of the couplings.

Clean connector with paper towel before connecting the lines.



Ensure a good seal of connectors.

Before connecting lines ensure the gasket seal is present in the connector.

- Connect the gas **supply** and **return** couplings to the cold head.
- Tighten the Aeroquip couplings using the Aeroquip spanners/wrenches. When tight, turn the coupling back **1/4 turn** to relieve strain.
- Check the **static pressure** on the compressor unit.
 - » **Record** the pressure in the **Cold Head Exchange Report 4K units** .
 - » If necessary, pressurize the **compressor unit** gas charge with helium. (Refer to chapter "Procedures" - Refrigerator - **Gas charging**)

Equalization (static) pressure at 20° C (68° F) for 12 to 20 m long gas lines:

Tab. 8 Compressor helium gas pressures

Cold head	Compressor	Static Pressure	Dynamic Pressure
4K - 50/60 Hz	F70	13.50 - 14.0 bar 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa
	CSW-71	16.0 - 16.5 bar 232 - 239 psi (G) 1600 - 1650 kPa	21 - 23 bar 304 - 333 psi (G) 2100 - 2300 kPa

- Check that the cold head vent valve is **closed**.
- Reactivate the pressure control to build up the magnet pressure:
 - remove the pressure tool.
 - **Or** switch on power to the MSUP (F2 / F3) in the **ACC**, or **ECA**.
- Manually build the magnet pressure to its designed operating pressure by:
 - setting the pressure heater to **100% duty cycle**, using the helium remote display unit..
 - Or - Turning the pressure heater on to **100%** at the MRC host computer.
- Check that the compressor unit starts when magnet pressure is **0.2 psi (G)**.
- Check that the cold head is running. The 2306 MSUP can be have up to 1 hr delay before the compressor unit starts.
- Turn the pressure heater back to auto mode, once the pressure is:
 - **15.3 psi (A)** for OR105/OR122/OR60
 - **15.5 psi (A)** for OR97/OR98/OR99/OR103/OR65
- Check the compressor unit dynamic pressure after 20 min
 - » Record the pressure in the **Cold Head Exchange Report 4K Units** .

3.1.3.1 Re-torque Cold Head



The cold head contracts and settle as it cools. To confirm that the 50K copper interface has made a good connection, after 2 hrs (minimum), or just before ramping:-

- Re-torque the 4 off M6 x 25 mm screws to 4 Nm (3 ft/lbs).
- Re-torque the 8 off M6 x 16 mm screws to 8 Nm (6 ft/ lbs).

3.1.3.2 Final Checks

- Re-boot the host computer.
- Warm up the cold head flange and dry with lint free paper towelling.

Leak check

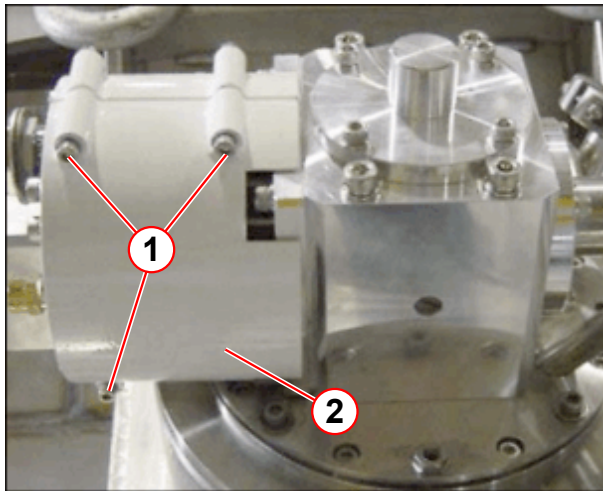
- When the system is back to normal operating pressure:
 - Leak check around the cold head side sock seal cold head flange.
 - Leak check around the high pressure NW16 vent line connection on the cold head.
- Complete the **Post Exchange Details** of the **Cold Head Exchange Report 4 K units**.
- Monitor the system pressure to check that the magnet starts to stabilize.



Wait a minimum of 2 hrs after the compressor unit has started before ramping the magnet.

- Re-fit the cold head iron shield

Fig. 119: Removing the iron shield

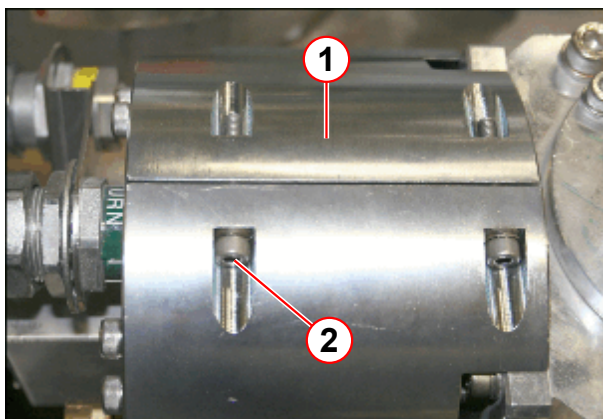


- (1) M5 x 70 mm skt head screws (3 off)
- (2) Iron shield

Type 1 shield

- To fit the iron shield:
 - **Carefully** place the 2 sections of the iron shield around the cold head motor.
 - Fit the (3 off) **M5 x 70 mm** socket head screws, Nordlock washers, washers and nuts.
 - Check there is a equal and parallel gap between the 2 shield sections.
 - Torque to 5 Nm.

Fig. 120: Universal Cold head shield



- (1) Shielding component
- (2) M5 x 35 mm screw (x3)

Type 2 shield

- To fit the iron shield:
 - **Carefully** place the 2 sections of the iron shield around the cold head motor.
 - Fit the (3 off) **M5 x 35 mm** socket head screws and Nordlock washers.
 - Check there is a equal and parallel gap between the 2 shield sections.
 - Torque to 5 Nm.

- Ramp the magnet to field.

- Download the MPS Ramp logs and MSUP logs.
- Remove the service platform.
- Refit the system covers.



If (SRS) remote access is available, the magnet performance can be monitored remotely.

After 24 hours

- Check that the magnet is maintaining its design pressure, (15.3 psi (A) or 15.5 psi (A)).
- Record the pressure heater average power.
- Check that the magnet shield temperatures are returning to normal.
- Complete the **Cold Head Exchange Report 4K units**
 - » **Email the report form and MSUP log files** to: (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

Packing and Return of Cold Head

- Remove the cold-head balance line from the old replaced cold head.
- Remove the cold head release screws from the removed cold head.
- Pack the old cold head (re-using the packaging).
- Return to CSML.

3.2 Replacing the compressor CSW-71

3.2.1 Required time and tools

3.2.1.1 Time

60 min

3.2.1.2 Tools

- Standard tool kit

Fig. 121: Edwards Leak detector



- Use the Edwards hand-held leak detector (part no **10614850**).

This is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

- Ground test meter (Fluke 1503, or equivalent)
- Transportation Lifting bars (part no **5137190**)

WARNING

Risk of personal injury and damage to equipment.

The weight of the compressor is approx. 120kg (264lbs).

- » When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.
- » Moving the unit is a 2 or 4 man operation. **USE THE TRANSPORTATION BARS. DO NOT ATTEMPT TO MOVE THE UNIT WITHOUT ASSISTANCE.**



On receipt of the new compressor, ensure the STATIC pressure is positive.
DO NOT fit a compressor that is showing ZERO pressure.

3.2.2 Preliminary

- Press the **STOP** button on the Control panel. (→ 2/Fig. 134 Page 107)
- Turn **off** the compressor at the **main power switch**. (→ 3/Fig. 123 Page 101)
- Switch **off** the **F102 fuse** on top of the ACC cabinet.
 - The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - Press the **ACKNOWLEDGE** button to cancel the audible alarm.

3.2.3 Replacement procedure

3.2.3.1 Removing the Compressor

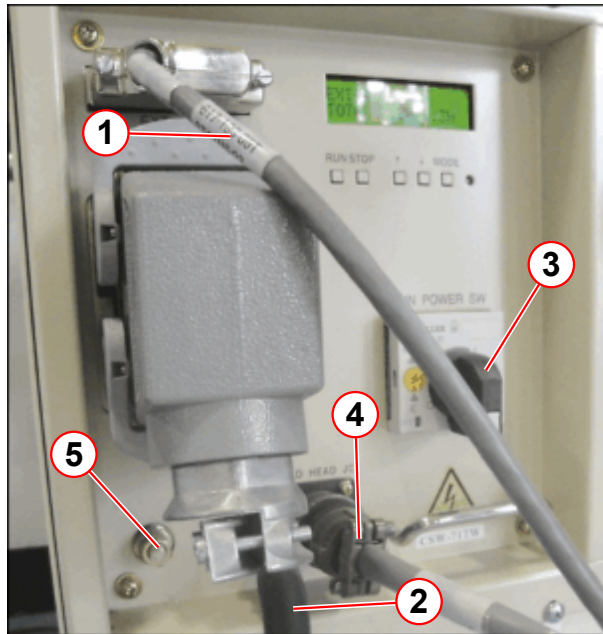
Fig. 122: Aeroquip gas supply/return flexline couplings



- (1) Hold (1" wrench)
- (2) Turn (1 3/16" wrench)

- Remove the self sealing helium flexlines in the following order:-
 - **Return** (low pressure)
 - **Supply** (high pressure)
- Hold nut (1) and loosen nut (2).

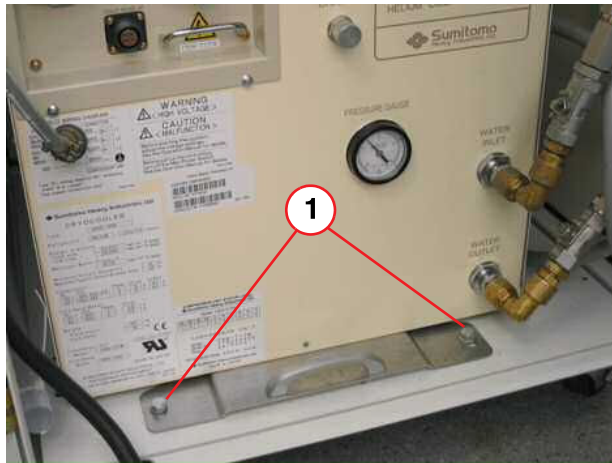
Fig. 123: Compressor control panel



- (1) MSUP cable (EXT.JR)
- (2) Mains supply cable
- (3) Main power switch
- (4) Cold head motor cable
- (5) Ground wire

- Disconnect the **Mains supply, Cold head motor** and **MSUP** electrical power cables.
- Disconnect the **ground wire** (pos 5).

Fig. 124: Compressor fixing plate/handle



- (1) M8 Hex hd screws (2 off)

- Remove the 2 off M8 hex hd screws securing the **fixing plate** to the cabinet.

**WARNING**

Risk of personal injury and damage to equipment.

The weight of the compressor is 120kg.

- » When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.
- » Moving the unit is a 2 or 4 man operation. **USE THE TRANSPORTATION BARS. DO NOT ATTEMPT TO MOVE THE UNIT WITHOUT ASSISTANCE.**

Fig. 125: Compressor - removal from cabinet



- Use the **fixing plate handle** to carefully pull the compressor onto the wooden support 'ramp'.

(1) Supporting 'ramp' made from wooden blocks

Fig. 126: Compressor water inlet/outlet shut off valves



(1) Shut off valve

- Close the **shut-off valves** on the inlet and outlet water lines.
- **Disconnect** the water lines from the compressor.



There is the potential for irreversible damage to the unit if the internal water pipes freeze during shipping.

Before returning a faulty compressor, use a gas line on either the inlet/outlet lines to expel any water.



CAUTION

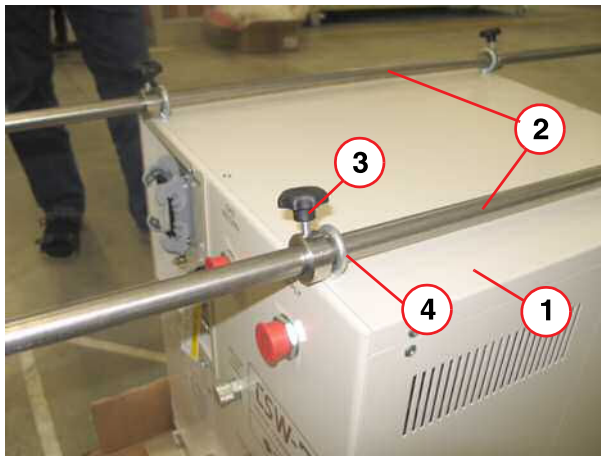
Chemical hazard.

If the chilled water contains ethylene glycol, it should be collected, in a suitable container, while draining.

- » The water should be returned to the water circuit after the new unit is installed.
- » If necessary, dispose of in line with local regulations for hazardous waste.

- Use the **transportation lifting bars** to move the compressor.

Fig. 127: Helium compressor



- (1) Helium compressor
- (2) Lifting bar
- (3) Locking device
- (4) Eye bolt

- Slide the lifting bars through the **eye bolts**.
- Securely tighten the **locking devices** (4 positions) to prevent the compressor from sliding on the bars.

Fig. 128: Helium compressor



- The compressor can be lifted by 2 or 4 persons.

3.2.3.2 Installing the new compressor

NOTICE

Prevent equipment failure.

Tilting the unit by more than 30 degrees may cause oil sealed in the unit to move - preventing normal operation.

- » When carrying or moving the compressor unit, do not tilt it by more than 30 degrees.

**WARNING**

Risk of personal injury and damage to equipment.

The weight of the compressor is approx. 120kg (264lbs).

- » When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.
- » Moving the unit is a 2 or 4 man operation. **USE THE TRANSPORTATION BARS. DO NOT ATTEMPT TO MOVE THE UNIT WITHOUT ASSISTANCE.**

- Use the **transportation lifting bars** to move the compressor onto the supporting ramp. (→ Fig. 127 Page 104)
- Connect the **inlet** and **outlet** water lines. (→ Fig. 126 Page 103)
- Open the water **shut off valves**. (→ 1/Fig. 126 Page 103)
- Carefully slide the new compressor into the cabinet.
- Secure the **fixing plate** to the cabinet using the 2 off M8 hex hd screws. (→ 1/Fig. 124 Page 101)

Fig. 129: Flex line connection



(1) Flat rubber gasket

- Check the **rubber gaskets** for dirt or dust and ensure it is correctly fitted.

NOTE: Incorrect fitment may cause a helium gas leak.

- Connect first the **supply**, and then the **return** helium flex lines.

- Connect the **Mains supply**, **Cold head motor** and **MSUP** electrical power cables. (→ Fig. 123 Page 101)

Compressor protective (ground) earth connection

- A **protective ground connection test** must be performed as a final step **before** switching on the compressor.

Fig. 130: Compressor ground connection



(1) Ground connection

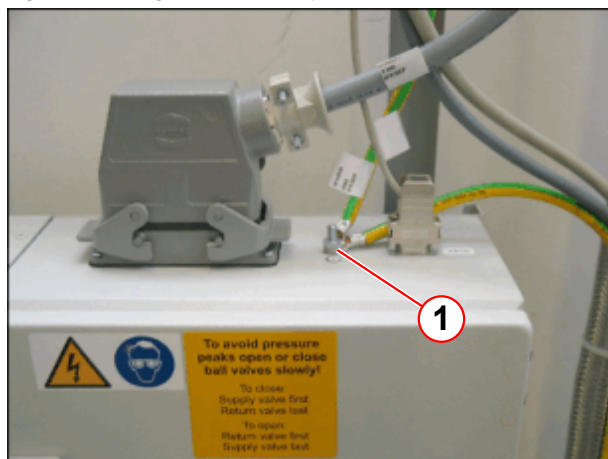
- Connect the **ground wire (1)** to the helium **compressor** terminal.

Test the protective earth connection

- Measure the **protective earth connection** resistance $\leq 100 \text{ m}\Omega$ with a **ground test meter** (Fluke 1503, or equivalent), between the **protective earth connector** on the compressor unit, and the **protective earth wire** connected to the **top of the SEP** (on the electronic box of the IFP).

Stand-alone option

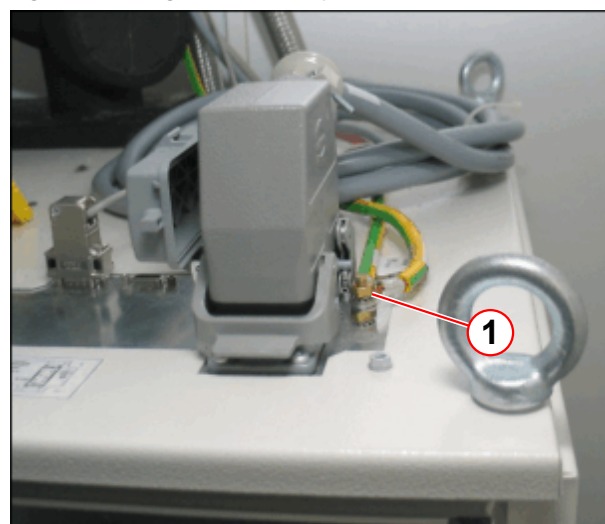
Fig. 131: IFP ground wire (optional)



(1) Ground wire

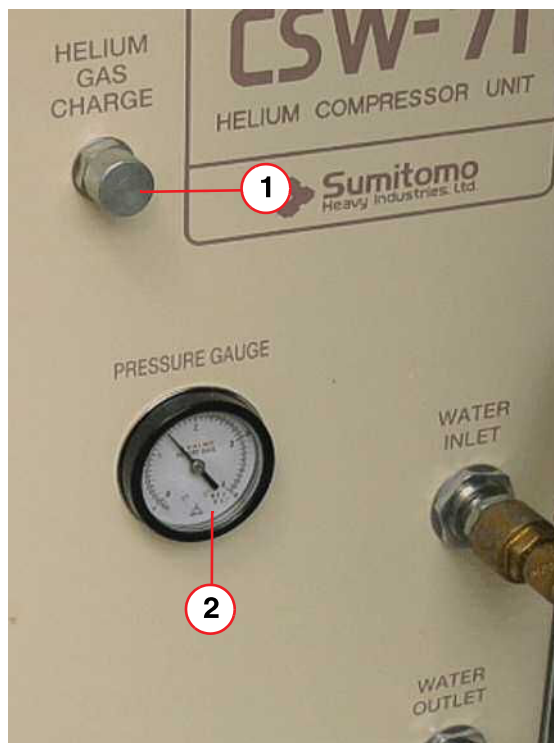
SEP option

Fig. 132: SEP ground wire (optional)



(1) Ground wire

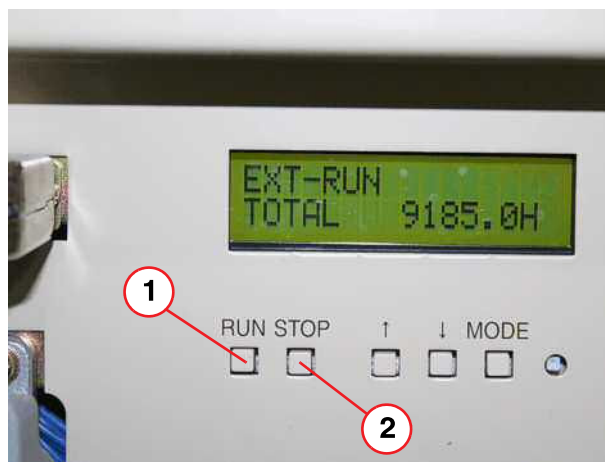
Fig. 133: Compressor He Gas charge & pressure gauge



- (1) Helium gas charge
(2) Helium pressure gauge

- Check the **static helium** pressure on the compressor pressure gauge is **16.0 - 16.5bar** (or 1.60 - 1.65 MPa, or 232-239 psig).
- Charge with helium gas if pressure is lower.
 - Refer to **chapter "Procedures"** - Helium Gas charging.
- Re-fit the **F102 fuse**.
- Check for water flow to the compressor.
- Turn **on** the compressor **main power switch**. (→ 3/ Fig. 123 Page 101)
 - The LCD will read '**EXT-READY**'.

Fig. 134: Compressor control panel LCD display



- (1) Press RUN to start the compressor
(2) Press STOP to switch off the compressor

- **Start** the compressor by pressing the **RUN** button.
 - The LCD will read '**EXT-RUN**'.



The magnet system design requires that the compressor must run in EXT mode only. If the LCD reads 'INT', refer to the manufacturer's Operating Manual.

3.2.4 Final checks

- Wait 20 minutes and check the **operating** (dynamic) helium pressure is **21.0 - 23.0 bar** (or 2.10 - 2.30 MPa or 304 - 333 psig) on the compressor pressure gauge.
- Leak check the flex line connections.
- Check for leakage at the water connections.
- Document the successful measurement in the service report (measured value, and measured equipment) as required by IEC 62353.

3.3 Replacing the Compressor F70

3.3.1 Tools and time required

3.3.1.1 Time

60 min

3.3.1.2 Tools

- Transportation lifting bars (part no. **5137190**).
- Ground test meter (Fluke 1503, or equivalent).
- Standard tool kit.

Fig. 135: Edwards Leak detector



- Use the Edwards hand-held leak detector (part no **10614850**).

This is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.



WARNING

Risk of personal injury and damage to equipment

The weight of the compressor is approx. 100 kg. (220 lbs)

- » When installing or removing from a cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.
- » Moving the unit is a 2 or 4-man operation. **DO NOT ATTEMPT TO MOVE THE UNIT WITHOUT ASSISTANCE. USE THE TRANSPORTATION LIFTING BARS.**

3.3.2 Removal

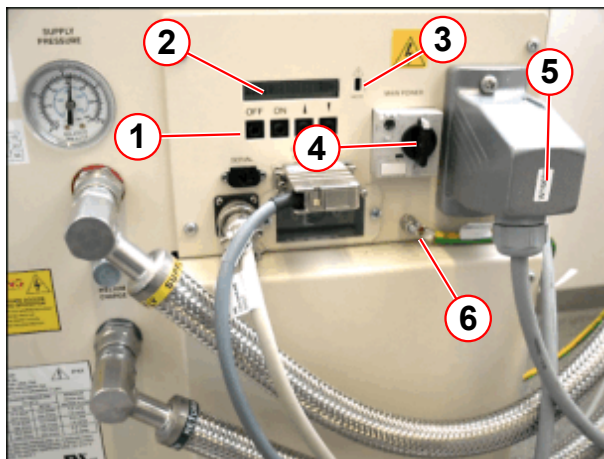
! CAUTION

High-pressure helium gas.

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves, and eye protection must be worn.

Fig. 136: F70 Compressor control buttons

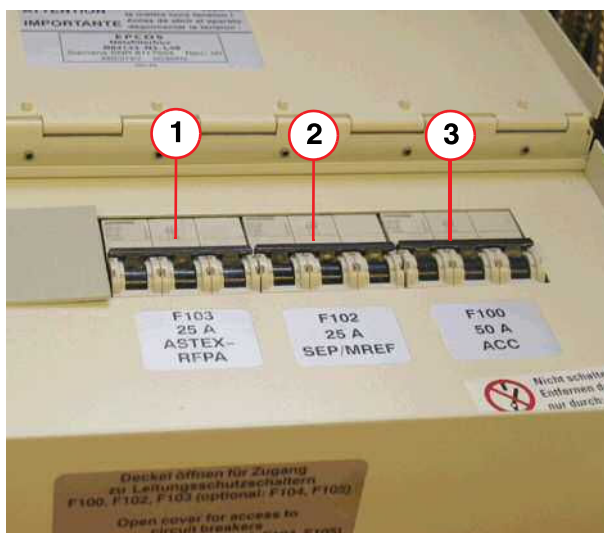


- (1) Control buttons
- (2) LCD display
- (3) Configuration selector switch
- (4) Main power switch
- (5) Mains supply cable
- (6) Ground wire

- Turn **off** the compressor at the **main power switch** (pos 4).

- The **ON/OFF** buttons are disabled in **mode 2**.

Fig. 137: ACC cabinet - Circuit breakers F100, F102, F103



- (1) F103 - isolates RFPA
- (2) F102 - isolates SEP or IFP (stand-alone compressor)
- (3) F100 - isolates ACC cabinet

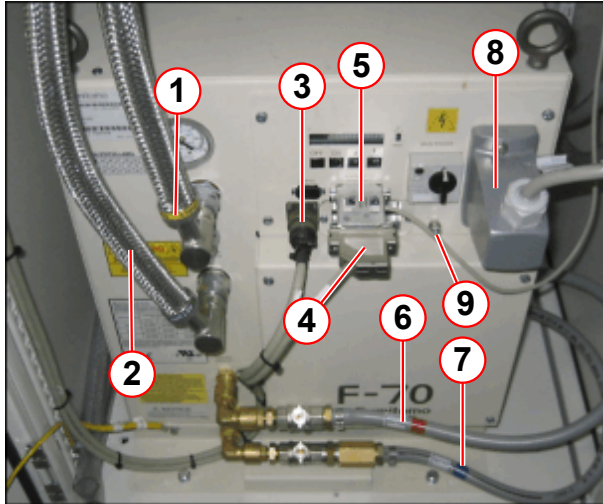
ACC cabinet

- **Isolate power to the compressor:**
 - Switch **OFF** circuit breaker **F102** (2).

- The **alarm box** will display a warning LED and an audible alarm will sound.
- Press the **ACKNOWLEDGE** button to cancel the audible alarm.

3.3.2.1 Disconnecting and removing the compressor

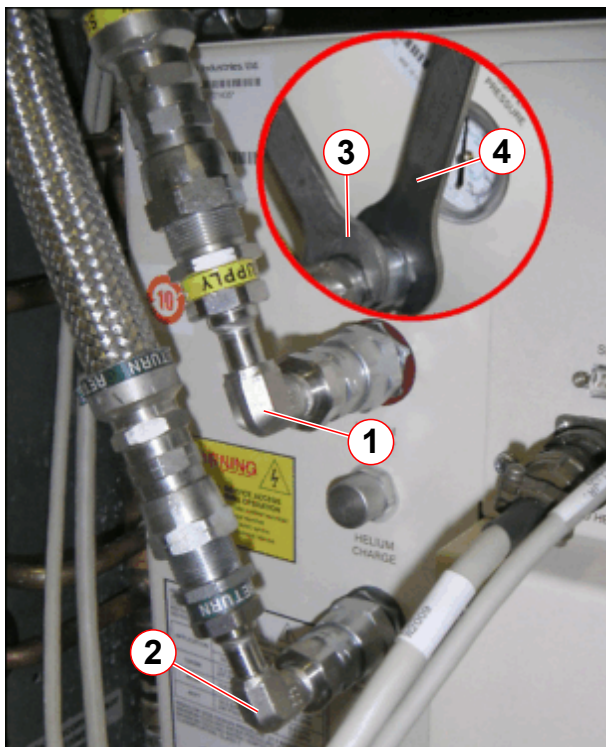
Fig. 138: F70 installation in SEP cabinet



- (1) Helium Supply (high pressure)
- (2) Helium Return (low pressure)
- (3) Cold head cable
- (4) Connector X7 (to ACC A93)
- (5) Interface plug
- (6) Water line (inlet)
- (7) Water line (outlet)
- (8) Mains power connection
- (9) Grounding wire

- The compressor may be:
 - Installed in an **SEP cabinet**.
 - A **stand-alone unit** (with fitted castors).
- The **base** of the compressor is fitted with 2 **nylon slide rails** (prefitted) to ease movement.

Fig. 139: F70 He compressor connections



- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

SEP cabinet option

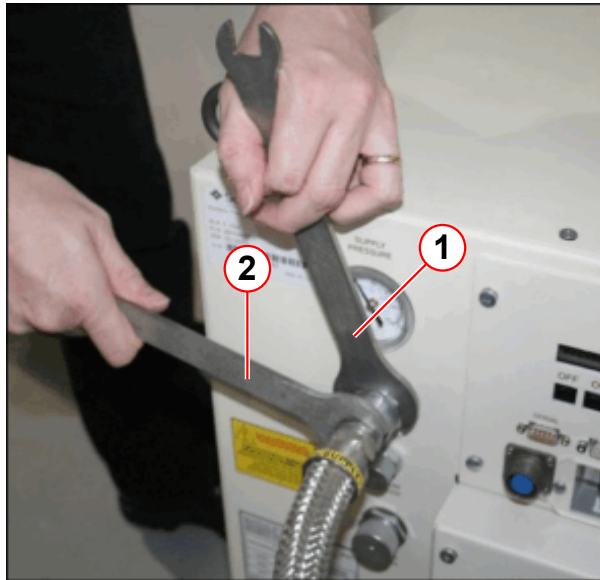
Each helium pressure line is connected to the compressor via a **90 degree elbow**.

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).



90° elbow connections are not required for a stand-alone unit.

Fig. 140: F70 He compressor connections (stand-alone unit)

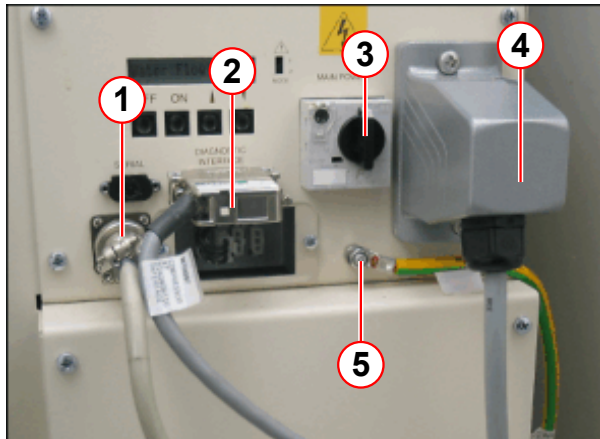


- (1) Turn nut (1 3/16" open-end crowfoot spanner/wrench)
- (2) Hold nut (1" spanner/wrench)

Stand-alone unit

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (2) and **turn** nut (1).

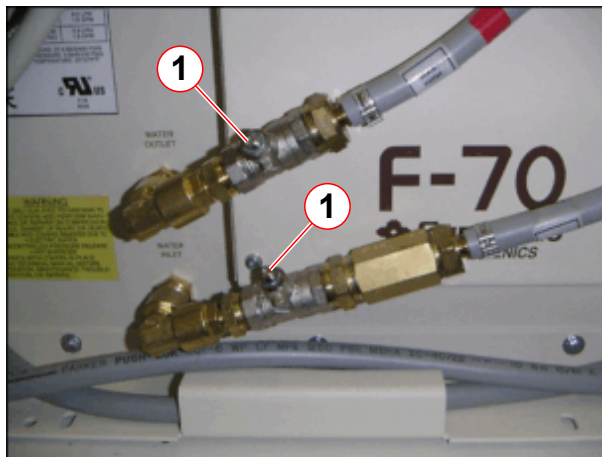
Fig. 141: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- Disconnect the **mains supply**, **cold head motor**, and **MSUP** (DB25 diagnostic interface) cable.
- Disconnect the **ground wire** (5).

Fig. 142: F70 Compressor water inlet/outlet shutoff valves



(1) Shutoff valve

- Close the **shut-off valves** on the **inlet** and **outlet** water lines.
- Disconnect the **water lines** from the compressor.



There is the potential for irreversible damage to the unit if the internal water pipes freeze during shipping.

Before returning a faulty compressor, use a gas line on either the inlet/outlet lines to expel any water.

Fig. 143: F70 compressor - draining cooling water notice



Always **thoroughly drain the coolant** from the cooling circuit if the compressor is to **be shipped** (or stored).

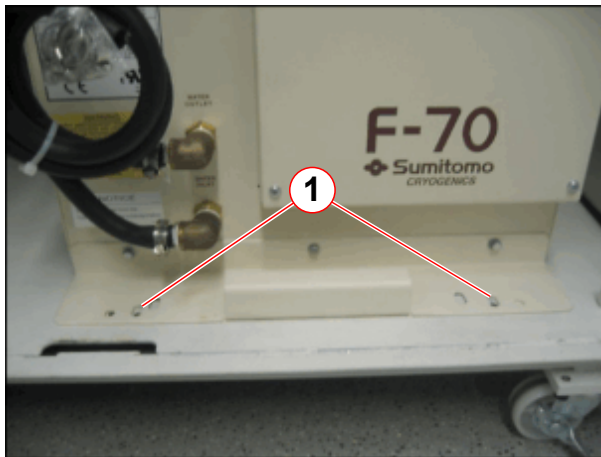
Refer to the disposal instructions for further information.

**CAUTION****Chemical hazard**

If the chilled water contains ethylene glycol, it should be collected in a suitable container while draining.

- » The water should be returned to the water circuit after the new unit is installed.
- » If necessary, dispose of in line with local regulations for hazardous waste.

Fig. 144: F70 Compressor fixing plate



(1) M8 Hex hd screws (2 off)

SEP cabinet

- Remove the 2 M8 hex head screws securing the **fixing plate** to the cabinet.
- Use the **fixing plate handle** to carefully pull the compressor onto a **wooden support 'ramp'**.

The unit is fitted with **nylon slide rails** to simplify movement.

- Use the **transportation lifting bars** when lifting and moving the compressor. (→ Fig. 152 Page 120)

**WARNING****Risk of personal injury and damage to equipment**

The weight of the compressor is 100 kg. (220 lbs)

- » When installing or removing from the cabinet, reduce the height difference by ensuring there is a sturdy support 'ramp' in position to support the unit.
- » Moving the unit is a 2 or 4-man operation. **USE THE LIFTING BARS. DO NOT ATTEMPT TO MOVE THE UNIT WITHOUT ASSISTANCE.**

3.3.3 Installation

3.3.3.1 Unpacking and preinstallation checks



AVOID CONTAMINATION. Do not tip the compressor more than 5 degrees from horizontal to avoid contamination.

When checking the compressor for shipping damage, do not connect gas lines or the cold head. Components may become contaminated with compressor oil.



There is the potential for irreversible damage to the unit if the internal water pipes freeze during shipping.

Before returning a faulty compressor, use a gas line on either the inlet/outlet lines to expel any water.

Before installation, the compressor **packing crate** should be checked for any shipping damage or mishandling.

Fig. 145: F70 Compressor - Packing crate



Fig. 146: F70 Compressor Tip-N-Tell indicator and pressure gauge 'peep hole'



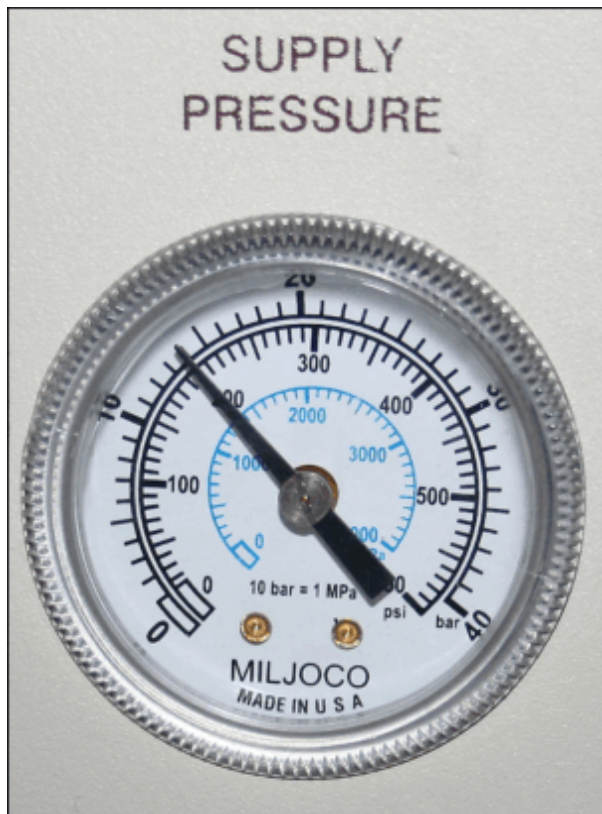
- (1) Tip-N-Tell indicator
- (2) Pressure gauge 'peep hole'

Fig. 147: F70 Compressor Tip-N-Tell indicators



- Inspect the **shipping container 'Tip-N-Tell'** sensor for signs of activation. The sensor turns blue if the unit has been tipped or mishandled during transport.

Fig. 148: F70 Static pressure reading



- Check that the **static pressure is positive**.
 - View the **pressure gauge** through the 'peep hole' on the container. (→ Fig. 146 Page 116)
 - If the gauge indicates **zero pressure** the compressor **cannot** be used. Ask **Siemens CSML** to arrange a replacement compressor.

Tab. 9 F70 static pressure on arrival

Pressure	Ambient temperature
207 - 212 psi (G)	20°C (68°F)
1430 - 1460 kPa	
13.5 - 14.0 bar	

- If the **static pressure** is **positive**, continue with removing the compressor from the packaging crate.
 - Retain all packaging materials for reuse if returning the compressor for replacement or repair.
- Proceed with **internal inspection**.

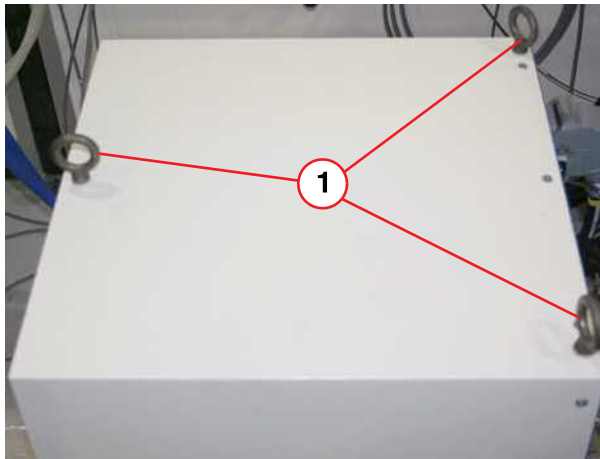
Fig. 149: F70 Compressor - rear panel Tip-N-Tell indicator



(1) Tip-N-Tell indicator

- Inspect the **Tip-N-Tell sensor** mounted on the **compressor** rear panel.
 - If the sensor **has** been activated (the arrow point is blue), and the pressure is within specifications, allow the compressor to stand upright for **2 hours**. Refer to (→ Test running the compressor after mis-handling / Page 127)
 - If the sensor has **not** been activated, continue with the replacement procedure.

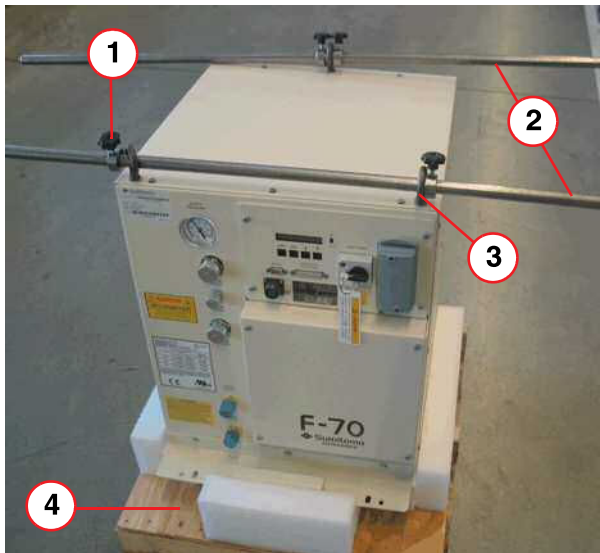
Fig. 150: F70 Lifting eyebolts



(1) Lifting eyebolts (3 positions)

- Fit the 3 lifting **eyebolts**.
- Carefully lift the compressor off the wooden base using the **transportation lifting bars**.

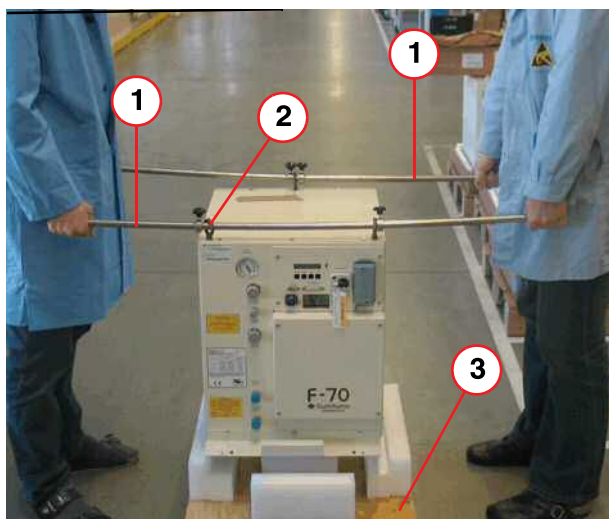
Fig. 151: F70 Fitting the transportation lifting bars



(1) Locking clamps
(2) Transportation lifting bars
(3) Lifting eyes
(4) Wooden pallet

- Align the **lifting eyebolts**.
- Slide the **transportation lifting bars** through the **eyebolts**.
- Position and securely tighten the **locking clamps** to prevent the compressor from sliding.

Fig. 152: F70 Lifting the compressor



- (1) Transportation lifting bars
- (2) Lifting eyes
- (3) Wooden support pallet

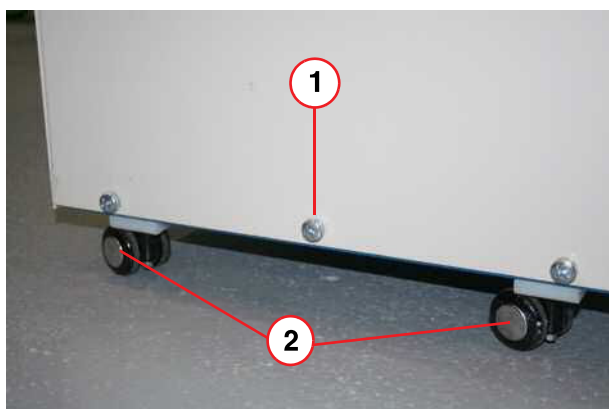
- The compressor can be lifted by 2 or 4 people.



AVOID CONTAMINATION. Do not tip the compressor more than 5 degrees from horizontal to avoid contamination.

3.3.3.2 Installing the new compressor

Fig. 153: F70 Back panel view

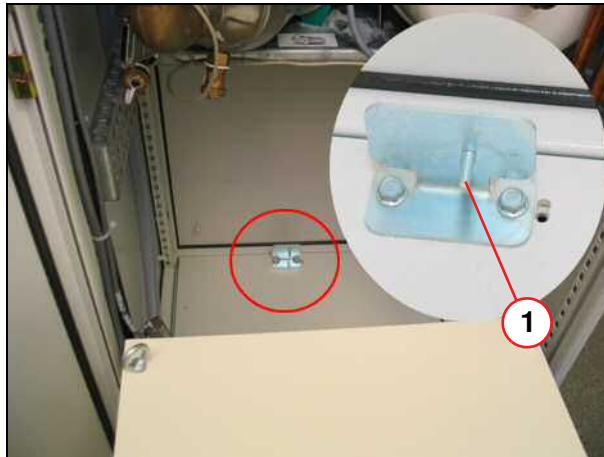


- (1) Center screw is removed for SEP cabinet installation
- (2) Casters x 4 (stand-alone units only)

- The unit is supplied with 4 **castors** (pre-fitted), and **nylon slide rails**.
- For installation into the **SEP cabinet**:
 - Elevate and support the compressor base. **Remove the 4 castors** using a 7mm open-ended wrench/spanner.
 - **Remove** the center **screw** (pos 1). (The screw hole will accommodate the **guide pin** at the back of the SEP cabinet).

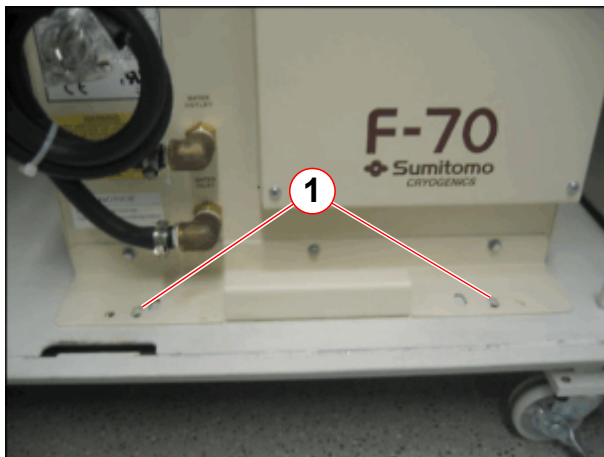
- Use the **transportation bars** to move the compressor onto the **supporting ramp** and carefully slide the new compressor into the cabinet.

Fig. 154: SEP - Compressor guide pin



- Place the (center) screw hole on the back panel of the compressor around the **guide pin** at the back of the cabinet.

Fig. 155: F70 Compressor fixing plate



- Secure the **fixing plate** to the cabinet using the 2 M8 hex head screws.

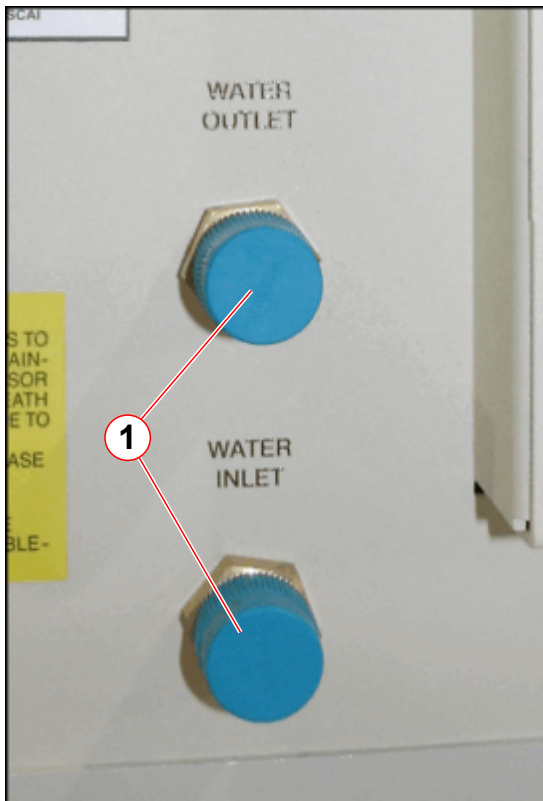
(1) M8 Hex hd screws (2 off)

3.3.3.3 Test running the compressor before start-up

The new compressor must be operated for **10 minutes** to clean up any oil migration incurred during transportation **before** the **helium lines** and **cold head** are connected. For this test, only the compressor **water** and **power** connections are required.

Connect water and power

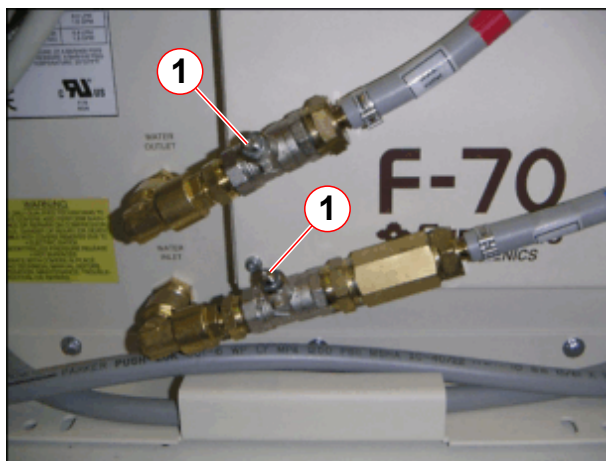
Fig. 156: F70 Water line protective caps



(1) Protective caps

- **Remove** the protective caps from the **water in** and **water out** fixings.

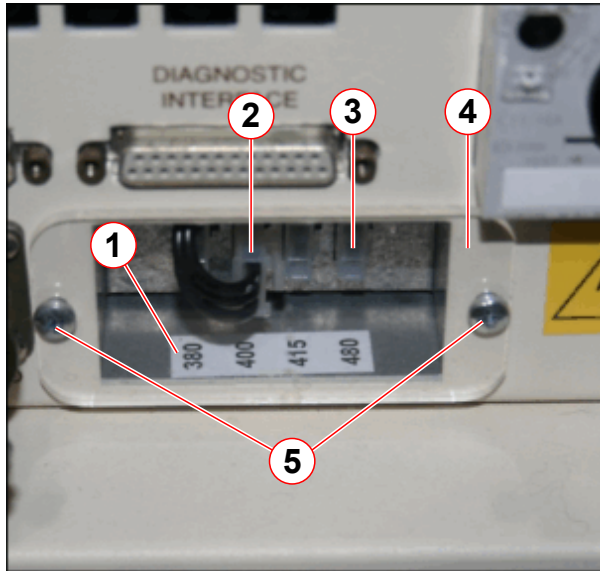
Fig. 157: F70 Compressor water inlet/outlet shutoff valves



(1) Shutoff valve

- **Connect** the **water lines** to the compressor.
 - Ensure the **water in** is connected to the **water supply line**, and the water is turned **on**.
 - **Open** the **shut-off valves**, turn on the coolant supply, and check for leaks.

Fig. 158: F70 Compressor voltage selection access panel



- (1) Voltage label
- (2) Voltage selection connectors
- (3) Voltage tap
- (4) Clear plastic cover
- (5) Screws (2 off)

- Check that the **transformer voltage taps** are correctly selected to match the supply voltage.
- If required, remove plastic cover (3), reposition the tap, and replace the cover.



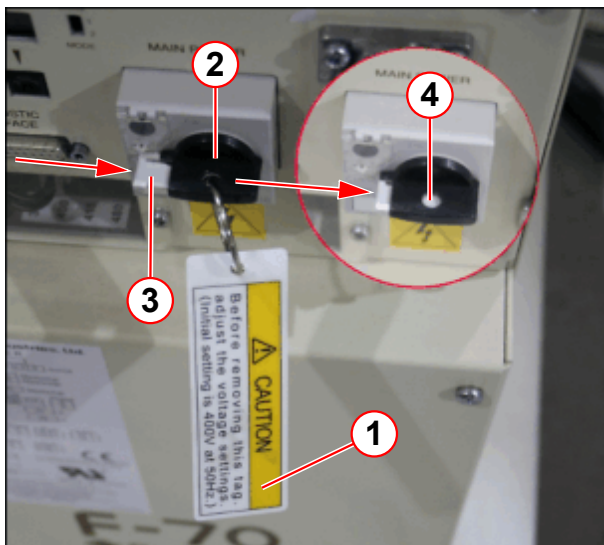
The unit is configured in the factory to 400 V/3~50 Hz.

If the compressor is not wired correctly, connecting to a higher voltage may damage the control circuit.



Check that the transformer voltage tap is correctly selected.

Fig. 159: F70 Main power switch - locking device



- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- Remove the **voltage caution label** (1) (fitted through the main circuit breaker).
- Engage the **white switch locking device** fully **before** attempting to turn on.
- Push in the white locking device fully.
 - » The **device** should be **visible** through the **blocking hole** (pos. 4).



Ensure that the white switch locking device is **FULLY** engaged. **THE SWITCH WILL BREAK IF FORCE IS USED.**

The compressor will become unusable and will have to be replaced.

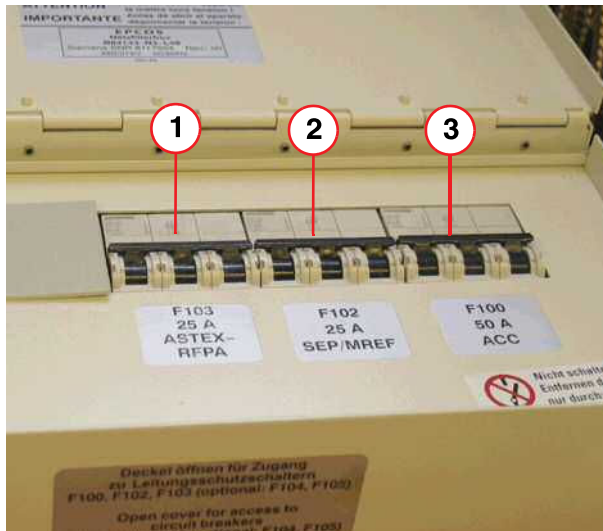
- Connect the **main power supply cable**.

Fig. 160: F70 Main power supply connection



- (1) Pull cover to remove
- (2) Connect power supply cable
- (3) Fasten the cover by attaching the screws

Fig. 161: ACC cabinet - Circuit breakers F100, F102, F103



- (1) F103 - isolates RFPA
- (2) F102 - isolates SEP or IFP (stand-alone compressor)
- (3) F100 - isolates ACC cabinet

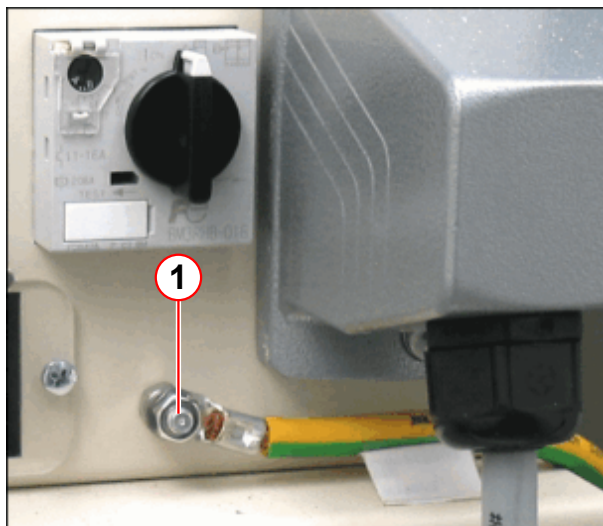
ACC cabinet

- Switch **ON** circuit breakers F102 (2).

Compressor protective (ground) earth connection

A **protective ground connection test** must be performed as a final step **before** switching on the compressor.

Fig. 162: Compressor ground wire connection



- (1) Ground wire connection

- Connect the **ground wire** (1) to the helium **compressor** terminal.

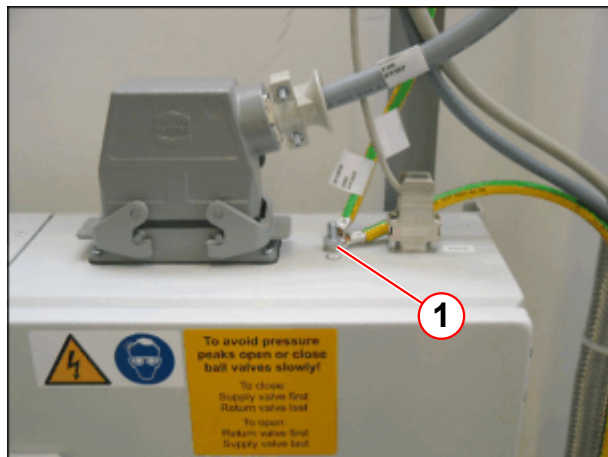
Test the protective earth connection

- Measure the **protective earth connection** resistance $\leq 100 \text{ m}\Omega$ with a **ground test meter** (Fluke 1503, or equivalent), between the **protective earth connector** on

the compressor unit, and the **protective earth wire** connected to the **top of the SEP** (on the electronic box of the IFP).

Stand-alone option

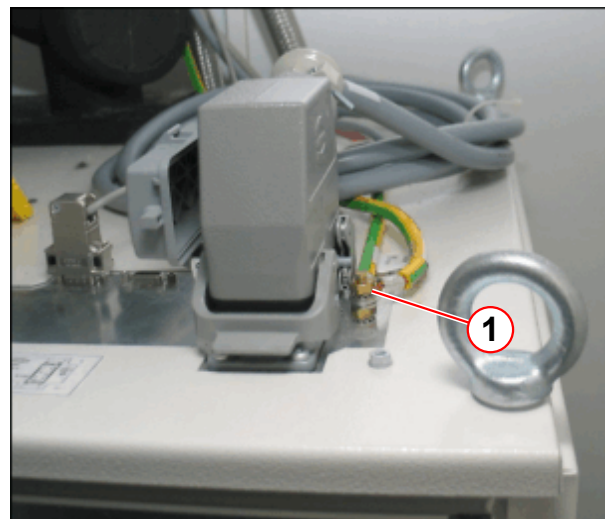
Fig. 163: IFP ground wire (optional)



(1) Ground wire

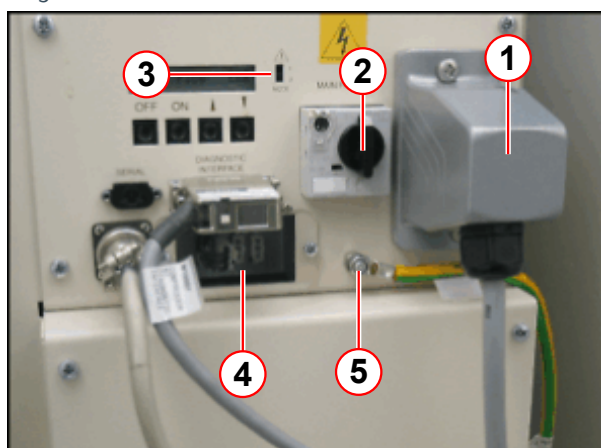
SEP option

Fig. 164: SEP ground wire (optional)



(1) Ground wire

Fig. 165: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Ensure that the **main power** switch is **OFF** before adjusting the selector switch.
- Check the configuration mode selector switch is set to **1**.

Use a flat-blade screwdriver to move the switch up and down.

- » The **ON/OFF** buttons will only work in configuration mode **1**.
- » The compressor must **NOT** be run in standby mode in configuration **2**.

In configuration **2**, both **ON** and **OFF** buttons are **disabled**. Operation is controlled remotely via the **DB25 diagnostic interface**.

- Turn the **main power** switch **ON**.
- Press the **ON** button to start the compressor. The display will read **RUN**.
- Check the **temperature** of the water supply (water in) and return (water out) lines:
 - » The **return** (water out) line should be **warmer** than the **supply** (water in).
 - » **Stop** the compressor and reverse water lines if the **return** line is **cooler**.
- Run the compressor for **10 minutes** and then press the **OFF** button.

- Turn **OFF** the **main power** switch.
- Reset the **configuration selector switch** to **2**.
- Switch **OFF F102** in the ACC cabinet.



DO NOT change the configuration selector switch position AFTER the main power has been switched ON. Loss of control and incorrect indications will result.



The compressor is protected from running in reverse. If it fails to start and the LCD displays 'Phase Seq - ERR'.



If the compressor starts but does not build pressure, TURN OFF IMMEDIATELY. Contact Siemens Magnet HSC.

3.3.3.4 Test running the compressor after mishandling



AVOID CONTAMINATION. Do not tip the compressor more than 5 degrees from horizontal to avoid contamination.

When checking the compressor for shipping damage, do not connect gas lines or the cold head. Components may become contaminated with compressor oil.

Before installation and connection, the **compressor** packing crate/SEP cabinet must be inspected for any **shipping damage** or **mishandling**.

This procedure should be followed if **either**:

1. The **SEP cabinet 'Tip-N-Tell'** sensor has been activated.

2. A stand-alone and/or replacement compressor 'Tip-N-Tell' sensor has been activated.

Fig. 166: F70 Compressor Tip-N-Tell indicators



The sensor turns blue if the unit has been tipped or mishandled during transport.

Fig. 167: F70 Static pressure reading

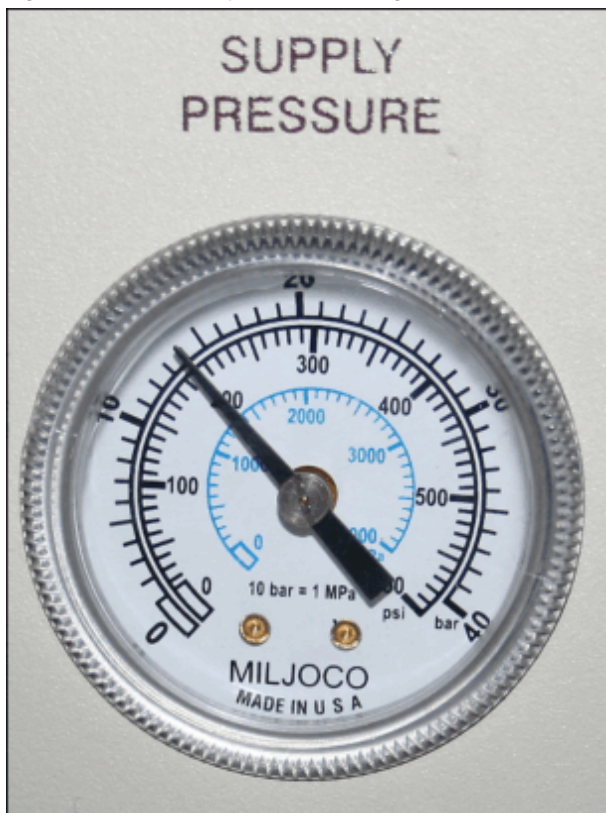


Fig. 168: F70 Compressor Tip-N-Tell indicator and pressure gauge 'peep hole'



- (1) Tip-N-Tell indicator
(2) Pressure gauge 'peep hole'

- Check that the **static pressure** is **positive**.
 - View the **pressure gauge** through the 'peep hole' on the container.
(→ 2/Fig. 168 Page 128)

Tab. 10 F70 static pressure on arrival

Pressure	Ambient temperature
207 - 212 psi (G)	20°C (68°F)
1430 - 1460 kPa	
13.5 - 14.0 bar	

Pressure within specification:

- If the compressor has been tipped, allow it to stand upright for **2 HOURS** before performing the following **test**.

Pressure outside specification:

- Notify the delivering carrier of the damage if:
 - The compressor has been on its side or upside down for an **extended period of time** (more than one hour).
 - There is physical damage to the unit.

NOTICE

If the compressor arrives at **ZERO BAR** - **DO NOT CONNECT THE HE PRESSURE LINES or COLD HEAD POWER CABLE** to the compressor.

Connecting the lines to the compressor at zero bar may cause contamination of the whole refrigeration system.

- » Check the compressor pressure before attempting to connect the He pressure lines.
- » Contact Siemens Magnet HSC for advice.

3.3.3.5 Performing the test

To clean up any oil migration incurred during transportation, **test run** the compressor for a **MINIMUM of 2 hours**.

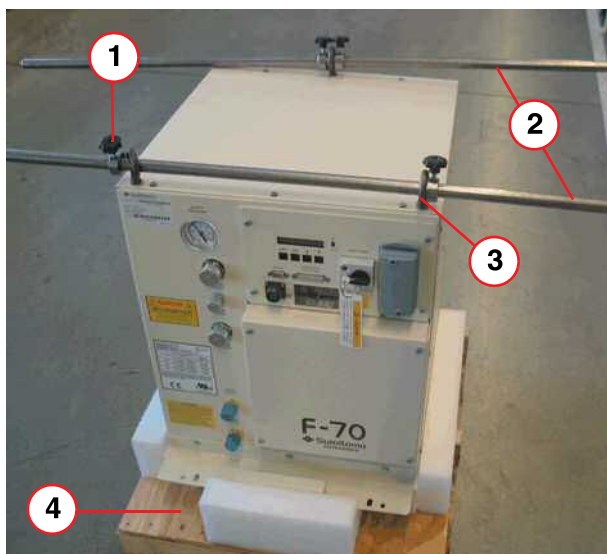
For this test, only compressor **water** and **power** connections are required. **Ensure all other lines/cable connections are disconnected from the compressor.**

Install the **replacement** compressor into the **SEP cabinet** before performing the test.

F70 stand-alone and/or replacement units

- **Remove** the compressor from the **packing crate**.
 - **Retain** all packaging materials for reuse if returning the compressor for replacement or repair.

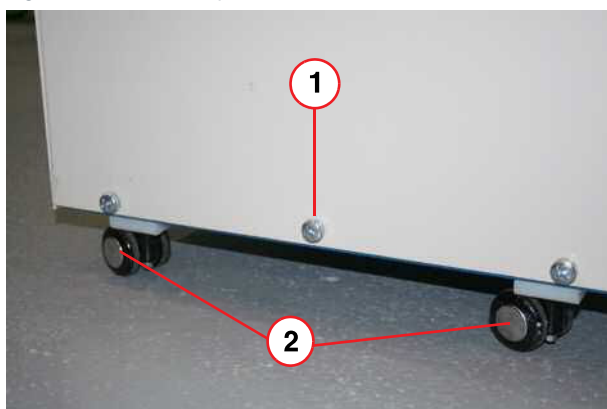
Fig. 169: F70 Fitting the transportation lifting bars



- (1) Locking clamps
- (2) Transportation lifting bars
- (3) Lifting eyes
- (4) Wooden pallet

- Fit and align the **lifting eyebolts**.
- Slide the **transportation lifting bars** through the eyebolts.
- **Position** and securely tighten the **locking clamps** to prevent the compressor from sliding.
- The compressor can be lifted by 2 or 4 people.

Fig. 170: F70 Back panel view

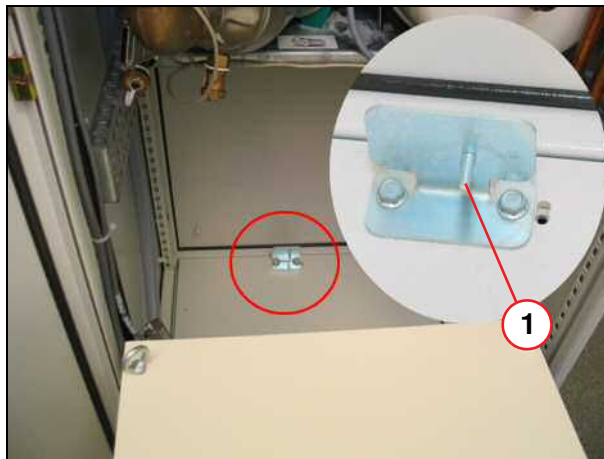


- (1) Center screw is removed for SEP cabinet installation
- (2) Casters x 4 (stand-alone units only)

- The unit is supplied with 4 **castors** (pre-fitted), and **nylon slide rails**.
- The **castors** must be **removed** before installation into a **SEP cabinet**.
 - Elevate and support the compressor base. **Remove the 4 castors** using a 7mm open-ended wrench/spanner.
 - **Remove the center screw** (pos 1). (The **screw hole** will accommodate the **guide pin** at the **back** of the **SEP cabinet**.)

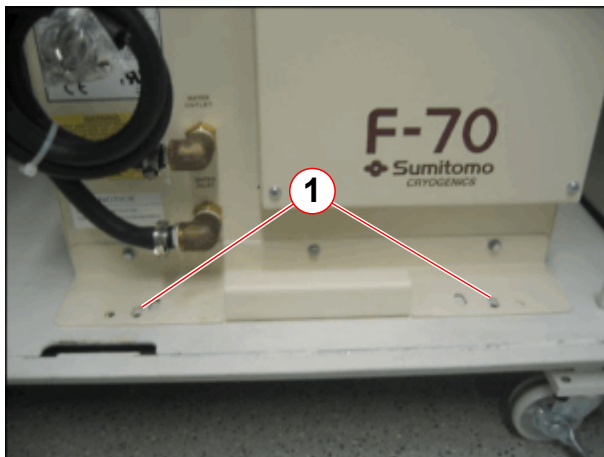
Installing the F70 (replacement) in the SEP cabinet

Fig. 171: SEP - Compressor guide pin



- Use the **transportation bars** to move the compressor onto a **supporting ramp**, and carefully slide the new compressor into the **cabinet**.
- Place the (center) **screw hole** on the back panel of the compressor around the **guide pin** at the back of the cabinet.

Fig. 172: F70 Compressor fixing plate

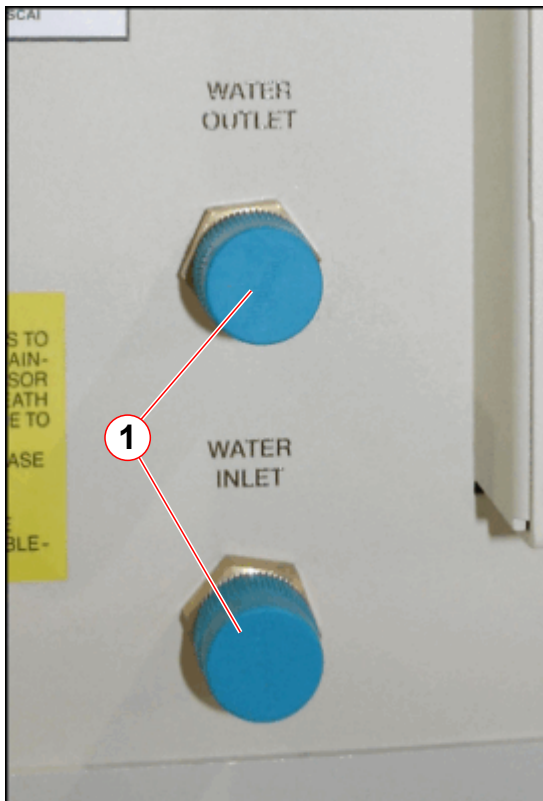


- Secure the **fixing plate** to the cabinet using the 2 M8 hex head screws.

(1) M8 Hex hd screws (2 off)

Connect water and power

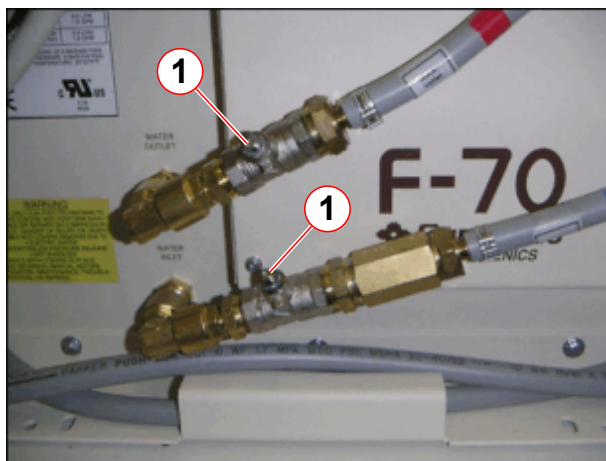
Fig. 173: F70 Water line protective caps



(1) Protective caps

- Remove the **protective caps** from the water in/out fixings.

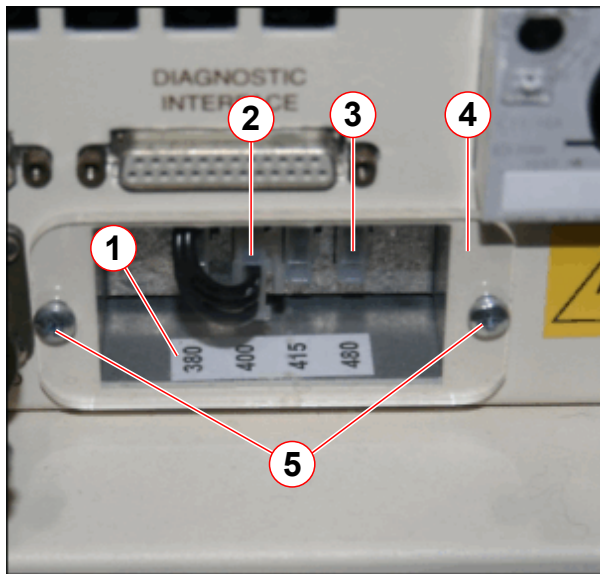
Fig. 174: F70 Compressor water inlet/outlet shutoff valves



(1) Shutoff valve

- Connect the **water lines** to the compressor.
 - Ensure that the water is connected to the **water supply line** and that the water is turned **on**.
 - **Open the shut-off valves**, turn on the coolant supply, and **check for leaks**.

Fig. 175: F70 Compressor voltage selection access panel



- (1) Voltage label
- (2) Voltage selection connectors
- (3) Voltage tap
- (4) Clear plastic cover
- (5) Screws (2 off)

- Check that the **transformer voltage taps** are correctly selected to match the supply voltage.
- If required, remove **plastic cover(3)**, **reposition** the tap, and **replace** the cover.



The unit is configured in the factory to 400 V/3~50 Hz.

If the compressor is not wired correctly, connecting to a higher voltage may damage the control circuit.



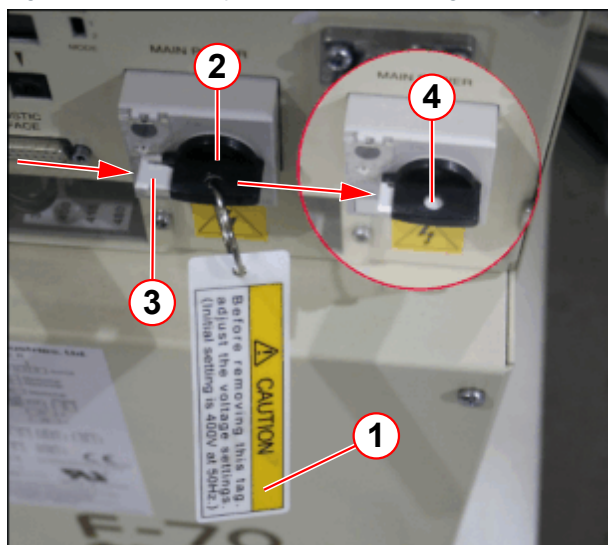
Ensure that the white switch locking device is FULLY engaged. THE SWITCH WILL BREAK IF FORCE IS USED.

The compressor will become unusable and will have to be replaced.



Check that the transformer voltage tap is correctly selected.

Fig. 176: F70 Main power switch - locking device



- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- Remove the **voltage caution label** (1) (fitted through the main circuit breaker).
- Engage the **white switch locking device** fully **before** attempting to turn on.
- Push in the white locking device fully.
 - » The **device** should be **visible** through the **blocking hole** (pos. 4).

- **Connect the mains power supply cable.**

Fig. 177: F70 Main power supply connection



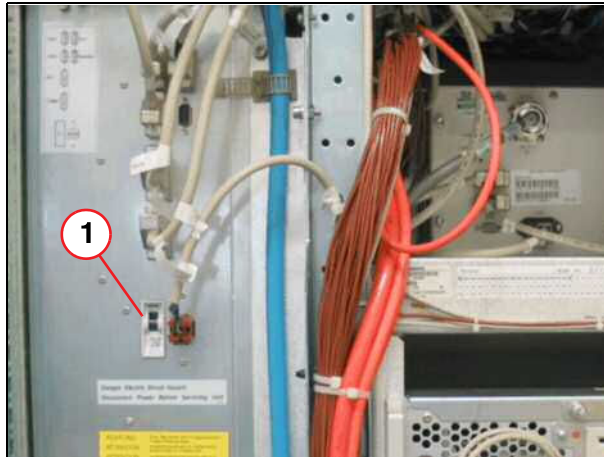
- (1) Pull cover to remove
- (2) Connect power supply cable
- (3) Fasten the cover by attaching the screws

F70 in SEP cabinet

- Ensure that the **SEP cabinet** is connected to the **water** supply and return and that the water is turned **on**.
- Ensure the compressor **water shut-off valves** are **open** and water is flowing through the compressor.
- Remove the **voltage caution label** (fitted through the main circuit breaker).
 - Engage the **white switch locking device** fully **before** attempting to **turn on**. (→ Fig. 176 Page 134)
- Perform the **ground connection test**. (→ Compressor protective (ground) earth connection / Page 125)

Switching on the compressor via the ACC cabinet

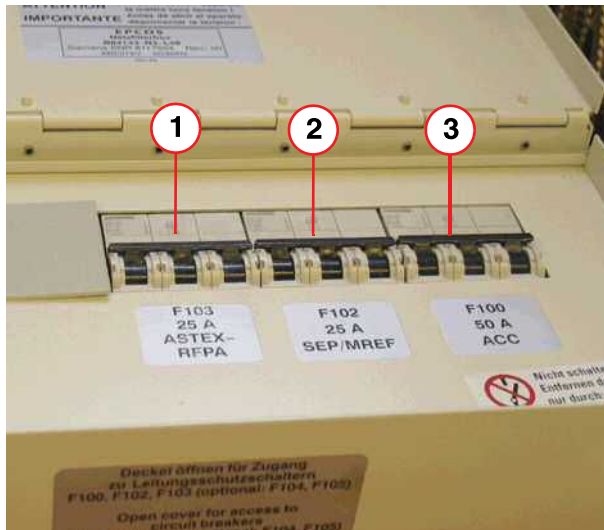
Fig. 178: Fuse F1 in the ACC



(1) F1

- Check that **circuit breaker F1 (1)** is switched **on**.

Fig. 179: ACC cabinet - Circuit breakers F100, F102, F103

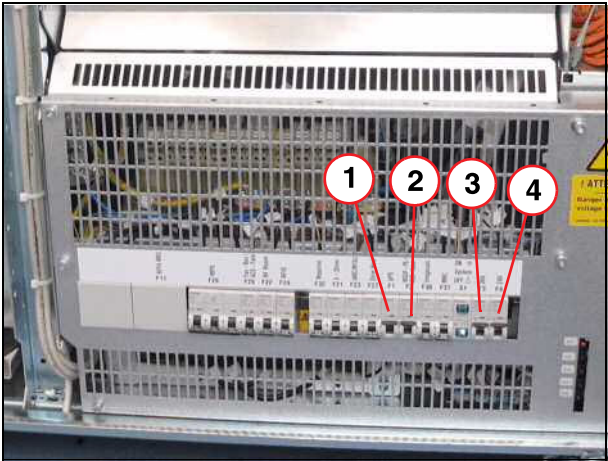


- (1) F103 - isolates RFPA
- (2) F102 - isolates SEP or IFP (stand-alone compressor)
- (3) F100 - isolates ACC cabinet

ACC cabinet

- Switch **ON** circuit breaker **F100 (3)**.
- Switch **ON** circuit breaker **F102 (2)**.

Fig. 180: ACC fuse (inside)



- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

ACC cabinet

- Switch ON circuit breakers **F1**, **F2**, **F3**, and **F4**.

Setting the configuration selector switch

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Configured in the factory for configuration 1.

Before the compressor’s **main power switch** is switched on, the **configuration mode selector switch** must be set to either configuration **1** or configuration **2**.
The **ON** and **OFF** buttons are **disabled** in configuration mode **2**, as the compressor is controlled via the pressure within the magnet.

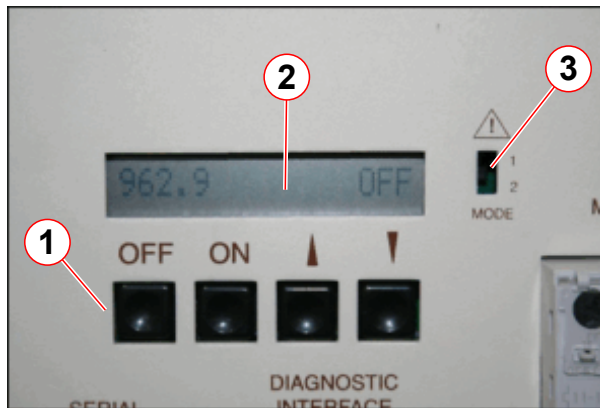
Tab. 11 Configuration mode switch

Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10 K shield cooling	Enabled
Mode 2	DOWN	4 K recondensing	Disabled

i

For this test, the configuration mode selector switch must be set to 1 (ON/OFF buttons enabled).

Fig. 181: F70 System status display



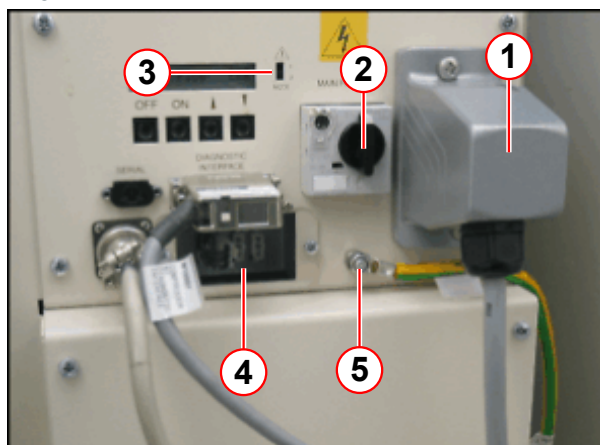
- (1) Control buttons
- (2) LCD display
- (3) Configuration selector switch

- Use a flat-blade screwdriver to move the switch (3) up or down.
 - Switch **up** = **configuration 1**.
 - Switch **down** = **configuration 2**.



Do not change the switch position after the main power has been switched on.

Fig. 182: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- **Ensure that the main power switch is OFF before adjusting the selector switch.**
- Check the configuration mode selector switch is set to **1**.
Use a flat-blade screwdriver to move the switch up and down.
 - » The **ON/OFF** buttons will only work in configuration mode **1**.
 - » The compressor must **NOT** be run in standby mode in configuration **2**.
In configuration **2**, both **ON** and **OFF** buttons are **disabled**. Operation is controlled remotely via the **DB25 diagnostic interface**.
- Turn the **main power switch ON**.
- Press the **ON** button to start the compressor. The display will read **RUN**.

- **Touch** the water supply (water in) and return (water out) lines.
 - » The **return** (water out) line should be **warmer** than the **supply** (water in).
- **Stop** the compressor and reverse the water lines if the **return** line is **cooler**.

- **Run** the compressor for a **MINIMUM of 2 HOURS**.



If the compressor shuts down during the test run, contact Siemens Magnet HSC.

- If there are **no problems** during the test run, stop and isolate the compressor.
 - **Press** the **OFF** button.
 - **Turn OFF** the **main power** switch.
 - Switch **OFF F100** in the ACC cabinet.
 - **The test run is complete.**
 - **Reset** the **configuration selector switch** to **2**.



DO NOT change the configuration selector switch position AFTER the main power has been switched ON. Loss of control and incorrect indications will result.



The compressor is protected from running in reverse. If it fails to start and the LCD displays 'Phase Seq - ERR'.



If the compressor starts but does not build pressure, **TURN OFF IMMEDIATELY**. Contact Siemens Magnet HSC.

3.3.4 Startup

3.3.4.1 Installation - connecting and starting the compressor



CAUTION

High-pressure helium gas

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves, and eye protection must be worn.

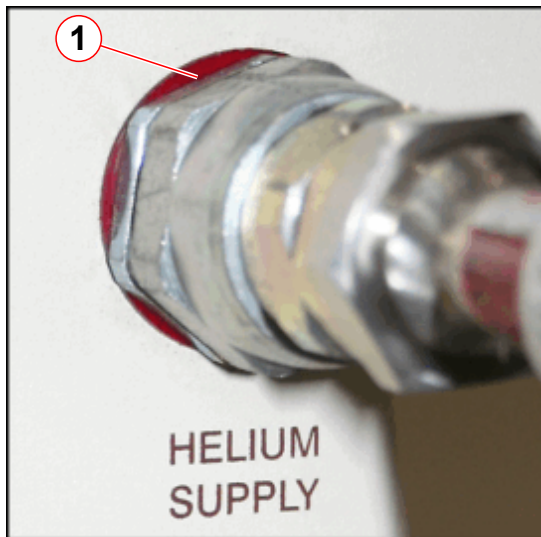
Helium gas lines

Do not confuse the **supply** (HP) and **return** (LP) lines when connecting them. The 4K helium lines are color-coded as shown in the table below.

Tab. 12 Gas line color connections

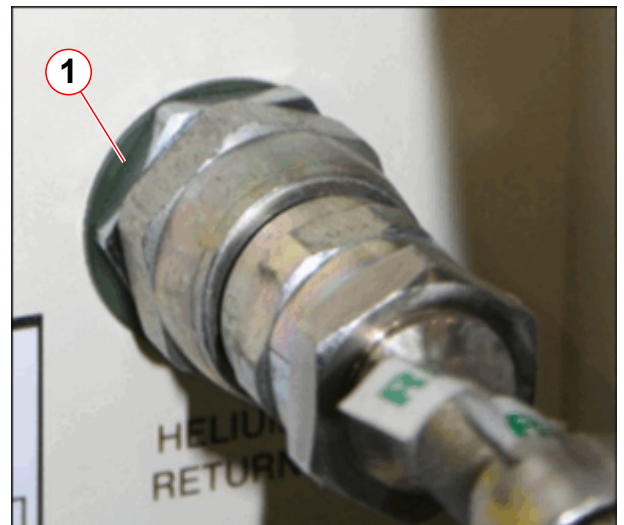
System	Gas line color	F70 He connection
4K supply (HP)	Yellow	Red
4K return (LP)	Green	Green

Fig. 183: F70 Helium supply connections - color coding



(1) Helium supply = RED

Fig. 184: F70 Helium supply connections - color coding



(1) Helium return = GREEN

Fig. 185: F70 Helium gas coupling- protective cap



(1) Protective cap

- **Remove** the protective transit covers from the **return** and **supply** connections.

Fig. 186: F70 High-pressure Aeroquip coupling



- (1) Locknut
(2) Flat rubber gasket

- Check the **rubber gasket** for dirt or dust and ensure it is correctly fitted.
NOTE: An incorrect fit may cause a helium gas leak.



Please clean surface of quick couplings.

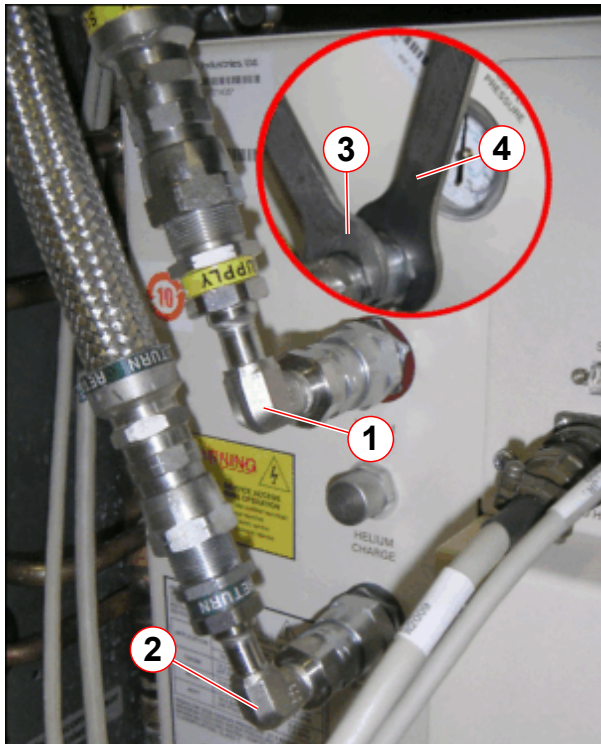
Clean connector with clean paper towel before connecting the lines.



Ensure a good seal of connectors.

Before connecting lines ensure the o ring is present in the connector.

Fig. 187: F70 He compressor connections



- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

SEP cabinet option

Each **helium pressure line** is connected to the compressor via a **90 degree elbow**.

- Fit the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).
- **Tighten down** all couplings as far as possible, and then turn them back by 1/4 turn to **relieve strain**.

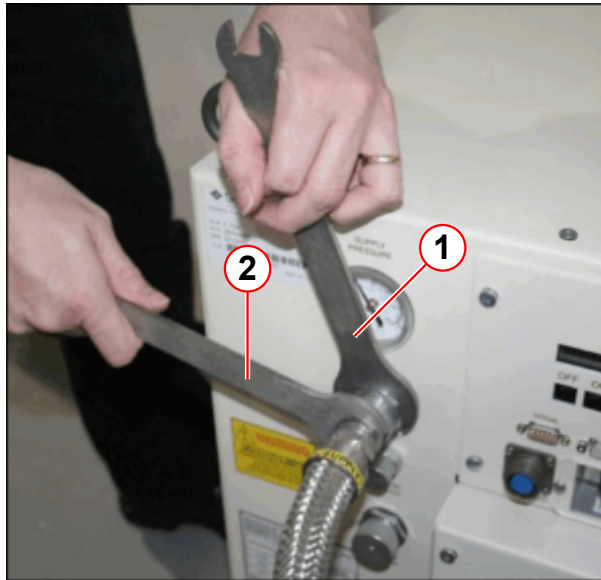


When connecting the He pressure lines, **DO NOT OVER-TIGHTEN** the couplings.



90° elbow connections are not required for a stand-alone unit.

Fig. 188: F70 He compressor connections (stand-alone unit)



- (1) Turn nut (1 3/16" open-end crowfoot spanner/wrench)
- (2) Hold nut (1" spanner/wrench)

Stand-alone unit

- **Fit** the self-sealing helium pressure lines in the following order:-
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (1) and **turn** nut (2).
- **Tighten down** all couplings as far as possible, and then turn them back by 1/4 turn to **relieve strain**.

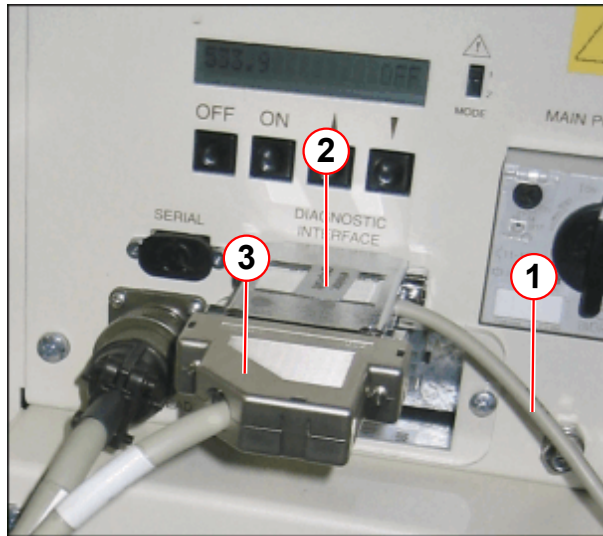
Fig. 189: F70 Cold head cable connection



- (1) Cold head cable (W2009)

- **Connect** the cold head power cable W2009 (1).

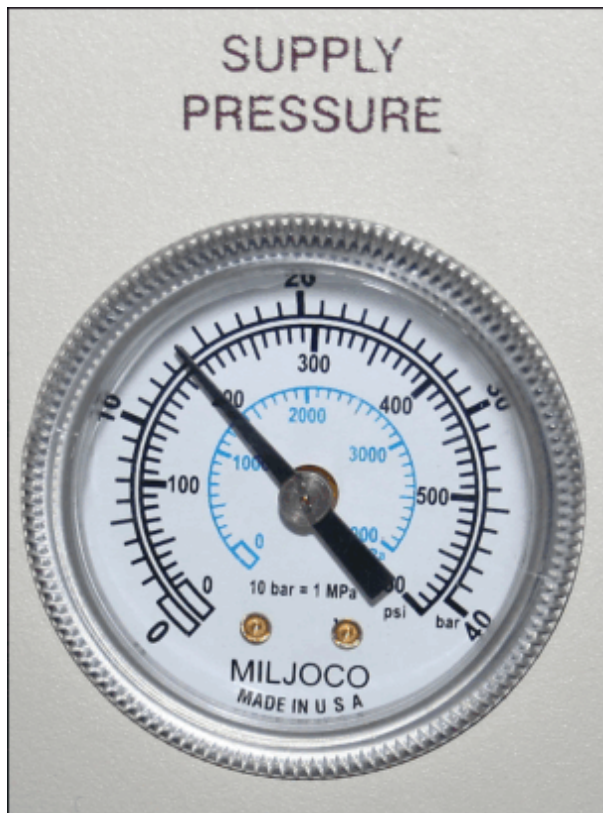
Fig. 190: F70 Diagnostic interface connector



- (1) Interface plug cable (compressor to SEP cabinet)
- (2) Interface plug (removed at newer systems)
- (3) Connector X7 (compressor to ACC via diagnostic cable W2007)

- Connect the **MSUP DB25 diagnostic interface (2)**.

Fig. 191: F70 Static pressure reading



- Check the **static helium pressure** on the compressor pressure gauge.
 - Refer to (→ Tab. 13 Page 144).
- Charge with helium gas if pressure is lower.
 - Refer to **Chapter Procedures - Helium Gas charging**.

Equalization (static) pressure at 20 °C (68 °F) for 12 to 20 meter long gas lines:

Tab. 13 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

- Check the **configuration switch** is set to **mode 2** (→ 3/Fig. 182 Page 137).
- Switch **ON F102** (ACC cabinet) (→ 2/Fig. 179 Page 135).
- Check for **water flow** to the compressor.
- Switch **ON** the **main power** switch (→ 3/Fig. 141 Page 113).
- Check the **alarm box** status and clear any faults.

3.3.5 Configuration for Mobile use

If the compressor is to be installed in a mobile trailer system¹:-

1. Turn Main Power switch **OFF**.
2. Press and hold down all four control buttons to enter '**Service Level**' mode. Keep buttons held and turn Main Power switch **ON**. Release the buttons.
3. Scroll until the display shows **Mobil?Y-ON N-OFF**.
4. Press the **ON** control button to select mobile use.
5. Turn Main Power switch **OFF** to leave service level.

To revert to static use, repeat steps 1 - 3 and press the **OFF** control button to de-select mobile option.

3.3.6 Final checks

- Wait **20 minutes** and check the operating (**dynamic**) helium pressure on the compressor pressure gauge.
 - - **Supply: 21-22 bar** (304-319 psi (G))



If the dynamic pressure is outside of this range, charge the compressor with helium gas.

- **Check** the helium line connections for **leaks**.
- **Check** for leakage at the **water connections**.
- **Document** the successful measurement in the **service report** (measured value, and measured equipment) **as required by IEC 62353**.

1. The mobile configuration is only a feature on early compressors.

3.4 Replacing the F70 Compressor with Tooling kit

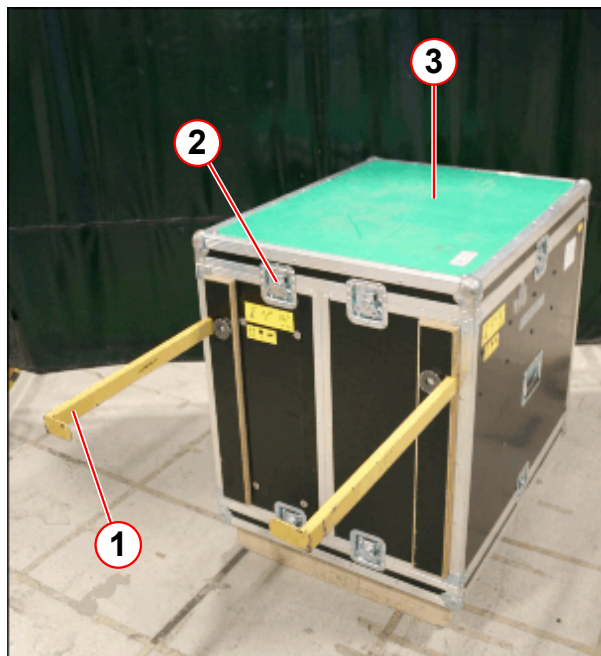
3.4.1 General

This procedure describes the replacement of the compressor unit when using the replacement tooling and transport kit (material number **11060700**)

The replacement compressor unit is shipped in a specially designed wooden transport box. The transport box contains the tooling and accessories required for a safe replacement of the compressor unit:

- It allows for the safe transport and replacement of the compressor unit
- The transport box allows 1 person to pack a damaged compressor unit.
- The shipping crate is suitable for worldwide transport and complies to shipping regulations IEC 60721-3-2.

Fig. 192: Transport box



- (1) Lifting handles (x 2)
- (2) top lid latch (x 6)
- (3) Lid with PTFE sheet covering

The transport box is fitted with 2 x lifting handles to allow movement. Built-in wheels allow movement over a short distance without requiring auxiliary lifting tools.



If necessary, the transport box can also be moved using a pallet truck.

3.4.1.1 Prerequisites

The document refers to compressor units identified by material no:

- 10121435 - Sumitomo F70
- 07461119 - compressor unit M/N HC-10
- 07758191 - SHI Compact compressor unit CSW71TW

- 07014827 - Robust Coolpack
- 04296518 - RCA - compressor unit Coolpak 6000 (MKI)
- 07120913 - Helium compressor unit. Coolpack. MKII

3.4.2 Safety Information

Carefully read the Operator Manual supplied in the transport box prior to using the tooling for the first time!

- The Operator Manual contains additional Safety Information. As well as general safety, transport, installation, operation, and maintenance.
- Also observe the special Safety Information and Notes included in the chapters!

NOTICE

Tripping hazard

Tripping hazard caused by edges, thresholds, cables and lines when moving the compressor unit on the casters.

- » Check for hazards before moving the compressor unit.



CAUTION

Unsafe lifting and movement of the tooling

Personal injury or equipment damage if safety procedures are not followed.

- » It is not permitted to lift or lower loads on inclines!
- » Do not park the shipping crate or tooling on inclines.
- » Only lift compressor units that the tooling is designed for.



CAUTION

Personal injury

When performing the replacement procedure personal injury can occur.

- » Safety shoes must be worn.
- » Wear protective gloves and suitable clothing when handling the equipment.
- » The operator must ensure that non-authorized personnel do not use the tooling.

**CAUTION**

High-pressure helium gas.

Follow the Safety Information provided to avoid injury.

- » Protective clothing, gloves, and eye protection must be worn.

NOTICE

Risk of injury and damage to equipment

The weight of the compressor unit is 100 kg. (220 lbs)

- » Use the supplied lifting equipment provided in the transport box.
- » Assemble the lifting equipment correctly to the instructions provided.

3.4.3 Completion Time and Number of Persons

Tab. 14 Completion time

Work steps	Time	Persons
Preparation	n.a.	1
Replacement	90 min	1
Startup/Tune-up/QA	n.a.	1
Final steps	30 min	1

3.4.4 Tools

- Transportation lifting kit (material number **11060700**, F70 compressor kit).
- 14-mm and 17-mm open-ended wrench.
- 17-mm socket.
- Torque wrench (0–20 Nm).
- 8-mm Allen (Hex) key.

- 7-mm socket and ratchet.

Fig. 193: Edwards Leak detector



- Use the Edwards hand-held leak detector (material no **10614850**).

The Edwards is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

3.4.5 Disconnecting and Removing the Compressor Unit



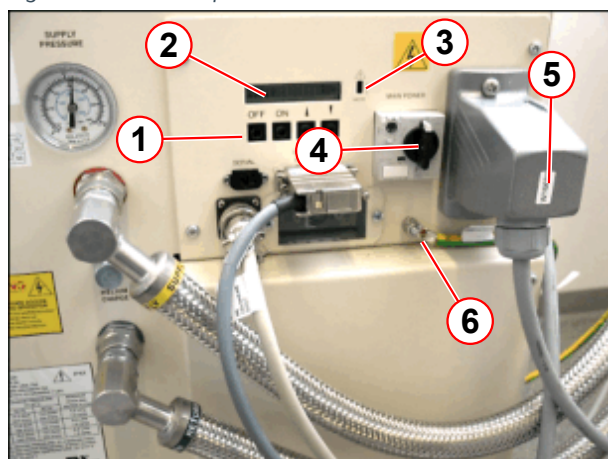
CAUTION

High-pressure helium gas.

Failure to observe the following Safety Information may result in injury.

- » Protective clothing, gloves, and eye protection must be worn.

Fig. 194: F70 Compressor control buttons



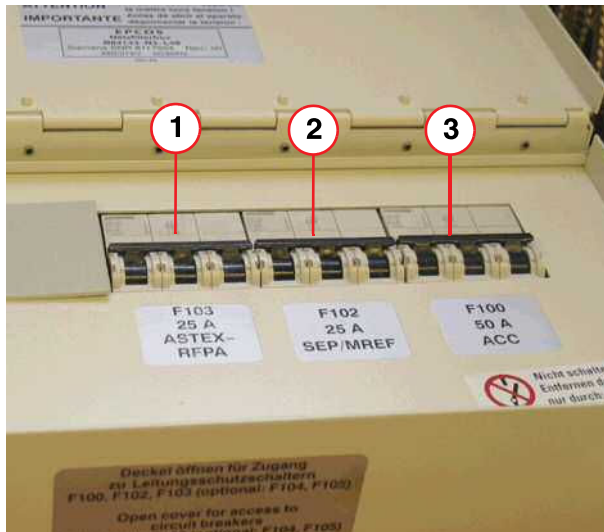
- (1) Control buttons
- (2) LCD display
- (3) Configuration selector switch
- (4) Main power switch
- (5) Mains supply cable
- (6) Ground wire

- Turn **off** the compressor unit at the **main power switch** (pos 4).
 - The **ON/OFF buttons** are disabled in **mode 2**.

- The **alarm box** displays a warning LED, and an audible alarm.

- Press the **AUDIO ALARM OFF** button to cancel the audible alarm.

Fig. 195: ACC cabinet - Circuit breakers F100, F102, F103

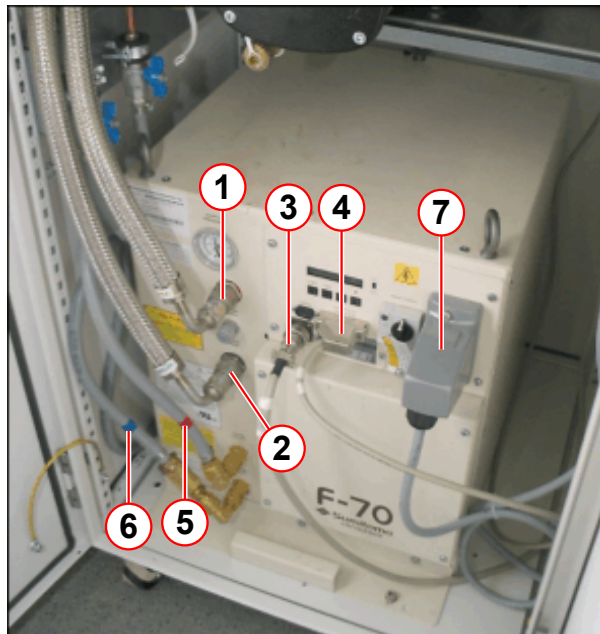


- (1) F103 - isolates RFPA
- (2) F102 - isolates SEP or IFP (stand-alone compressor)
- (3) F100 - isolates ACC cabinet

ACC cabinet

- Isolate power to the compressor:
 - Switch **OFF** circuit breaker F102 (2).

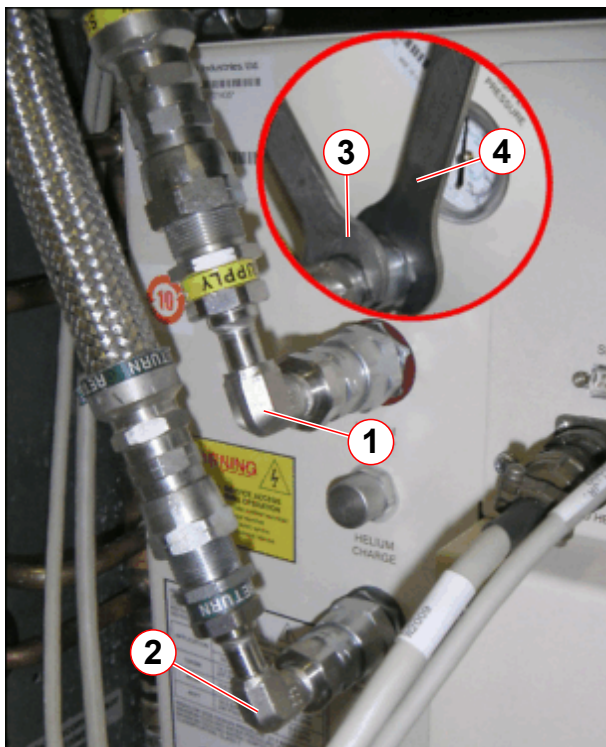
Fig. 196: F70 installation in SEP cabinet



- (1) Helium supply (high pressure)
- (2) Helium return (low pressure)
- (3) Cold head cable (to filter plate)
- (4) Diagnostic interface
- (5) Water line (inlet)
- (6) Water line (outlet)
- (7) Mains power connections

- The compressor unit may be:
 - installed in a **SEP cabinet**
 - a **stand-alone unit** (with fitted casters)
- The base of the compressor unit is fitted with 2 nylon slide rails (pre-fitted) to ease movement.

Fig. 197: F70 He compressor connections



- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

SEP cabinet option

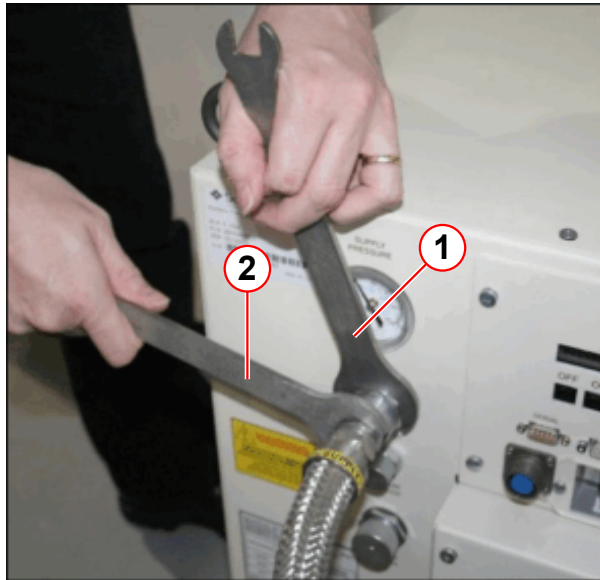
Each helium pressure line is connected to the compressor unit via a **90 degree elbow**.

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).



90° elbow connections are not required for a stand-alone unit.

Fig. 198: F70 He compressor connections (stand-alone unit)

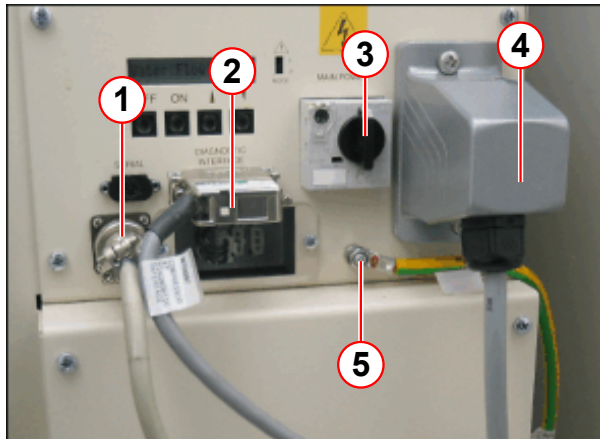


- (1) Turn nut (1 3/16" open-end crowfoot spanner/wrench)
- (2) Hold nut (1" spanner/wrench)

Stand-alone unit

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (2) and **turn** nut (1).

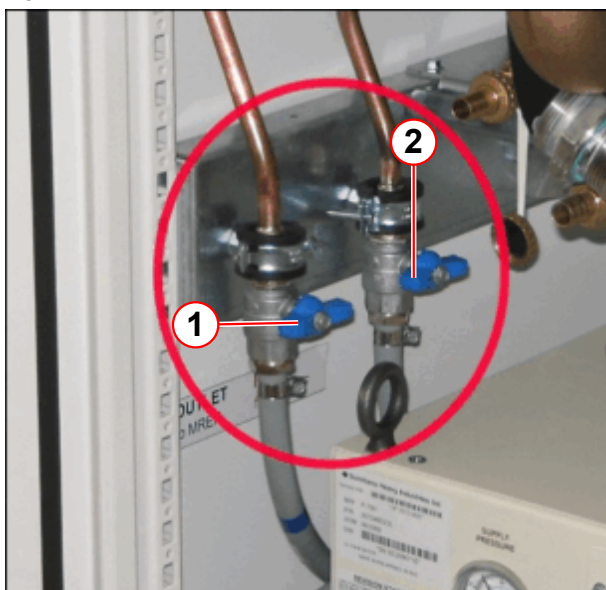
Fig. 199: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- Disconnect the **mains voltage**, **cold head motor**, and **MSUP** (DB25 diagnostic interface) cable.
- Disconnect the **ground wire** (5).

Fig. 200: Water inlet/outlet shut-off valves



- (1) Return (outlet) valve
(2) Supply (inlet) valve

- Close the **shut-off valves** on the **inlet** and **outlet** water pipes.



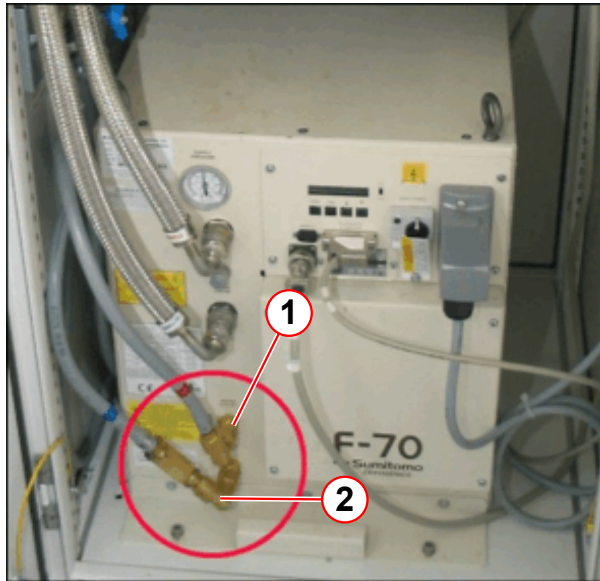
CAUTION

Chemical hazard

If the chilled water contains ethylene glycol, it must be collected in a suitable container while draining.

- » Return the collected water back to the water circuit after the new unit is installed.
- » If necessary, dispose of in line with local regulations for hazardous material.

Fig. 201: F70 connect the water lines



- (1) Water line (inlet)
(2) Water line (outlet)

- **Disconnect the water pipes** from the compressor unit.



There is the potential for irreversible damage to the unit if the internal water pipes freeze during shipping.

Before returning a faulty compressor unit, use a gas line on either the inlet and outlet pipes to expel any water.

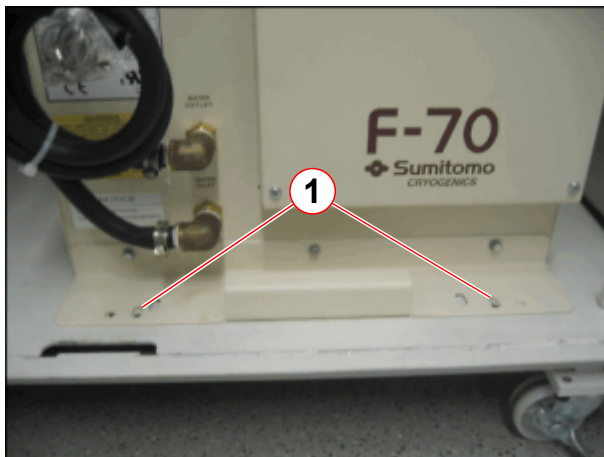
Fig. 202: F70 compressor - draining cooling water notice



Always **drain the coolant** from the cooling circuit when the compressor unit is shipped (or stored).

Refer to the disposal instructions for further information.

Fig. 203: F70 Compressor fixing plate



(1) M8 Hex hd screws (2 off)

SEP cabinet

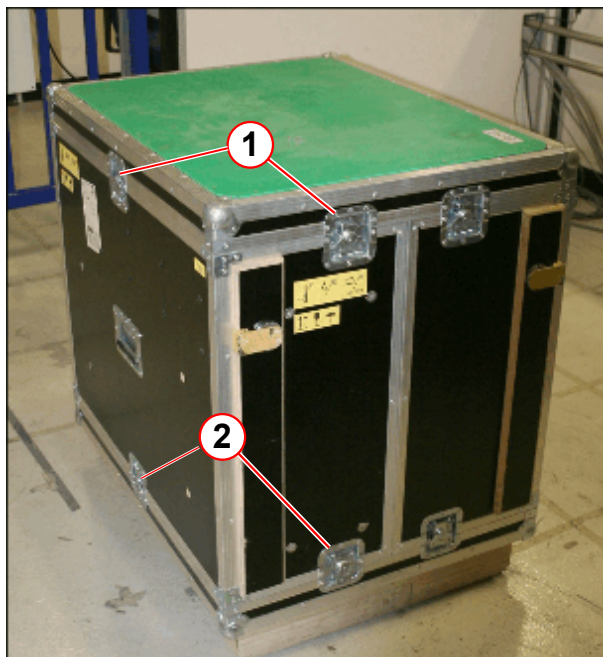
- **Remove** the 2 x M8 hex head screws securing the **fixing plate** to the cabinet.
- Leave the compressor unit in the cabinet until the replacement compressor unit is ready to install.

3.4.6 Unpacking and Pre-installation Checks

Before installation, check the compressor transport box for any transport damage or mishandling.

- Check the shock indicators when the transport box covers are removed.

Fig. 204: Transport box



(1) Top retaining latch x 6
 (2) Bottom retaining latch x 6

- Unclip the 6 x latches securing the top cover.
- Lift the top cover away from the transport box.

Fig. 205: Cover lid of transport crate



- The transport box cover has x 4 casters fitted to the underside. The cover lid is used as a support ramp when removing the defective F70 compressor unit.
 - » The upper side includes a PTFE plate with gliding ability.

Fig. 206: Transport lid castors



When not being used stand the transport lid upright in a safe area against a wall.

- » Do not place the cover on the floor! Danger of rolling away.
- » Do not stand on the cover!

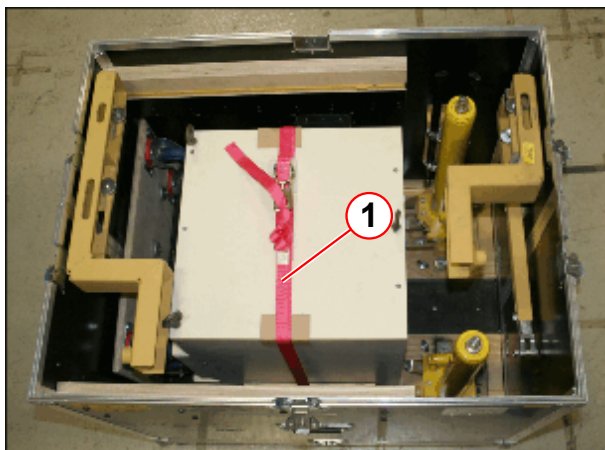
**CAUTION**

Storing the cover:

Risk of accident! Do not step on the cover!

- » When not in use, place cover upright in a safe area.

Fig. 207: Top view of compressor kit

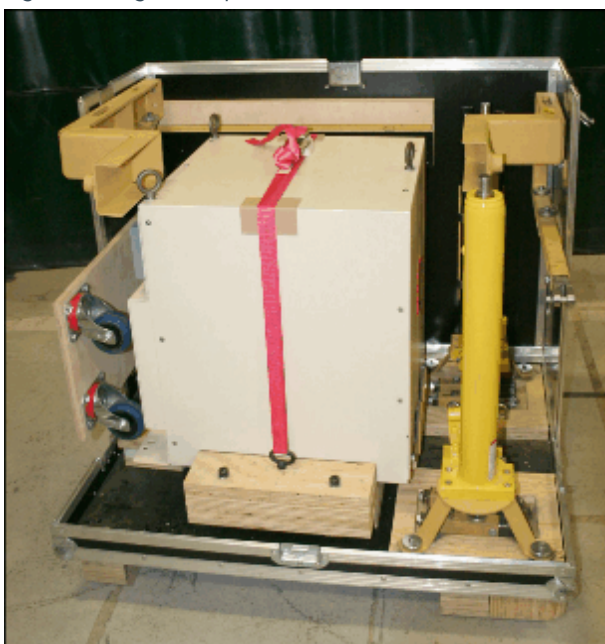


(1) Securing strap

Top view of the transport box.

- Check that the compressor unit has remained strapped securely to the transport base.

Fig. 208: Right side panel removed



All tooling components remain attached in position inside the transport box when removing the right side cover.

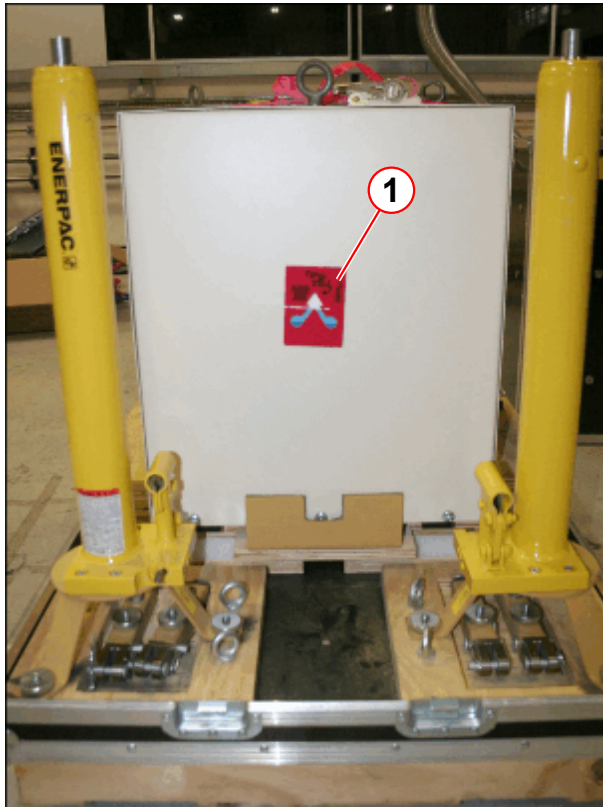
- Unclip the 3 latches and remove the right side cover of the transport box.
- Place the cover in a safe location to the side.



Do NOT tip the compressor unit more than 10 degrees from horizontal to avoid contamination.

DO NOT connect gas lines to the compressor unit if Tip-N-Tell indicator shows that mishandling has occurred.

Fig. 209: Crate sides removed



(1) Shock indicator

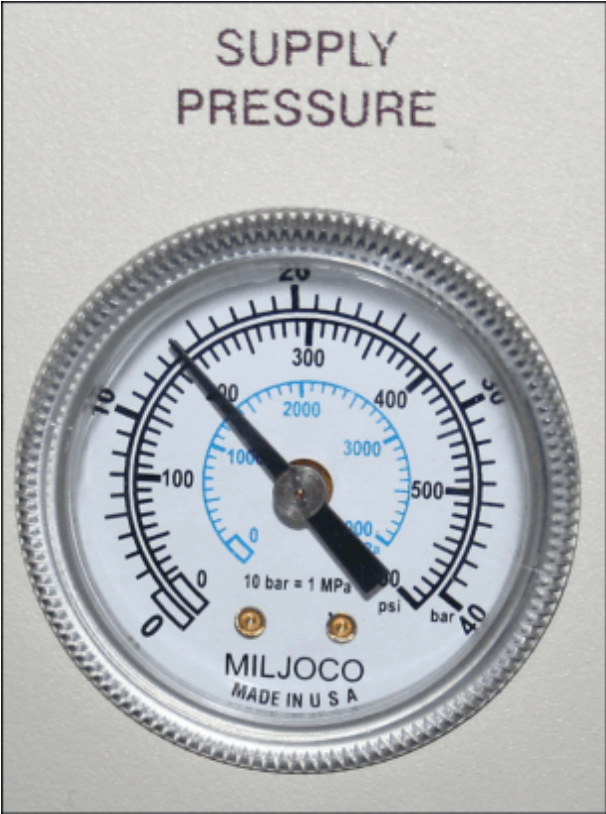
- Check the Tip-N-Tell indicator on the compressor unit.
- The sensor is blue (above the line) when the unit has been tipped or mishandled during transport.

Fig. 210: F70 Compressor Tip-N-Tell indicators



- If the sensor has been activated (the arrow point is blue), and the pressure is within specifications, allow the compressor unit to stand upright for **2 hours**. Refer to (→ Testing the Compressor Unit after Mishandling / Page 179)
- If the sensor has **not** been activated, continue with the replacement procedure.

Fig. 211: F70 Static pressure reading



- Check that the **static pressure** is **positive**.
 - If the gauge indicates **zero pressure** the compressor unit **cannot** be used.
 - Arrange a replacement compressor unit.

Tab. 15 F70 static pressure on arrival

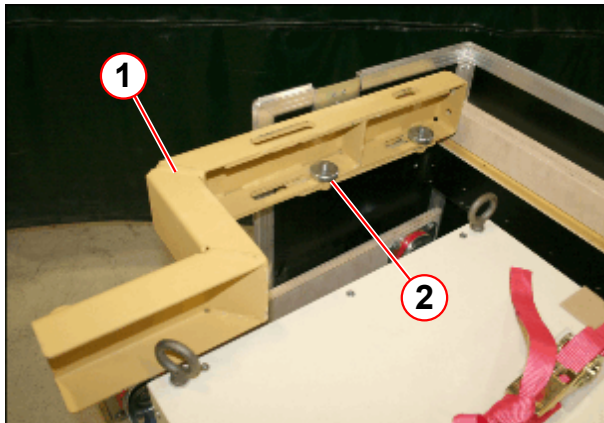
Pressure	Ambient temperature
207 - 212 psi (G)	20°C (68°F)
1430 - 1460 kPa	
13.5 - 14.0 bar	

Continue with the replacement procedure if:

- The sensor has **not** been activated
- The static compressor pressure is positive.

Remove the tooling components from the transport kit and assemble.

Fig. 212: Long Cross beam



- (1) Cross beam
(2) Knurled head fastener

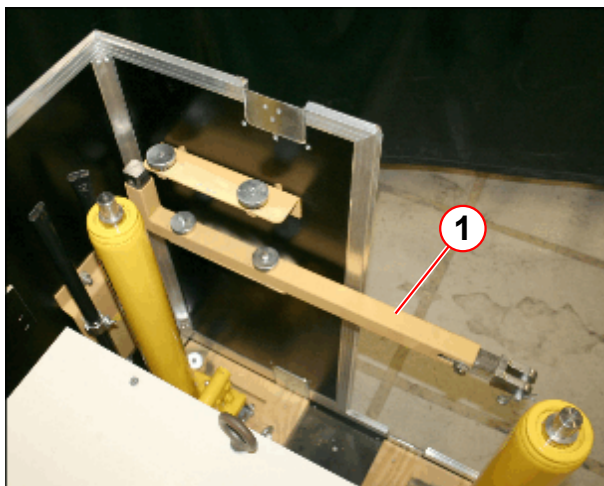
- Unscrew the 2 x knurled head fasteners securing the long 'cross beam'.
- Refit the fasteners to the case.

Fig. 213: Short cross beam



- Unscrew the 2 knurled head fasteners and remove the short cross beam.
- Refit the fasteners to the case.

Fig. 214: Vertical lifting beam

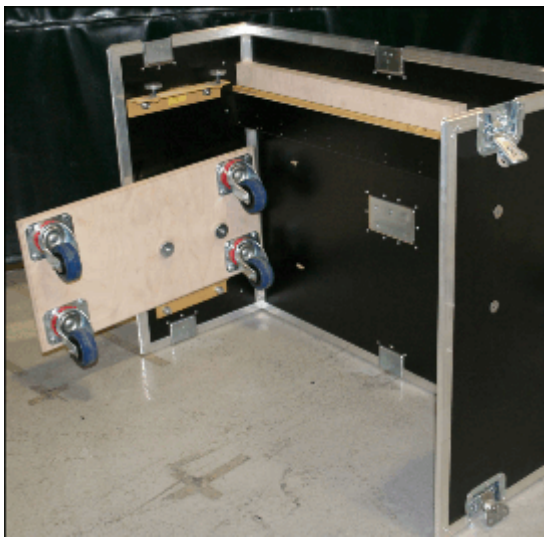


- (1) Vertical lifting beam tool

If there are only 2 x lifting eyes available for lifting the compressor unit:

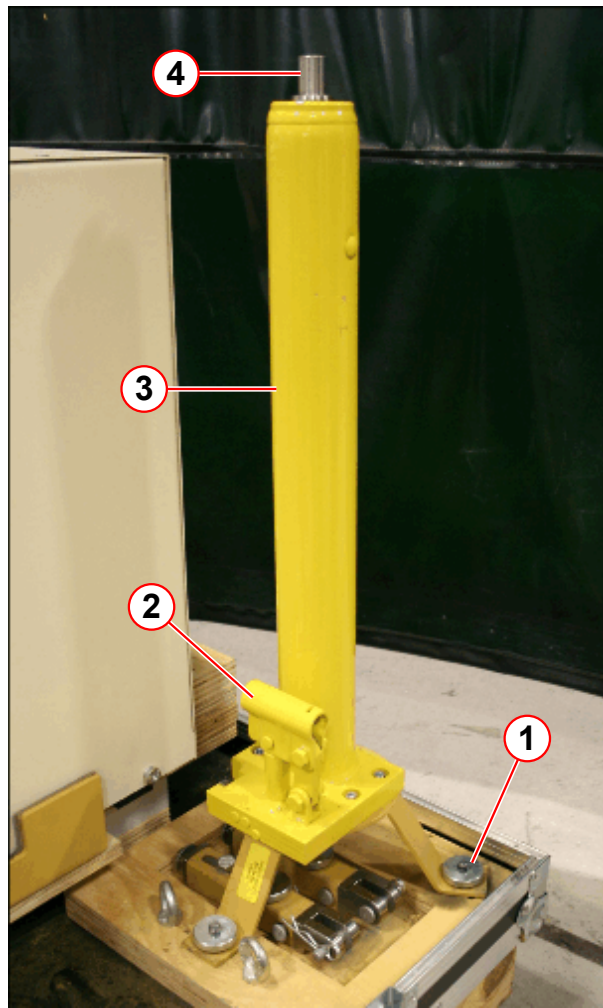
- Unscrew the 2 knurled head fasteners and remove the vertical support arm.
- Replace the fasteners.

Fig. 215: Board with castors



- Remove the knurled head fasteners and detach the moveable board with castors from the transport box side.
 - » Do not leave the board unattended.
 - » When not in use, place it upright in a safe area.

Fig. 216: Lifting Jack x 2



- Remove the knurled head fasteners x 3 from each of the 2 jacks.
- Remove the 2 lifting jacks from the base of the transport box.

- (1) Knurled fastner x 3
- (2) Hydraulic pump handle attachment
- (3) Cylinder body
- (4) Hydraulic lifter

**CAUTION****Hydraulic steel jack**

Non-observance of these warning notices may lead to damage or error of the steel jacks. The results are property damage and serious injuries.

- » Immediately support the compressor unit.
- » Working on a compressor unit when supported by the 2 x steel jacks is not permitted
- » Use only a hard, even floor area with sufficient load bearing capacity for the lifting device.
- » Prevent excess weight. The load bearing capacity cannot be exceeded!
- » Do not open the oil filler rubber plug under any circumstances. Exception: the jack has been fully lowered.
- » The operator has to monitor all movements of the load and the configuration.

Fig. 217: Support arms x 4



- Below the 2 jacks there are 4 x horizontal support arms.

The quantity required will correspond to the number of lifting eyes fitted to the compressor unit.

Fig. 218: Removing straps

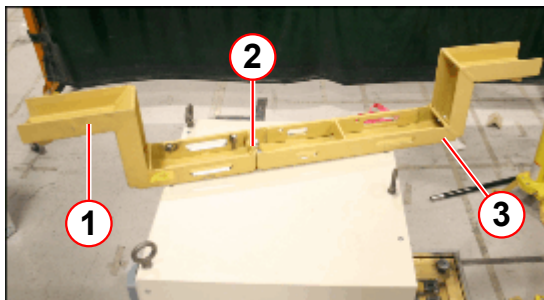


- Carefully open the restraining strap securing the compressor unit.
- Remove the edge protection pieces and the strap.



During assembly of the lifting frame place some cloth / cardboard on top to avoid paint damage to the paint.

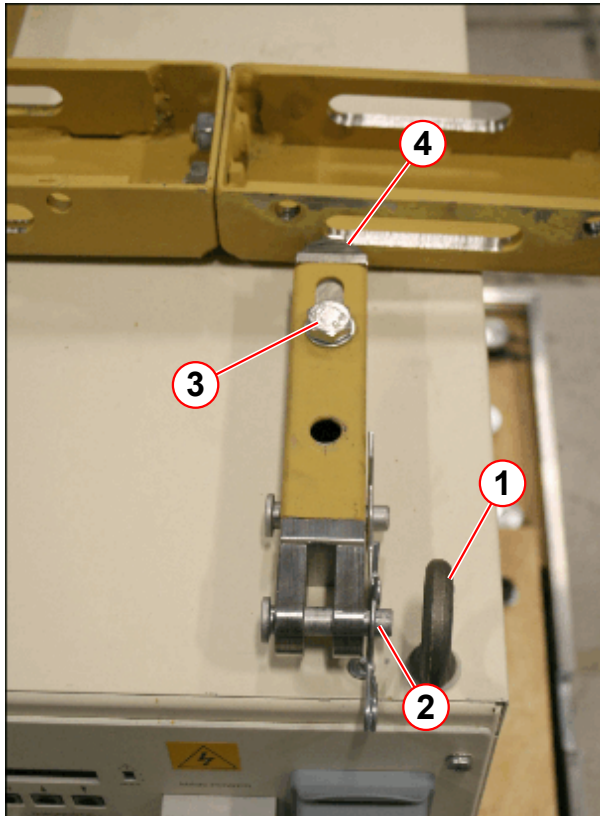
Fig. 219: Assemble lifting beams



- (1) Short beam
- (2) Bolts x 2
- (3) Long beam

- Screw both cross beams together with M10 nuts and washers.
- Torque screws to 23 Nm (17 Ft/lbs).

Fig. 220: Fitting the Support arms (x3)



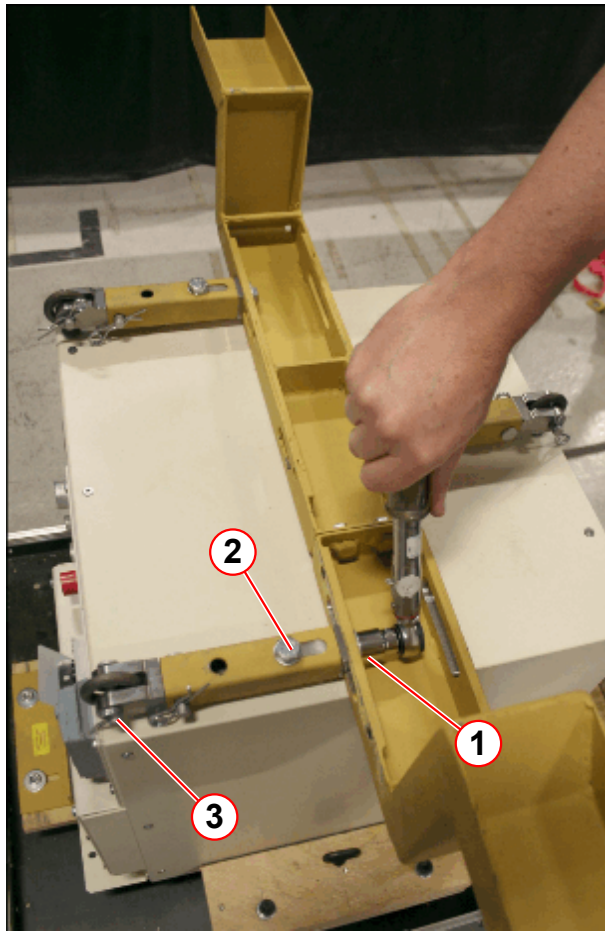
- (1) Lifting eye
- (2) Bolt and Split pin
- (3) Adjustment screw M10
- (4) Cross beam securing screw M10

Before attaching the support arms check that the lifting eyes are orientated at 90 degrees to fit into the lifting arms.

Attach the support arms:

- If there are no lifting eyes fitted on compressor unit, fit the lifting eyes supplied with the tooling.
- Insert the bolt and secure with the split pin, (pos: 2).
- For fine movement undo the adjustment screw, (pos: 3).
- Remove the cross beam securing bolt, (pos: 4).

Fig. 221: Secure support arms



- (1) Support arm & Lifting beam screw
- (2) Adjustment screw
- (3) Split pins

- Attach the support arms to the cross beams M10 x 20 screws and washers.
- Torque screws to 23 Nm (17 Ft/lbs).
- Center the complete cross beam assembly and tighten all screws with the 17-mm wrench.
- Torque the M10 adjustment screws to 23 Nm (17 Ft/lbs).
- Check that the split pins are all attached.

The cross beam lifting assembly is ready for the hydraulic jacks to be attached.

Fig. 222: Alternative lifting method



(1) Lifting eye

Alternative Lifting method

- If only 2 x lifting eyes are available on the upper side of the compressor unit:
- Attach the additional vertical support arm with the supplied crane lifting eye to the back of the compressor unit.

NOTE: The horizontal support arm seen in the figure is omitted here.

Fig. 223: Front F70 fixing adjustable plate



- Loosen the 2 x screws at the front angle adjustable plate.
- Move the angle plate away from the F70 cabinet housing.

3.4.7 Installing the Compressor Unit



Prevent stress situations caused by shocks generated by abruptly opening or closing the actuating valve while lowering the piston.

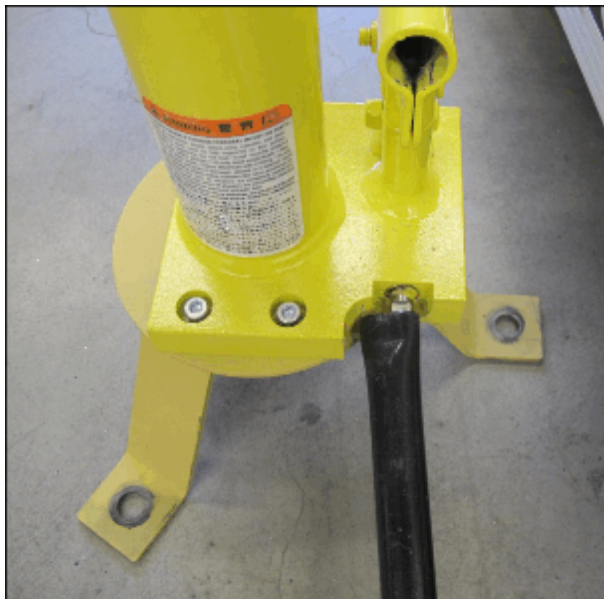
Shock loads can lead to hydraulic circuit overloads that damage the jack.

When the jack is not in use, the piston must be in the lowest position.



Before using, check the jack and the cross beam tooling for damage and oil leaks. If faulty, do not use the tooling.!

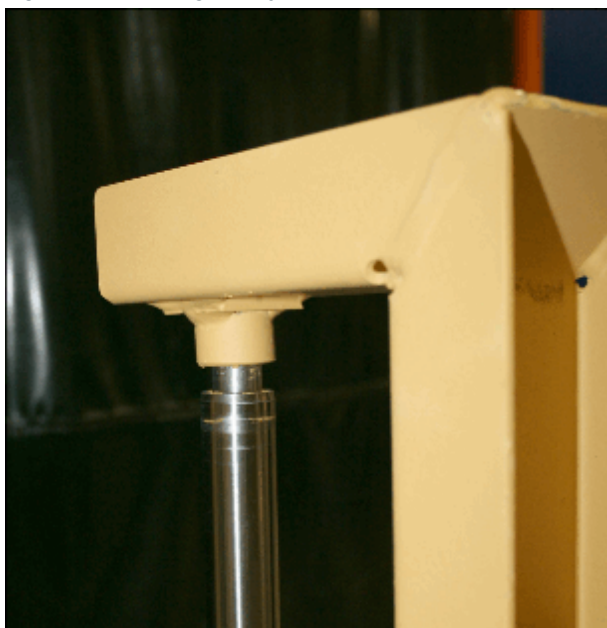
Fig. 224: Pressure relief valve



Lifting operation

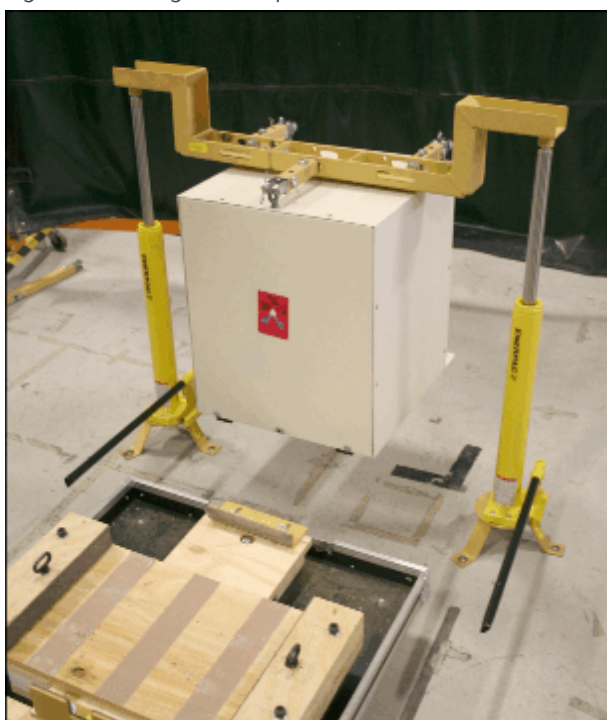
- Fit the rod onto the pressure relief valve screw.
- Turn it clockwise until the valve is firmly closed.
- Remove the rod and insert into the hydraulic pump bracket.
- Repeat this operation for the 2nd hydraulic jack.

Fig. 225: Locating the Hydraulic arms



- Slide hydraulic supports underneath the each lifting arm.
- Using the rod, pump each hydraulic arm up to the socket on the lifting beam.
- Check that the hydraulic arm is located into the socket on the lifting beam before taking the full weight of the compressor unit.

Fig. 226: Lifting the compressor



- Continue pumping the hydraulic jack rods **at the same time** to ensure a uniform lift of the compressor unit.
 - » Keep the compressor unit in a horizontal position. It must not be tipped at more than a 10 degree angle.
- Move the lower assembly of the transport box away from the compressor unit.

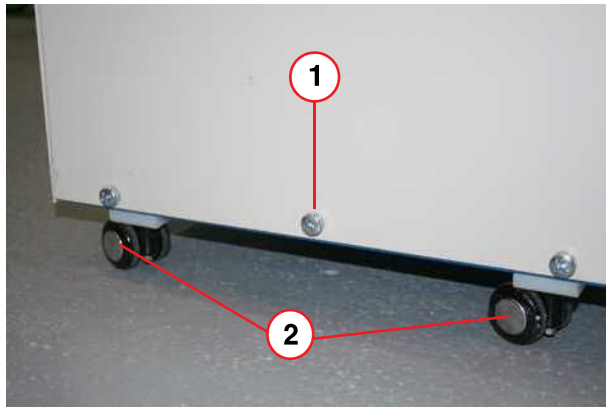


The new compressor unit is supplied fitted with 4 x caster wheels. When the compressor unit is held off the ground on the hydraulic jacks remove the caster wheels.

Compressor caster Removal

- The casters are to be removed for fitting the compressor into a SEP cabinet.
- For Standalone and IFP systems the caster wheels can remain fitted.

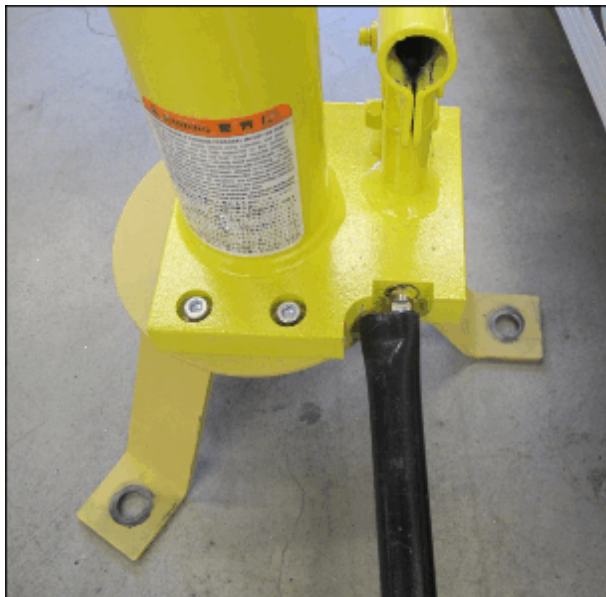
Fig. 227: F70 Back panel view



- (1) Center screw is removed for SEP cabinet installation
 (2) Casters x 4 (stand-alone units only)

- To remove the 4 casters from the compressor unit.
 - Use a 7 mm ratchet and socket and a 14 mm open ended wrench.
 - **Remove** the center screw (pos 1). (The screw hole aligns with the **guide pin** at the back of the SEP cabinet).

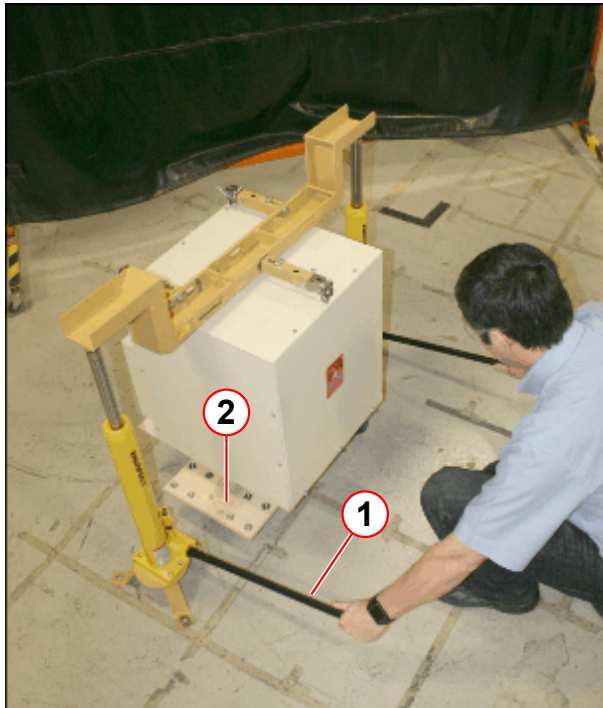
Fig. 228: Pressure relief screw



Lowering operation

- Remove the 2 x rods from the hydraulic pump bracket.
- Fit the rods over each pressure relief valve screws
- Slowly turn it counter-clockwise by not more than 1 full turn.
- Turn it slowly to be able to check the lowering speed.

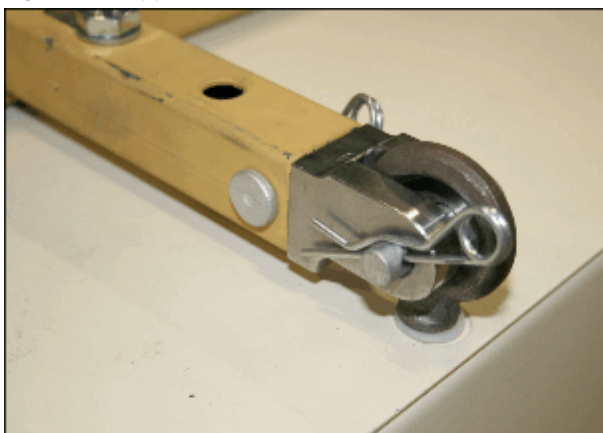
Fig. 229: Lowering compressor



- (1) Hydraulic jack rods x2
- (2) Moveable board

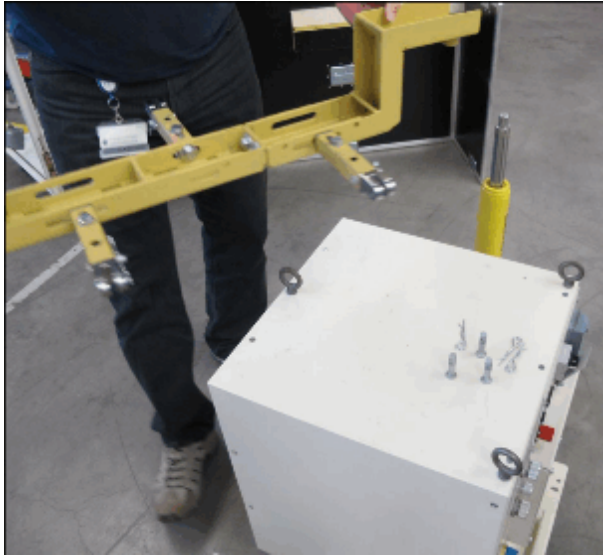
- Place the moveable board below the compressor unit.
- Turn both pressure relief valves slowly counter-clockwise to lower the compressor unit onto the moveable board.

Fig. 230: Support arm removal



- Remove the splints and bolts from the support arms fitted to the new F70 compressor unit.

Fig. 231: Removing lifting beam



- Remove the cross beam assembly.
- Carefully move the compressor unit to the final position near to the cabinet.



The moveable board casters are fixed. Movement will only be in a straight line. Check that the assembly tool is in the direct vicinity of the cabinet.



Uneven flooring leads to the hazard of tipping over!

Fig. 232: Removing the defective compressor



- Position the top cover board of the transport box in front of the cabinet.
- Carefully slide the defective compressor unit out of the cabinet onto the board.

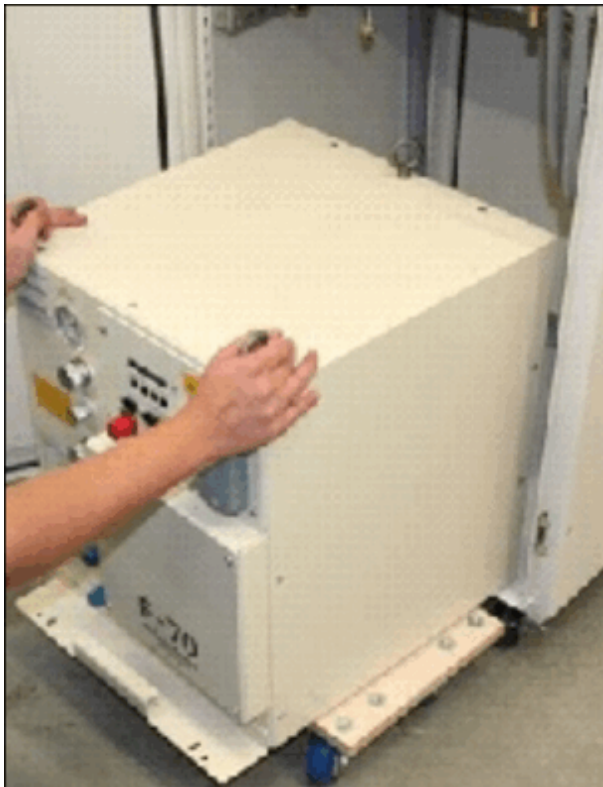
Fig. 233: Remove defective compressor (2)



When the defective compressor unit has been removed, move it away from the cabinet.

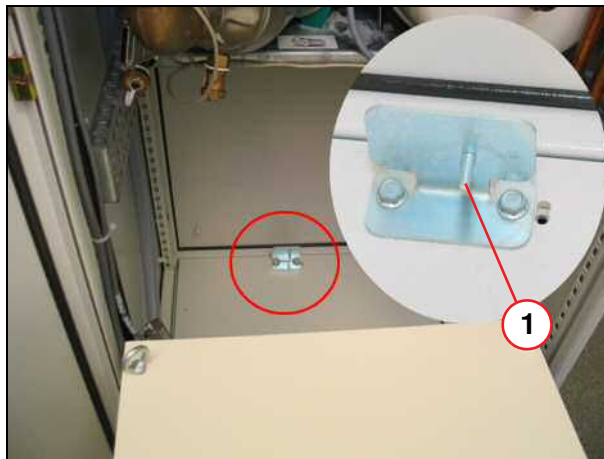
The cross beam and support arm assembly can be fitted later. The hydraulic jacks will be used to lift the defective compressor unit onto the transport box.

Fig. 234: Moving in the compressor



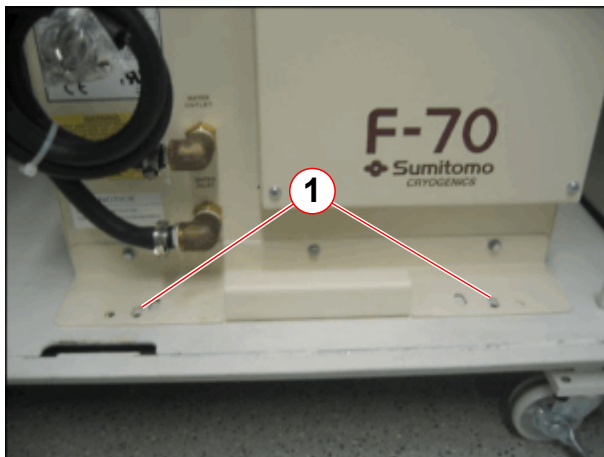
- When fitting the new compressor unit:
 - » Check that the compressor unit is directly in line to the front of the cabinet.
- Slide the new compressor unit into the cabinet.

Fig. 235: SEP - Compressor guide pin



- The (center) screw hole on the rear panel of the compressor unit should align with the **guide pin** at the back of the cabinet.

Fig. 236: F70 Compressor fixing plate



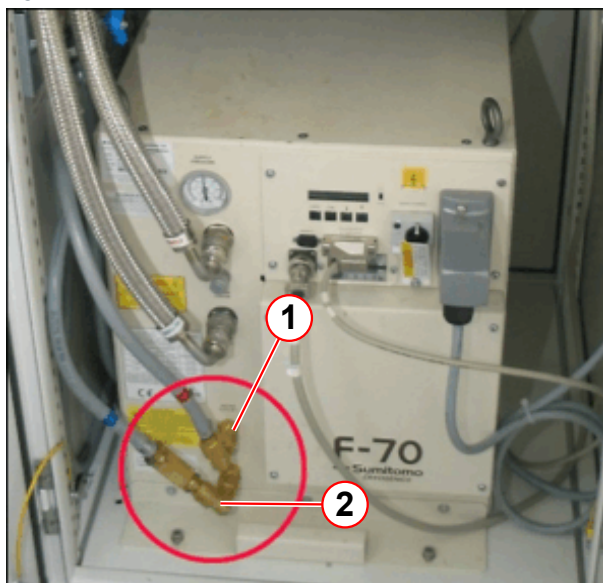
- Secure the **fixing plate** to the cabinet using the 2 x M8 hex head screws.

(1) M8 Hex hd screws (2 off)

3.4.8 Compressor Unit Configuration and Connections

The new compressor unit must be operated for **10 minutes** to clean up any oil contamination incurred during transportation. Run the compressor unit **before** the **helium pressure lines** and **cold head** are connected. For this test, only the **water** and **power** connections are required.

Fig. 237: F70 connect the water lines

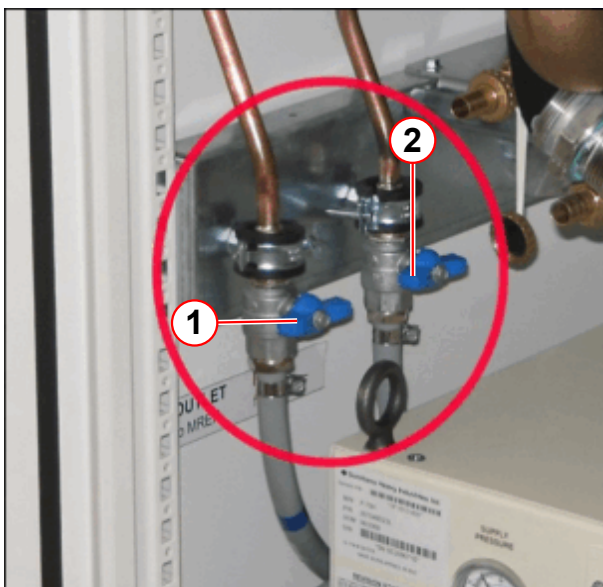


- (1) Water line (inlet)
(2) Water line (outlet)

Connecting the water:

- Remove the protective caps from the water in and water out fixings.
- Connect the water pipes to the compressor unit.
 - Ensure that the water is connected to the water supply line and that the water is turned on..

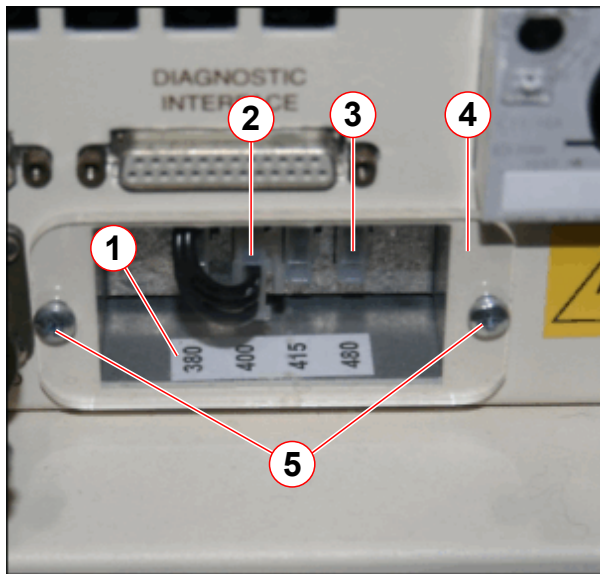
Fig. 238: Water inlet/outlet shut-off valves



- (1) Return (outlet) valve
(2) Supply (inlet) valve

- **Open** the water shutoff valves.
- Check for leaks.

Fig. 239: F70 Compressor voltage selection access panel



- (1) Voltage label
- (2) Voltage selection connectors
- (3) Voltage tap
- (4) Clear plastic cover
- (5) Screws (2 off)

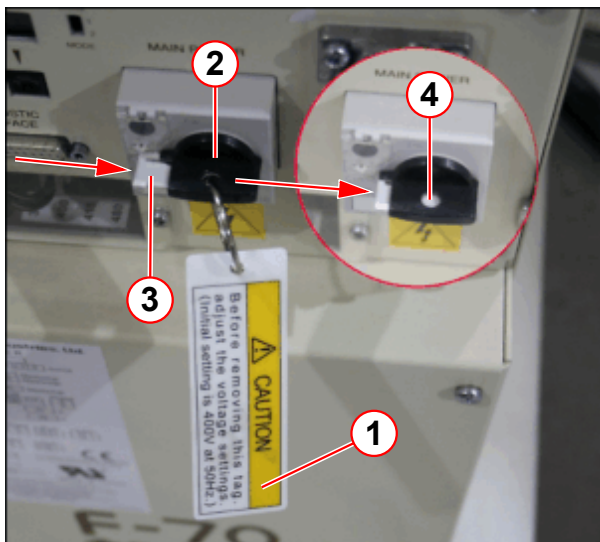
- Check that the **transformer voltage taps** are correctly selected to match the supply voltage.
- If necessary, remove the plastic cover (3), reposition the tap, and replace the cover.



The unit is configured in the factory to 400 V/3~50 Hz.

If the compressor unit is not wired correctly, connecting to a higher voltage may damage the control circuit.

Fig. 240: F70 Main power switch - locking device



- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- **Remove** the voltage caution label (pos. 1) (fitted through the main circuit breaker).
- Engage the white switch locking device fully **before** attempting to turn on.
- **Push in** the white locking device fully.
 - » The device should be visible through the blocking hole (pos. 4).



Ensure that the white switch locking device is **FULLY** engaged. **Do not use too much force on the switch.** A new compressor unit will be required if damaged.

- Connect the mains voltage cable.

Fig. 241: F70 Main power supply connection

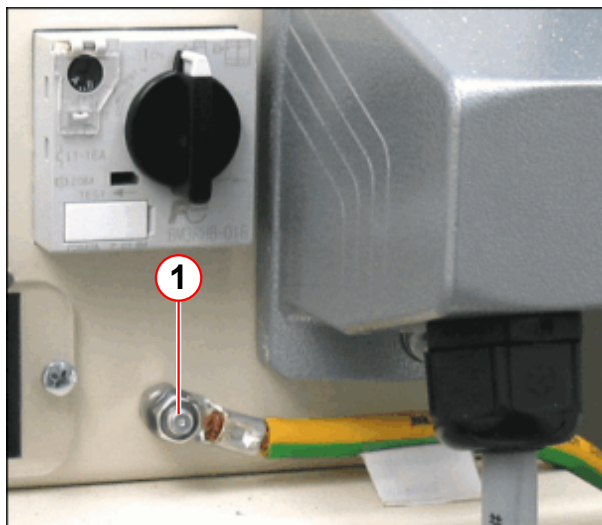


- (1) Pull cover to remove
- (2) Connect power supply cable
- (3) Fasten the cover by attaching the screws

3.4.9 Compressor Unit (Ground) Earth Connection

A protective ground connection test must be performed as a final step **before** switching on the compressor unit.

Fig. 242: Compressor ground wire connection

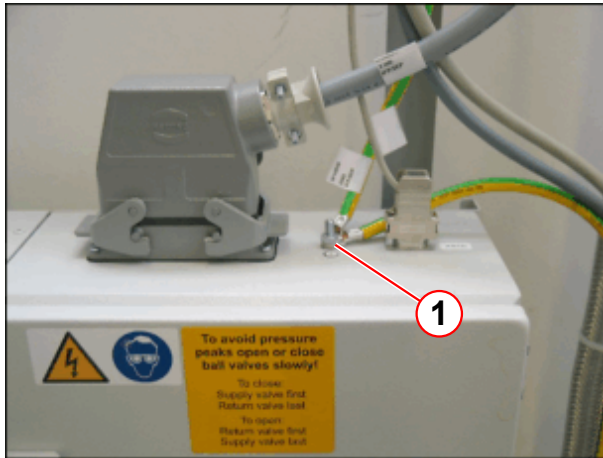


- (1) Ground wire connection

- Connect the ground wire to the helium compressor terminal.

Stand-alone option

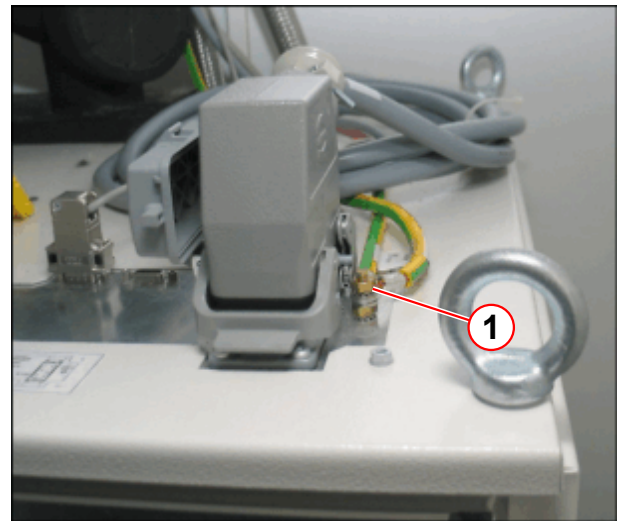
Fig. 243: IFP ground wire (optional)



(1) Ground wire

SEP option

Fig. 244: SEP ground wire (optional)



(1) Ground wire

Test the protective earth connection

- Measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent). Check between the protective earth connector on the compressor unit, and the protective earth wire connected to the top of the SEP (on the electronic box of the IFP).

3.4.9.1 Compressor Unit Run Mode



Compressor unit is supplied in configuration **mode 1**.

Before the compressor unit is switched on, the **mode selector switch** must be set to either mode **1** or **2**.

Tab. 16 Configuration mode switch

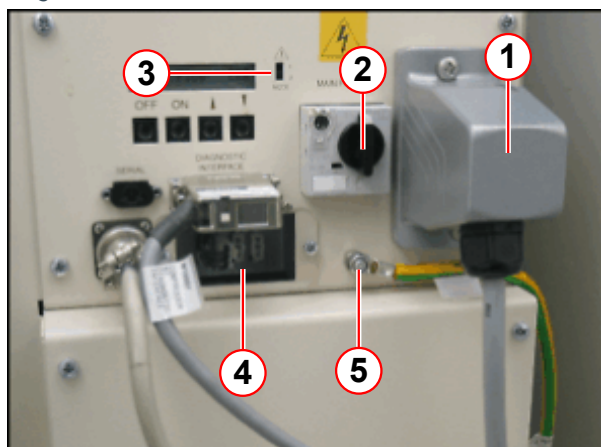
Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10-K shield cooling	Enabled
Mode 1	UP	F70 Startup checks	Enabled
Mode 2	DOWN	4-K recondensing	Disabled



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. Loss of control and incorrect errors will result.

3.4.9.2 Testing the Compressor Unit on Startup

Fig. 245: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Ensure that the **main power** switch is **OFF** before adjusting the selector switch.
- Check that the configuration mode switch is set to **1**.

Use a flat-blade screwdriver to move the switch up and down.

- » The **ON/OFF** buttons only work in configuration mode **1**.
- » The compressor unit must **NOT** be run in stand-by in mode **2**.

In **mode 2** both **ON** and **OFF** buttons are disabled. Operation is controlled remotely via the DB25 diagnostic interface.

- Switch on the circuit breaker F102 (EPC cabinet).
- Turn the main power switch.
- Press the ON button.
- Check the temperature of the water supply (water in) and return (water out) lines:-
 - » The **return** (water out) pipe should be **warmer** than the **supply** (water in).
 - » **Stop** the compressor unit and reverse water pipes if the **return** line is **cooler**.
- Run the compressor unit for **10 minutes** and then press the **OFF** button.
- Turn **OFF** the main power switch.
- Reset the configuration mode switch to **2**.
- The test run is now complete.



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. The result is loss of control and incorrect indications.



The compressor unit is protected from running in reverse. If it fails to start and the LCD displays 'Phase Seq - ERR'.



If the compressor unit starts but does not build pressure, **TURN OFF IMMEDIATELY**. Contact Siemens Magnet HSC.

3.4.10 Testing the Compressor Unit after Mishandling

Follow this procedure if:

1. The Tip-N-Tell indicator has been activated.
2. The compressor pressure is lower than specification.

Fig. 246: F70 Compressor Tip-N-Tell indicators



Pressure within specification:

- If the Tip-N-Tell has been activated but the pressure is in specification:
 - Stand compressor unit upright for **2 hours**.
 - Perform the test run.

Pressure outside specification:

- Notify the delivering carrier of the damage if:
 - The compressor unit has been delivered on its side or upside down.
 - There is physical damage to the unit.

NOTICE

If the compressor unit arrives at zero pressure - **DO NOT CONNECT THE HE PRESSURE LINES or COLD HEAD POWER CABLE** to the compressor unit.

Connecting the lines to the compressor unit at zero pressure may cause contamination of the whole refrigeration system.

- » Check the compressor pressure before attempting to connect the helium pressure lines.
- » Contact Siemens Magnet HSC for advice.

- **Remove** the compressor unit from the **transport box**.
 - **Retain** all packaging materials for reuse if returning the compressor unit for replacement or repair.

Install the replacement compressor unit into the SEP cabinet before performing the test.

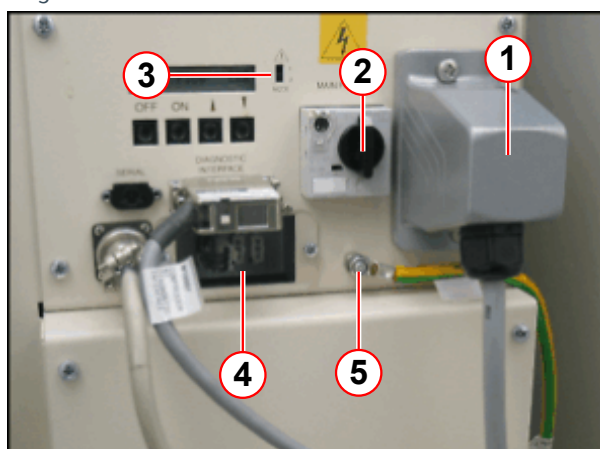
For the test, only compressor water and power connections are required.

Ensure all other lines and cable connections are disconnected from the compressor unit.



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. Loss of control and incorrect indications will result.

Fig. 247: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Check that the configuration mode selector switch is set to **1**.
Use a flat-blade screwdriver to move the switch up and down.
 - » The **ON/OFF** buttons only work in configuration mode **1**.
 - » The compressor unit must **NOT** be run in stand-by in mode **2**.
In configuration **2**, both **ON** and **OFF** buttons are **disabled**. Operation is controlled remotely via the **DB25 diagnostic interface**.
- Switch **ON** the circuit breaker F102 (EPC).
- Turn on the main power switch.
- Press the **ON** button to start the compressor unit. The LED displays **RUN**.

- Check the water supply and return line temperatures.
 - » The **return** line should be **warmer** than the **supply**.
- **Stop** the compressor unit and reverse the water pipes if the **return** line is **cooler**.
- Run the compressor unit for a **MINIMUM of 2 HOURS**.



If the compressor unit shuts down during the test run, contact Siemens Magnet HSC.

- If there are no problems during the test run, stop the compressor unit.
 - **Press** the **OFF** button.
 - **Turn OFF** the **main power** switch.
 - The test run is complete.

- Reset the configuration mode switch to 2.



The compressor unit is protected from running in reverse. If it fails to start and the LCD displays 'Phase Seq - ERR'.



If the compressor unit starts but does not build pressure, **TURN OFF IMMEDIATELY.** Contact Siemens Magnet HSC.

3.4.11 Cold Head and Compressor Unit Start up



CAUTION

High-pressure helium gas

Failure to observe the following Safety Information may result in injury.

- » Protective clothing, gloves, and eye protection must be worn.

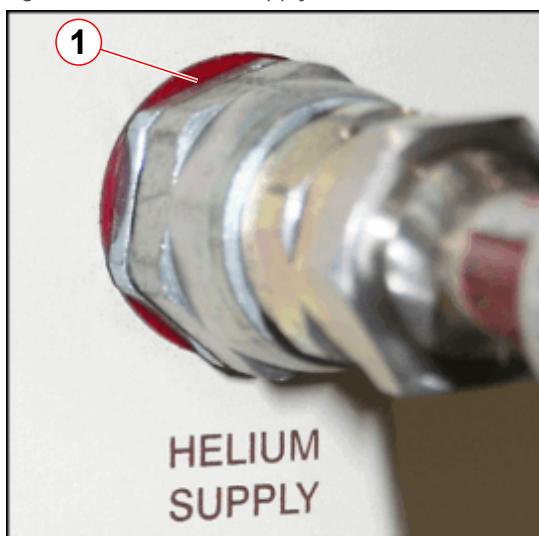
Helium gas lines

Ensure that the **Supply** (HP) and **Return** (LP) lines are connected correctly. The 4K helium pressure lines are colour coded as shown in the table below.

Tab. 17 Gas line connections

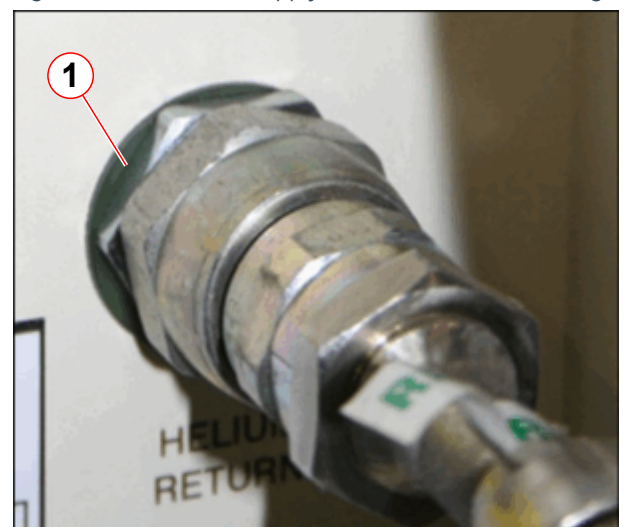
System	Gas line color	F70 He connection
4K supply (HP)	Yellow	Red
4K return (LP)	Green	Green

Fig. 248: F70 Helium supply connections - color coding



(1) Helium supply = RED

Fig. 249: F70 Helium supply connections - color coding



(1) Helium return = GREEN

Fig. 250: F70 High-pressure Aeroquip coupling



- (1) Locknut
(2) Flat rubber gasket

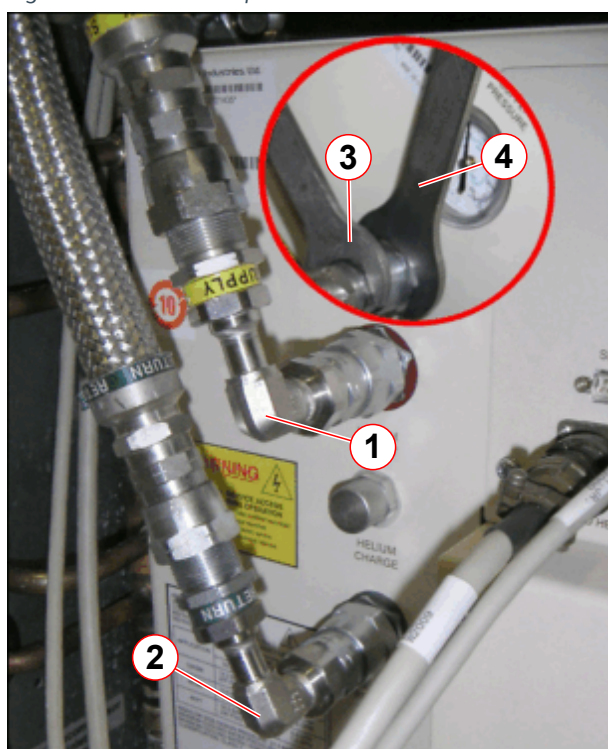
- Remove the protective transport covers from the return and supply connections.
- Check the **rubber gasket** seal for dirt or dust and check that it is correctly fitted.

NOTE: An incorrect fit may cause a helium gas leak.



Incorrect fitting of components may cause a helium gas leak.

Fig. 251: F70 He compressor connections



- (1) SUPPLY (high pressure) 90-degree elbow
(2) RETURN (low pressure) 90-degree elbow
(3) Hold nut (1" spanner/wrench)
(4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

SEP cabinet option

Each **helium pressure line** is connected to the compressor unit via a **90 degree elbow**.

- Fit the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).
- **Tighten down** all couplings as far as possible, and then turn them back by 1/4 turn to **relieve strain**.

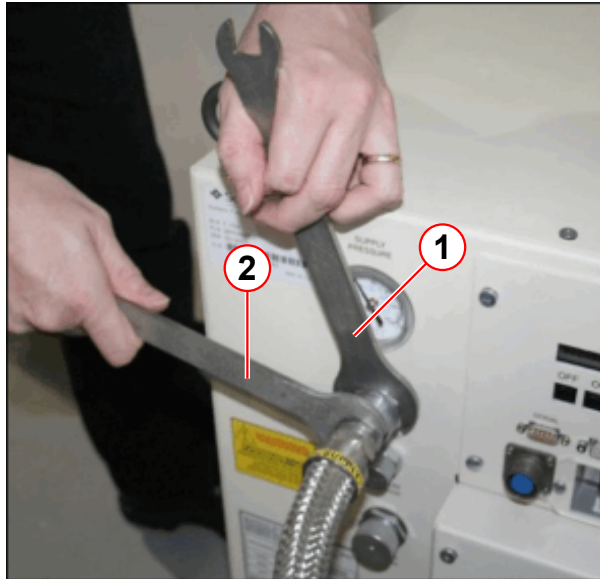


When connecting the Helium pressure lines, **DO NOT OVER-TIGHTEN** the couplings.



90° elbow connections are not required for a stand-alone unit.

Fig. 252: F70 He compressor connections (stand-alone unit)

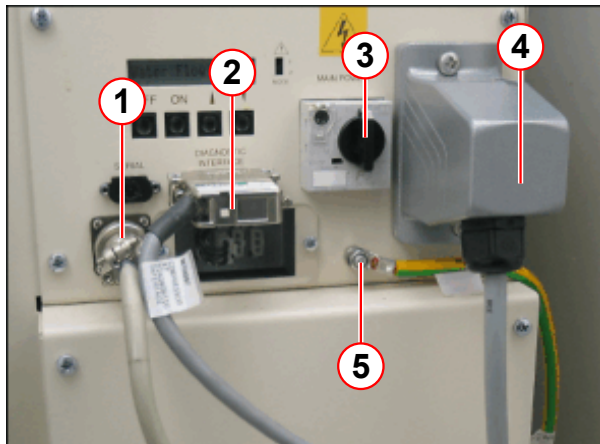


- (1) Turn nut (1 3/16" open-end crowfoot spanner/wrench)
- (2) Hold nut (1" spanner/wrench)

Stand-alone unit

- **Fit** the self-sealing helium pressure lines in the following order:-
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (1) and **turn** nut (2).
- **Tighten down** all couplings as far as possible, and then turn them back by 1/4 turn to **relieve strain**.

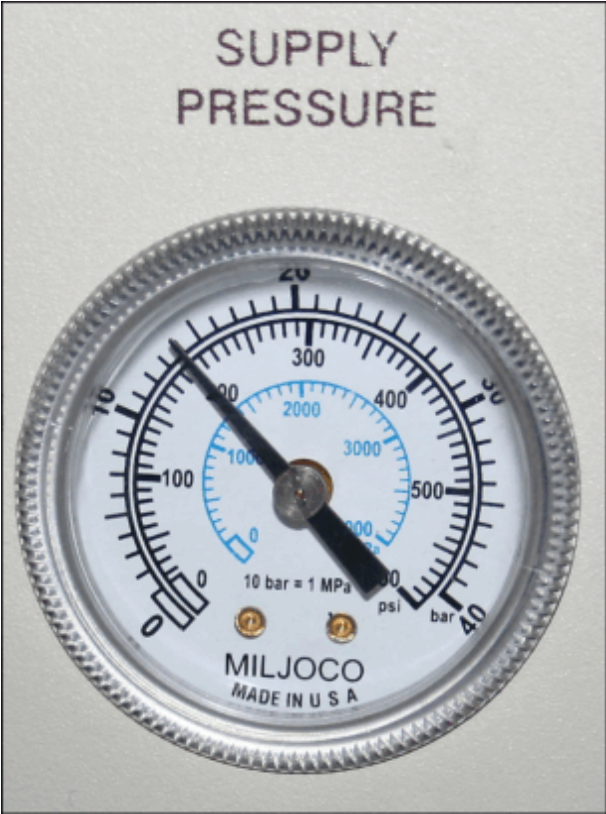
Fig. 253: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- **Connect** the **cold head cable** (1) and **MSUP cable** (DB25 diagnostic interface) (2).

Fig. 254: F70 Static pressure reading



- Check the **equalization (static)** pressure on the pressure gauge.
 - » **Gas charging** will be required if the static pressure is lower than the specification.

Tab. 18 Compressor gas pressure

Cold head	Static pressure at 20 °C (680 F)	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

- Check that the **configuration switch** is set to **mode 2**.
- Check for **water flow** to the compressor unit.
- Switch **ON** the **main power** switch.
- Check the **alarm box** status and clear any faults.

3.4.12 Final Checks

- Wait **20 minutes** and check the operating (**Dynamic**) helium gas pressure on the pressure gauge.
 - **Supply: 21-22 bar (304-319 psi (G))**.

If the dynamic pressure is outside of this range, refer chapter: Procedures / Helium gas charging.

- **Check** the helium pressure line and connectors for **leaks**.

- Check for leakage at the water connections.
- Document the measurement in the **service report**. The (measured value, and measured equipment) is required by IEC 62353.

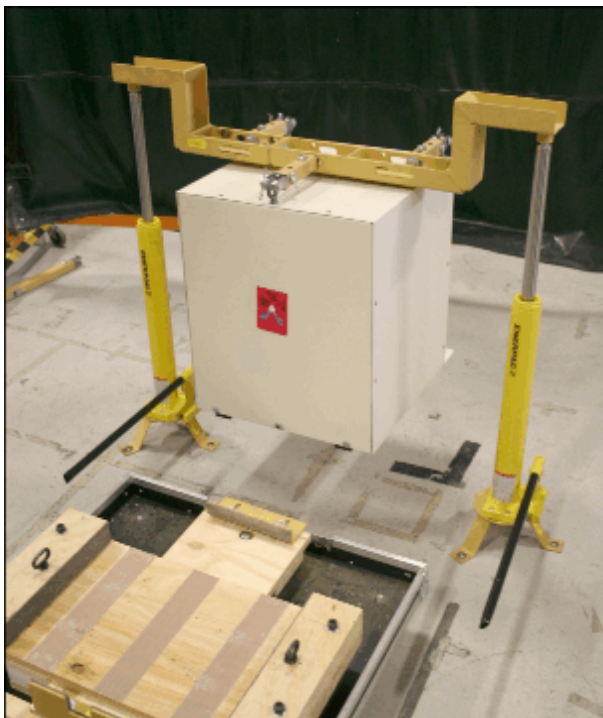
3.4.13 Packing the Compressor Unit

Fig. 255:



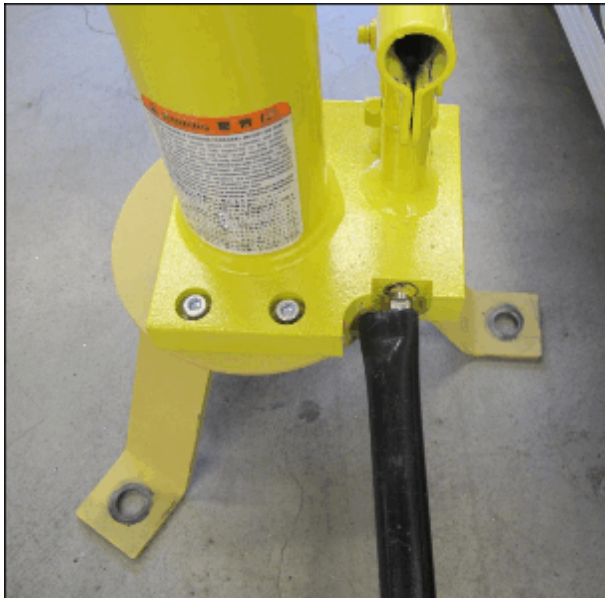
- Move the failed compressor unit on the moveable board next to the lifting jacks.
- Fit the lifting beam assembly.
- Pumping the 2 x hydraulic jacks at the same time lift the compressor unit.
- Remove the moveable board.

Fig. 256: Lifting the compressor



- Keep the compressor unit in a horizontal position
- Move the lower assembly of the transport box under the compressor unit.

Fig. 257: Pressure relief screw

**Lowering operation**

- Remove the 2 x rods from the hydraulic pump bracket.
- Fit the rods over each pressure relief valve screws
- Slowly turn it counter-clockwise by not more than 1 full turn.
- Turn it slowly to be able to check the lowering speed.

Fig. 258: Back F70 fixing plate



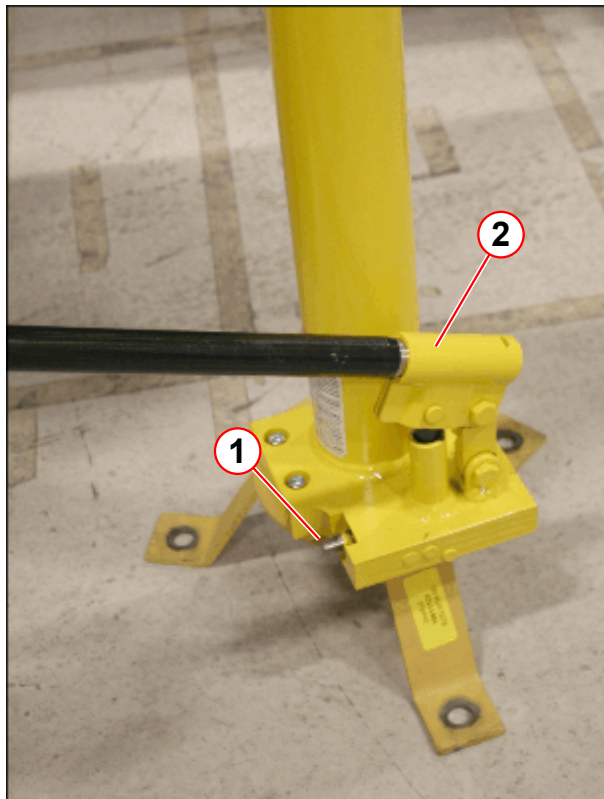
- Lower and place the defective F70 compressor unit onto the transport box in the location provided.
- Check that the F70 cabinet fits flush with the back angle fixing plate.

Fig. 259: Front F70 fixing adjustable plate



- Loosen the 2 x screws at the front angle adjustable plate.
- Move the angle flush and tight with the F70 cabinet housing and retighten the 2 x screws.

Fig. 260: Packing the hydraulic jacks



- (1) Pressure relief valve screw
(2) Hydraulic pump bracket

To removing air from the hydraulic jack

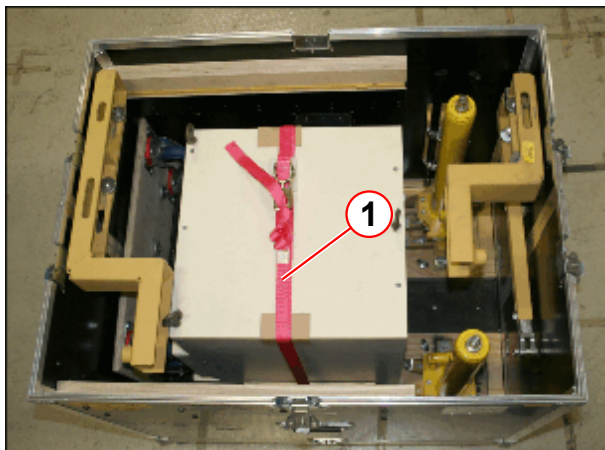
- Place the rod onto the pressure relief valve screw.
- Turn it counterclockwise by not more than 3 turns.
- Insert rod into the neck of the rod.
- Perform between 6 to 8 complete pump strokes.
- Repeat if necessary.

Fig. 261: Manual press piston



- If the piston has not fully lowered, manually press down.
- Repeat for the 2nd jack.

Fig. 262: Top view of compressor kit



- For the return shipment, pack all the tooling parts back into the transport box in the correct locations and tighten all screws.

(1) Securing strap

Fig. 263: Transport crate



- (1) Lifting handles (x 2)
- (2) top lid latch (x 6)
- (3) Lid with PTFE sheet covering

- Replace the top lid of the transport box.
- Return the compressor transport box to CSML.

4.1 MSUP for Avanto / Avanto Dot Upgrade / Espree

4.1.1 Replacing the Magnet Supervisory Unit (MSUP 2657)

4.1.1.1 General

This procedure is only applicable to Avanto and Espree systems.

4.1.1.2 Tools and time required

Time

60 min

Tools

- ERDU test load kit (part no **03861197**)
- Non-magnetic tool kit

4.1.1.3 Preliminary

Data from the old MSUP must be transferred to the replacement unit .

- Download the magnet specific values from the MSUP to the HOST.
 - Open **SeSO/Magnet&Cooling/Initialisation**.
 - If control and communication of the old MSUP still works, item 1,2 and 3 can be downloaded to the host.
- Note the MSUP parameters from the data locations detailed below.

Tab. 19 MSUP Configurable parameters

	Parameter	Data Location
1	0% helium level calibration He Probe 1	Acceptance document/or MSUP download
2	100% helium level calibration He Probe 2	Acceptance document/or MSUP download
3	Pressure heater resistance	Acceptance document/or MSUP download
4	Magnet serial number	Acceptance document and Magnet ID plate
5	Magnet revision	Acceptance document and Magnet ID plate
6	Supervisory serial number	Label on MSUP
7	Supervisory revision	Label on MSUP
8	Operating current	Last ramp current saved on MSUP

	Parameter	Data Location
9	Date	Current date
10	Time	Current time

**CAUTION**

The replacement procedure disables the magnet emergency shut down system at the MSU.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » Connect the ERDU Test Load.
- » The magnet may only be quenched using the MAGNET STOP button on the ERDU Test Load
- » Exclude all non-service personnel and all magnetic materials / tools from the magnet area.
- » Close the door of the RF room

4.1.1.4 Connecting the ERDU Test Load (ETL)

**WARNING**

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorised personnel must be excluded from the imaging suite.

- » Secure the room and prevent unauthorised access.

**WARNING**

It is not possible to quench the magnet using the normal system MAGNET STOP buttons while the ERDU Test Load is connected to the MSUP and magnet.

- » The magnet may be quenched only by using the MAGNET STOP button on the ERDU Test Load.
- » Restrict access to the magnet accordingly.

**CAUTION**

The ERDU Test Load is slightly ferromagnetic.

The ETL is attracted by the magnetic force and will be projected into the magnet bore, resulting in injury or damage.

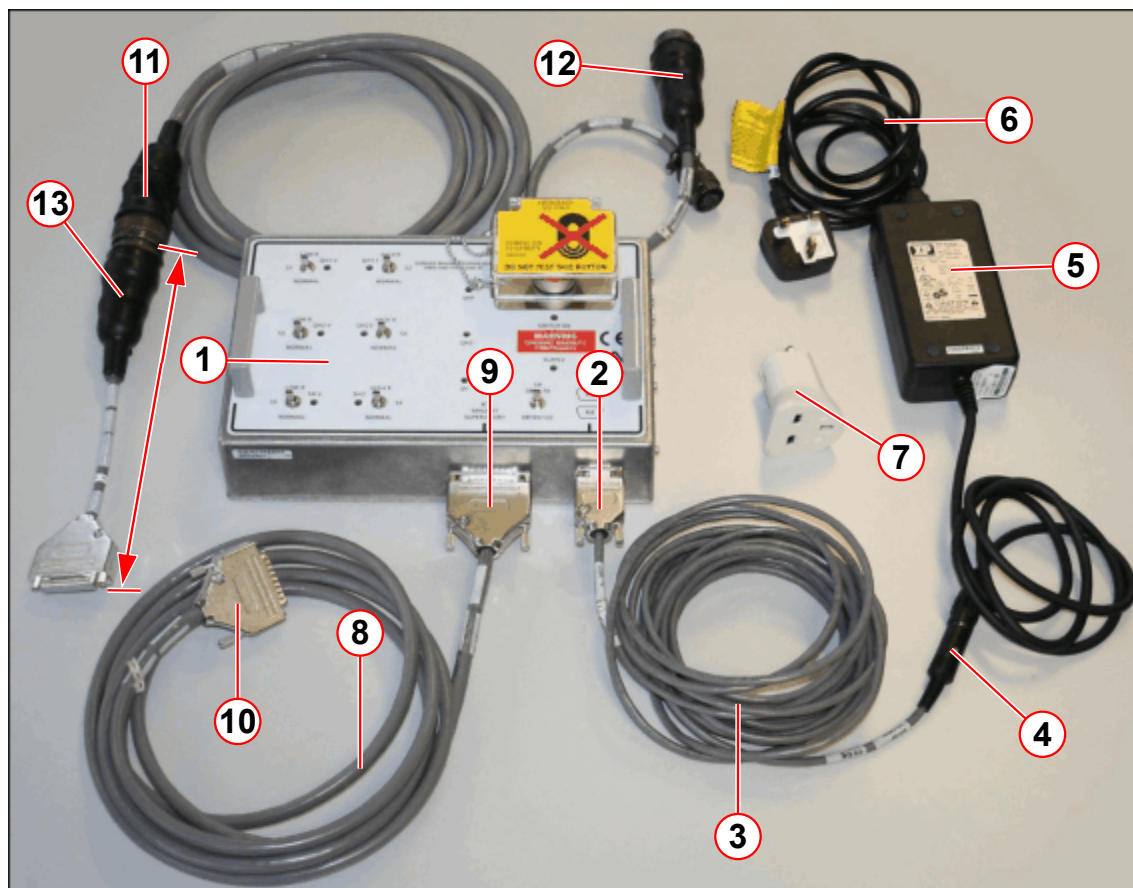
- » Keep the ETL as far away as possible from the magnet bore.

- Check that the **ERDU Test Load Kit** is complete with all the listed parts present:



The Adaptor cable B is available from CSML as a spare (part no. 10129824)

Fig. 264: ERDU Test load kit (2661)



- (1) ERDU Test load 3 (ETL)
- (2) PSU output extension lead to ETL X3 or X4
- (3) PSU output extension lead
- (4) PSU output extension lead connector to PSU
- (5) Power Supply Unit (PSU)
- (6) AC supply lead
- (7) International adapter - AC supply
- (8) MSUP - ETL cable
- (9) 25-pin connector to ETL X1
- (10) 25-pin connector to MSUP X1
- (11) 41-pin connector, magnet turret
- (12) Adapter for OR64 / OR70, 41-pin connector to 10-pin magnet turret connector 'B' (optional ETL3 2661)
- (13) Adaptor cable B (optional for OR105, OR122, OR97)

- Connect the **ETL PSU** to a convenient AC outlet **outside of the magnet room**, and switch on. (→ 5/ Fig. 264 Page 193)



Ensure that the **PSU ETL switch mode** is as far away from the magnet as possible (magnetic field < 50 Gauss).

- The ETL will sound a **continuous** buzzing alarm, until the ETL is disconnected.
- Connect the ETL switch mode PSU to the ETL **X3** or **X4** using the extension cable (→ 3/Fig. 264 Page 193)
 - » Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.
- Connect the "<MSUP-ETL>" cable (→ 8/Fig. 264 Page 193) to the 25-pin connector **X1** on the ERDU Test Load.

WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

- » Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**
- » **DO NOT** operate the **MAGNET STOP** button on the ERDU Test Load unless an emergency magnet quench is required.

- Set **SH** for correct magnet type. (→ Fig. 265 Page 194).

Fig. 265: ERDU Test Load (front panel)



- Set all **HIGH R / LOW R** switches on the ETL to **NORMAL**.

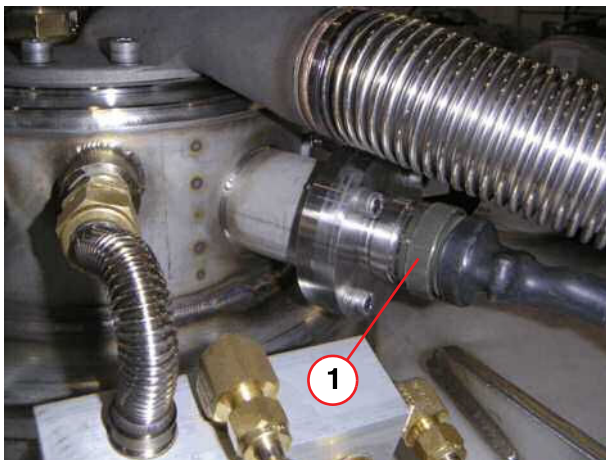
**WARNING**

The magnet cannot be quenched!!

If the circuit between the 41 way connector on the magnet to X1 connector on the MSUP is disconnected, THERE IS NO ERDU FACILITY.

- » Reconnect X1 (or the 41-pin turret connector), WITHOUT DELAY.
- » As soon as the ETL is connected to the magnet, the test ERDU function on the ETL is operational.

Fig. 266: Mk2 Turret - 41pin connector



(1) 41pin connector



Keep the ETL as far away as possible from the magnet bore.

Workflow 1 - Turret connection	Workflow 2 - Adaptor cable B connection
Disconnect the existing lead from the 41-pin connector on the magnet turret .	Connect the adaptor cable B to the 41-pin turret connector cable on the ETL.
IMMEDIATELY connect the 41-pin connector from the ETL to the magnet turret connection . (→ 11/Fig. 264 Page 193)	Disconnect X1 from the MSUP and connect X1 cable to the adaptor cable B on the ETL. (→ 13/Fig. 264 Page 193)

- Check that the alarm sounds and that the **MAG STOP** supply indicator lights up on the **Alarm Box** (→ Fig. 267 Page 196).
 - » Press **ACKNOWLEDGE** to cancel the audible alarm.
- Check that the **QH1, QH2, SH** and **SWITCH OK** and **SUPPLY** indicators glow on the ETL.

Fig. 267: Alarm Box (buttons and controls)

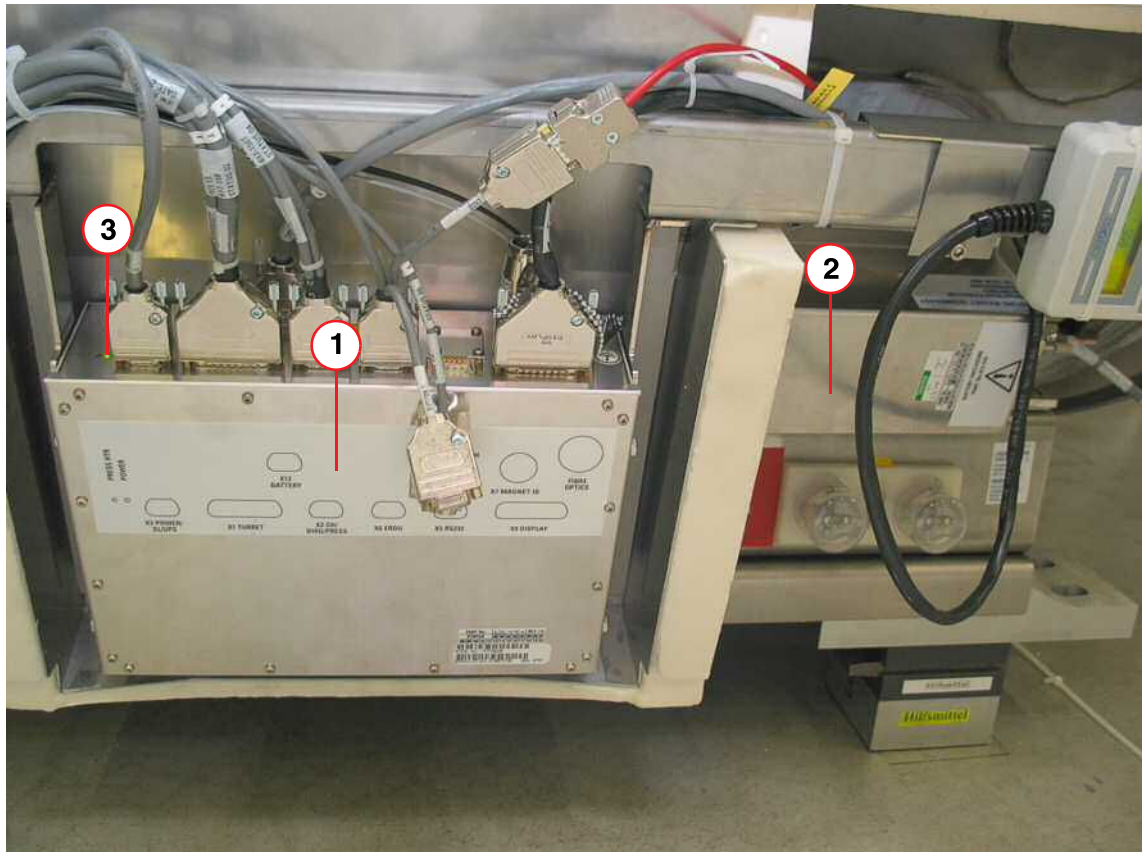


4.1.1.5 MSUP Positioning

Avanto - Static and Mobile Systems

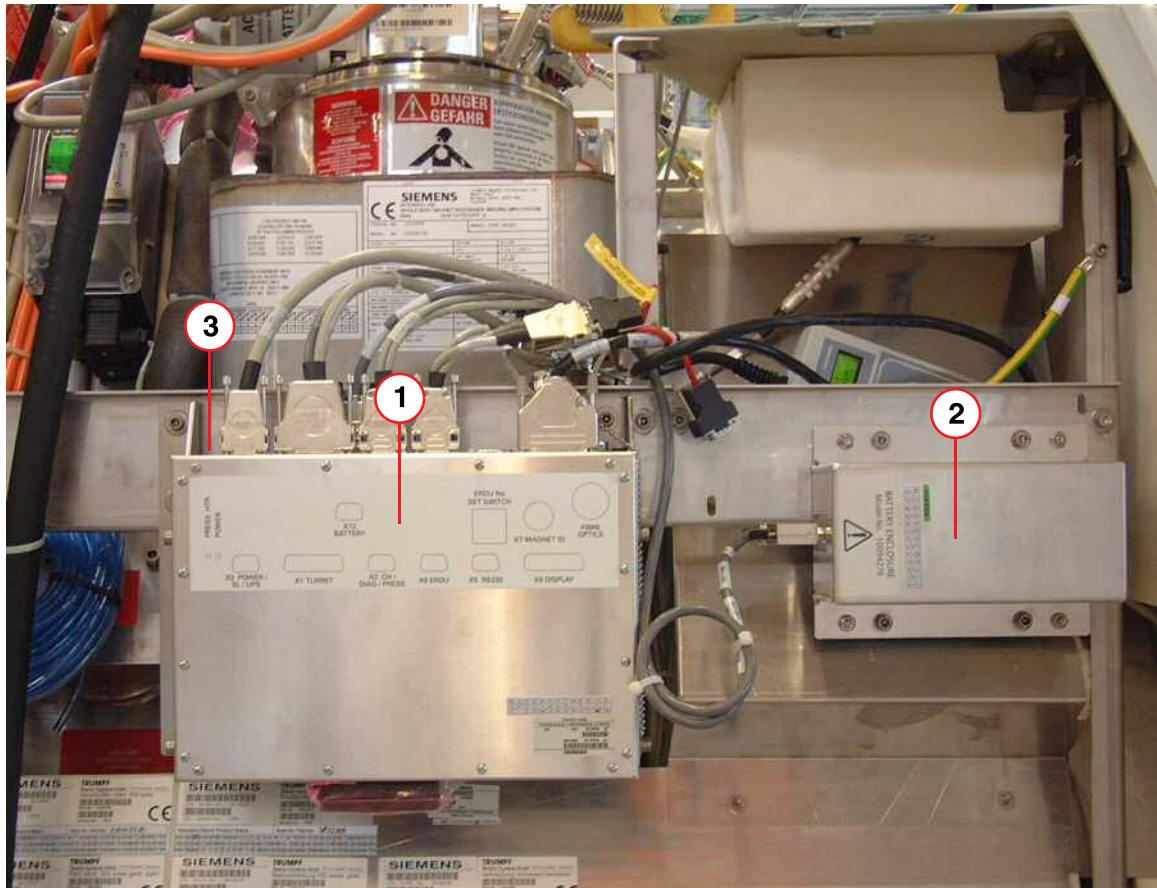
- Early Avanto systems - see (→ Fig. 268 Page 197)
- Current Avanto system see (→ Fig. 269 Page 198)
- Mobile Avanto systems - see (→ Fig. 270 Page 199)

Fig. 268: MSUP Avanto (early systems)



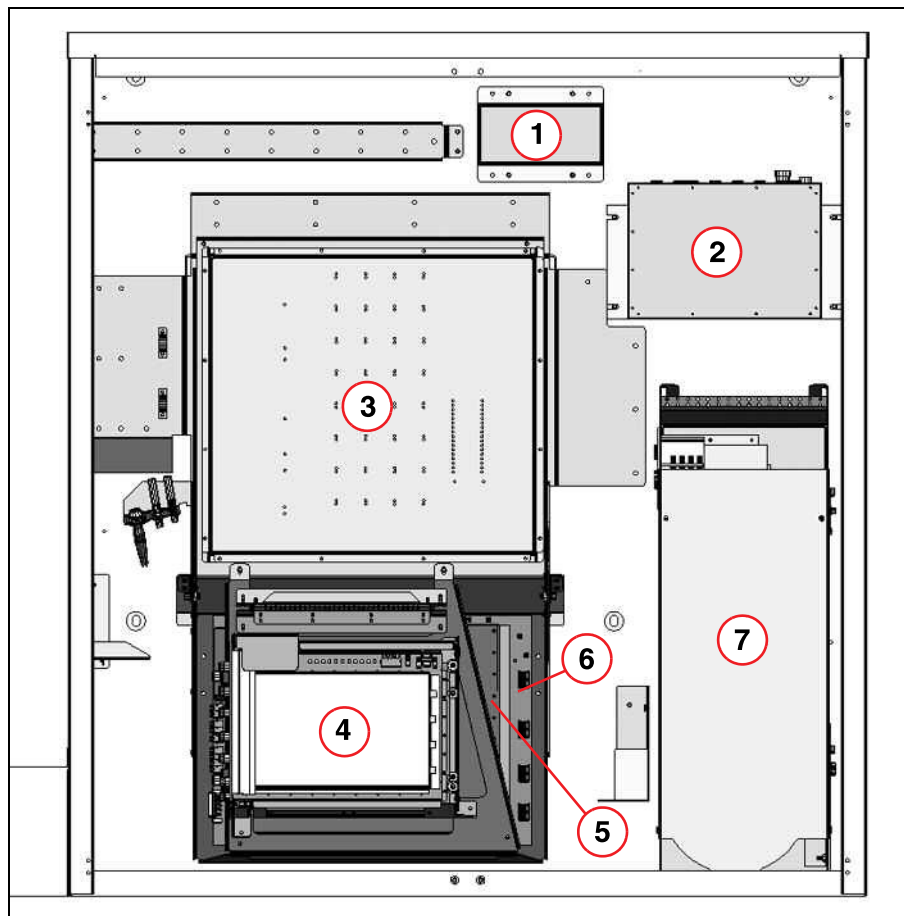
- (1) MSUP
- (2) MSUP battery
- (3) LEDs (green = 36 VDC POWER ON, yellow = Pressure Heateron)

Fig. 269: MSUP Avanto



- (1) MSUP
- (2) MSUP battery
- (3) LEDs (green = 36 VDC POWER ON, yellow = Pressure Heater on)

Fig. 270: Overview Electronics Cabinet - Mobile

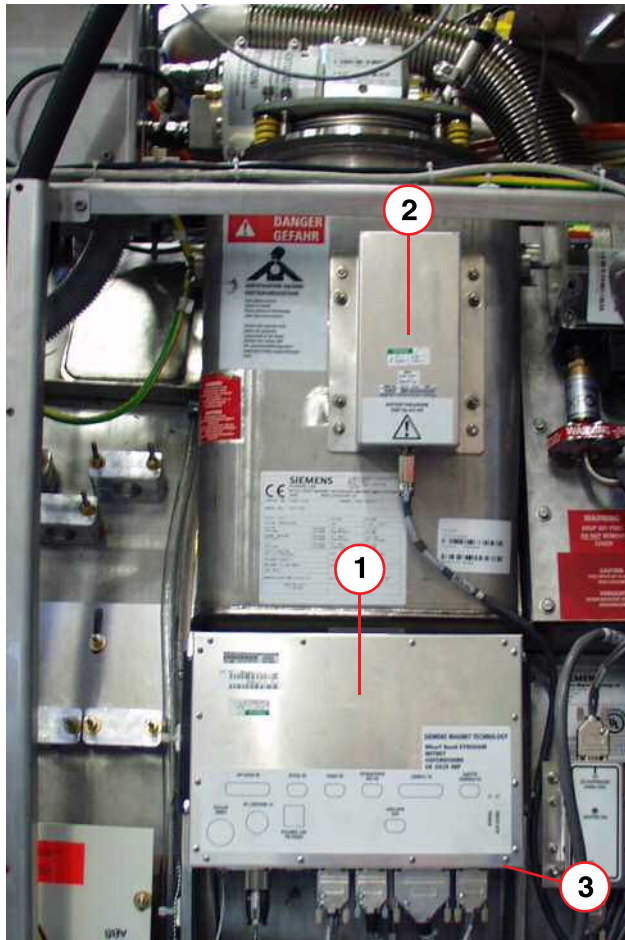


- (1) MSUP battery
- (2) MSUP
- (3) RFIS
- (4) TALEs
- (5) BCCS
- (6) RCCS
- (7) Control Electronics (Patient Table)

Espreo - Static and Mobile Systems

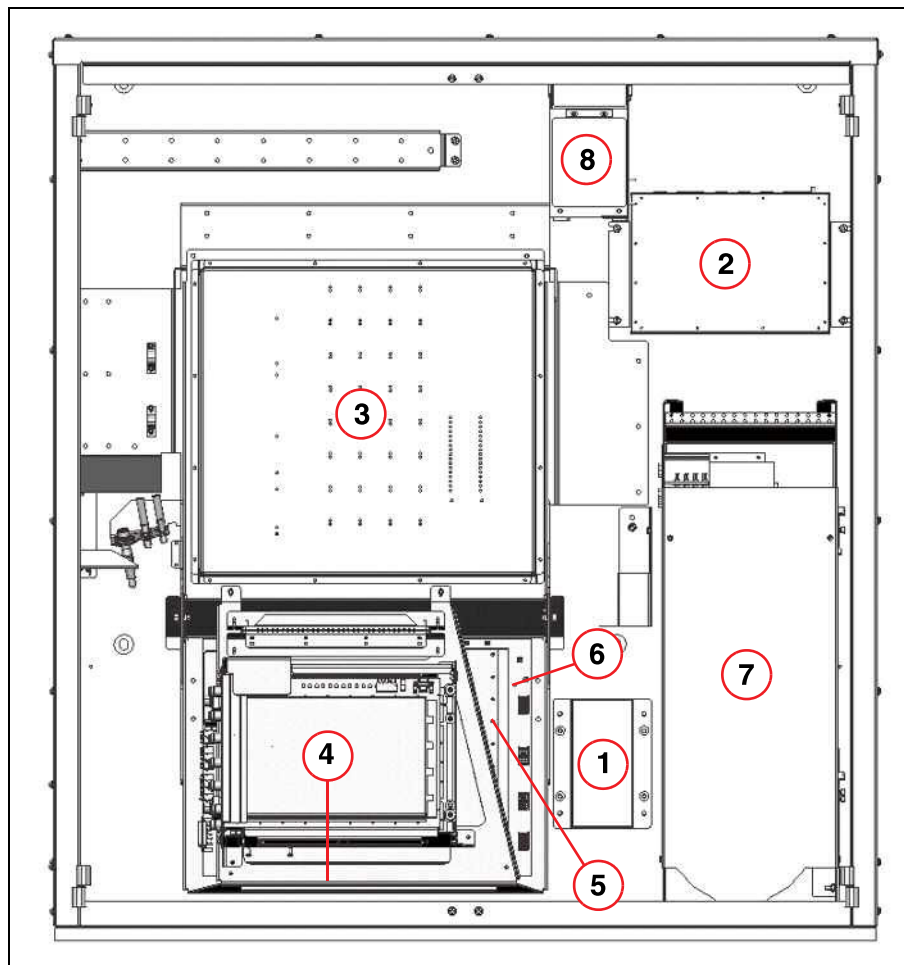
- Current Espreo System - see (→ Fig. 271 Page 200)
- Mobile Espreo System - see (→ Fig. 272 Page 201)

Fig. 271: MSUP Espree



- (1) MSUP
- (2) MSUP battery
- (3) LEDs (green = 36 VDC POWER ON, yellow = Pressure Heater on)

Fig. 272: Overview Electronics Cabinet, Mobile



- (1) MSUP battery
- (2) MSUP
- (3) RFIS
- (4) TALES
- (5) BCCS
- (6) RCCS
- (7) Control Electronics (Patient Table)
- (8) Z2-Shim

4.1.1.6 Replacement Procedure

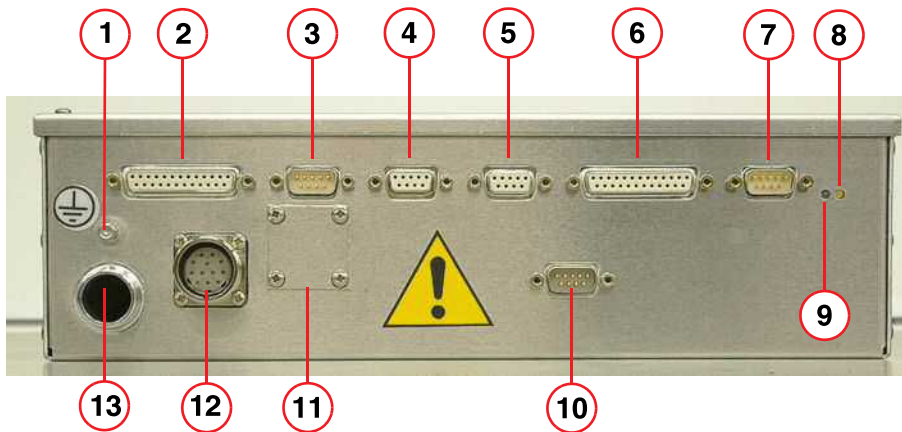
Removing the old MSUP

- Remove **fuse F3** (ACC) or turn off **F3** on the ACC.
 - » The power for the MPSU is now **off**.



Use the ESD kit during the following steps.

Fig. 273: MSUP 2657 Indicators and input/output connectors

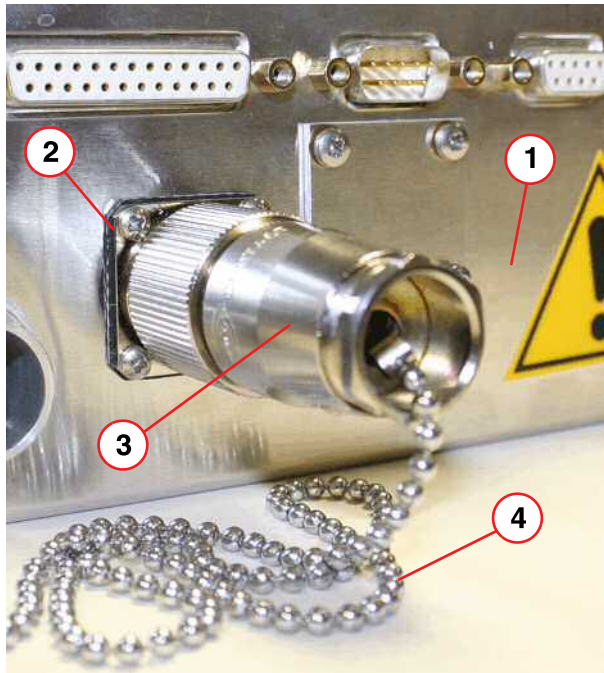


- (1) Earth stud (6mm)
- (2) X9 Display
- (3) X5 RS232
- (4) X6 ERDU/Field
- (5) X2 CH/DIAG/PRESS
- (6) X1 Turret
- (7) X3 Power/SL/UPS
- (8) Pressure heater LED (yellow)
- (9) Power ON LED (green)
- (10) X12 Battery
- (11) ERDU switch cover
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

■ **Disconnect** from the MSUP:-

- Turret cable **X1** (→ 6/Fig. 273 Page 202)
- Power cable **X3** (→ 7/Fig. 273 Page 202)
- Battery cable **X12** (→ 10/Fig. 273 Page 202)

Fig. 274: MSUP Magnet ID



- (1) MSUP connector panel
- (2) Connector X7
- (3) Magnet ID connector
- (4) Securing chain

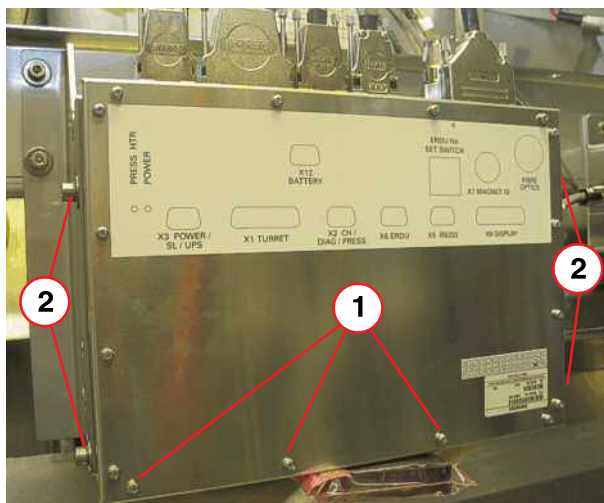
- **Disconnect** the **Magnet ID connector** from **X7** (pos 2).

The **Magnet ID** contains information about this **specific** magnet.

Ensure that the Magnet ID connector is transferred to the replacement MSUP unit.

- **Disconnect** the remaining 'D' type electrical connectors (→ Fig. 273 Page 202).

Fig. 275: MSUP Mounting bolts and screws

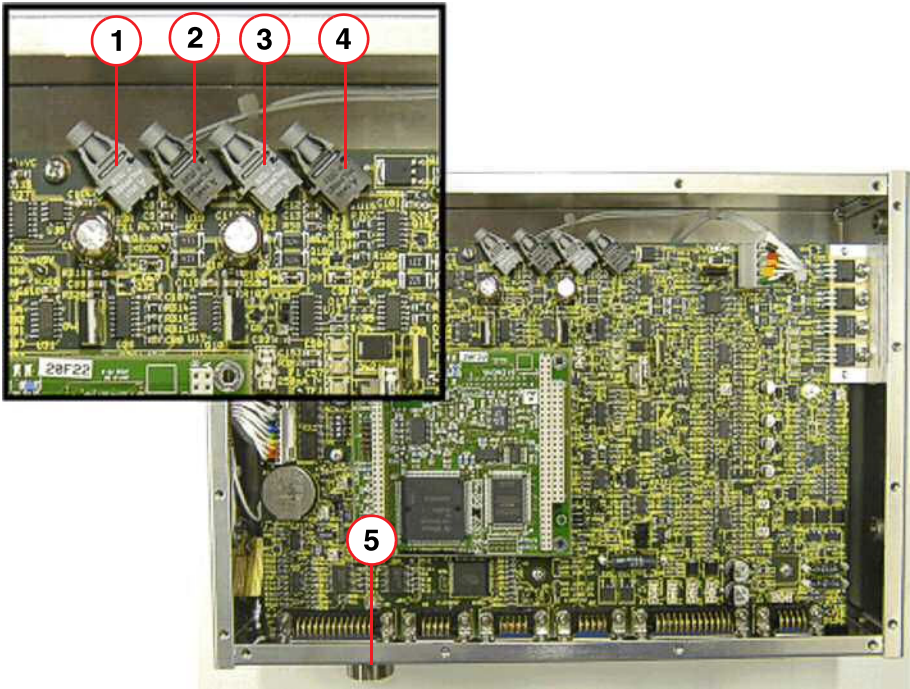


- (1) Retaining screws (16 off)
- (2) Securing bolts and washers (2 off each end)

- **Remove** the lid of the MSUP by removing the screws (16 positions).

- Carefully **disconnect** fiber optic cables **U38, U39, U40, U41** from the connectors on the **MSUP PCB**, and withdraw the **fiber optic cables** through the **wave guide**.
(→ 5/ Fig. 276 Page 204)

Fig. 276: MSUP Fiber Optic cable connectors



- (1) U38
- (2) U39
- (3) U40
- (4) U41
- (5) CAN Fiber optics wave guide

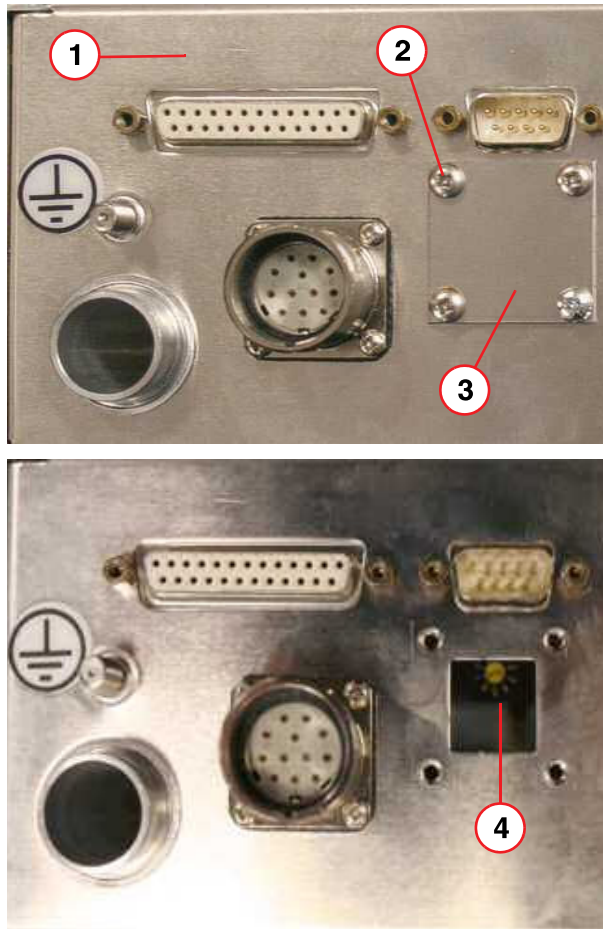
- Remove** the (4 off) bolts securing the MSUP to the cradle (→ 2/ Fig. 275 Page 203).
- Carefully lift out the MSUP unit.
- Re-fit** the lid using the (16 off) screws. (→ 1/ Fig. 275 Page 203)
- Return the defective unit to CSML.

Installing the new MSUP

Tab. 20 ERDU Switch Settings

Number of ERDU Buttons	ERDU Selection Switch
Alarm Box	1
Alarm Box + 1 ERDU Button	2
Alarm Box + 2 ERDU Buttons	3

Fig. 277: MSUP ERDU selection switch

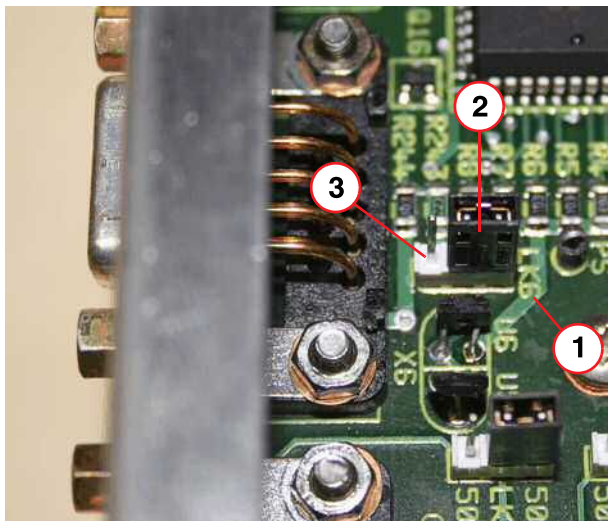


- (1) MSUP connector panel
- (2) Screws (4 positions)
- (3) Switch cover plate
- (4) ERDU selection switch

- **Remove** the (4 off) screws (pos 2) securing the **switch cover** (pos 3).
- Set the **ERDU switch** as required (pos 4)- refer to (→ Tab. 20 Page 204).
- **Re-fit** the switch cover using the (4 off) screws.

- Remove the **MSUP lid** by removing the (16 off) screws (→ 1/Fig. 275 Page 203).

Fig. 278: Setting the internal/external field sensor



- (1) Location of LK6 on PCB
 (2) LK6 - jumper set in INT position (STATIC systems)
 (3) Jumper position for EXT (Mobile systems)

- Set the **external field sensor LK6** in the correct position on the PCB.
 - » On **static** Avanto and Espree systems, **LK6** must be in the **INT** position.
 - » On **mobile** Avanto and Espree systems, **LK6** must be in the **EXT** position.

- Carefully insert the fiber optic cables **U38, U39, U40, U41** through the **waveguide** and into their correct connectors on the PCB (→ Fig. 276 Page 204).
- **Secure** the new MSUP to the support cradle using the (4 off) bolts (→ 2/Fig. 275 Page 203).
- **Refit** the MSUP lid using the (16 off) screws (→ 1/Fig. 275 Page 203).



Do not mix up the lids of the MSUP. Check the serial number labels.

- **Re-connect to the MSUP:-**
 - External connectors **X2, X6** (→ Fig. 273 Page 202).
 - **Magnet ID** connector to **X7** (→ 2/Fig. 274 Page 203).
 - Battery connector **X12** (→ 10/Fig. 273 Page 202).
 - Power connector **X3** (→ 7/Fig. 273 Page 202).
- Connect the **ETL-MSUP lead** (→ 8/Fig. 264 Page 193) to **X1** on the MSUP.
- Insert fuse **F3** (ACC), (or switch on F3).
 - » Check that the **green LED** on the MSUP is illuminated (→ 9/Fig. 273 Page 202).
 - » The Alarm box may sound an **audible alarm** - press the **ACKNOWLEDGE** button to cancel the alarm.
- Press the **ACKNOWLEDGE** button on the **Alarm Box** to silence all alarms.

- Perform an **ERDU test** using the ERDU test load - (refer to **Maintenance, Safety-Related Tests- ERDU**).
 - » Begin the procedure at **“Heater resistance monitor tests”**, and follow the steps to the end of the procedure.

4.1.1.7 Concluding checks

Upload old data from HOST to MSUP

- **Reboot** the Host PC to load the correct loadware.
- **Check** that all alarm and fault indications show that the system is OK.
- **Open** SeSo / Magnet&Cooling / Initialization.
 - **Check** the magnet specific values and compare with the **acceptance document**.
 - Press the **UPLOAD** button to transfer the magnet-specific values from the **HOST** to the **new MSUP**.
- **Exit** the Magnet&Cooling menu.

Direct input to MSUP

If it was not possible to upload the old data to the host, refer to the **acceptance documents**, (refer to (→ Tab. 19 Page 190)) and then do the following:-

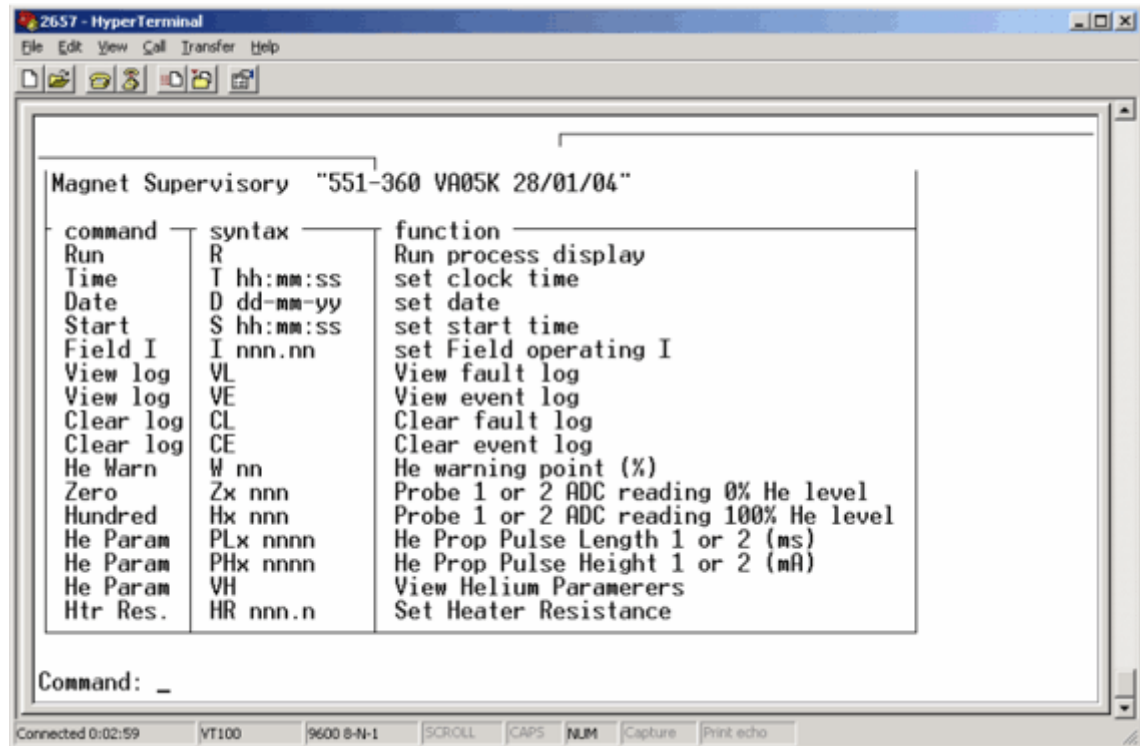
- **Open** the **HyperTerminal** application and select the appropriate **Com Port**.
- Select the following settings:

Baud rate	9600
Bits	8
Parity	None
Stop Bits	1
Flow	None
Terminal Emulation	VT100

- **Connect** the laptop via a **RS232** interface cable to **X5** (→ 3/ Fig. 273 Page 202).

- Press **Escape < R>** - the **Command Input** screen is displayed.

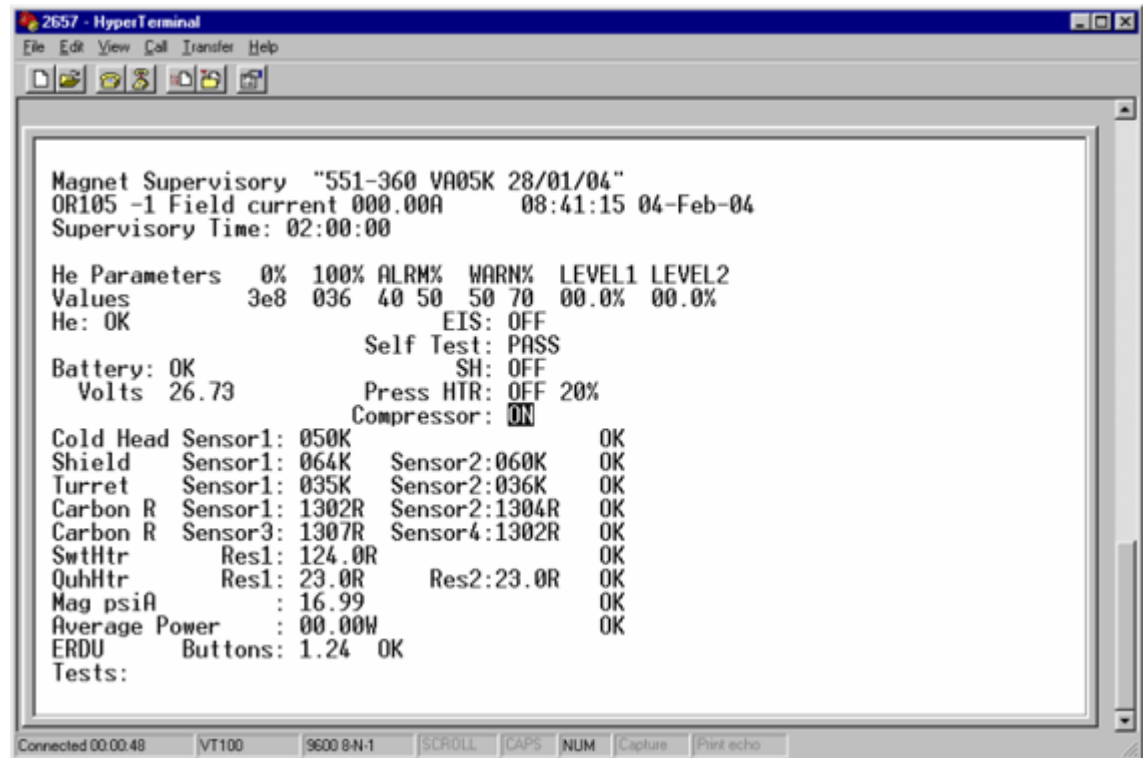
Fig. 279: MSUP Command Input screen



- Enter the required values into the **Command: _** line (refer to (→ Tab. 21 Page 210)).

- Press **R <R>** to run the **Process Display**. The updated data will be displayed.

Fig. 280: MSUP Process Display Screen



Calculating Helium Probe Calibration at early sites

The latest version of the acceptance document provided by SMT lists both the resistance and the converted values.

For earlier sites, (where the calculated value may not be available) use the formula below to calculate the calibration values.

$R \times 0.25 \times 36.27 = \text{Input Value}$, where R = Helium Probe Resistance.

- Enter the resulting value into the MSUP as a **whole number**. Example:-
 - » Resistance = 88.8 Ohms
 - » $88.8 \times 0.25 \times 36.27 = 805.194$
 - » Value used to calibrate is **805**

All values are entered from the **command menu** (→ Fig. 279 Page 208).

Tab. 21 MSUP Input Commands

Command	Input Data	Value 'n'	Example
0% Calibration	Z1 nnnn<R>	He Probe 1	Probe1: 88.8 ohms -enter as Z1 805<R>
	Z2 nnnn<R>	He Probe 2	Probe2: 88.8 ohms - enter as Z2 805<R>
100% Calibration	H1 nnnn<R>	He Probe 1	Probe1: 1.8 ohms - enter as H1 16<R>
	H2 nnnn<R>	He Probe 2	Probe2: 1.8 ohms - enter as H2 16<R>
Pressure Heater Resistance	*L 9912<R> HR nnn.n<R>	Pressure Heater Res	19.4 ohms - enter as HR 19.4<R>
Magnet Serial No.	*L 9912<R> MSN nnnnnnnn<R>	Magnet Serial No	Enter as MSN12345678<R>
Magnet Revision No.	*L 9912<R> MRN nnn<R>	Mag Rev No	Magnet Rev 4 - enter as MRN4<R>
Supervisory Serial No.	*L 9912<R> SSN nnnnnnnn<R>	Supervisory Serial No.	Supervisory Serial No 26570001 - enter as SSN 26570001<R>
Supervisory Revision No	L 9912<R> SRN nnn<R>	Supervisory Revision No	Supervisory Revision No 14 - enter as SRN14<R>
Magnet Operating Current	I nnn.nn<R>	Mag Operating Current	Operating current 508.23 - enter as I 508.23<R> This parameter is also automatically updated when the magnet is next ramped to field.
Date	D dd-mm-yy<R>	n/a	1st Jan 2006 - enter as D 01-01-06<R>
Time	T hh:mm:ss<R>	n/a	1 hour 30mins 30secs - enter as T 13:30:30<R>

Command	Input Data	Value 'n'	Example
Notes: ■ <R> = 'Enter' or 'Return' ■ *L 9912<R> 'unlocks' the option (otherwise it is not possible to change the value of the parameter). » Only settable from supervisory loadware version VA06D onwards.			

4.1.1.8 Final steps

- Perform a manual sample by pressing SAMPLE on the MSUP-Display Unit.
- Check that all alarm and fault indications show that the system is OK.

4.1.2 Replacing the Magnet Supervisory Unit (MSUP 2657B)

4.1.2.1 Tools and time required

Time

60 min

Tools

- ERDU test load kit (part no **03861197**)
- Non-magnetic tool kit

Spare Parts

LVQD Functionality

The MSUP 2657B is **not compatible** with some older Avanto/Espree magnets.

To use the MSUP 2657B, the loadware version must be at least **VA07B**. Therefore the minimum **system software** must be **VB13A + SP2**.

Order the correct replacement MSUP according to the spare part list.

Tab. 22 MSUP 2657B Compatibility

Magnet	System status	X14 connected
Avanto OR105	**34 and below	No
	35 and onwards	Yes
Espree OR122	**10 and below	No
	11 and onwards	Yes



MSUP 2657B can be used on any TIM system, as long as the system software has been upgraded to VB13A + SP2 or higher.

X14 should NOT be used on older systems** due to the LVQD circuit invalidating UL approval

4.1.2.2 Preliminary

Data from the old MSUP must be transferred to the replacement unit .

- Download the magnet specific values from the MSUP to the HOST (items 1,2 and 3 in table below (→ Tab. 23 Page 213)).
 - Open **SeSO > Magnet&Cooling > Initialisation**.
 - If control and communication of the old MSUP still works, item 1,2 and 3 can be downloaded to the host.
- Note the MSUP parameters from the data locations detailed below.

Tab. 23 MSUP 2657B Configurable parameters

	Parameter	Data Location
1	0% helium level calibration He Probe 1,2	Acceptance document/or MSUP download
2	100% helium level calibration He Probe 1,2	Acceptance document/or MSUP download
3	Pressure heater resistance	Acceptance document/or MSUP download
4	Magnet serial number	Acceptance document and Magnet ID plate
5	Magnet revision	Acceptance document and Magnet ID plate
6	Supervisory serial number	Label on MSUP
7	Supervisory revision	Label on MSUP
8	Operating current	Last ramp current saved on MSUP
9	Date	Current date
10	Time	Current time
11	Link and Bore Shield limits	Link and Bore Shield limits
12	CCR Calibration (as required)	Magnet Acceptance Document, or MSUP download.

**CAUTION**

The replacement procedure disables the magnet emergency shut down system at the MSU.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » Connect the ERDU Test Load.
- » The magnet may only be quenched using the MAGNET STOP button on the ERDU Test Load
- » Exclude all non-service personnel and all magnetic materials / tools from the magnet area.
- » Close the door of the RF room

4.1.2.3 Connecting the ERDU Test Load (ETL)

**WARNING**

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorised personnel must be excluded from the imaging suite.

- » Secure the room and prevent unauthorised access.

**WARNING**

It is not possible to quench the magnet using the normal system MAGNET STOP buttons while the ERDU Test Load is connected to the MSUP and magnet.

- » The magnet may be quenched only by using the MAGNET STOP button on the ERDU Test Load.
- » Restrict access to the magnet accordingly.

**CAUTION**

The ERDU Test Load is slightly ferromagnetic.

The ETL is attracted by the magnetic force and will be projected into the magnet bore, resulting in injury or damage.

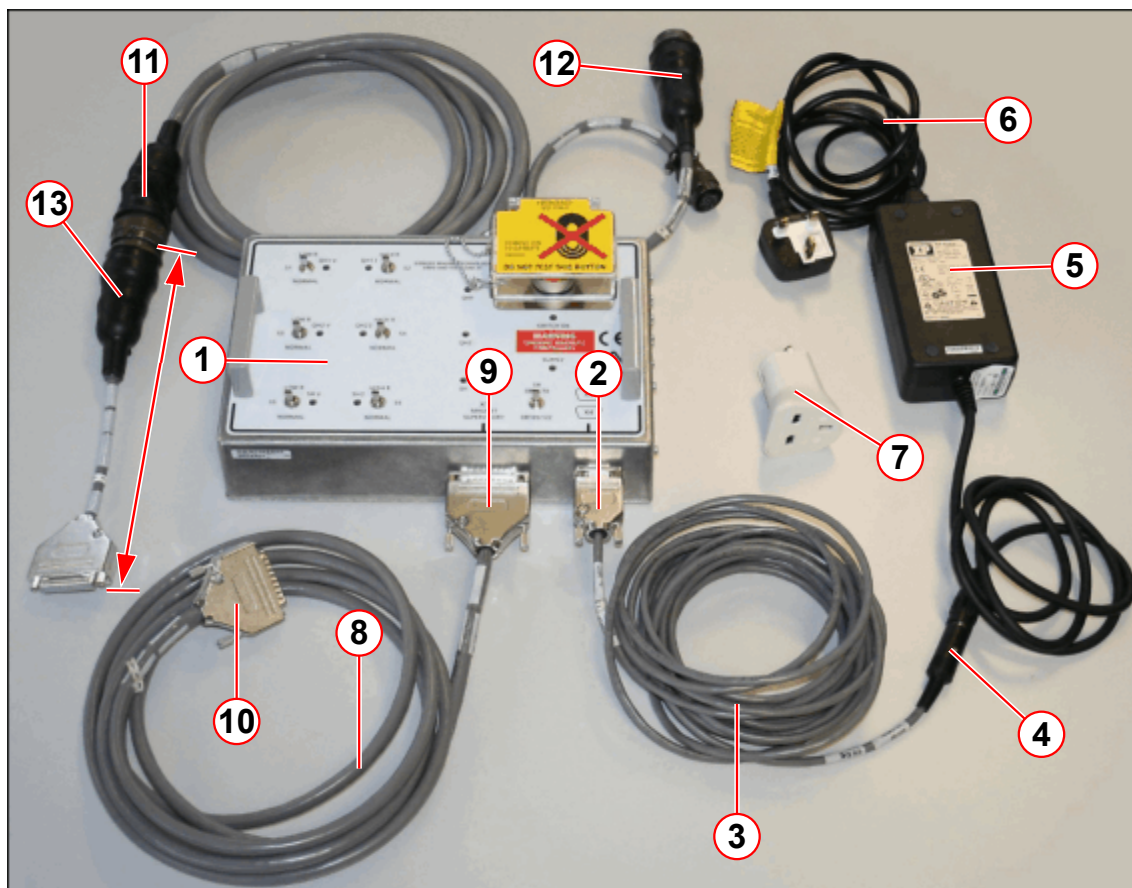
- » Keep the ETL as far away as possible from the magnet bore.

- Check that the **ERDU Test Load Kit** is complete with all the listed parts present:



The Adaptor cable B is available from CSML as a spare (part no. 10129824)

Fig. 281: ERDU Test load kit (2661)



- (1) ERDU Test load 3 (ETL)
- (2) PSU output extension lead to ETL X3 or X4
- (3) PSU output extension lead
- (4) PSU output extension lead connector to PSU
- (5) Power Supply Unit (PSU)
- (6) AC supply lead
- (7) International adapter - AC supply
- (8) MSUP - ETL cable
- (9) 25-pin connector to ETL X1
- (10) 25-pin connector to MSUP X1
- (11) 41-pin connector, magnet turret
- (12) Adapter for OR64 / OR70, 41-pin connector to 10-pin magnet turret connector 'B' (optional ETL3 2661)
- (13) Adaptor cable B (optional for OR105, OR122, OR97)

- Connect the **ETL PSU** to a convenient AC outlet **outside of the magnet room**, and switch on. (→ 5/ Fig. 281 Page 215)



Ensure that the **PSU ETL switch mode** is as far away from the magnet as possible (magnetic field < 50 Gauss).

- The ETL will sound a **continuous** buzzing alarm, until the ETL is disconnected.
- Connect the ETL switch mode PSU to the ETL **X3** or **X4** using the extension cable (→ 3/ Fig. 281 Page 215)
 - » Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.
- Connect the "<MSUP-ETL>" cable (→ 8/ Fig. 281 Page 215) to the 25-pin connector **X1** on the ERDU Test Load.

WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

- » Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**
- » **DO NOT** operate the **MAGNET STOP** button on the ERDU Test Load unless an emergency magnet quench is required.

- Set **SH** for correct magnet type. (→ Fig. 282 Page 216).

Fig. 282: ERDU Test Load (front panel)



- Set all **HIGH R / LOW R** switches on the ETL to **NORMAL**.

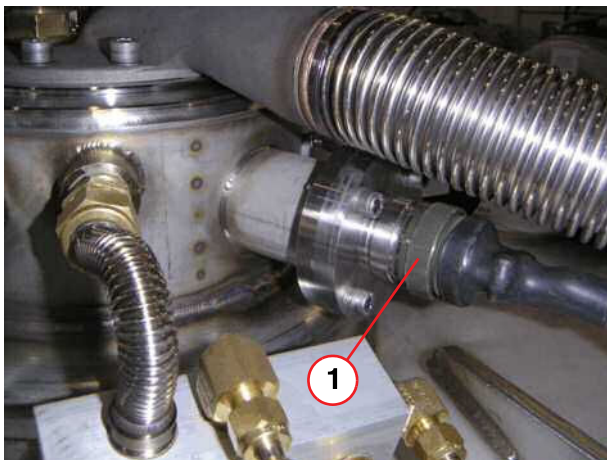
**WARNING**

The magnet cannot be quenched!!

If the circuit between the 41 way connector on the magnet to X1 connector on the MSUP is disconnected, THERE IS NO ERDU FACILITY.

- » Reconnect X1 (or the 41-pin turret connector), WITHOUT DELAY.
- » As soon as the ETL is connected to the magnet, the test ERDU function on the ETL is operational.

Fig. 283: Mk2 Turret - 41pin connector



(1) 41pin connector



Keep the ETL as far away as possible from the magnet bore.

Workflow 1 - Turret connection	Workflow 2 - Adaptor cable B connection
Disconnect the existing lead from the 41-pin connector on the magnet turret .	Connect the adaptor cable B to the 41-pin turret connector cable on the ETL.
IMMEDIATELY connect the 41-pin connector from the ETL to the magnet turret connection . (→ 11/Fig. 281 Page 215)	Disconnect X1 from the MSUP and connect X1 cable to the adaptor cable B on the ETL. (→ 13/Fig. 281 Page 215)

- Check that the alarm sounds and that the **MAG STOP** supply indicator lights up on the **Alarm Box** (→ Fig. 284 Page 218).
 - » Press **ACKNOWLEDGE** to cancel the audible alarm.
- Check that the **QH1, QH2, SH** and **SWITCH OK** and **SUPPLY** indicators glow on the ETL.

Fig. 284: Alarm Box (buttons and controls)



4.1.2.4 Replacement Procedure

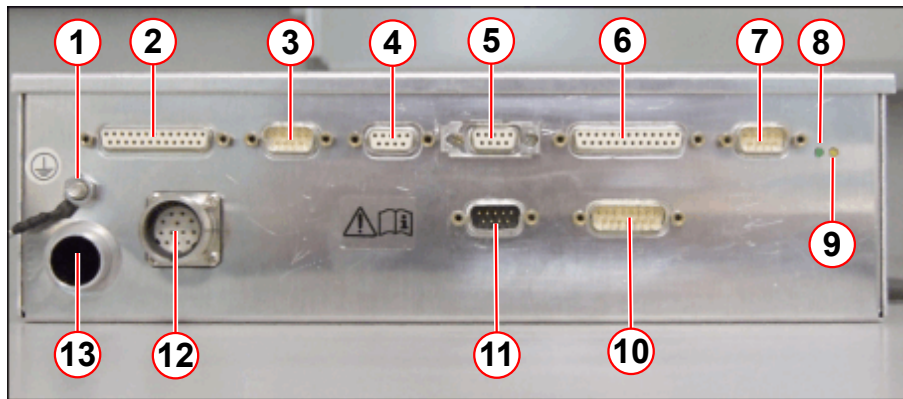
Removing the old MSUP

- Remove **fuse F3** (ACC) or turn off **F3** on the ACC.
 - » The power for the MPSU is now **off**.



Use the ESD kit during the following steps.

Fig. 285: MSUP 2657B - Indicators and input/output connections

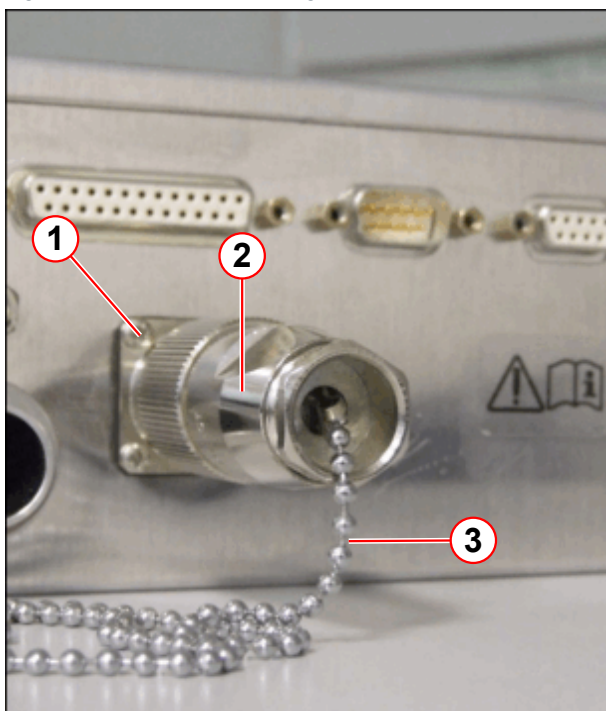


- (1) Earth stud (6mm)
- (2) X9 DISPLAY
- (3) X5 RS232
- (4) X6 ERDU
- (5) X2 CH/DIAG/PRESS
- (6) X1 TURRET
- (7) X3 POWER/SL/UPS
- (8) Power ON LED (green)
- (9) Pressure heater LED (yellow)
- (10) X14 LVQD
- (11) X12 BATTERY
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

■ **Disconnect from the MSUP:-**

- Turret cable **X1**. (→ 6/Fig. 285 Page 219)
- Power cable **X3**. (→ 7/Fig. 285 Page 219)
- Battery cable **X12**. (→ 11/Fig. 285 Page 219)

Fig. 286: MSUP 2657B - Magnet ID



- (1) Connector X7
- (2) Magnet ID connector
- (3) Securing chain

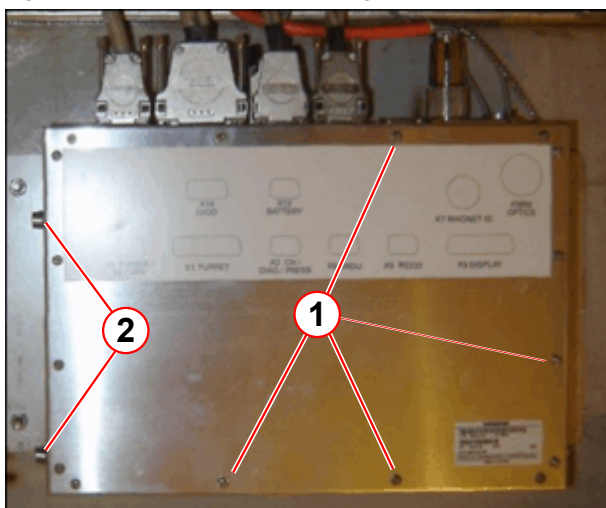
- **Disconnect** the **Magnet ID connector** from **X7** (pos 1).

The **Magnet ID** contains information about this **specific** magnet.

Ensure that the Magnet ID connector is transferred to the replacement MSUP unit.

- **Disconnect** the remaining 'D' type electrical connectors. (→ Fig. 285 Page 219)

Fig. 287: MSUP 2657B - Mounting bolts and screws

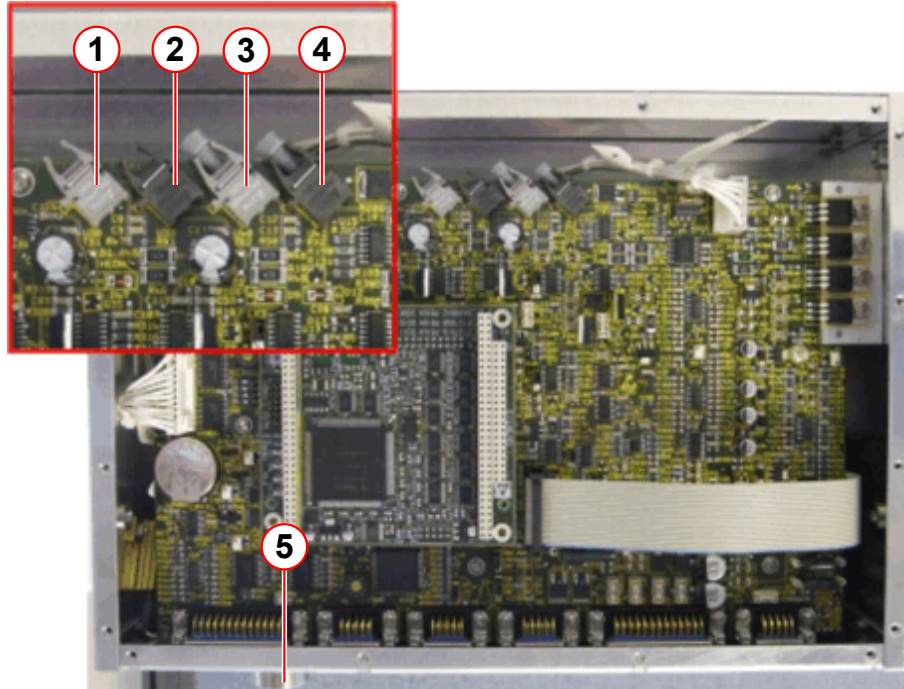


- (1) Retaining screws (16 off)
- (2) Securing bolts and washers (2 off each end)

- **Remove** the lid of the MSUP by removing the screws (16 positions).

- Carefully **disconnect** fiber optic cables **U38, U39, U40, U41** from the connectors on the **MSUP PCB**, and withdraw the **fiber optic cables** through the **wave guide**.
(→ 5/Fig. 288 Page 221)

Fig. 288: MSUP 2657B - Fiber Optic cable connectors



- (1) U38
- (2) U39
- (3) U40
- (4) U41
- (5) CAN Fiber optics wave guide

- **Remove** the (4 off) bolts securing the MSUP to the cradle.
- Carefully lift out the MSUP unit.
- **Re-fit** the lid using the (16 off) screws.
- Return the defective unit to CSML.

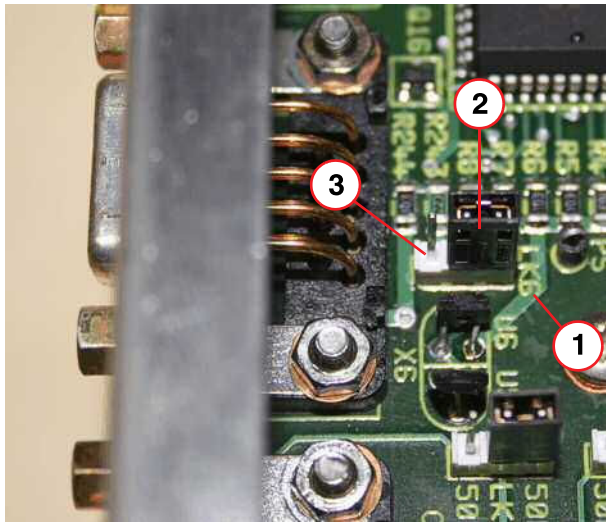
Installing the new MSUP 2657B

Tab. 24 ERDU Switch Settings

Number of ERDU Buttons	ERDU Selection Switch
Alarm Box	1
Alarm Box + 1 ERDU Button	2
Alarm Box + 2 ERDU Buttons	3

- **Remove** the lid of the MSUP by removing the screws (16 positions).

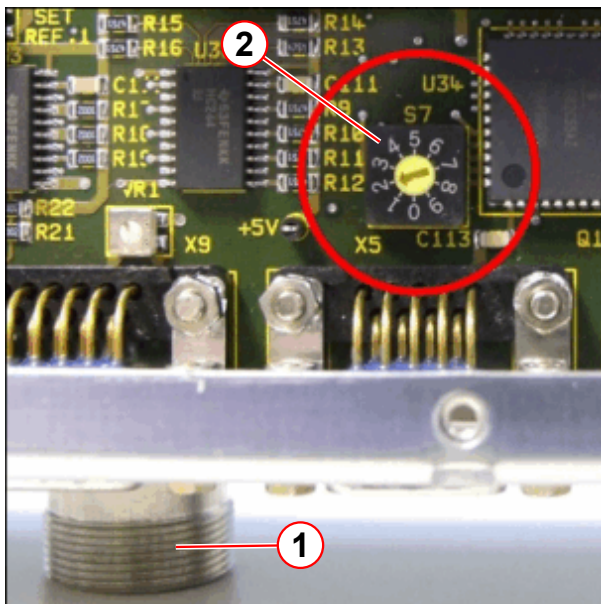
Fig. 289: Setting the internal/external field sensor



- (1) Location of LK6 on PCB
- (2) LK6 - jumper set in INT position (STATIC systems)
- (3) Jumper position for EXT (Mobile systems)

- Set the **external field sensor LK6** in the correct position on the PCB.
 - » On **static** Avanto and Espree systems, **LK6** must be in the **INT** position.
 - » On **mobile** Avanto and Espree systems, **LK6** must be in the **EXT** position.

Fig. 290: MSUP - 2657B ERDU Selection switch



- (1) Magnet ID connector
- (2) ERDU selection switch

- Set the **ERDU switch** as required refer to (→ Tab. 24 Page 221).

- Carefully insert the fiber optic cables **U38, U39, U40, U41** through the **waveguide** and into their correct connectors on the PCB. (→ 5/ Fig. 288 Page 221)
- **Secure** the new MSUP to the support cradle using the (4 off) bolts.
- **Refit** the MSUP lid using the (16 off) screws.



Do not mix up the lids of the MSUP. Check the serial number labels.

- **Re-connect to the MSUP:-**
 - External connectors **X2, X6.** (→ Fig. 285 Page 219)
 - **Magnet ID** connector to **X7.** (→ 1/ Fig. 286 Page 220)
 - Battery connector **X12.** (→ 11/ Fig. 285 Page 219)
 - Power connector **X3.** (→ 7/ Fig. 285 Page 219)
- Insert fuse **F3** (ACC), (or switch on F3).
 - » Check that the **green LED** on the MSUP is illuminated. (→ 8/ Fig. 285 Page 219)
 - » The Alarm box may sound an **audible alarm** - press the **ACKNOWLEDGE** button to cancel the alarm.



Even though the MSUP is connected to the system, the ERDU buttons will NOT be able to quench the magnet.

- **Re-boot** the scanner.
 - » this will ensure the correct software is installed on the supervisory.
- **Open** SeSo > Magnet&Cooling > Initialization.
 - **Check** the magnet specific values and compare with the **acceptance document.**
 - Press the **UPLOAD** button to transfer the magnet-specific values from the **HOST** to the **new MSUP.**
- **Exit** the Magnet&Cooling menu.



If the Host PC is not available, or unable to complete this task then refer to the section 'Calibration of MSUP using a Laptop' (→ Calibration of MSUP using a Laptop / Page 223)

4.1.2.5 Concluding Checks

- Connect the **ETL-MSUP lead** (→ 8/ Fig. 281 Page 215) to **X1** on the MSUP.
- Perform an **ERDU test** using the ERDU test load - (refer to **Maintenance, Safety-Related Tests- ERDU**).
 - » Begin the procedure at "**Heater resistance monitor tests**", and follow the steps to the end of the procedure.

4.1.2.6 Calibration of MSUP using a Laptop

Direct input to MSUP

If it was not possible to upload the old data to the host, refer to the **acceptance documents**, (refer to (→ Tab. 23 Page 213)) and then do the following:-

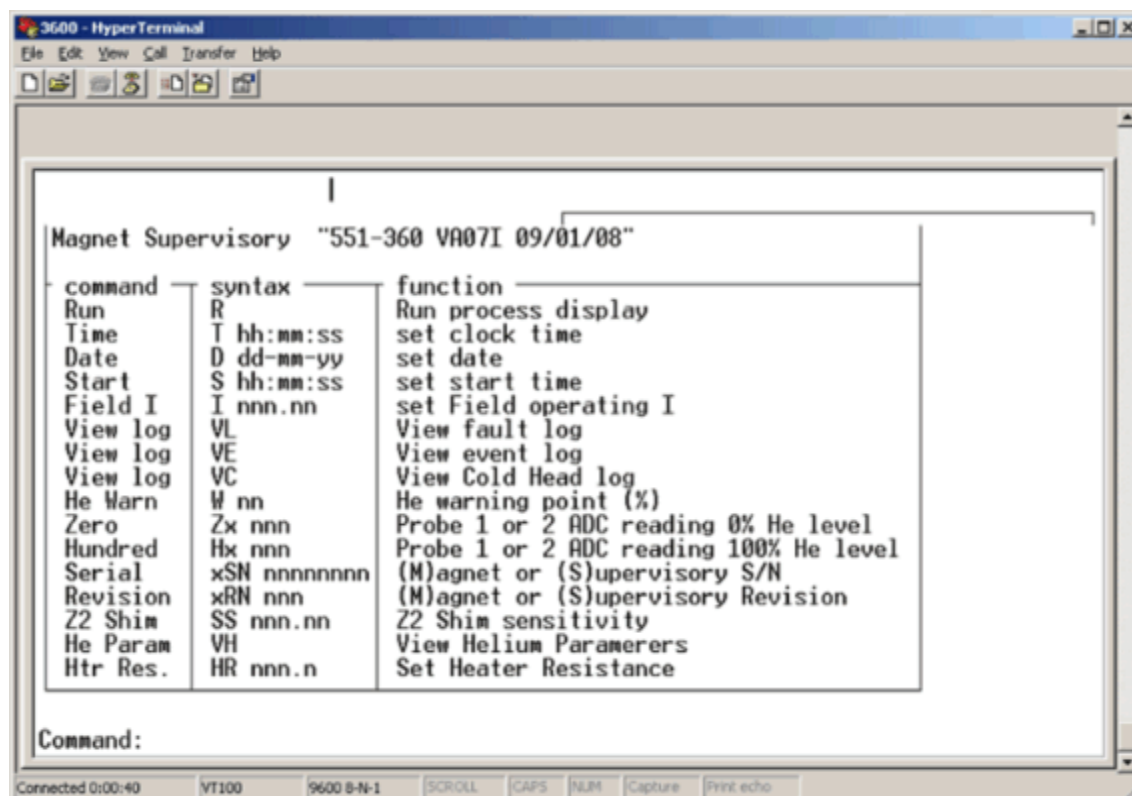
- **Open** the **HyperTerminal** application and select the appropriate **Com Port.**
- Select the following settings:

Baud rate	9600
Bits	8

Parity	None
Stop Bits	1
Flow	None
Terminal Emulation	VT100

- **Connect** the laptop via a **RS232** interface cable to **X5**. (→ 3/ Fig. 285 Page 219)
- Press **Escape** - the **Command Input** screen is displayed.

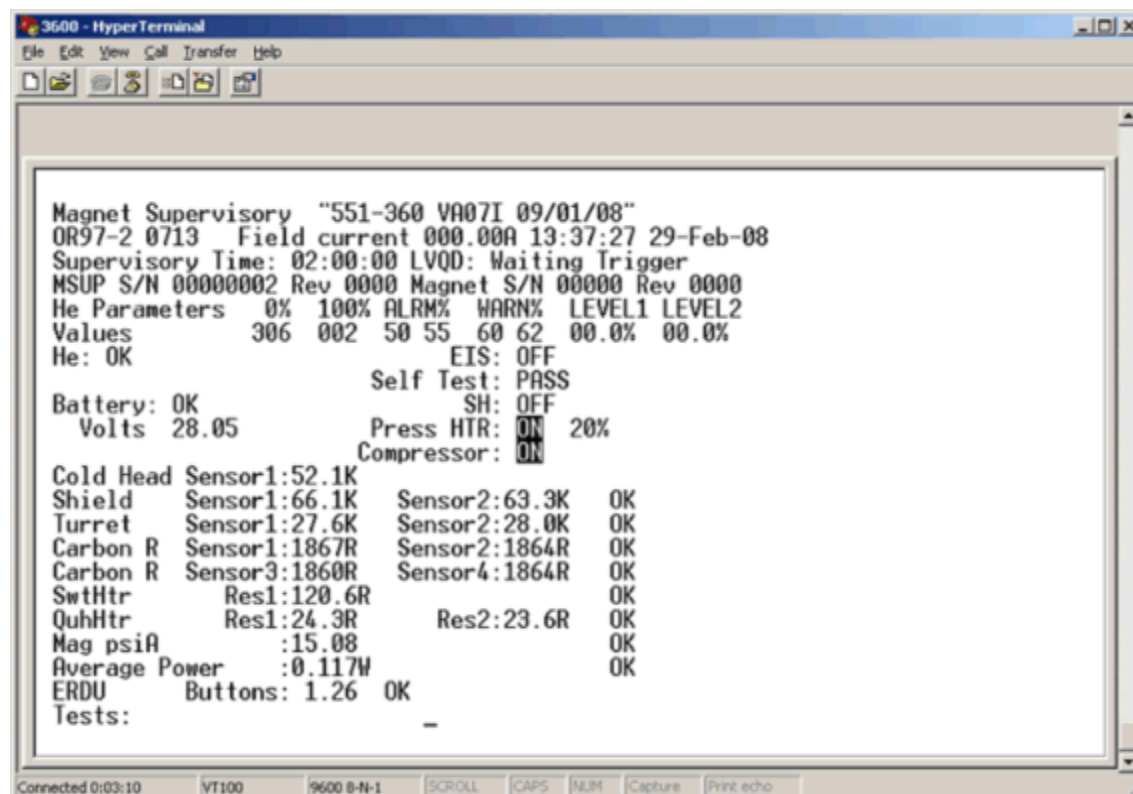
Fig. 291: MSUP 2657B - Command Input screen (typical)



- Enter the required values into the **Command: _** line (refer to (→ Tab. 25 Page 230)).

- Press **R <R>** to run the **Process Display**. The updated data will be displayed.

Fig. 292: MSUP 2657B - Process Display screen (typical)



Calculating Helium Probe Calibration at early sites

The latest version of the acceptance document provided by SMT lists both the resistance and the converted values.

For earlier sites, (where the calculated value may not be available) use the formula below to calculate the calibration values.

$R \times 0.25 \times 36.27 = \text{Input Value}$, where **R = Helium Probe Resistance**.

- Enter the resulting value into the MSUP as a **whole number**. Example:-
 - » Resistance = 88.8 Ohms
 - » $88.8 \times 0.25 \times 36.27 = 805.194$
 - » Value used to calibrate is **805**

All values are entered from the **command menu**. (→ Tab. 25 Page 230)

Calibration of CCRs

On systems fitted with CCRs, the MSUP will require calibration to ensure the temperatures monitored are at the correct value.



On systems without CCRs fitted, the MSUP will NOT require calibration.

The value for **RT, 77K** and **4.2K** can be found on the **Magnet Acceptance document** (TDC Calibration Values)

All values are entered from the **command menu**.

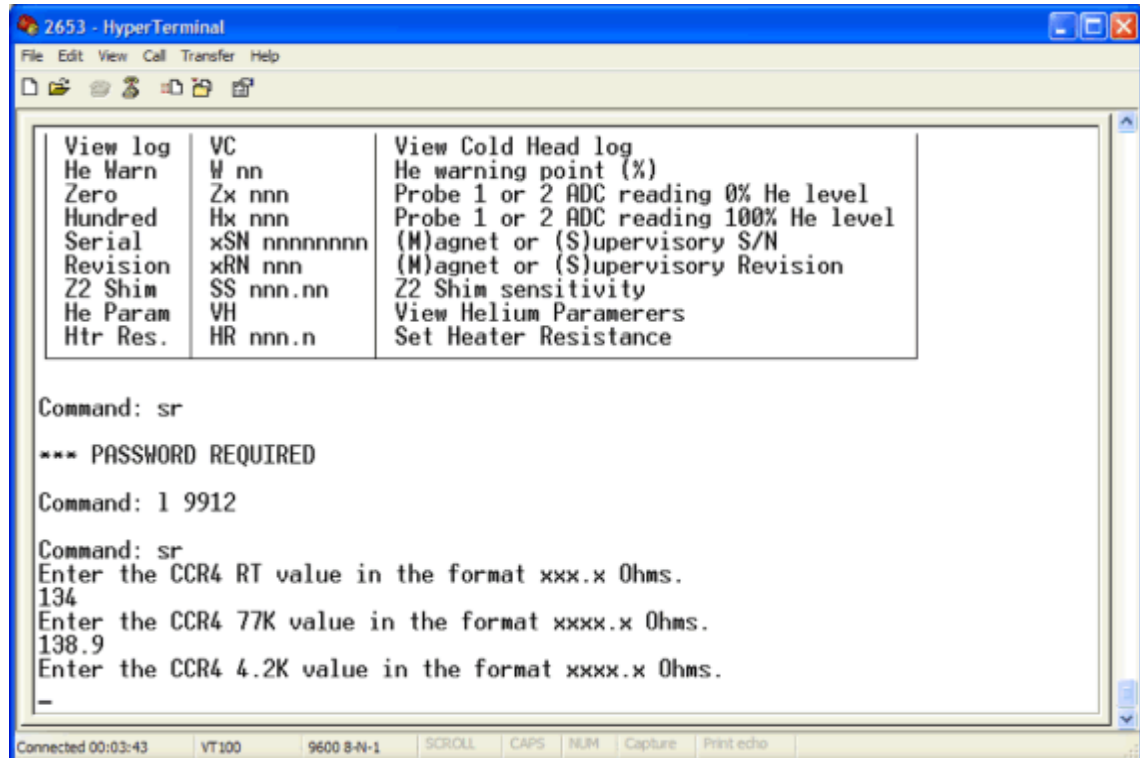
Fig. 293: OR105/122 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation Data	
MR Magnet Technology Quality Regulations			
Cryostat Number	21243234	SMT Reference	21245901 06.10
System Description		OR105	
1.494T Actively Shielded Superconducting Twin Turret Magnet System			
Measured Magnet Centre (Offset)		-2 (S) mm	
If magnetic Offset is outside the spec (+.5mm) the magnet will be marked (+.5mm) and Siemens Magnet Technology Ltd will be required to Shim the Magnet			
Operating Parameters			
Description	Specification value	Measured value	
Operating Current @ 1.494T	> 600 AMPS	510.05	AMPS
Field/Current Ratio	N/A	12.48	kHz/O.1 Amp
Carbon Resistor Values			
POSITION	Room Temperature	Nitrogen(77K)	Helium Temperature
TDC d - e	878		2567
Top a - e	926		2938
Mid b - e	985		2715
Bottom c - e	877		2613
TDC Calibration Values:	850.4	1122.9	2731.7
Helium Level Probe Resistance (When using a 250 mA supply)			
		Probe 1	Probe 2
@ 0 Resistance	Ohms	85.36	85.36
0 % Calibration		774	774
100 % Resistance	Ohms	0.22	0.22
100 % Calibration		2	2
Diode Table No		-1	
Pressure Heater Res	Ohms	19	
Final Temperature Sensor Recon Values			
50kK	50kK Lin	Coldhead	Recon Margin
53	47	44 K	530 mW
Checked by	K Suraina	Signature	
Date	02/06/2010		
Page 1 of 1		25-AD EPS 000 01	

- At the **Command** line:
 - Enter **SR <R>**
 - **PASSWORD REQUIRED** is displayed

- Enter **L 9912** <R>
- Enter **SR** <R>
- Enter the CCR **RT, 77K**, and **4.2K** values

Fig. 294: MSUP2657B CCR Calibration screen (example)



- Enter **VH** <R> to view, and verify data is correct.

Fig. 295: MSUP2657B View VH download screen (example)

```
2653 - HyperTerminal
File Edit View Call Transfer Help

Command:
Helium Parameters
Probel: 0% = 0774 100% = 0002
       Sample Height = 256 = 250 mA

Probe2: 0% = 0774 100% = 0002
       Sample Height = 256 = 250 mA

Heater Resistance = 19.0 Ohms

Z2 Shim Sensitivity = 19.00 ppm/A

CCR4 Calibration
CCR4 RT Cal = 134.0 Ohms
CCR4 77K Cal = 138.9 Ohms
CCR4 4.2K Cal = 1104.0 Ohms

Shield Limits
Link1 Alarm = 1024mV = 94.6K
Link1 Warn  = 1050mV = 76.8K
Bore1 Alarm = 1012mV = 102.5K
Bore1 Warn  = 1039mV = 84.4K

Command: _

Connected 00:04:53 VT100 9600 8-N-1 SCROLL CAPS NUM Capture Print echo
```

- Enter **R** <R> to return to the **Process Display**.

Tab. 25 MSUP 2657B Input Commands

Command	Input Data	Value 'n'	Example
0% Calibration	Z1 nnnn<R>	He Probe 1	Probe1: 88.8 ohms -enter as Z1 805<R>
	Z2 nnnn<R>	He Probe 2	Probe2: 88.8 ohms - enter as Z2 805<R>
100% Calibration	H1 nnnn<R>	He Probe 1	Probe1: 1.8 ohms - enter as H1 16<R>
	H2 nnnn<R>	He Probe 2	Probe2: 1.8 ohms - enter as H2 16<R>
Pressure Heater Resistance	*L 9912<R> HR nnn.n<R>	Pressure Heater Res	19.4 ohms - enter as HR 19.4<R>
Magnet Serial No.	*L 9912<R> MSN nnnnnnnn<R>	Magnet Serial No	Enter as MSN12345678<R>
Magnet Revision No.	*L 9912<R> MRN nnn<R>	Mag Rev No	Magnet Rev 4 - enter as MRN4<R>
Supervisory Serial No.	*L 9912<R> SSN nnnnnnnn<R>	Supervisory Serial No.	Supervisory Serial No 26570001 - enter as SSN 26570001<R>
Supervisory Revision No	L 9912<R> SRN nnn<R>	Supervisory Revision No	Supervisory Revision No 14 - enter as SRN14<R>
Magnet Operating Current	I nnn.nn<R>	Mag Operating Current	Operating current 508.23 - enter as I 508.23<R> This parameter is also automatically updated when the magnet is next ramped to field.
Date	D dd-mm-yy<R>	n/a	1st Jan 2006 - enter as D 01-01-06<R>
Time	T hh:mm:ss<R>	n/a	1 hour 30mins 30secs - enter as T 13:30:30<R>
Shield Bore Warning	L4 2947<R> SBW nnn<R>	Shield Bore Warning	Shield Bore warning of 77K - enter as SBW 77<R>
Shield Link Warning	L4 2947<R> SLW nnn<R>	Shield Link Warning	Shield Link warning of 80K - enter as SLW 80<R>

Command	Input Data	Value 'n'	Example
CCR Calibration	L 9912 <R> SR <R> XXX.X <R> YYYY.Y <R> ZZZZ.Z <R>	XXX.X = RT YYYY.Y = 77K ZZZZ.Z = 4.2K	L 9912 <R> SR <R> Enter CCR values as shown in (→ Fig. 294 Page 228)
Notes: ■ <R> = 'Enter' or 'Return' ■ Where a password is required, type L 9912<R> before the command is input. This will 'unlock' the option (otherwise it is not possible to change the value of the parameter) » Only settable from supervisory loadware version VA06D onwards ■ CCR Calibration - The MSUP will prompt the user to input values after the command SR has been used. - Refer to the Magnet Acceptance Document for TDC CCR4 Calibration Values.			

4.1.3 ERDU battery

Required time: 40 min

Required tools: Screwdriver (**non-magnetic**), 10 mm wrench, 5 mm Allen key

SI Replacing the ERDU battery



Do not replace the battery for a new installation.



It is possible to disconnect the battery plug without turning off power to the MSUP. While the ERDU function is still provided, it will not be available if a MSUP power failure occurs when the battery is disconnected.

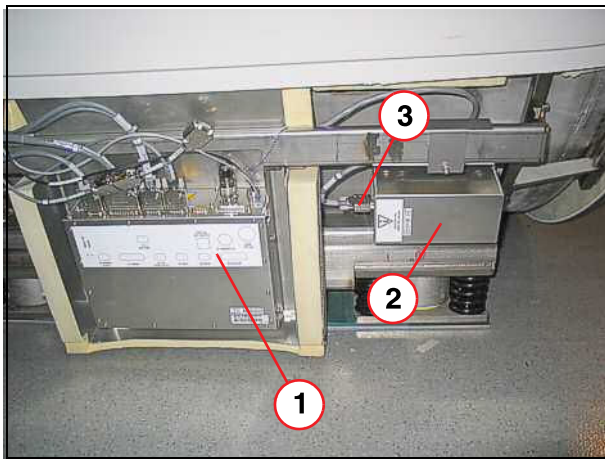
4.1.3.1 ERDU battery location

Looking from the Patient End, the ERDU battery is located on the **LEFT** for Avanto and Espree. (Earlier Avanto systems were on the **RIGHT**).

Avanto (early systems)

- Remove the **right** side cover.
- Remove the ERDU battery plug (3).

Fig. 296: MSUP-battery (early Avanto systems)

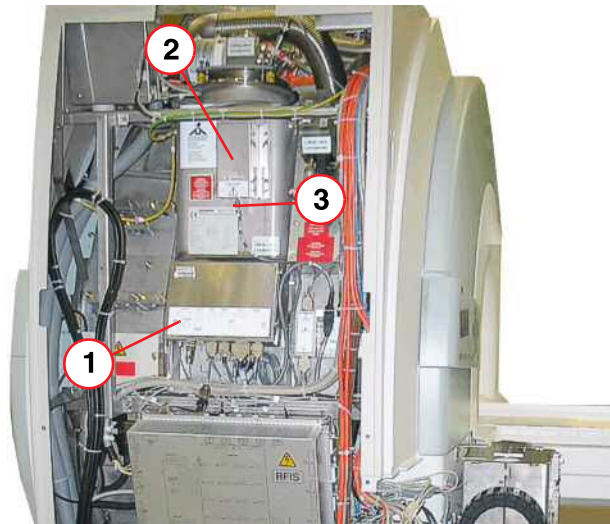


- (1) MSUP
- (2) MSUP-battery
- (3) MSUP-battery plug

Avanto/Espree

- Remove the service access cover on the **left** side cover.
- Remove the ERDU battery plug (3).

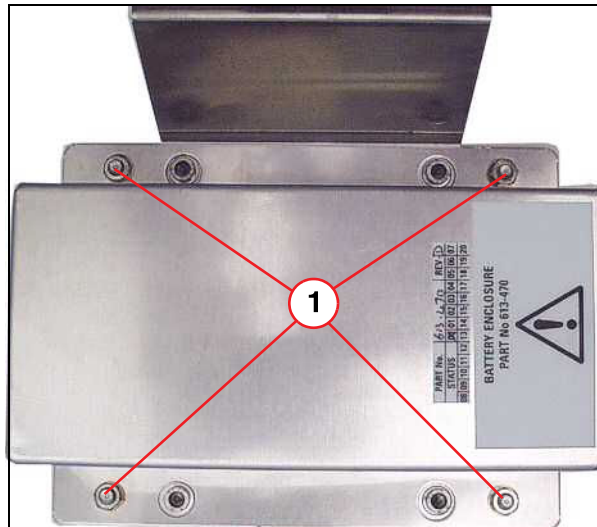
Fig. 297: MSUP battery (Espree system)



- (1) MSUP
- (2) MSUP battery
- (3) MSUP battery plug

4.1.3.2 Disassembling / Assembling

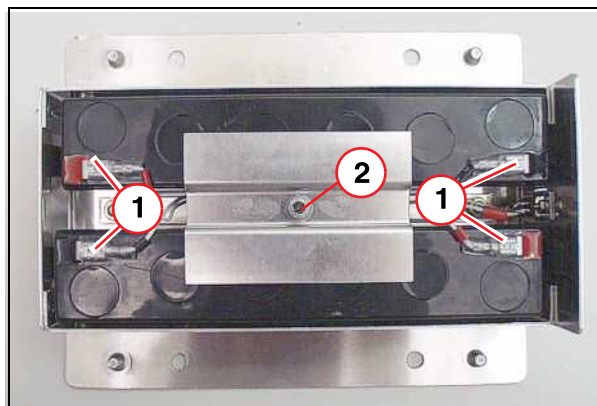
Fig. 298: ERDU-battery



(1) 10 mm nut

- Unscrew the 4 x 10mm nuts (1) and remove the cover.

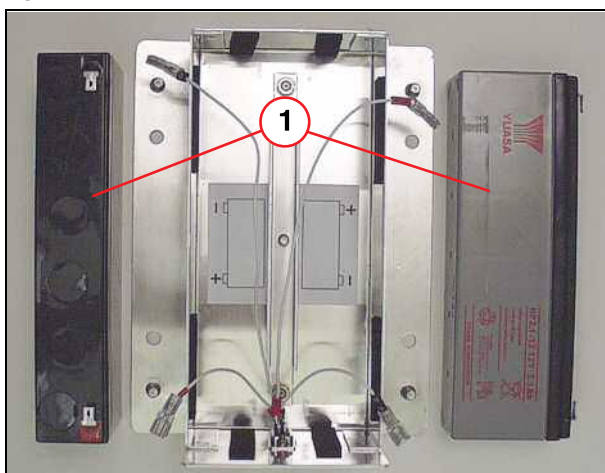
Fig. 299: ERDU battery



(1) Battery plugs
(2) Allen screw

- Remove the Allen screw (2) and remove the battery holder plate.
- Disconnect the four battery terminals (1).

Fig. 300: ERDU batteries



- Remove the batteries.
 - The batteries contain lead.
 - Turn the batteries over to a disposal company in accordance with local regulations.

1. Fit the new batteries into the tray-ensure correct orientation.
2. Connect the battery terminals (→ 1/Fig. 298 Page 233). Check the polarity once more (→ 1/Fig. 299 Page 233).
3. Re-fit the holder plate (→ 2/Fig. 299 Page 233).
4. Re-mount the cover and tighten the 4 x 10mm nuts (→ 1/Fig. 298 Page 233).
5. Connect the ERDU battery lead.

4.2 MSUP for Avanto, Espree (VB19) / Avanto Dot Ex-factory

4.2.1 Replacing the Magnet Supervisory Unit (MSUP 2306)

4.2.1.1 Requirements

This procedure is applicable to **OR105** and **OR122** systems.

Safety information



This procedure can be performed with the magnet at field.



CAUTION

The replacement procedure disables the magnet emergency shutdown system at the MSUP 2306.

Failure to adhere to safety precautions can result in serious bodily injury or property damage.

- » Connect the ERDU Test Load.
- » The magnet can only be quenched using the Magnet Stop button on the ERDU Test Load.
- » Exclude all non-service personnel and all magnetic materials / tools from the magnet area.
- » Close the door of the RF room.

Estimated completion time and number of persons

Tab. 26 Completion time

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	30 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

Tools

- Non-magnetic tool kit.
- ERDU test load kit (part no **03861197**).

- Laptop for data confirmation.

Spare parts

- MSUP 2306 tested (part no 10523587).
- Tamper-proof seal (part no 8100831).

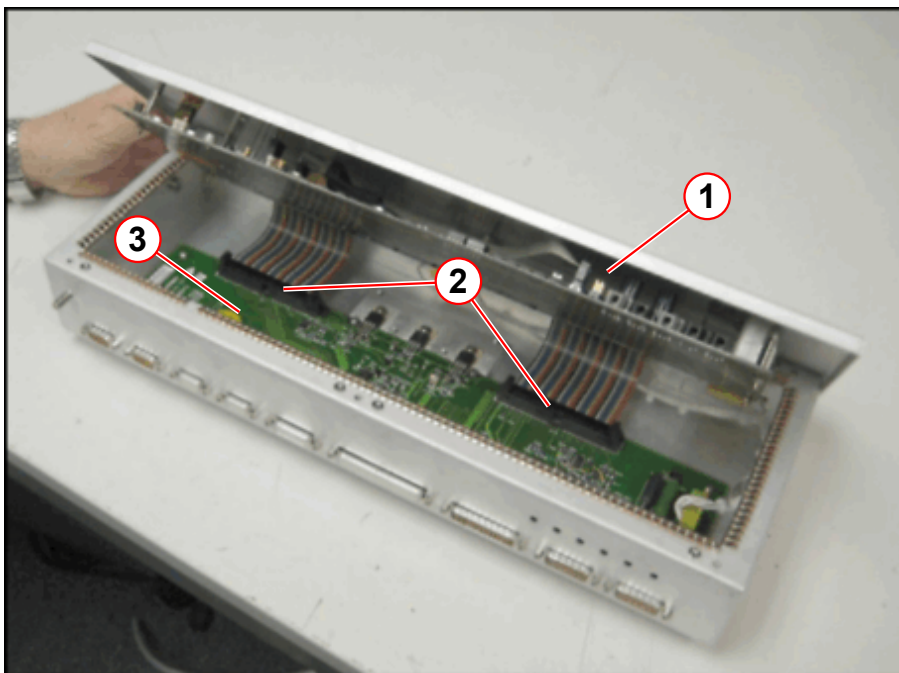
4.2.1.2 Preparatory Work Steps

General

The **MSUP 2306** is a **Field Replaceable Unit**. It contains 2 assemblies:

- The **MSUP 2306 controller assembly** is attached to the underside of the lid and is a **replaceable** spare part. (→ 1/ Fig. 301 Page 236)
- The **ERDU Test Load assembly** is mounted in the base of the MSUP and is **not separately replaceable**. If faulty, the **complete MSUP 2306 unit** must be replaced. (→ 3/ Fig. 301 Page 236)

Fig. 301: MSUP 2306 complete unit



- (1) MSUP controller assembly (replaceable part)
- (2) Ribbon cables
- (3) ERDU assembly (not replaceable)

Setting up the new MSUP for mobile systems

**CAUTION**

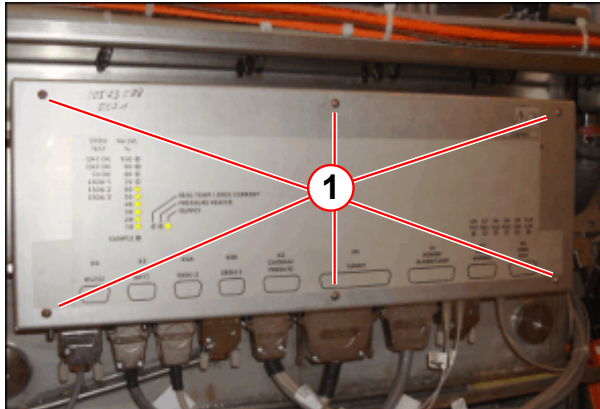
Component failure.

ESD precautions must be followed to stop the risk of damaging or reducing the life of the component.

- » Use ESD ground straps when handling ESD components.

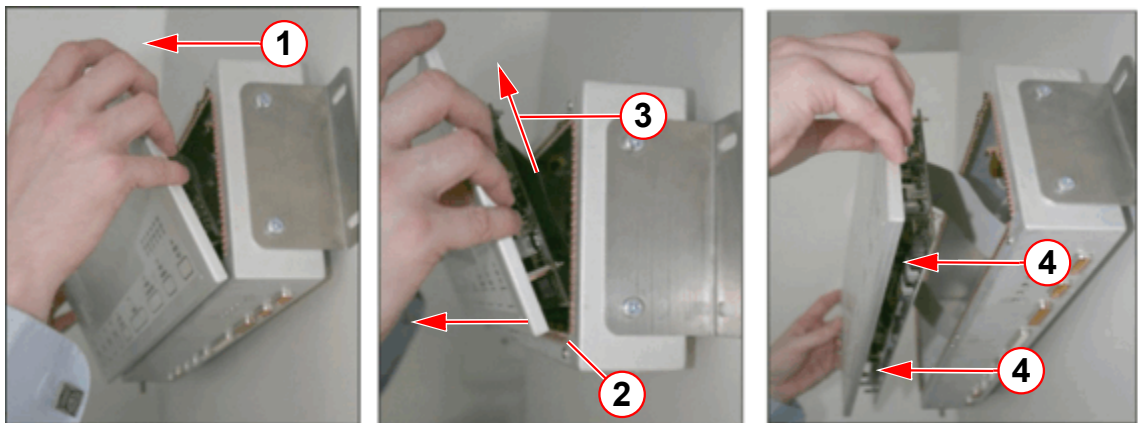
Fig. 302: MSUP 2306 fit/remove the controller assembly

- Remove the (6 off) **M3 screws** securing the MSUP controller assembly.



(1) M3 screws and washers (6 off)

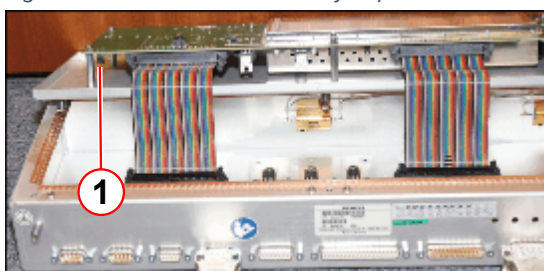
Fig. 303: MSUP 2306 removing the controller assembly



- (1) Lift the lid at 90 degrees
- (2) Lip
- (3) Lift over the lip
- (4) Lift assembly horizontally

- Tilt the controller assembly at an 90 degree angle (pos 1).
- **Lift** over the **lip** (pos 2,3).
- **Lift** the assembly **horizontally** (pos 4).

Fig. 304: External field sensor jumper location.



(1) External field sensor jumper

Locate the **external field sensor jumper** on the controller PCB. Move away from the pillar nearest to the ribbon cable for mobile systems. Or nearest to the pillar for static systems.

- Check the ribbon cables.
 - Ensure the connectors are seated correctly.
 - Lock in place with the connector clips.

Fig. 305: MSUP 2306 feed through the ribbon cables



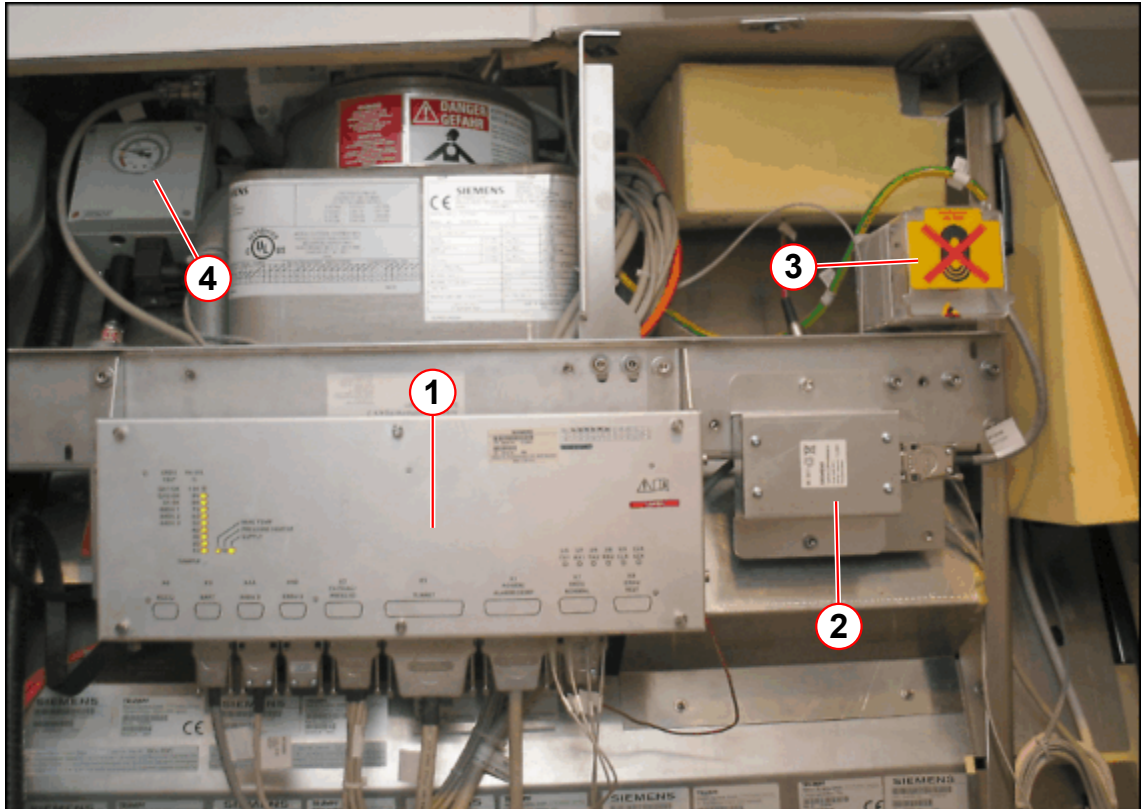
- Ensure the ribbon cables are not trapped as the controller assembly is re-fitted.
 - » Carefully tuck the ribbon cables under the PCB.

- Refit the controller assembly using the (6 off) screws.

MSUP 2306 positioning

- Remove the appropriate magnet cover panels.

Fig. 306: MSUP 2306 position (static systems)



- (1) MSUP
- (2) ERDU Battery
- (3) Service ERDU Button
- (4) Pressure display

The **MSUP 2306** is located behind the magnet service access covers. The **MSUP**, **battery box** and **Service ERDU button** are each secured to a mounting plate.

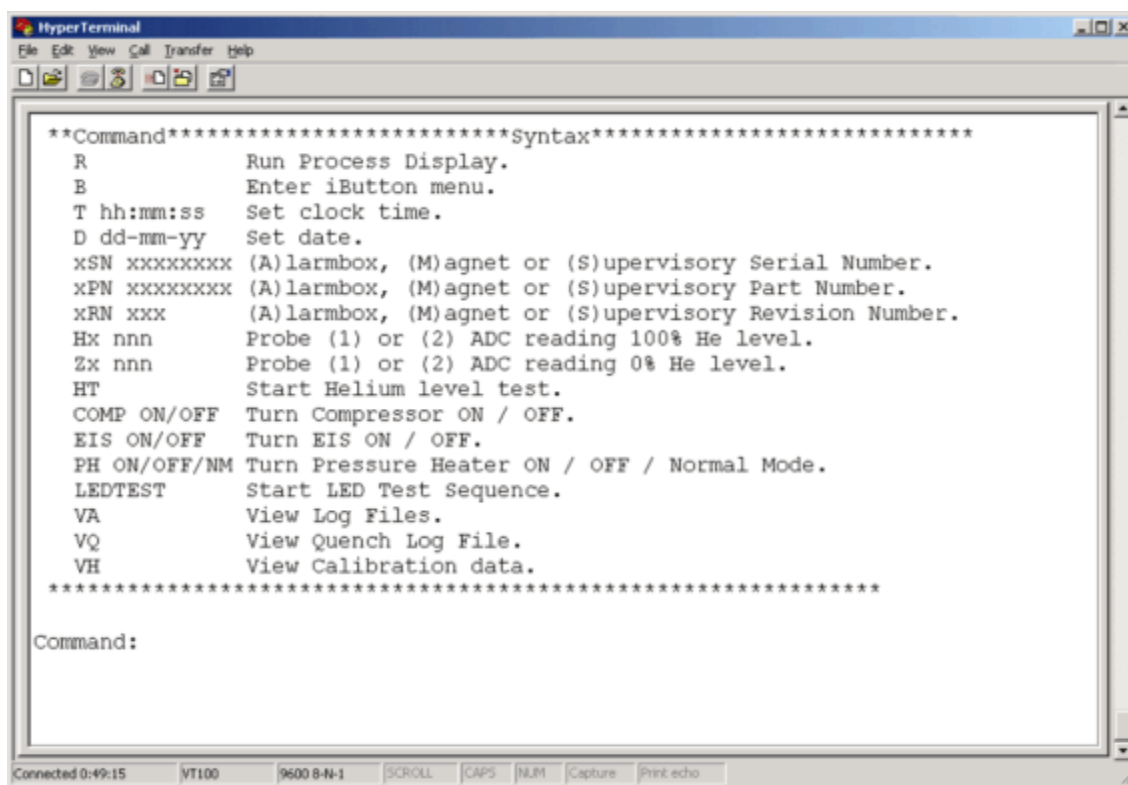
4.2.1.3 Data confirmation after replacement

Ensure the **i-Button** and **magnet data** are captured from the **original MSUP**. This data is then checked if it reads correctly on the new **MSUP**.

Capturing the iButton menu on the MSUP using a laptop

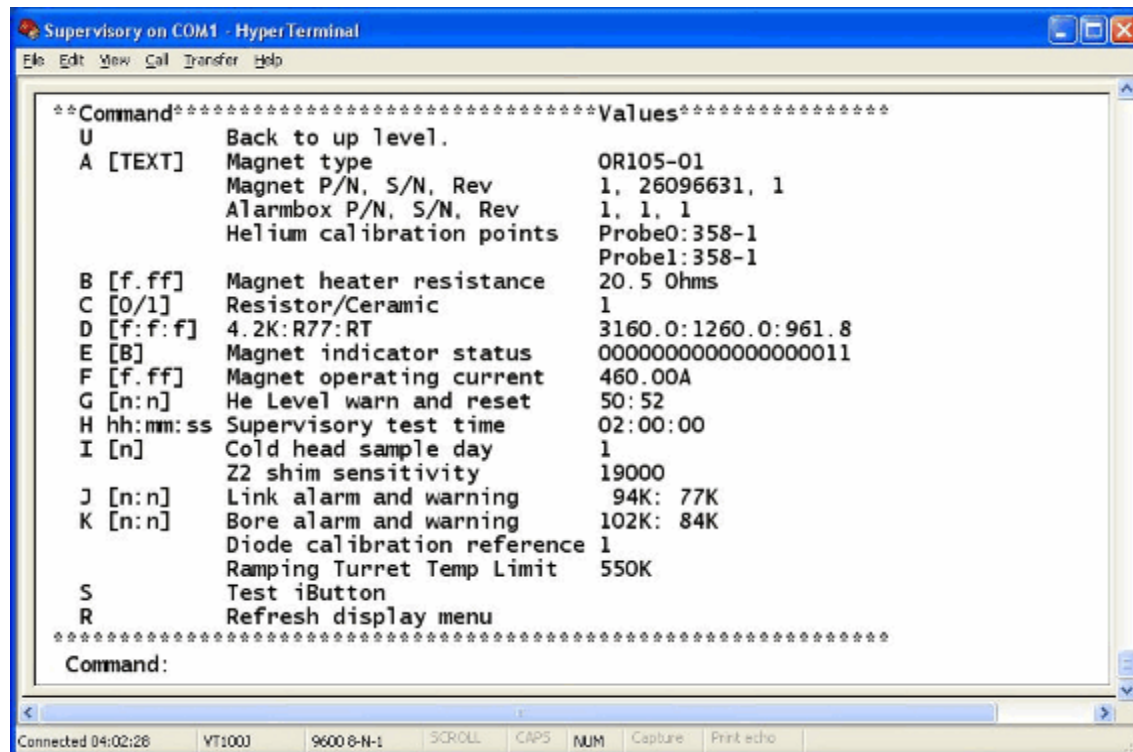
1. Turn on the laptop and open the **Terminal emulation software**, (eg HyperTerminal or PuTTY).
 - » Set bits per second to 9600.
 - » Set data bits to 8.
 - » Set parity to 1.
 - » Set flow control to none.
 - » Set Emulation to VT100 (File:Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP via RS232 cable.
3. Press **ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>.
6. Press **ESC** to display the **MSUP Command input screen**.

Fig. 307: MSUP Command Input screen (example)



7. Type **B** <enter> this displays the **iButton** menu.

Fig. 308: MSUP 2306 iButton menu (example)



8. **Capture the screenshot.** (→ Capturing a screenshot / Page 241)

9. Type **U** <enter>.

10. Type **R** <enter>.

Capturing a screenshot

- Capture a **screenshot** of the **MSUP Terminal emulation screen**:

- a) Press **Ctrl:Alt:Prt Sc**.
- b) Go to: **Start:Programs:Accessories**:and open Paint.
- c) Press **Ctrl:V** to paste screenshot.
- d) Name and save as a **.jpg** or **.bmp**.
Name the file DD (**Day**), MM (**Month**), YYYY (**Year**), TTTT (**Time:24hour format**).
i.e. iButtonDDMMYYYYTTTT: MSUPDDMMYYYYTTTT.

4.2.1.4 Connecting the ERDU Test Load (ETL)

**WARNING**

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorised personnel must be excluded from the imaging suite.

- » Secure the room and prevent unauthorised access.

**WARNING**

It is not possible to quench the magnet using the normal system MAGNET STOP buttons while the ERDU Test Load is connected to the MSUP and magnet.

- » The magnet may be quenched only by using the MAGNET STOP button on the ERDU Test Load.
- » Restrict access to the magnet accordingly.

**CAUTION**

The ERDU Test Load is slightly ferromagnetic.

The ETL is attracted by the magnetic force and will be projected into the magnet bore, resulting in injury or damage.

- » Keep the ETL as far away as possible from the magnet bore.

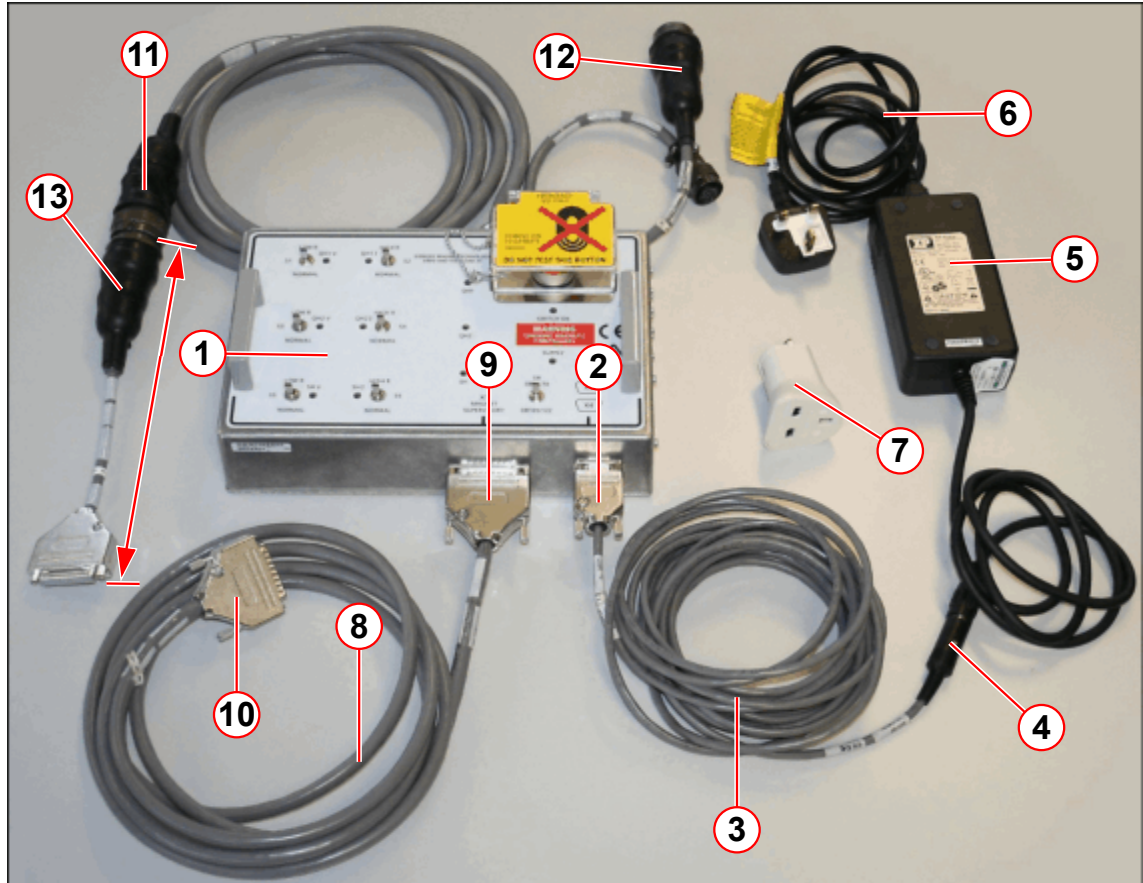
- Check that the **ERDU Test Load Kit** is complete with all the listed parts present:

i

For this procedure, items 1 - 6, and 11 are required. Item 7 may be required depending on outlet socket.

When connected, the ETL acts as an ERDU button.

Fig. 309: ERDU Test load kit (2661)



- (1) ERDU Test load 3 (ETL)
- (2) PSU output extension lead to ETL X3 or X4
- (3) PSU output extension lead
- (4) PSU output extension lead connector to PSU
- (5) Power Supply Unit (PSU)
- (6) AC supply lead
- (7) International adapter - AC supply
- (8) MSUP - ETL cable
- (9) 25-pin connector to ETL X1
- (10) 25-pin connector to MSUP X1
- (11) 41-pin connector, magnet turret
- (12) Adapter for OR64 / OR70, 41-pin connector to 10-pin magnet turret connector 'B' (optional ETL3 2661)
- (13) Adaptor cable B (optional for OR105, OR122, OR97)

- Connect the **ETL PSU** to a convenient AC outlet **outside of the magnet room**, and switch on. (→ 5/Fig. 309 Page 243)



Ensure that the PSU ETL switch mode is as far away from the magnet as possible (magnetic field < 50 Gauss).



The ETL will sound a continuous buzzing alarm, until the ETL is disconnected.

- Connect the ETL switch mode PSU to the ETL **X3** or **X4** using the extension cable. (→ 3/Fig. 309 Page 243)
 - » Check that the **SUPPLY** and **SWITCH OK** indicators glow on the ETL.



WARNING

Magnet quench!

A magnet quench is caused by pressing the ETL Magnet Stop button.

- » Ensure that the **MAGNET STOP** button on the ETL is **NOT OPERATED!**
- » **DO NOT** operate the **MAGNET STOP** button on the ERDU Test Load unless an emergency magnet quench is required.

- Set **SH** for correct magnet type. For this magnet, set the load to (**OR105/122**).
(→ Fig. 310 Page 245)

Fig. 310: ERDU Test Load (front panel)



- Set all **HIGH R / LOW R** switches on the ETL to **NORMAL**.

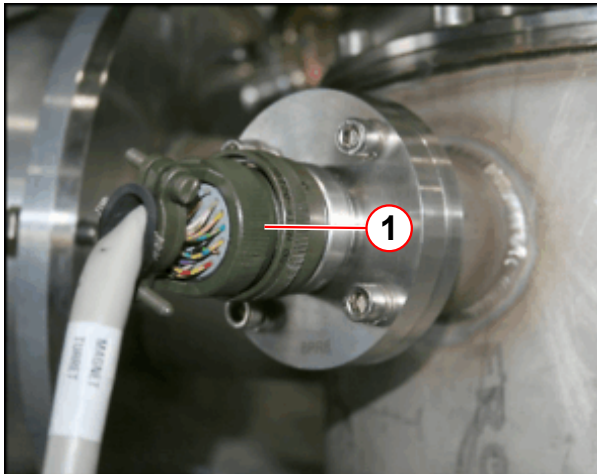
WARNING

The magnet cannot be quenched!!

If the circuit between the 41 way connector on the magnet to the MSUP is disconnected, THERE IS NO ERDU FACILITY.

- » Reconnect X1 (or the 41-pin turret connector), WITHOUT DELAY.
- » As soon as the ETL is connected to the magnet, the test ERDU function on the ETL is operational.

Fig. 311: Turret - electrical connector



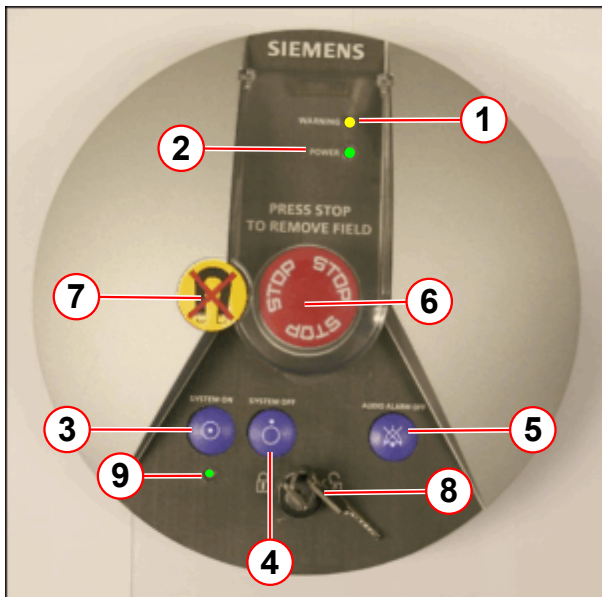
(1) 41 pin connector

- **Disconnect** the existing lead from the **41-pin connector** on the **magnet turret**.
- **IMMEDIATELY** connect the **41-pin connector** from the **ETL** to the **magnet turret** connection.
(→ 11/Fig. 309 Page 243)



Keep the ETL as far away as possible from the magnet bore.

Fig. 312: Alarm box - LEDs and buttons



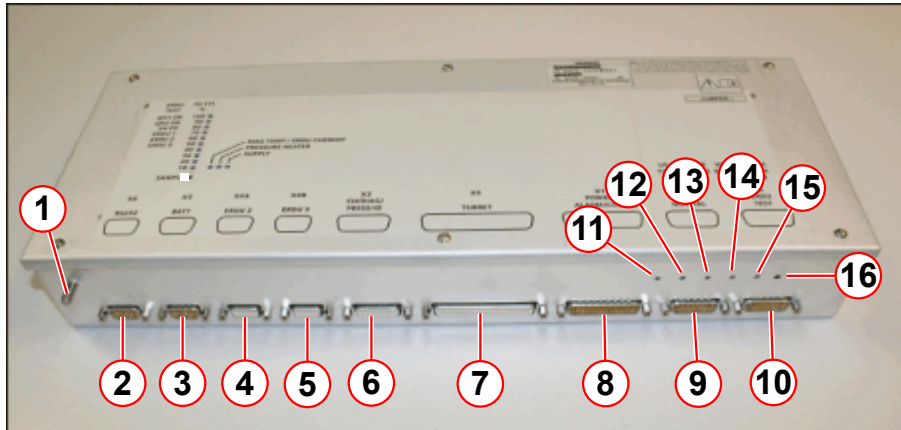
- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

- Check that the alarm sounds and that the **WARNING** supply indicator lights up on the **alarm box**.
 - » Press **audio alarm OFF** to cancel the audible alarm.

- Check that the **QH1, QH2, SH** and **SWITCH OK** and **SUPPLY** indicators **glow** on the ETL.

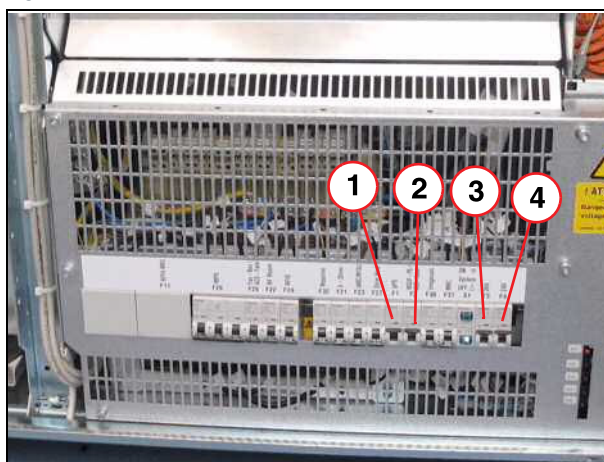
4.2.1.5 Removal

Fig. 313: MSUP 2306 indicators and input/output connections



- (1) Earth stud
- (2) X6 RS232
- (3) X3 Battery
- (4) X4A ERDU 2
- (5) X4B ERDU 3
- (6) X2 CH/DIAG/PRESS/ID
- (7) X5 TURRET
- (8) X1 POWER/ALARM/COMP
- (9) X7 ERDU NORMAL
- (10) X8 ERDU TEST (not connected when system is in use)
- (11) F/O TX1 U5
- (12) F/O RX1 U7
- (13) F/O TX2 U6
- (14) F/O RX2 U8
- (15) F/O CLK U9
- (16) LED CLK LCK

Fig. 314: ACC fuse (inside)



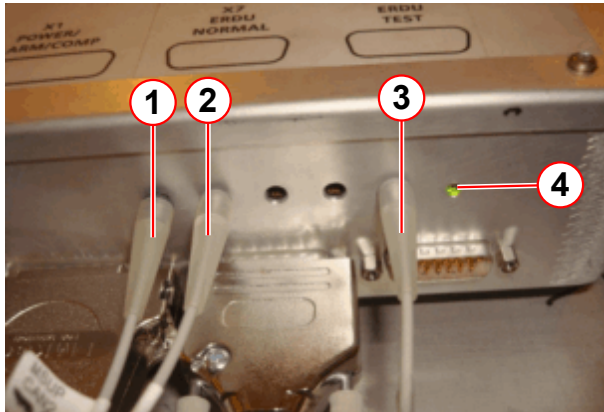
- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

ACC cabinet

- Switch **OFF** circuit breaker **F3**.
 - » **Power** to the MSUP is now **off**.
 - » the alarm box **audible alarm** will sound.
 - » the alarm box **yellow warning LED** flashes.
- Press **AUDIO ALARM OFF** to silence the alarm.

- Disconnect the 'D' type electrical connectors.
 - » the cold head switches **OFF**, if **X1** is **disconnected**.
- Disconnect the ERDU battery cable from **X3**. (→ 3/ Fig. 313 Page 247)

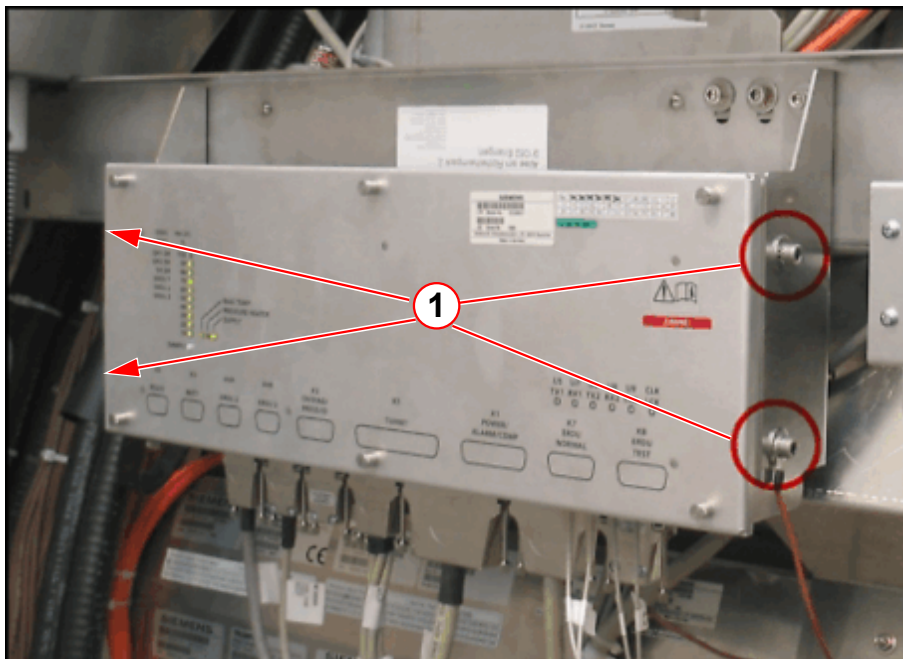
Fig. 315: MSUP 2306 fiber optic cables



- (1) F/O TX1 U5
- (2) F/O RX1 U7
- (3) F/O CLK U9
- (4) LED CLK LCK

- Carefully disconnect the (3 off) fiber optic cables.

Fig. 316: MSUP 2306 - replacing the complete assembly



- (1) 2 off M6 screws, plain and shake proof washers (both ends).

The MSUP is secured in position using the M6 screws, plain and shake proof washers (each end).

- Remove the screws and washers and carefully lift the MSUP from the mounting bracket.

4.2.1.6 Installation

Installing the MSUP 2306 complete unit

- **Fit** the brackets on the OVC studs.
- **Secure** the MSUP using the (4 off) M6 nuts, spring and flat washers.

**Do not Short pins on the battery connector.**

When connecting the battery cable ensure the pins on the connector are not shorted as this will blow the fuses in the battery pack. The fuses are not replaceable, hence a new battery pack will need to be ordered and fitted.

Connect the 'D' type electrical connectors:

» the cold head reactivates.

- Carefully **reconnect** the (3 off) fiber optic cables. (→ Fig. 315 Page 248)
- **Reconnect** the ERDU battery cable to **X3**.
- Turn **ON** the circuit breaker **F3** at the ACC.
- On the **ETL**, ensure the following **LEDs** are all **OFF**.

ERDU Test Load	Alarm box
QH1 V LED OFF	WARNING OFF
QH2 V LED OFF	Buzzer OFF
SH V LED OFF	
QH1 I LED OFF	
QH2 I LED's OFF	
SH I LED OFF	

4.2.1.7 Startup

- Perform a **"Re-start Software"** of **"Meas&Recon"** to download the correct loadware.



If a download is necessary it will take about 20 min

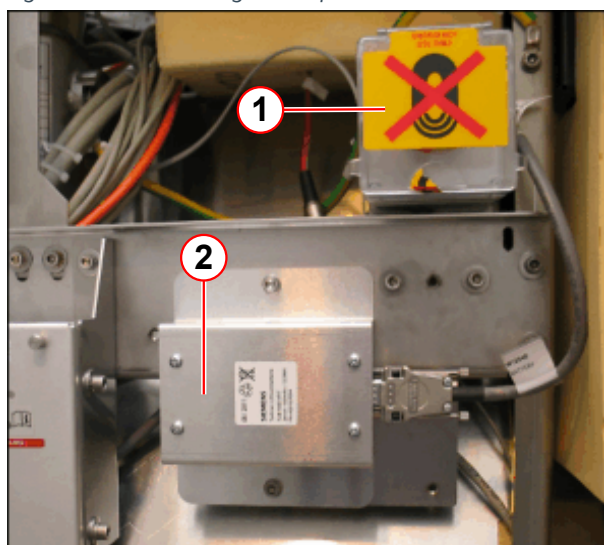
Testing the MSUP 2306 after replacement



The Magnet Stop button locks when depressed.
Turn the button clockwise to unlock.

- Ensure the **ERDU test connector** is connected to **X7**.

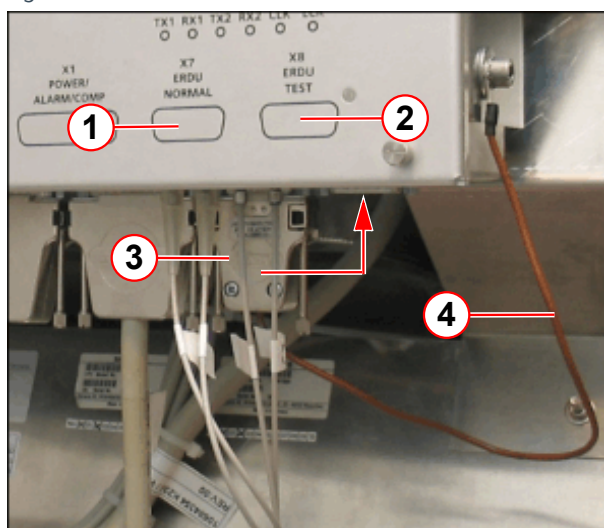
Fig. 317: Service Magnet Stop button



- (1) Service Magnet Stop button
(2) ERDU battery

- The **Service Magnet Stop button** is mounted above the MSUP battery.
- Check that the button is **not activated**.
- **Test the Service Magnet Stop button**.
 - » Press the **button** to ensure the **MSUP audible alarm** sounds.
 - » **Reset** the button - ensure the alarm **stops**.

Fig. 318: ERDU test connector

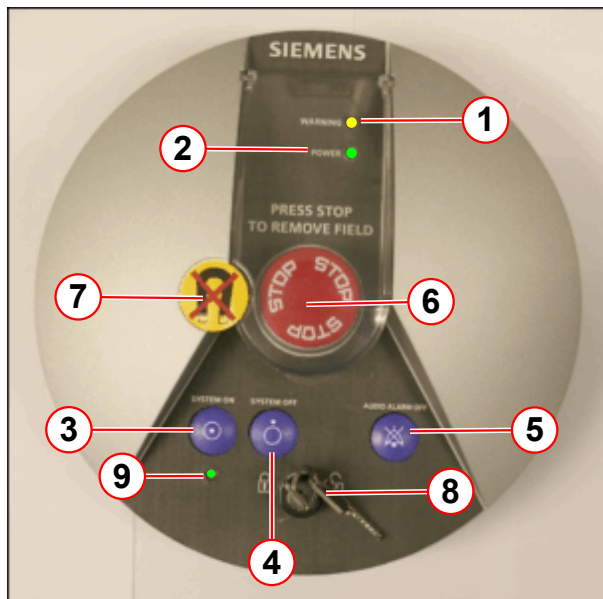


- (1) ERDU NORMAL
(2) ERDU TEST
(3) ERDU test connector
(4) ERDU connector strap

The **ERDU test connector** is fitted to **X7** during normal operation.

- Disconnect the test connector from **X7** (ERDU NORMAL).
 - » **MSUP audible alarm** is activated.
- Connect it to **X8** (ERDU TEST).
 - » On the **alarm box**, check the **audible alarm** sounds.
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- Check that the **ERDU TEST LEDs** are **NOT illuminated**.

Fig. 319: Alarm box - LEDs and buttons

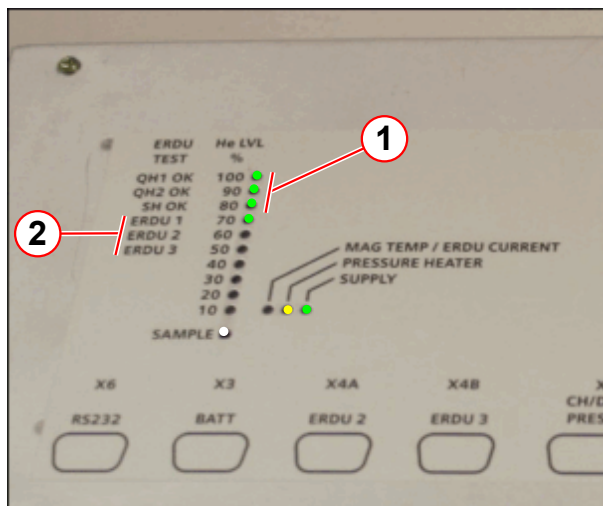


- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
 - » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
 - » Press AUDIO ALARM OFF to silence the alarm.

Fig. 320: ERDU test - LED indicators (typical)



- (1) QH1, QH2, SH = green LED (all ON)
- (2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
 - » QH1, QH2, SH and ERDU 1 are NOT illuminated.

In the **Magnet Room**:

- Press the **Magnet Stop button**.
 - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.
 - Press **AUDIO ALARM OFF** to silence the alarm.

- At the **MSUP**:
 - Check **QH1, QH2, SH LEDs** are illuminated.
 - Check **ERDU 2 LED** is illuminated.
- **Reset the Magnet Stop button** in the Magnet Room.
- **Check** the LEDs return to **normal**.

An optional **3rd Magnet Stop button** may be fitted. Follow the steps below:

- Press the **Magnet Stop button**.
 - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.
 - Press **AUDIO ALARM OFF** to silence the alarm.
- At the **MSUP**:
 - Check **QH1, QH2, SH LEDs** are illuminated.
 - Check **ERDU 3 LED** is illuminated.
- **Reset the 3rd Magnet Stop button**.
- **Check** the LEDs return to **normal**.
- Check **ALL Magnet Stop buttons** are **not activated**.
- **Disconnect** the ERDU test connector **from X8 (ERDU TEST)** and reconnect it to **X7 (ERDU NORMAL)**.
 - » On the **alarm box**, check the **audible alarm** sounds.
- **At the MSUP**:
 - » Check the **MSUP audible alarm** is **deactivated**.
 - » Check the **MSUP LEDs** return to **normal**.
- **At the alarm box**:
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- **Check** the **warning LED** is not illuminated '**ON**'.



If the LED is on, or flashing, the fault must be checked.
Do not disconnect the ERDU battery.

Downloading the i-Button MSUP 2306 data to the MRC after replacement

- Ensure that the old data is reading correctly on the new MSUP.
- **Check** the Serial, Revision, and Part numbers of the MSUP.
 - Update the numbers if necessary.

(Refer to MSUP download instructions).

The **MRC (MRAWP) Host** must be updated with the new **Serial, Revision, and Part numbers**.

- Access the **Service software** at the **MRC (MRAWP) Host**.

- Select menu item “Magnet & Cooling”.
- Select “Magnet Setup”.

Fig. 321: Updating system Identification numbers

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

Actions

- Cooling Status
- Magnet Status
- Compressor Status
- Cooling Setup
- Magnet Setup

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		Probe 1	
		Probe 2	
		Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87	A

MSUP Data

Downloads the magnet setup/calibration data to the host

Uploads the magnet setup/calibration data from the host to the MSUP

Download

Upload

Save values as default

Ready

Save | Default | Update

- (1) Download button (MSUP data stored on MRC)
- (2) Save button (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- Click “**DOWNLOAD**” to transfer calibration values to the **MRC**. (→ 1/ Fig. 321 Page 253)
 - » This takes the **saved data** from the **i-Button**, and ensures the **MRC data matches the MSUP**.

- Click **"UPDATE"**. (→ 3/Fig. 321 Page 253)
 - » This **refreshes** the on-screen values.

Downloading the i-Button MSUP 2306 data to the MRC

Refer to **Data Confirmation after replacement** (→ Data confirmation after replacement / Page 239)

The **MRC (MRAWP) Host** must be updated with the new **Serial, Revision, and Part numbers**.

- Access the **Service software** at the **MRC (MRAWP) Host**.
- Select menu item **"Magnet & Cooling"**.
- Select **"Magnet Setup"**.

Fig. 322: Updating system Identification numbers

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup
 - Magnet Setup

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☒ Probe 1 ☐ Probe 2 ☐ Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87	A

MSUP Data

1 Downloads the magnet setup/calibration data to the host

2 Uploads the magnet setup/calibration data from the host to the MSUP

3

Download

Upload

Save values as default

Ready. | Save | Default | Update

- (1) Download button (MSUP data stored on MRC)
- (2) Save button (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- Click **"UPDATE"**. (→ 3/ Fig. 322 Page 255)
 - » This **refreshes** the on-screen values.
- Update the **serial/revision/part** numbers (if required).
- Click **"SAVE"** to transfer calibration values **to the MSUP**. (→ 2/ Fig. 322 Page 255)

- Click **"DOWNLOAD"**(→ 1/Fig. 322 Page 255)
 - » This takes the **MSUP data** and stores it in a file on the **MRC**.

Restore to normal operation

- **Disconnect** the **ETL flying lead** from the magnet turret (**41-pin connector**) (→ 11/Fig. 309 Page 243).
- **Reconnect** the lead from the **MSUP (X5)** to the magnet turret (**41-pin connector**).
- Perform a **He test sample**.
 - press the **sample button** on the **MSUP**.
 - **wait for 30 secs**.
 - ensure the **LEDs** displays the **correct He level**.
- **Clear** all posted **alarms** on the **MRC**.
- Press the **ACKNOWLEDGE** button on the alarm box. Check that all indicators are now normal.
- The ERDU test generates some error / warning messages on the MRC. These error messages can be ignored.
- Click **< OK >** to confirm the messages.

4.2.1.8 Final steps

- **Disconnect** the **ETL PSU** from the AC supply.
- **Disconnect** the **ERDU Test Load** from its PSU.
- Re-pack all items in the **ERDU Test Load Kit** to prevent loss.
- **Replace** the appropriate **cover panels**.
- Affix the **tamper-proof seal (Part no.8100831)** to the **alarm box**. (→ 7/Fig. 319 Page 251)

4.2.2 Replacing the Magnet Supervisory Unit Controller Assembly (MSUP 2306)

4.2.2.1 Requirements

This procedure is applicable to **OR105 and OR122 systems**.

Safety information

**WARNING**

Strong magnetic fields.

Failure to observe the following safety warning can result in physical injury and equipment damage.

- » Only non-magnetic tools and equipment can be taken into the magnet room.
- » Close the door of the RF room.

Estimated completion time and number of persons

Tab. 27 Completion time

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	30 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

Tools

- Non-magnetic tool kit.

Spare parts

- **MSUP 2306 controller assembly** (part no **10128921**).

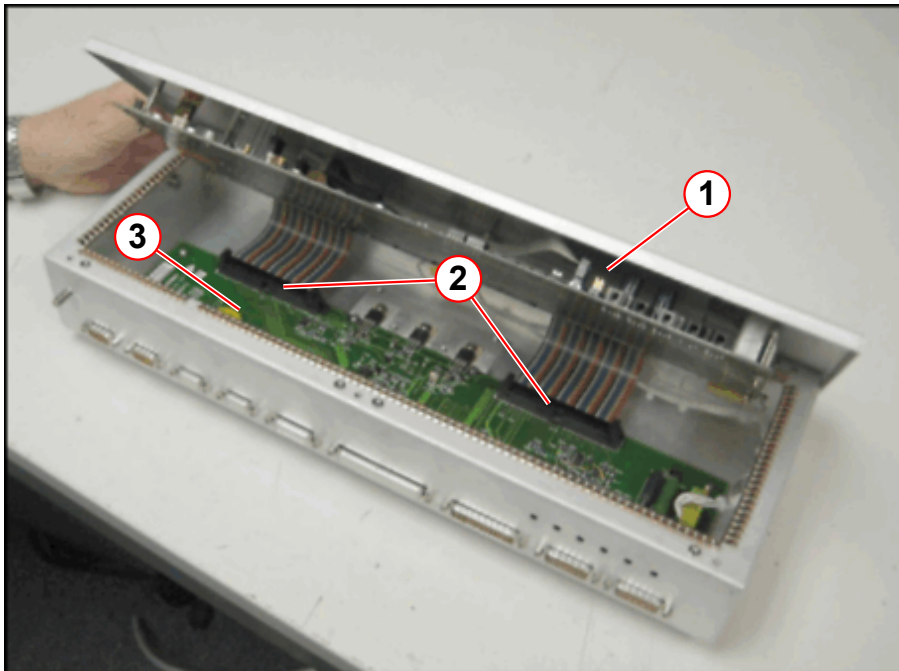
4.2.2.2 Preparatory Work Steps

General

The **MSUP 2306** is a **Field Replaceable Unit**. It contains 2 assemblies:

- The **MSUP 2306 controller assembly** is attached to the underside of the lid and is a **replaceable** spare part. (→ 1/Fig. 323 Page 258)
- The **ERDU assembly** is mounted in the base of the MSUP and is **not separately replaceable**. If faulty, the **complete MSUP 2306 unit** must be replaced. (→ 3/Fig. 323 Page 258)

Fig. 323: MSUP 2306 complete unit



- (1) MSUP controller assembly (replaceable part)
- (2) Ribbon cables
- (3) ERDU assembly (not replaceable)

Setting up the new MSUP for mobile systems



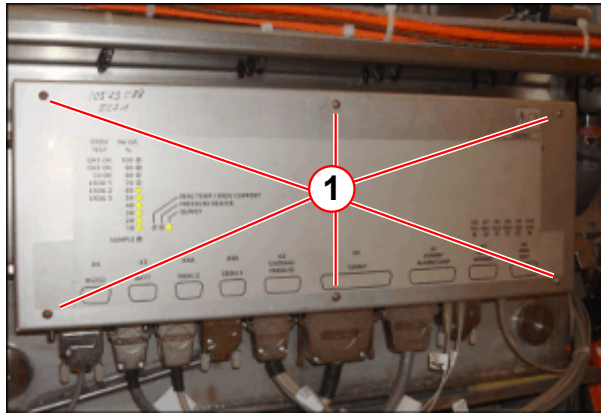
CAUTION

Component failure.

ESD precautions must be followed to stop the risk of damaging or reducing the life of the component.

- » Use ESD ground straps when handling ESD components.

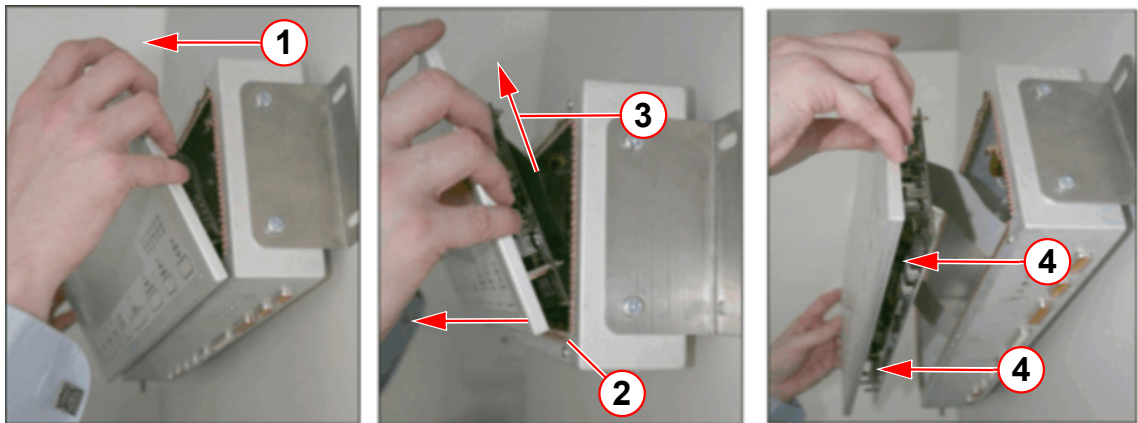
Fig. 324: MSUP 2306 fit/remove the controller assembly



(1) M3 screws and washers (6 off)

- **Remove** the (6 off) **M3 screws** securing the MSUP controller assembly.

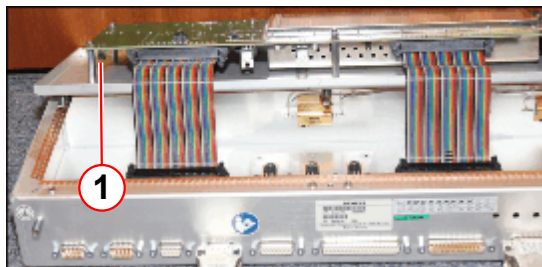
Fig. 325: MSUP 2306 removing the controller assembly



- (1) Lift the lid at 90 degrees
- (2) Lip
- (3) Lift over the lip
- (4) Lift assembly horizontally

- Tilt the controller assembly at an 90 degree angle (pos 1).
- **Lift** over the **lip** (pos 2,3).
- **Lift** the assembly **horizontally** (pos 4).

Fig. 326: External field sensor jumper location.



(1) External field sensor jumper

Locate the **external field sensor jumper** on the controller PCB. Move away from the pillar nearest to the ribbon cable for mobile systems. Or nearest to the pillar for static systems.

- Check the ribbon cables.
 - Ensure the connectors are seated correctly.
 - Lock in place with the connector clips.

Fig. 327: MSUP 2306 feed through the ribbon cables

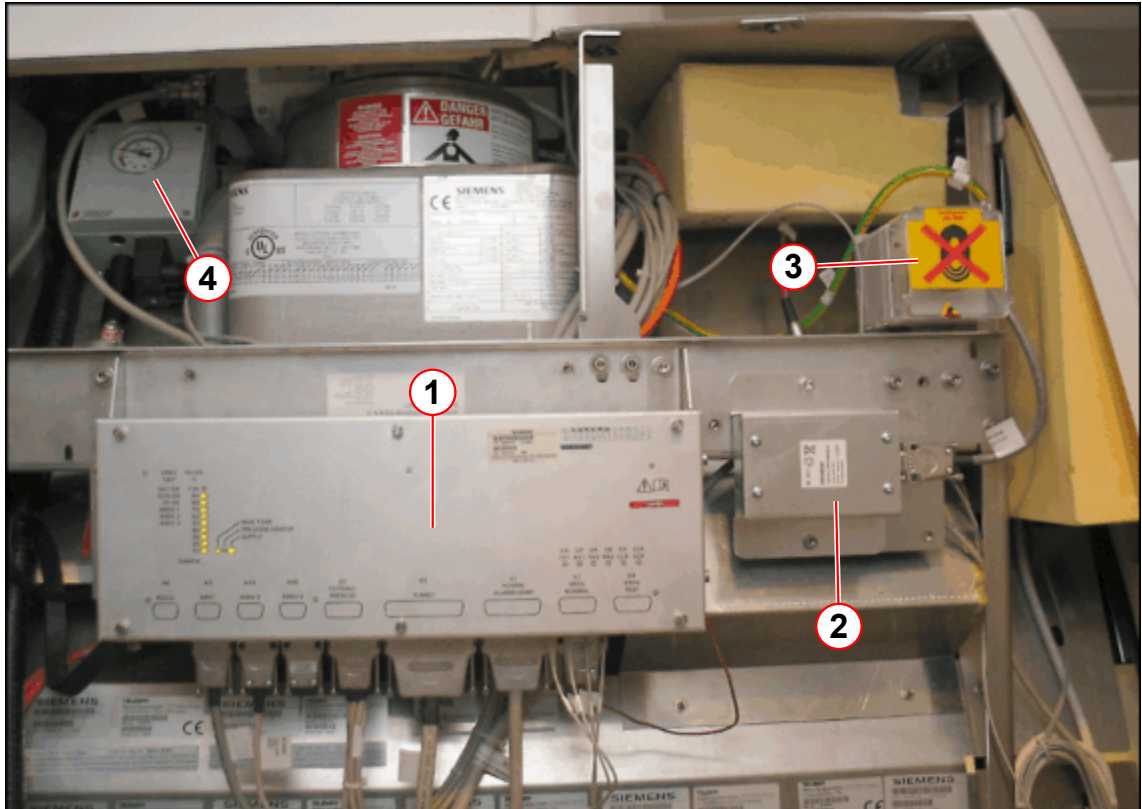


- Ensure the ribbon cables are not trapped as the controller assembly is re-fitted.
 - » Carefully tuck the ribbon cables under the PCB.
- Refit the controller assembly using the (6 off) screws.

MSUP 2306 Positioning

- Remove the appropriate cover panels.

Fig. 328: MSUP 2306 position (static systems)

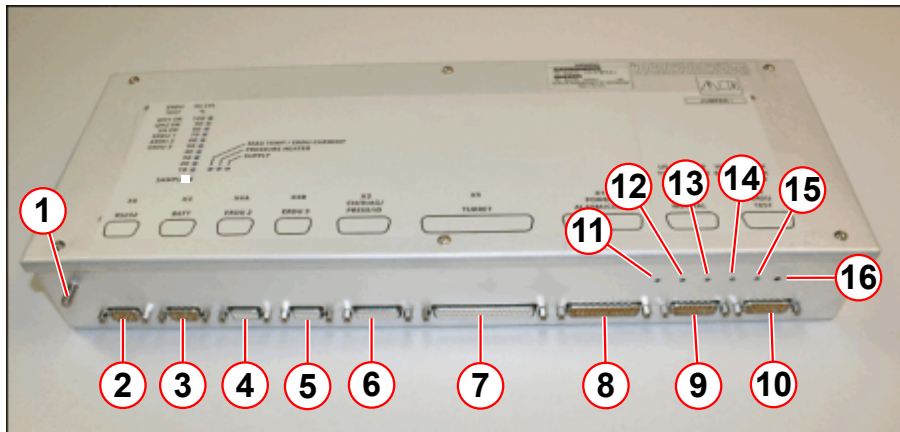


- (1) MSUP
- (2) ERDU Battery
- (3) Service ERDU Button
- (4) Pressure display

The **MSUP 2306** is located behind the magnet service access covers. The **MSUP** and **battery box** are each secured to a mounting plate.

4.2.2.3 Removal

Fig. 329: MSUP 2306 indicators and input/output connections



- (1) Earth stud
- (2) X6 RS232
- (3) X3 Battery
- (4) X4A ERDU 2
- (5) X4B ERDU 3
- (6) X2 CH/DIAG/PRESS/ID
- (7) X5 TURRET
- (8) X1 POWER/ALARM/COMP
- (9) X7 ERDU NORMAL
- (10) X8 ERDU TEST (not connected when system is in use)
- (11) F/O TX1 U5
- (12) F/O RX1 U7
- (13) F/O TX2 U6
- (14) F/O RX2 U8
- (15) F/O CLK U9
- (16) LED CLK LCK

Removing the MSUP 2306 controller assembly



This procedure can be performed with the magnet at field.



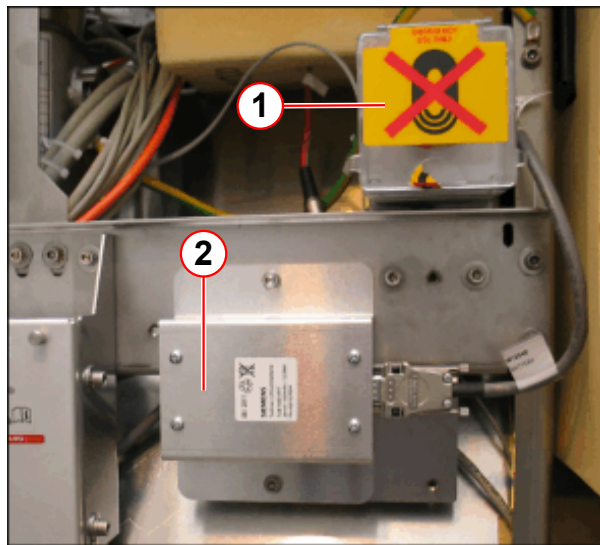
When the ERDU test connector is disconnected from BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

- Ensure the **ERDU test connector** is connected to **X7**.

Fig. 330: Service Magnet Stop button



- (1) Service Magnet Stop button
(2) ERDU battery

- The **Service Magnet Stop button** is mounted above the MSUP battery.
- Check that the button is **not activated**.
- **Test the Service Magnet Stop button.**
 - » Press the **button** to ensure the **MSUP audible alarm** sounds.
 - » **Reset** the button - ensure the alarm **stops**.

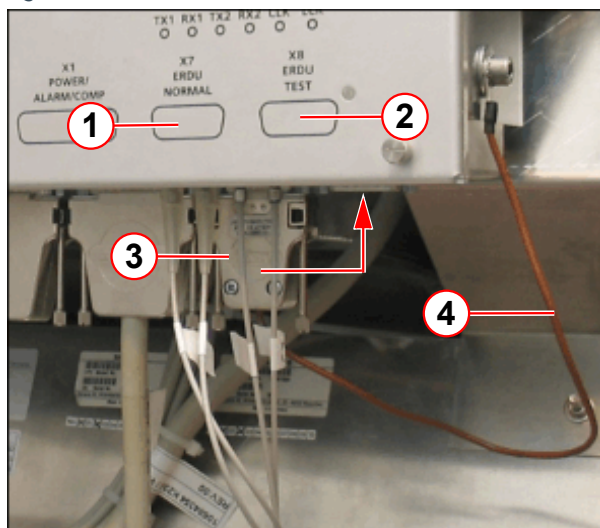
**WARNING**

Risk of quench.

The magnet can quench if the ERDU test connector is not fitted to X8.

- » Check ERDU connection on the MSUP.

Fig. 331: ERDU test connector



- (1) ERDU NORMAL
(2) ERDU TEST
(3) ERDU test connector
(4) ERDU connector strap

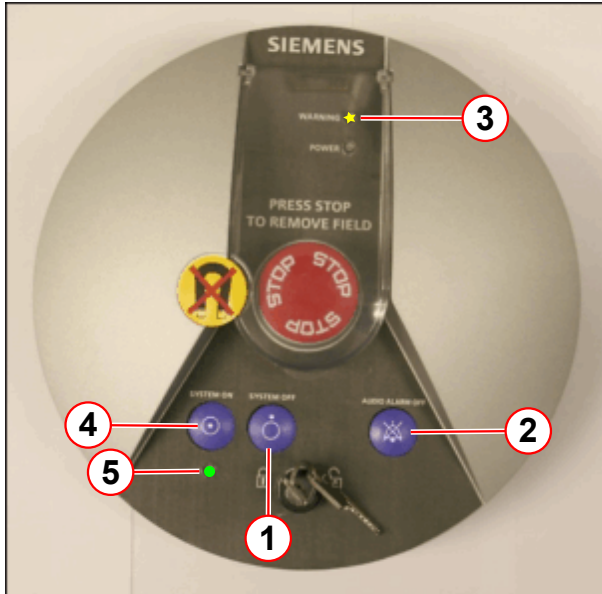
The **ERDU test connector** is fitted to **X7** during normal operation.

- Disconnect the test connector from **X7 (ERDU NORMAL)**.
 - » **MSUP audible alarm** is activated.
- Connect it to **X8 (ERDU TEST)**.
 - » On the **alarm box**, check the **audible alarm** sounds.
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- Check that the **ERDU TEST LEDs** are **NOT illuminated**.



If the MSUP audible warning is not sounding, **DO NOT CONTINUE**.
Investigate the cause of the problem.

Fig. 332: Alarm box

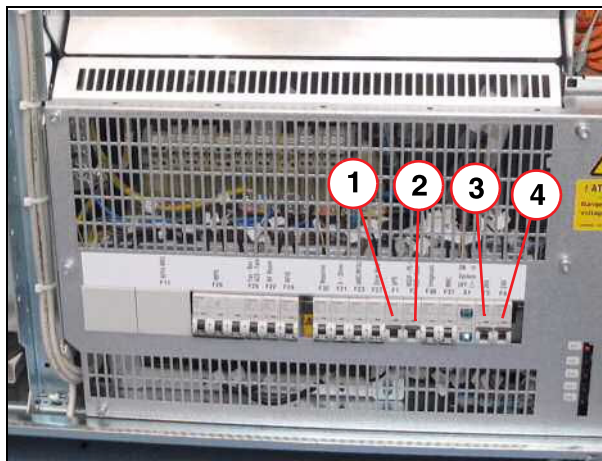


- (1) Button SYSTEM OFF
- (2) Button AUDIO ALARM OFF
- (3) LED WARNING = yellow
- (4) Button SYSTEM ON
- (5) LED SYSTEM ON = green

Shut down the MRC:

- Wait until the **MRC** has **shut down**.
- Press **SYSTEM OFF** button on the alarm box (pos 1).

Fig. 333: ACC fuse (inside)

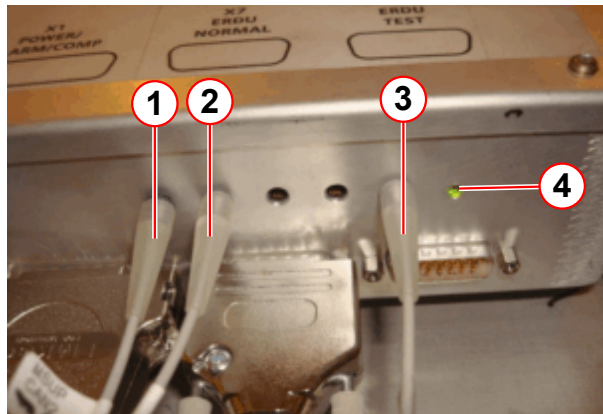


- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

ACC cabinet

- Switch **OFF** circuit breaker **F3**.
 - » the alarm box **audible alarm** will sound.
 - » the **yellow warning LED** flashes.
- Press **AUDIO ALARM OFF** to silence the **alarm**.
- The **MSUP audible alarm** is **deactivated**.
- The **supply LED** on the MSUP is **OFF**.

Fig. 334: MSUP 2306 fiber optic cables



- (1) F/O TX1 U5
- (2) F/O RX1 U7
- (3) F/O CLK U9
- (4) LED CLK LCK

- Carefully disconnect the (3 off) **fiber optic cables**.

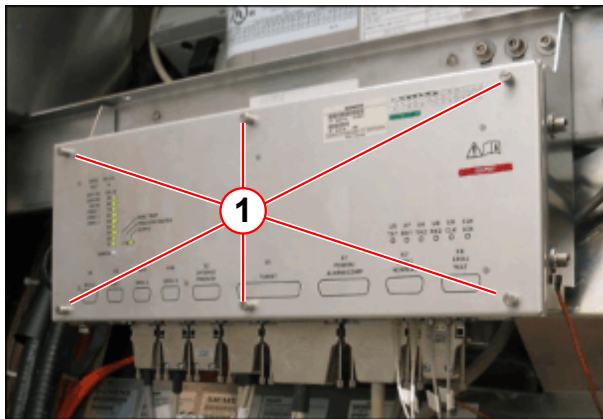
**CAUTION**

Component failure.

ESD precautions must be followed to stop the risk of damaging or reducing the life of the component.

- » Use ESD ground straps when handling ESD components.

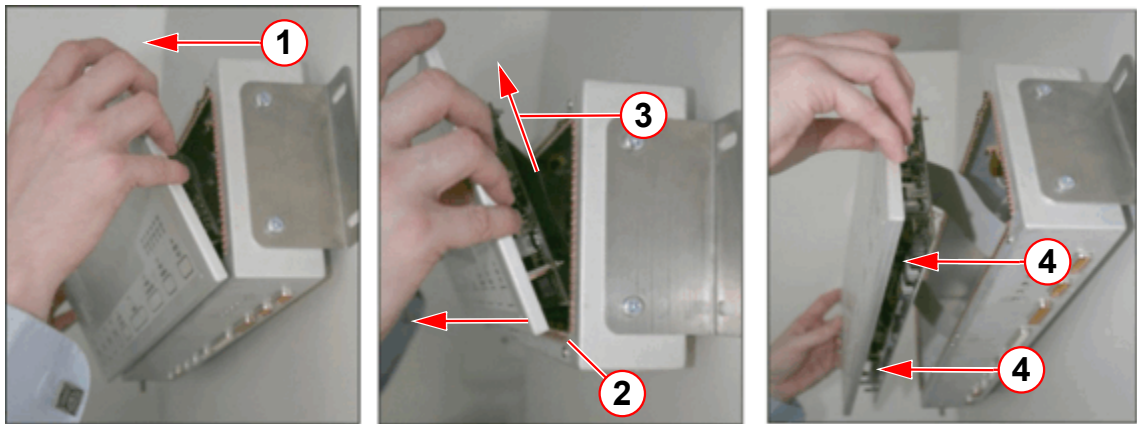
Fig. 335: MSUP 2306 fit/remove the controller assembly



- (1) M3 screws and washers (6 off)

- **Remove** the (6 off) **M3 screws** securing the MSUP controller assembly.

Fig. 336: MSUP 2306 removing the controller assembly



- (1) Lift the lid at 90 degrees
- (2) Lip
- (3) Lift over the lip
- (4) Lift assembly horizontally

- Tilt the **controller assembly** at an 90 degree angle (pos 1).
- **Lift** over the **lip** (pos 2,3).
- **Lift** the assembly **horizontally** (pos 4).

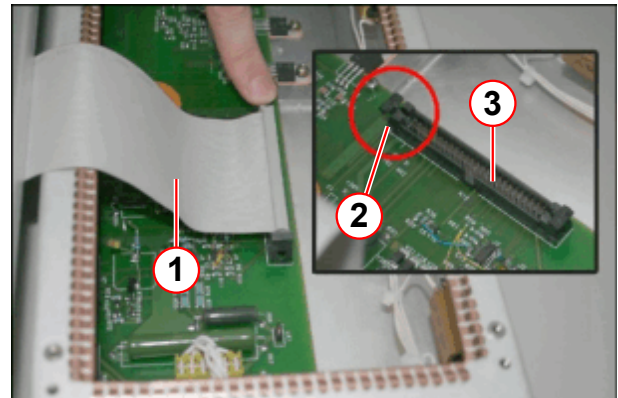
Fig. 337: MSUP 2306 controller assembly



Fig. 338: MSUP 2306 ribbon connections



Fig. 339: MSUP 2306 ribbon cables



- (1) Ribbon cable
- (2) Connector clip
- (3) Ribbon cable connector

- Release the **connector clips** (→ 2/Fig. 339 Page 266) and disconnect the ribbon cables. (→ 3/Fig. 339 Page 266)
 - » the alarm box **audible alarm** sounds.
 - » the **cold head** switches **OFF**.
 - » The **MSUP alarm** reactivates.

4.2.2.4 Installation

Fitting the MSUP controller assembly

- Fit the **ribbon cables**. (→ 1/Fig. 339 Page 266)
 - Ensure the **connectors** are seated correctly.
 - **Lock** in place with the **connector clips**. (→ 2/Fig. 339 Page 266)
 - » the alarm box **audible alarm** stops.
 - » the **cold head** reactivates.
 - » The **MSUP alarm** stops.

Fig. 340: MSUP 2306 feed through the ribbon cables



- Ensure the **ribbon cables** are not **trapped** as the controller assembly is re-fitted/
 - » Carefully tuck the ribbon cables **under the PCB**.

- **Refit** the controller assembly using the (6 off) screws. (→ 1/Fig. 335 Page 265).
- Carefully **reconnect** the (3 off) **fiber optic cables**. (→ Fig. 334 Page 265)

- Turn **ON** the circuit breaker **F3** at the **ACC**.
 - » the **supply LED** on the **MSUP** is **on**.
 - » the **MSUP alarm** sounds.
 - » on the **alarm box**, check the **audible alarm** sounds.
 - » press **AUDIO ALARM OFF** to silence the alarm. (→ 5/Fig. 341 Page 269)



The cold head switches off for 1 sec before reactivating.

- **Test** the replacement MSUP. (→ Testing the MSUP 2306 after replacement / Page 268)

4.2.2.5 Startup

- Perform a **“Re-start Software”** of **“Meas&Recon”** to download the correct loadware.



If a download is necessary it will take about 20 min

Testing the MSUP 2306 after replacement



WARNING

Risk of quench.

The magnet can quench if the ERDU test connector is not fitted to X8.

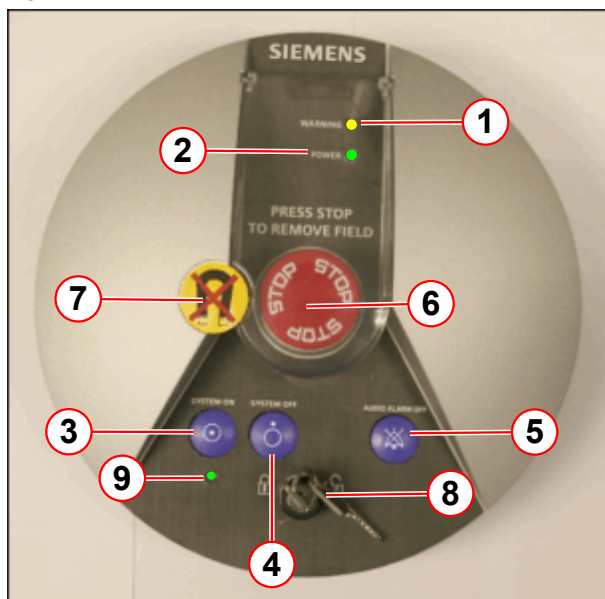
- » Check ERDU connection on the MSUP.



The Magnet Stop button locks when depressed.
Turn the button clockwise to unlock.

- Connect the **ERDU test connector** from **X7** to **X8**.
- Check that the **ERDU TEST LEDs** are **NOT illuminated**.

Fig. 341: Alarm box - LEDs and buttons

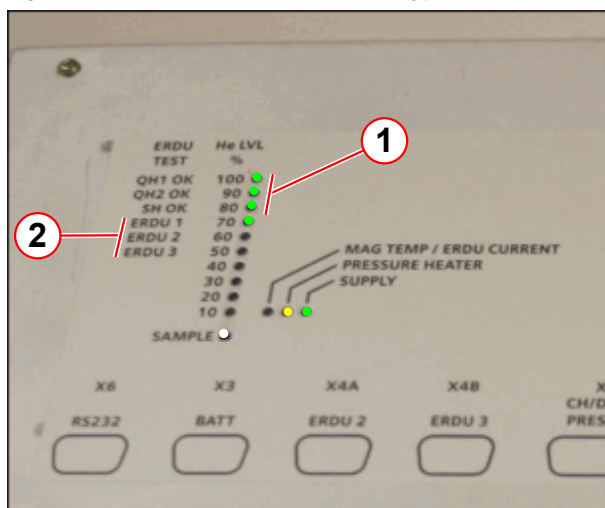


- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
 - » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
 - » Press AUDIO ALARM OFF to silence the alarm.

Fig. 342: ERDU test - LED indicators (typical)



- (1) QH1, QH2, SH = green LED (all ON)
- (2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
 - » QH1, QH2, SH and ERDU 1 are NOT illuminated.

In the **Magnet Room**:

- Press the **Magnet Stop button**.
 - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.
 - Press **AUDIO ALARM OFF** to silence the alarm.

- At the **MSUP**:
 - Check **QH1, QH2, SH LEDs** are illuminated.
 - Check **ERDU 2 LED** is illuminated.
- **Reset the Magnet Stop button** in the Magnet Room.
- **Check** the LEDs return to **normal**.

An optional **3rd Magnet Stop button** may be fitted. Follow the steps below:

- Press the **Magnet Stop button**.
 - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.
 - Press **AUDIO ALARM OFF** to silence the alarm.
- At the **MSUP**:
 - Check **QH1, QH2, SH LEDs** are illuminated.
 - Check **ERDU 3 LED** is illuminated.
- **Reset the 3rd Magnet Stop button**.
- **Check** the LEDs return to **normal**.
- Check **ALL Magnet Stop buttons** are **not activated**.
- **Disconnect** the ERDU test connector **from X8 (ERDU TEST)** and reconnect it to **X7 (ERDU NORMAL)**.
 - » On the **alarm box**, check the **audible alarm** sounds.
- **At the MSUP**:
 - » Check the **MSUP audible alarm** is **deactivated**.
 - » Check the **MSUP LEDs** return to **normal**.
- **At the alarm box**:
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- **Check** the **warning LED** is not illuminated '**ON**'.



If the LED is on, or flashing, the fault must be checked.
Do not disconnect the ERDU battery.

- **Boot up the MRC**:
 - At the **alarm box**, press **SYSTEM ON**.
 - The **green LED** will light on the **alarm box**. (→ 5/ Fig. 332 Page 264)
- Perform a He test **sample**.
 - press the **sample button** on the **MSUP**.
 - **wait** for **30 secs**.
 - ensure the **LEDs** displays the **correct He level**.
- **Clear** all posted **alarms** on the **MRC**.

Downloading the i-Button MSUP 2306 data to the MRC

The **MRC (MRAWP) Host** must be updated with the new **Serial, Revision, and Part numbers**.

- Access the **Service software** at the **MRC (MRAWP) Host**.
- Select menu item **"Magnet & Cooling"**.
- Select **"Magnet Setup"**.

Fig. 343: Updating system Identification numbers

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup
 - Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☒ Probe 1 ☐ Probe 2 ☐ Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87	A

MSUP Data

Downloads the magnet setup/calibration data to the host

Uploads the magnet setup/calibration data from the host to the MSUP

Download

Upload

☐ Save values as default

Ready. | Save | Default | Update

- (1) Download button (MSUP data stored on MRC)
- (2) Save button (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- Click **"UPDATE"** (→ 3/Fig. 343 Page 272)
 - » This **refreshes** the on-screen values.
- Update the **serial/revision/part** numbers (if required).
- Click **"SAVE"** to transfer calibration values **to the MSUP**. (→ 2/Fig. 343 Page 272)

- Click **"DOWNLOAD"**(→ 1/Fig. 343 Page 272)
 - » This takes the **MSUP data** and stores it in a file on the **MRC**.

4.2.2.6 Final steps

- Replace the appropriate **cover panels**.

4.2.3 Replacing the ERDU battery box

4.2.3.1 Safety information



WARNING

Strong magnetic fields.

Failure to observe the following safety warning can result in physical injury and equipment damage.

- » Only non-magnetic tools and equipment can be taken into the magnet room.
- » Close the door of the RF room.



The ERDU function can still be maintained if the battery is disconnected, as long as the 36V power input to the MSUP 2306 is maintained.



This procedure can be performed with the magnet at field.

4.2.3.2 Estimated completion time and number of persons

Tab. 28 Completion time

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	15 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

4.2.3.3 Aids or tools

Tools

- Screwdriver (**non-magnetic**).
- 10mm wrench (**non-magnetic**).

Spare parts

- ERDU battery box (part no **10128925**).

4.2.3.4 Preparatory Work Steps

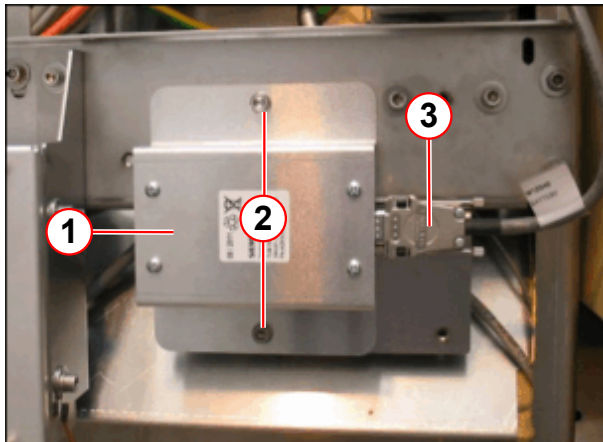
- **Remove** the appropriate **cover panels**.

4.2.3.5 Removal



Disconnecting/connecting the battery cable can cause the alarm box to alarm. Press **AUDIO ALARM OFF** to cancel the alarm.

Fig. 344: ERDU battery box location



- (1) ERDU battery box
- (2) ERDU battery cable
- (3) Nut and washer (2 off)

- **Disconnect** the **battery cable**.
 - » This may cause the **alarm box** to sound.
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- **Remove** the nuts and washers (2 off).
- **Remove** the old battery box.

4.2.3.6 Installation

- **Fit** the new **battery box** with the nuts and washers.



Do not Short pins on the battery connector.

When connecting the battery cable ensure the pins on the connector are not shorted as this will blow the fuses in the battery pack. The fuses are not replaceable, hence a new battery pack will need to be ordered and fitted.

- **Connect** the **battery cable**.

4.2.3.7 Startup

- Ensure there are **no alarms**.
- Fault find as required.

4.2.3.8 Final Steps

- **Replace** the appropriate **cover panels**.

5.1 Alarm box for Avanto/Avanto Dot Upgrade / Espreo

5.1.1 Replacing the Alarm box and cables with the Magnet at Field

5.1.1.1 Required time and tools

Time

60 min

Tools

- Non-magnetic tool kit.
- ERDU test load kit (3861197).

Spare parts

- Alarm box (part no 10096765).

5.1.1.2 Preliminary



CAUTION

The replacement procedure will disable the magnet emergency shut down system for short intervals.

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » For safety, set the MSUP to the ERDU test setting.
- » Exclude all non service personnel and all magnetic materials/tools from the magnet area.
- » Close the door of the RF room.

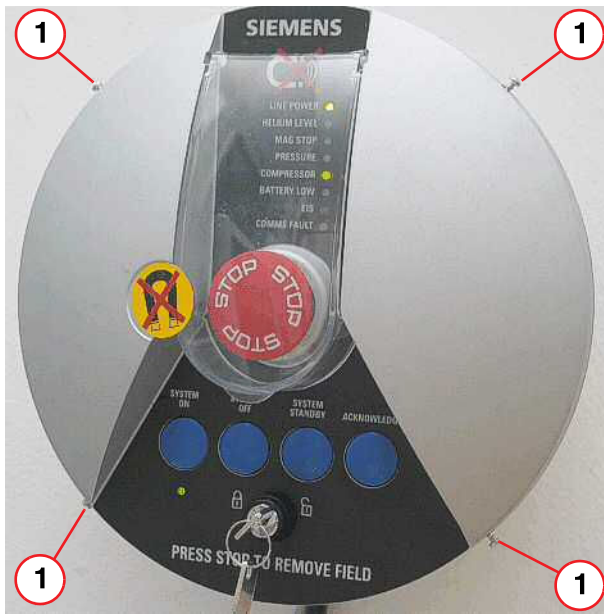
5.1.1.3 Replacement Procedure



It is important to replace the Alarm box along with the base plate, in order to keep the proper serial number information.

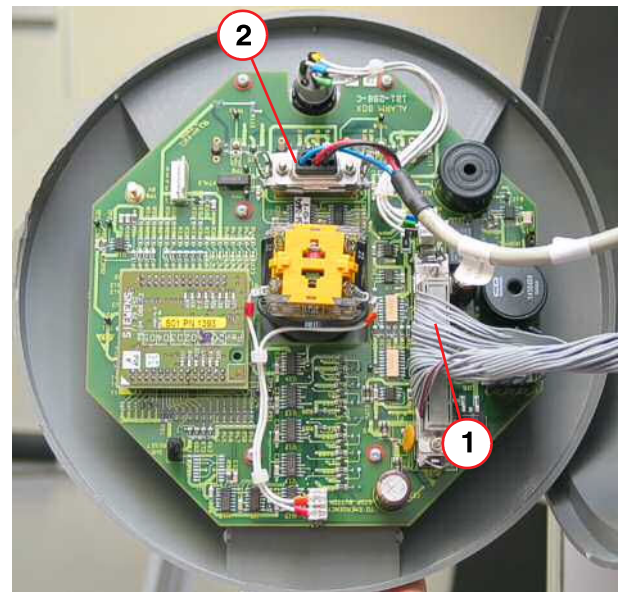
- **Isolate power** to the alarm box:
 - Switch **OFF** circuit breaker **F3** at the **ACC**.
- **Connect the ERDU Test Load** to the MSUP. Refer to the ERDU safety related test.

Fig. 345: Alarm box



(1) Securing screws

Fig. 346: Alarm box connectors

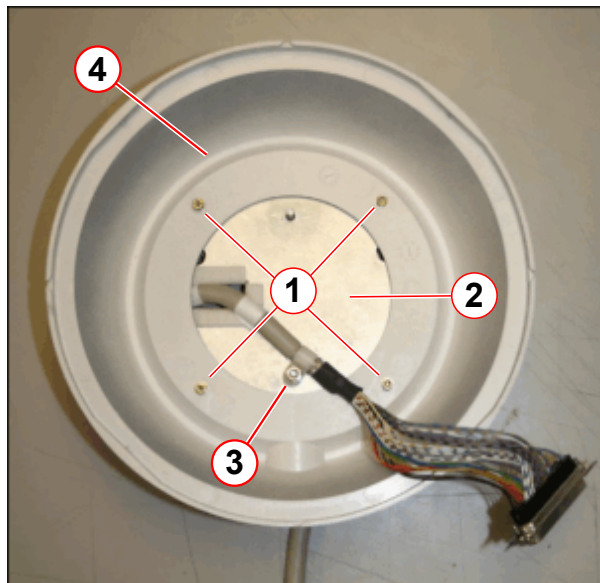


(1) Connector X7
(2) Connector X5

Removal

- **Remove** the 4 screws around the outer rim of the alarm box.
- **Separate** the 2 halves of the alarm box.
- **Disconnect** the control and RS232 communications cables (→ Fig. 346 Page 277).

Fig. 347: Back of Alarm Box (inside)

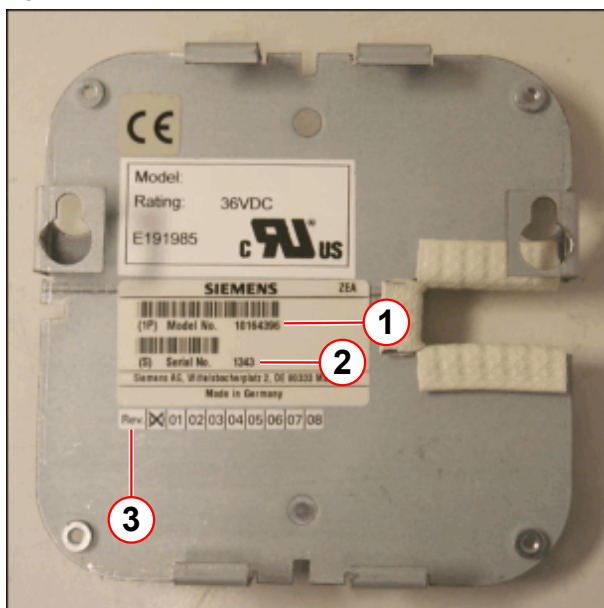


(1) Screws (4 off)
(2) Fixing plate
(3) 'P' clip and retaining nut
(4) Plastic base

- **Remove** the (4 off) mounting screws (1).
- Lift the **plastic base** clear (4).
- **Remove** the retaining **nut** and '**P**' **clip** (3).
- Loosen the (2 off) **fixing plate mounting screws**.
- **Remove metal fixing plate**.

Installation

Fig. 348: Alarm box identification

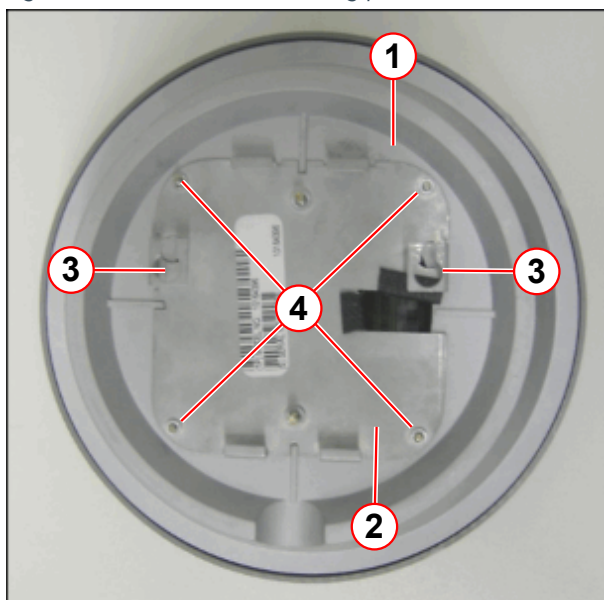


Note the alarm box **identification numbers** of the new unit.

- (1) Revision number
- (2) Serial number
- (3) Part number

- **Remove** the (4 off) screws around the outer rim.
- **Separate** the 2 halves of the alarm box.

Fig. 349: Alarm box base - fixing plate



- (1) Alarm box base
- (2) Fixing plate
- (3) Mounting slots (95mm apart)
- (4) Fixing plate screws (4 off)

- **Remove** the **metal fixing plate** from the plastic base.
 - **Unscrew** the (4 off) hex head socket screws.

- Feed the **control cable** through the slot in the fixing plate. (→ 2/Fig. 349 Page 278)
- **Locate** the fixing plate **mounting slots** over the **wall screws**, and tighten screws. (→ 3/Fig. 349 Page 278).
- Remove the **'P' clip** from the metal fixing plate (→ 3/Fig. 347 Page 277).
- **Secure the cable** through the **'P' clip** and refit to the **metal fixing plate**.
 - Ensure it makes **good grounding contact**.
- Ensure correct **cable channel orientation** for the mounting of the alarm box base.
- **Fit** the **plastic base** of the alarm box to the **fixing plate** with (4 off) screws.
- **Reconnect** the control and RS232 communications cables to **X7** and **X5**.
- **Refit** the alarm box **cover** using the (4 off) fixing **screws**.
- Turn **ON** circuit breaker **F3** at the **ACC**.



The compressor may take up to 20 minutes to restart.

5.1.1.4 Cable Removal

When **replacing cables** to the alarm box:

- **Isolate power** to the alarm box:
 - Switch **OFF** circuit breaker **F3** at the **ACC**.
- **Connect** the **ERDU Test Load** to the system. **Refer to the ERDU safety-related test**.
- Follow the **alarm box replacement procedure** when removing the covers.
- **Remove** the **'P' clip** on the fixing plate which secures the cable.
- **Remove** any other cable ties from the cables.
- **Carefully** remove the **cable** through the fixing plate. (→ 2/Fig. 347 Page 277)
- **Replace** the cable and **refit** the alarm box cover.

5.1.1.5 Final Checks

After replacing the alarm box, or one of the cables, an **operational check** must be performed on the new equipment.

- Turn **ON** circuit breaker **F3** at the **ACC**.
- Check the **LED functions** on the alarm box.
- Test the **ERDU circuit function** using the ERDU test load. **Refer to the ERDU safety-related test**.
- **Remove** ERDU test load from the 41 way turret connector and **refit the MSUP cable**.

5.2 Alarm Box for Avanto, Espree (VB19)/ Avanto Dot Ex-factory

5.2.1 Replacing the alarm box (Mat. number 10523563) and cables with the magnet at field

5.2.1.1 Requirements

Safety information



CAUTION

The replacement procedure disables the magnet emergency shut down system for short intervals.

Safety precautions **MUST** be followed to avoid serious bodily injury or property damage.

- » Exclude all non-service personnel and all magnetic materials/tools from the magnet area.
- » Close the door of the RF room.

Estimated completion time and number of persons

Tab. 29 Completion time

Worksteps	Time	Persons
Preparation	10 min	1
Removal/Installation	30 min	1
Startup/Tune-up/QA	10 min	1
Final steps	10 min	1

Aids or tools

Tools

- Non-magnetic tool kit.

Spare parts

- Alarm box (part no **10523563**).

5.2.1.2 Preparatory Work Steps

Check the Magnet Stop button



WARNING

Restrict Access.

Unauthorised personnel must be excluded from the imaging suite.

- » Secure the room and prevent unauthorised access.



WARNING

Strong Magnetic Fields.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only NON-MAGNETIC equipment may be taken into the magnet room.



CAUTION

Magnet quench!

A magnet quench is caused by pressing the service magnet stop button (ERDU).

- » Do not operate the Service Magnet Stop button unless an emergency magnet quench is required.

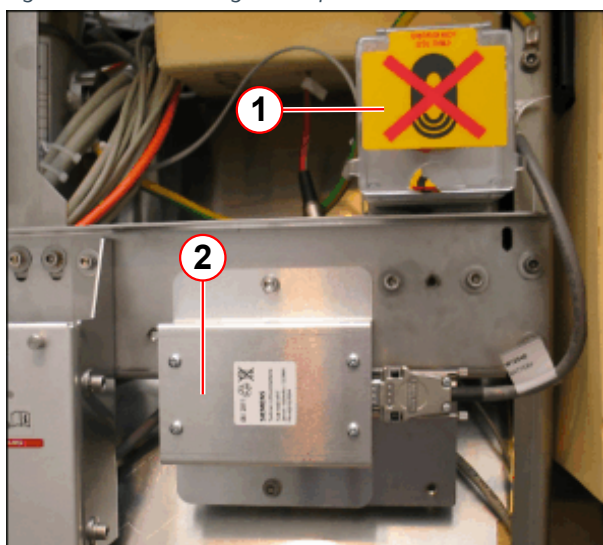
- **Restrict access** to the magnet if it is **at field** when the test is being performed.



Ensure the test connector is connected to X7.

- **Remove** the appropriate covers to gain access to the **MSUP** and **Service Magnet Stop button**.

Fig. 350: Service Magnet Stop button



- (1) Service Magnet Stop button
(2) ERDU battery

- The **Service Magnet Stop button** is mounted above the MSUP battery.
- Check that the button is **not activated**.
- **Test the Service Magnet Stop button**.
 - » Press the **button** to ensure the **MSUP audible alarm** sounds.
 - » **Reset** the button - ensure the alarm **stops**.

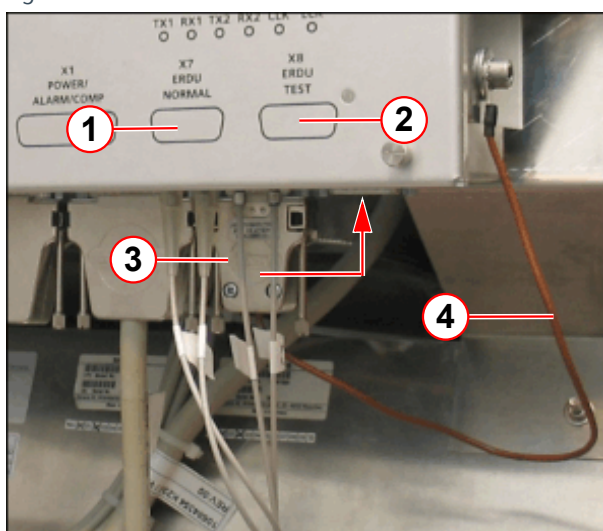
i

When the ERDU test connector is disconnected from BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

Fig. 351: ERDU test connector



- (1) ERDU NORMAL
(2) ERDU TEST
(3) ERDU test connector
(4) ERDU connector strap

The **ERDU test connector** is fitted to **X7** during normal operation.

- Disconnect the test connector from **X7** (ERDU NORMAL).
 - » **MSUP audible alarm** is activated.
- Connect it to **X8** (ERDU TEST).
 - » On the **alarm box**, check the **audible alarm** sounds.
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- Check that the **ERDU TEST LEDs** are **NOT illuminated**.



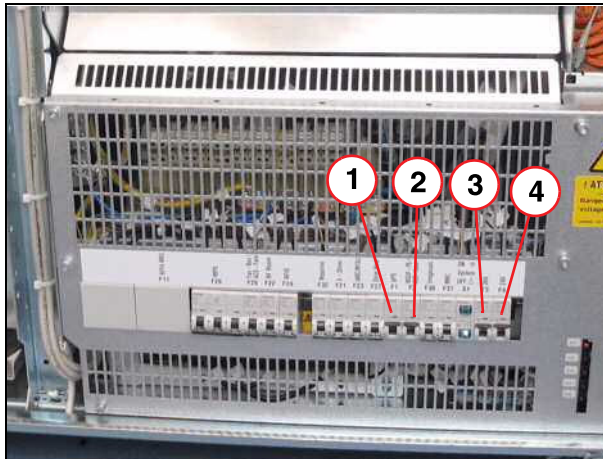
If the MSUP audible warning is not sounding, DO NOT CONTINUE.
Investigate the cause of the problem.



The Magnet Stop button locks when depressed.
Turn the button clockwise to unlock.

5.2.1.3 Removal

Fig. 352: ACC fuse (inside)

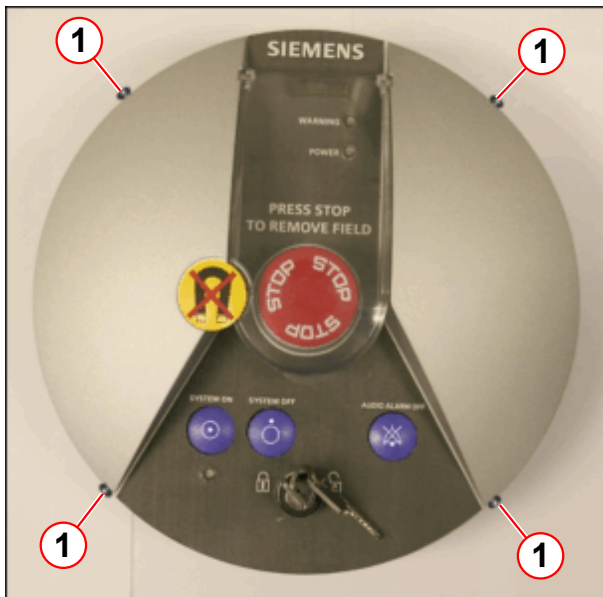


- (1) Fuse F1
- (2) Fuse F2
- (3) Fuse F3
- (4) Fuse F4

ACC cabinet

- Isolate power to the alarm box:
 - Switch **OFF** circuit breaker F3.

Fig. 353: Alarm box securing screws

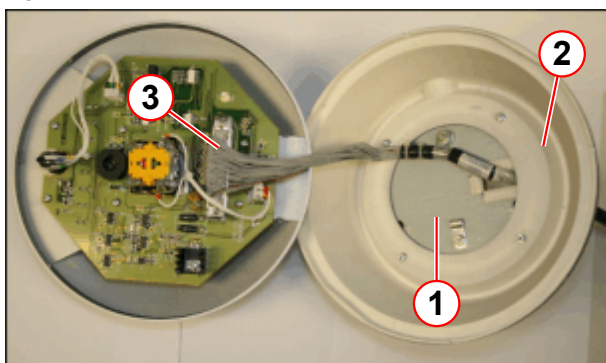


- (1) Securing screws (4off)

At the old alarm box:

- Remove the 4 off screws around the **outer rim** of the alarm box.

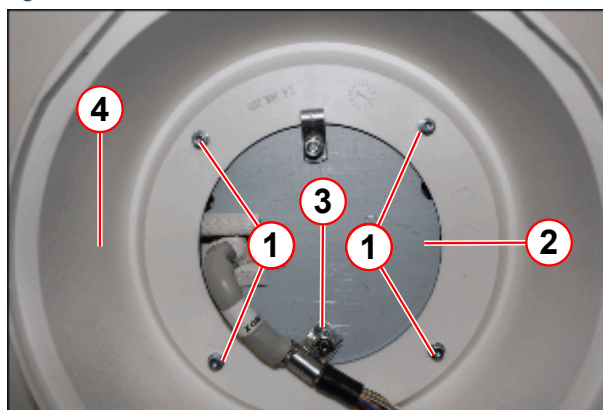
Fig. 354: Alarm box connector X1



- (1) Metal fixing plate
- (2) Plastic base
- (3) Connector X1

- **Separate** the 2 halves of the alarm box.
- Disconnect **connector X1**.

Fig. 355: Back of Alarm Box (inside)



- (1) Screws (4 off)
- (2) Fixing plate
- (3) 'P' clip facing outwards
- (4) Plastic base

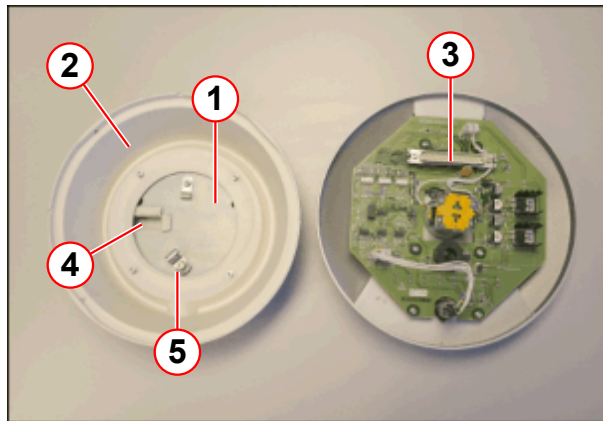
- **Remove** the (4 off) mounting screws (1).
- Lift the **plastic base** clear (4).
- **Remove** the retaining **nut** and '**P**' clip (3).
- Loosen the (2 off) **fixing plate mounting screws**.
- Remove **metal fixing plate** from the wall.

5.2.1.4 Installation



It is important to replace the Alarm box along with the base plate, in order to keep the proper serial number information.

Fig. 356: Alarm box

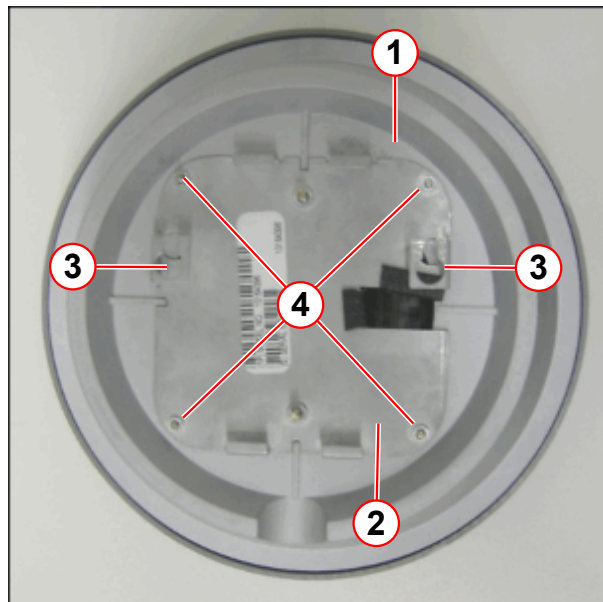


- (1) Metal fixing plate
- (2) Plastic base
- (3) Connector X1
- (4) Cable slot
- (5) P clip

At the new alarm box:

- **Remove** the (4 off) screws around the outer rim. (→ Fig. 353 Page 283)
- **Separate** the 2 halves of the alarm box.

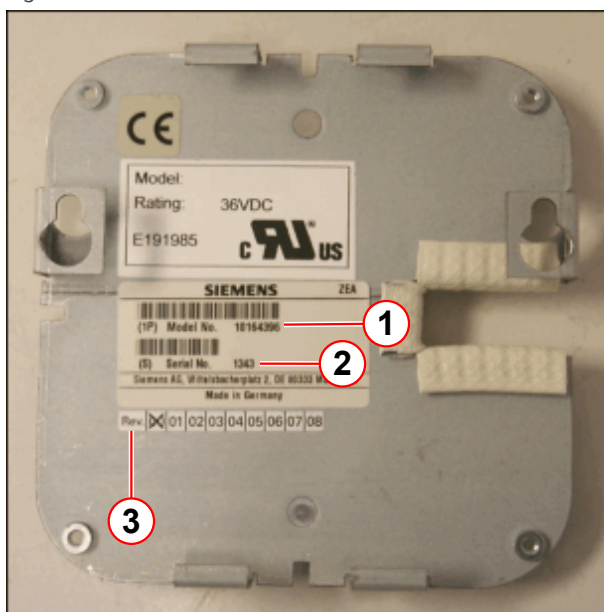
Fig. 357: Alarm box base - fixing plate



- (1) Alarm box base
- (2) Fixing plate
- (3) Mounting slots (95mm apart)
- (4) Fixing plate screws (4 off)

- **Remove** the **metal fixing plate** from the plastic base.
 - **Unscrew** the (4 off) hex head socket screws.

Fig. 358: Alarm box identification



- (1) Revision number
- (2) Serial number
- (3) Part number

Record the **identification numbers** of the **new unit**.

This information **must be updated** via the **MRC (MRAWP) Host, OR RS232 connection**.

1. **Revision** number.
2. **Serial** number.
3. **Part** number.

- Feed the **control cable** through the slot in the fixing plate (if required).
(→ 4/Fig. 356 Page 285).
- **Locate** the fixing plate **mounting slots** over the **wall screws**, and tighten screws.
(→ 3/Fig. 357 Page 285)
- Remove the '**P**' clip and retaining nut from the metal fixing plate.
(→ 5/Fig. 356 Page 285)
- **Secure the cable** through the '**P**' clip and refit to the **metal fixing plate**.
 - Ensure it makes **good grounding contact**.
- **Fit** the **plastic base** of the alarm box to the **fixing plate** with (4 off) screws.
- **Ensure** the alarm box ERDU button **is not pressed**.
- **Reconnect** cable to **X1** on the alarm box. (→ 3/Fig. 356 Page 285)
 - secure **fastening clips** on D-type connector.
- **Refit** the alarm box **cover** using the (4 off) fixing **screws**.
- Turn **ON** circuit breaker **F3** at the **ACC**.



The compressor may take up to 20 minutes to restart.

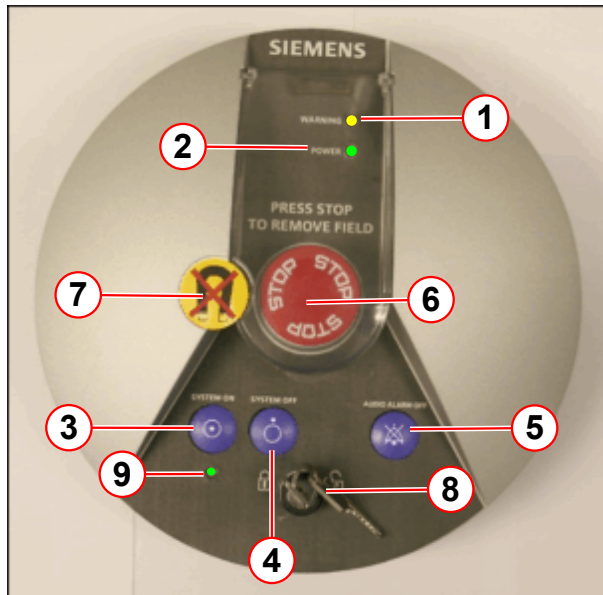
After replacing the alarm box, an **operational check** must be performed on the new equipment.

- Check the **green POWER LED** is lit.
- Turn the locking key to '**unlock**'.

- Press the **SYSTEM ON** button.
 - » Check the **green SYSTEM ON LED** is lit.

Magnet Stop button test (ERDU)

Fig. 359: Alarm box - LEDs and buttons

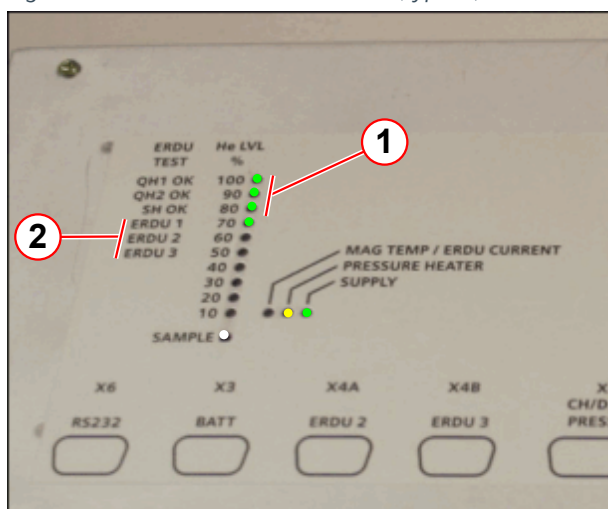


- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
 - » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
 - » Press AUDIO ALARM OFF to silence the alarm.

Fig. 360: ERDU test - LED indicators (typical)



- (1) QH1, QH2, SH = green LED (all ON)
 (2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
 - » QH1, QH2, SH and ERDU 1 are NOT illuminated.

Final steps

- Check **ALL Magnet Stop buttons** are **not activated**.
- **Disconnect** the ERDU test connector **from X8 (ERDU TEST)** and reconnect it to **X7 (ERDU NORMAL)**.
 - » On the **alarm box**, check the **audible alarm** sounds.
- **At the MSUP:**
 - » Check the **MSUP audible alarm** is **deactivated**.
 - » Check the **MSUP LEDs** return to **normal**.
- **At the alarm box:**
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- Check the **warning LED** is not illuminated '**ON**'.



If the LED is on, or flashing, the fault must be checked.
 Do not disconnect the ERDU battery.

5.2.1.5 Start up

Updating the alarm box identification numbers

The system must be **updated** with the new:

- **Serial** number
- **Revision** number
- **Part** number

This can be actioned via the **MRC (MRAWP) Host**, or by **direct RS232 connection** if the Host is unavailable.

Downloading MSUP 2306 data to the MRC

The **MRC (MRAWP) Host** must be updated with the new **Serial, Revision, and Part numbers**.

- Access the **Service software** at the **MRC (MRAWP) Host**.
- Select menu item **"Magnet & Cooling"**.
- Select **"Magnet Setup"**.

Fig. 361: Updating system Identification numbers

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
 - Cooling Status
 - Magnet Status
 - Compressor Status
 - Cooling Setup
 - Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☒ Probe 1 ☐ Probe 2 ☐ Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87	A

MSUP Data

1 Downloads the magnet setup/calibration data to the host

2 Download

3 Upload

Uploads the magnet setup/calibration data from the host to the MSUP

☐ Save values as default

Ready. | Save | Default | Update

- (1) Download button (MSUP data stored on MRC)
- (2) Save button (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- Click **"UPDATE"** (→ 3/ Fig. 361 Page 290)
 - » This **refreshes** the on-screen values.
- Update the **serial/revision/part** numbers (if required).
- Click **"SAVE"** to transfer calibration values **to the MSUP.**(→ 2/ Fig. 361 Page 290)

- Click **"DOWNLOAD"**(→ 1/Fig. 361 Page 290)
 - » This takes the **MSUP data** and stores it in a file on the **MRC**.

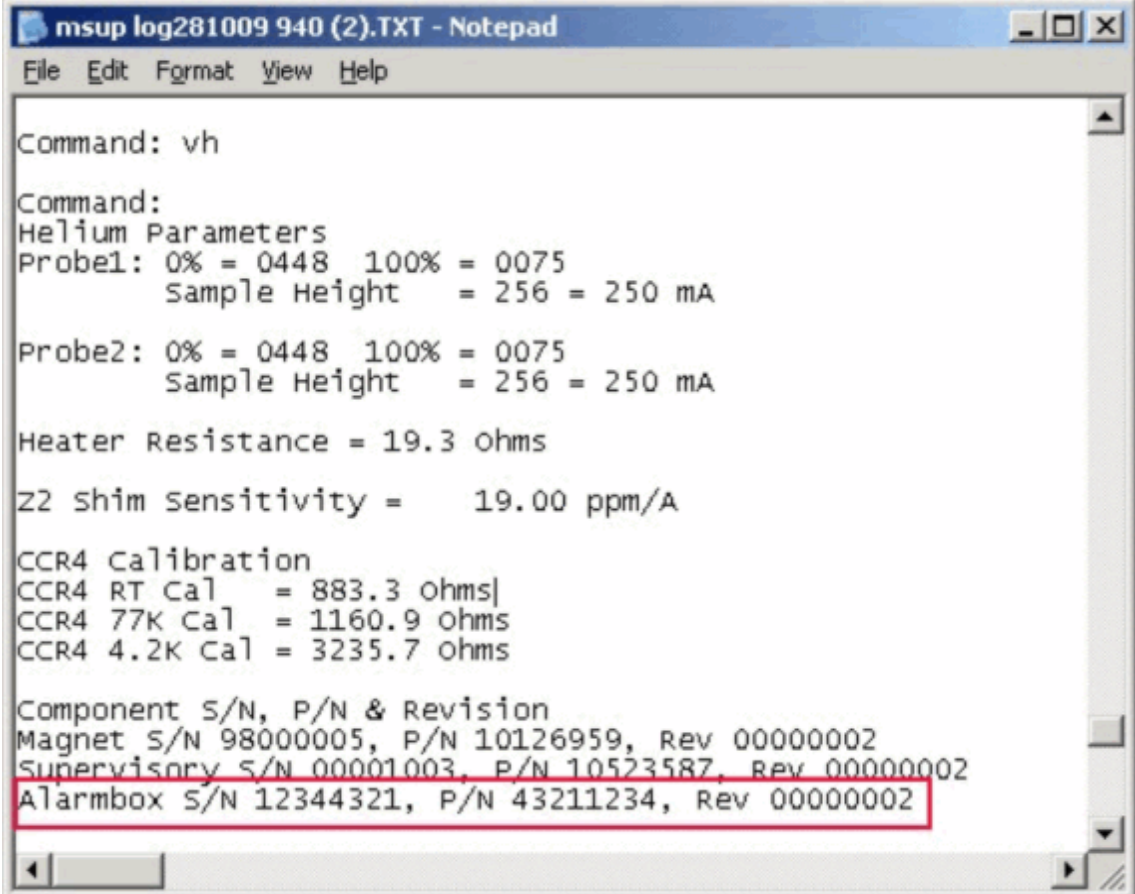
Downloading the MSUP 2306 data via RS232 connection

If the **MRC** is not available, follow the **RS232 procedure**.

- Select the appropriate **Com port**.
- **Connect** the Laptop to the MSUP (**X6**) with an RS232 cable.
- Press **ESC** at the **Command** prompt:
 - Input screen is displayed.
 - Type **L0 6932** <enter>
 - Type **R** <enter>
 - Press **ESC**
 - Input screen is displayed.
 - Type **ASN** <Space> **Alarm box Serial Number (nnnn)** <Enter> (e.g. ASN 1234)
 - Type **ARN** <Space> **Alarm box Revision Number (nn)** <Enter> (e.g. ARN 12).
 - Type **APN** <Space> **Alarm box Part Number (nnnnnnnn)** <Enter> (e.g. APN 12345678).

- Type <VH> <Enter>.
 - **Confirm** alarm box data is **updated**.

Fig. 362: Check the alarm box data is updated (command VH)



```
msup log281009 940 (2).TXT - Notepad
File Edit Format View Help

Command: vh

Command:
Helium Parameters
Probe1: 0% = 0448 100% = 0075
        Sample Height   = 256 = 250 mA

Probe2: 0% = 0448 100% = 0075
        Sample Height   = 256 = 250 mA

Heater Resistance = 19.3 ohms

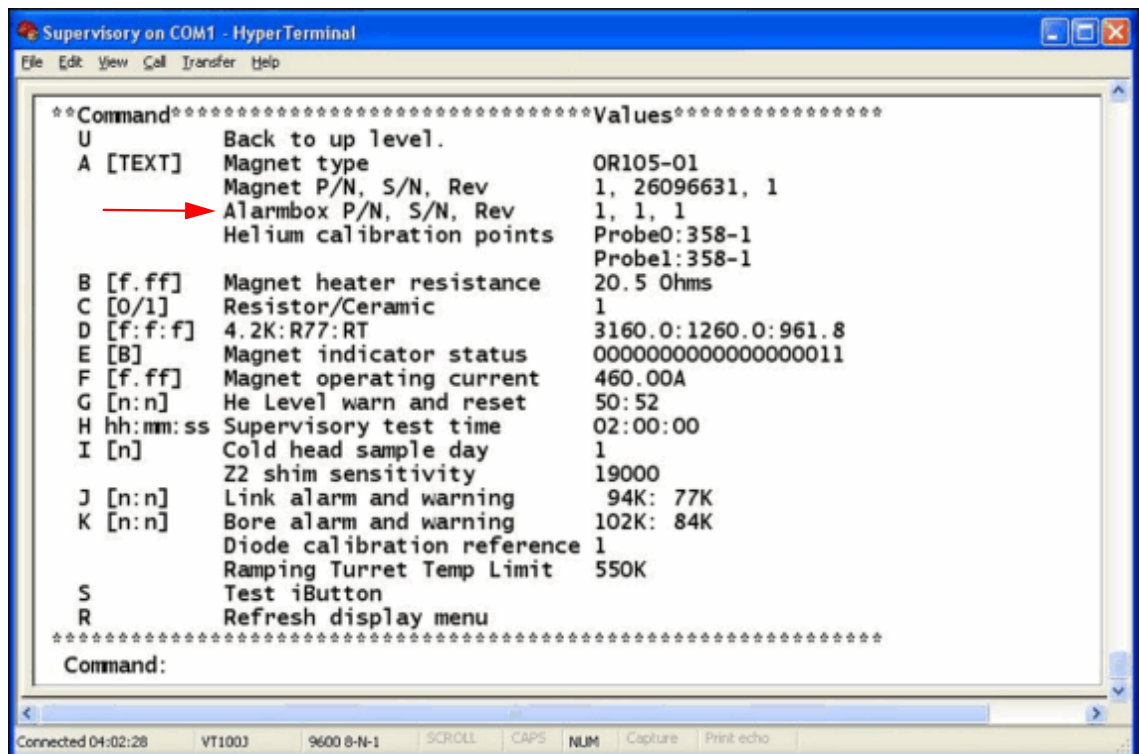
Z2 shim sensitivity = 19.00 ppm/A

CCR4 Calibration
CCR4 RT Cal   = 883.3 ohms|
CCR4 77K Cal  = 1160.9 ohms
CCR4 4.2K Cal = 3235.7 ohms

Component S/N, P/N & Revision
Magnet S/N 98000005, P/N 10126959, Rev 00000002
Supervisory S/N 00001003, P/N 10523587, Rev 00000002
Alarmbox S/N 12344321, P/N 43211234, Rev 00000002
```

- Type **B** <Enter>.
 - Check the data has been stored (**iButton**).

Fig. 363: Check alarm box data is stored on the iButton (command B)



- Type **U** <Enter>.
- Type **R** <Enter>.

5.2.1.6 Final Steps

- Seal the ERDU button with a **tamper-proof seal**.

6.1 Replacing the Shim Power Supply (SPSU)

6.1.1 Tools and time required

6.1.1.1 Time

60 min

6.1.1.2 Tools

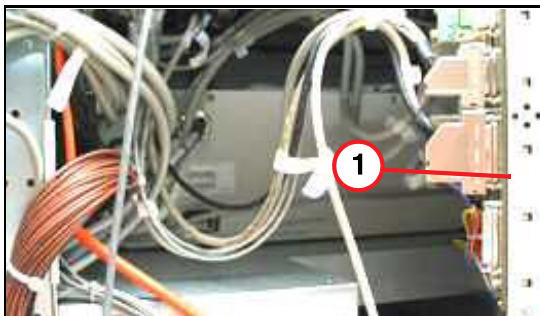
- Standard tool kit

6.1.2 Preliminary

- MR system is switched off

6.1.3 Replacement procedure

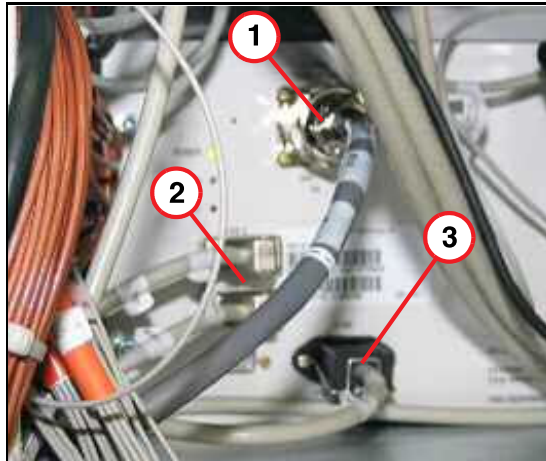
Fig. 364: Shim power supply



(1) AMC frame

- Switch off F21 in the ACC cabinet.
- Swing out the AMC frame.

Fig. 365: Shim power supply



- (1) Output shim cable
- (2) Can1 / Can2
- (3) X700

- Disconnect X700.
- Disconnect the Can 1 and Can 2 cables.
- Disconnect the output shim cable.
- Remove and retain the four rack screws securing the shim power supply unit into the cabinet.
- Lift the SPSU clear of the cabinet.
- Place the new SPSU in the cabinet and secure it.
- Connect the cables at the front of the SPSU.
- Switch on F21.
- Switch on the MR system.

6.1.4 Final checks

- Check the function of the SPSU.
- Perform "Phantom Shim" under Tune up
- Perform "Shim Check" under Quality Assurance

6.2 Replacing the Shim Filter Unit

6.2.1 Tools and time required

6.2.1.1 Time

60 min, additional ramping time

6.2.1.2 Tools

- Standard tool kit

6.2.2 Preliminary

- MR system is switched off



CAUTION

Magnetic components in the shim filter unit

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » The magnet has to be ramped down before removing the shim filter unit.

- Magnet is ramped down



CAUTION

Weight hazard.

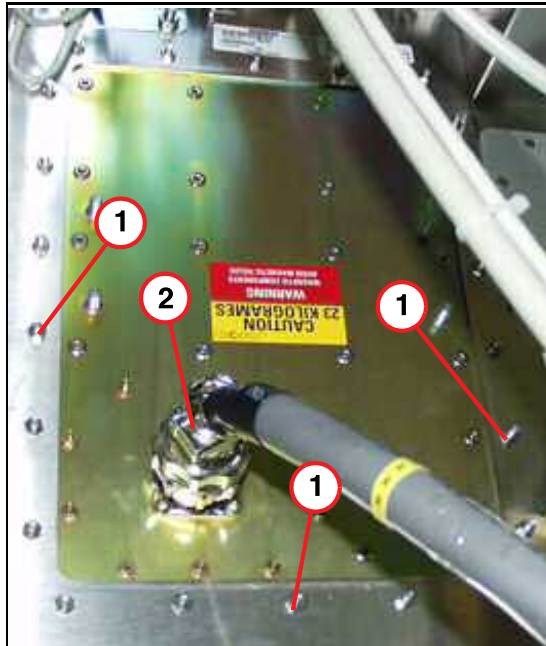
The filter unit weighs 23 kg (50.6 lbs.)

Failure to adhere to safety precautions may result in serious bodily injury or property damage.

- » Ensure sufficient people are available.

6.2.3 Replacement procedure

Fig. 366: Shim filter - equipment room



- (1) Studs
(2) Cable

- Switch off F21 in the ACC cabinet.
- Disconnect cable from the filter unit on the RF room side.
- Disconnect cable from the filter unit on the equipment room side.
- Support the filter unit at the RF room side; this requires a second person.
- Remove the the nuts and washers from each of the 22 studs at the equipment room.
- Carefully lift the filter unit.



Before attempting to fit the filter unit in place, inspect the RF gasket. If any damage is determined, the gasket has to be replaced.

- On the RF room side, move the shim filter up to the opening in the RF filter panel and locate the 22 studs in the holes in the RF filter panel.
- Push the filter unit firmly against the gasket.
- At the equipment room, place the washer and nut on each of the 22 studs and tighten nuts to secure the filter unit in place.
- Connect the two cables to the filter unit.
- Switch on F21.
- Switch on the MR system.

6.2.4 Final checks

- Perform "Phantom Shim" under Tune up.
- Perform "Shim Check" under Quality Assurance.

7.1 MSUP

7.1.1 MSUP 2306 download instructions



Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Log files can be downloaded via either:

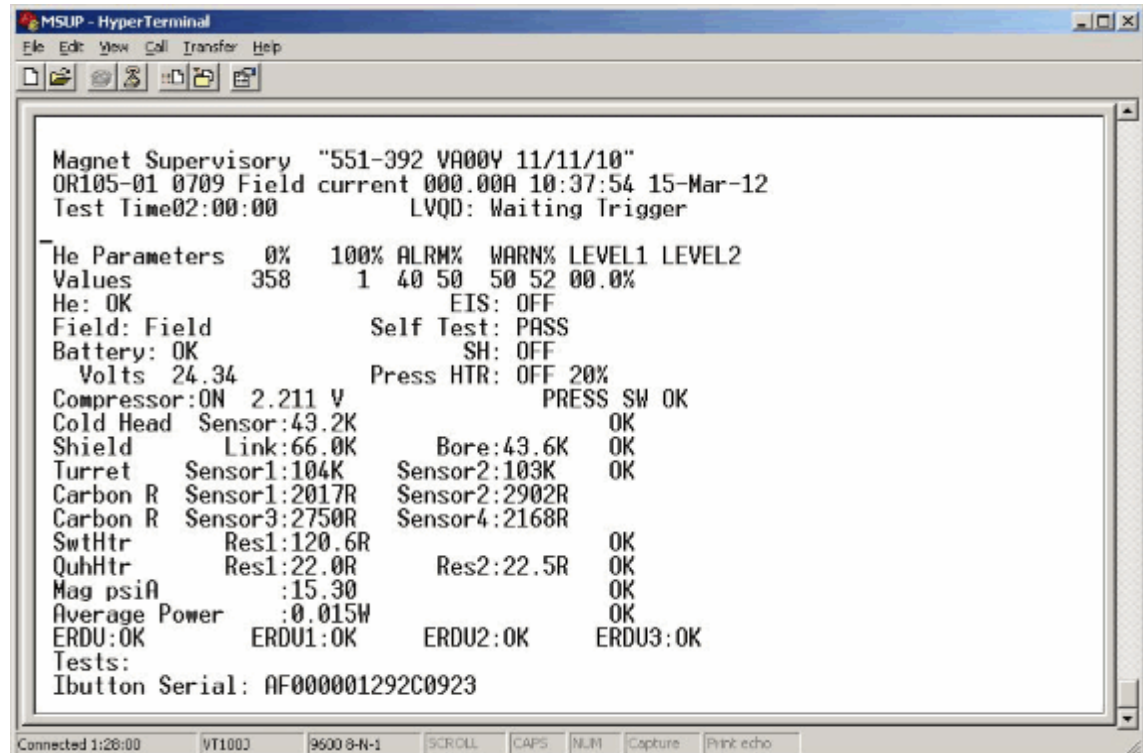
- **Direct connection** to the MSUP via RS232 (**X6**) connection.
- Service software at the **MRC (MRAWP) Host computer**.

7.1.1.1 Downloading via the RS232 connection

1. Turn on the **laptop** and open the **MSUP Terminal Emulation Software**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File: Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP via **RS232 cable**.
3. **Press ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>

6. Press **ESC** to display the **MSUP Process display screen**.

Fig. 367: MSUP Process display screen (Example)



7. **Capture the screenshot.** (→ Capturing a screenshot / Page 303)
8. **Press ESC** to display the **MSUP Command input screen**.
9. Click **Transfer** on the **MSUP Terminal Emulation Screen** tool bar.
10. Click **Capture text**.
11. Click **Browse** and select a directory and a name for the file to be stored. The **Name** should contain the **magnet serial number**.
 - e.g. C:\templmsn 12345678log.txt
12. Click **Start**.
13. Type the following:
 - **VA** <enter>
 - **VQ** <enter>
 - **VM** <enter>
 - **VP** <enter>
 - **VX** <enter>
14. Wait until data has been captured.
15. Click **Transfer, capture text, Stop**.
16. Type **R** <enter>.
17. **Email** the saved files to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)



After the download is complete, the MSUP must be reset.

This can be done via

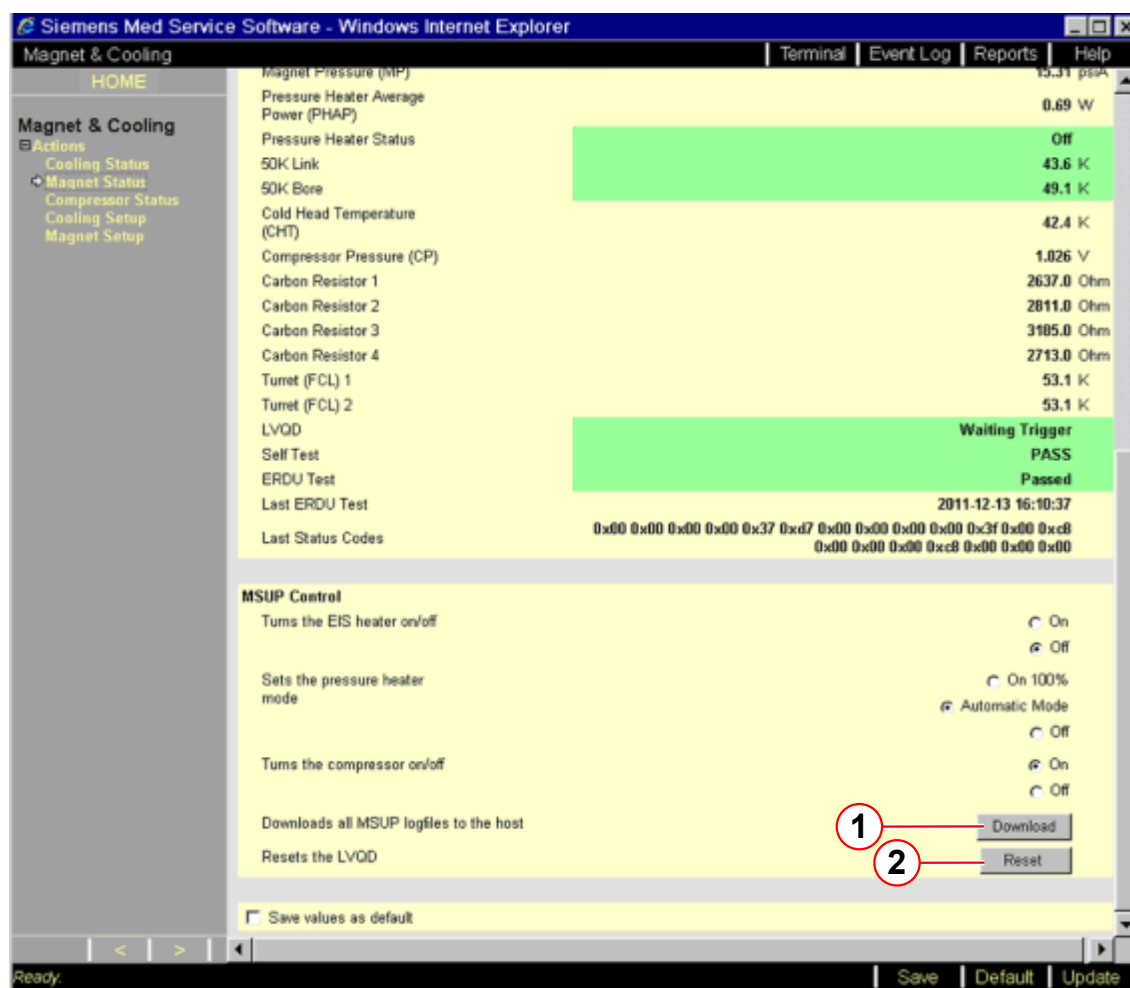
- rebooting the MRC (MRAWP) Host.
- cycling the power to the MSUP.
- allow the MSUP to complete an overnight sample.

7.1.1.2 Downloading the MSUP 2306 log file data to the MRC via SESO

Access the **Service software** at the **MRC (MRAWP) Host**

- Select menu item **"Magnet & Cooling."**
- Select **"Magnet Status"** in the **"Action"** menu.
- Click **"Download"** to download all the **MSUP logfiles** to the **MRC**.
 - » logfiles are **saved** to the MRC 'C' Drive, under **C:\MEDCOM.LOG** (→ Fig. 369 Page 301)

Fig. 368: Downloading MSUP logfiles to the Host

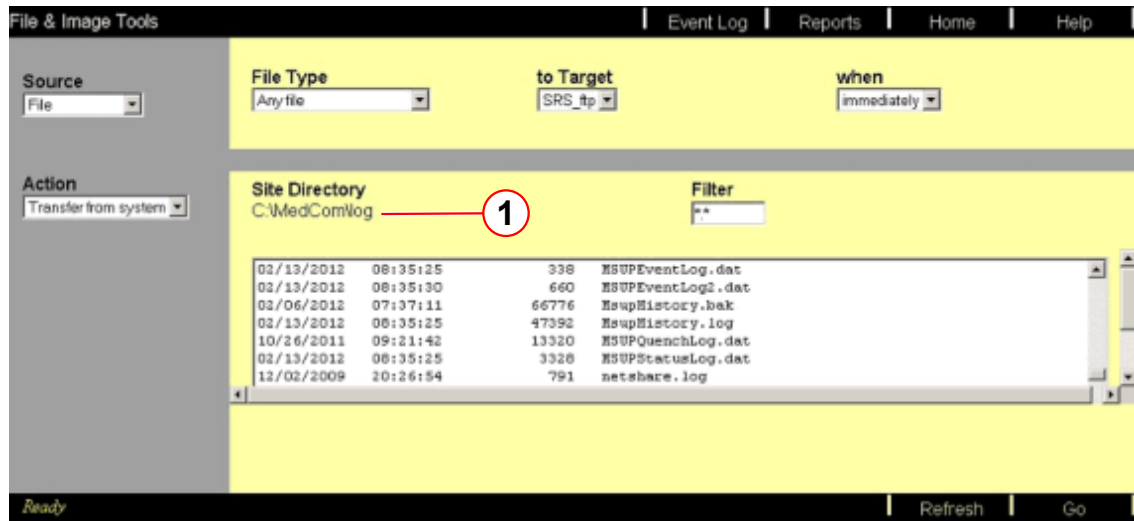


(1) Download button

(2) Reset button

- Click **"Reset"** (pos 2).

Fig. 369: Host MRC MSUP logfiles location



(1) Site Directory

- **Email** the downloaded log files to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

7.1.1.3 Capturing the iButton menu on the MSUP using a laptop



Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

1. Turn on the laptop and open the **MSUP Terminal Emulation Software**.
 - » Set bits per second to 9600
 - » Set data bits to 8
 - » Set parity to 1
 - » Set flow control to none
 - » Set Emulation to VT100 (File:Properties:Settings)
2. **Connect** the laptop to connector **X6** on the MSUP **via RS232 cable**.
3. Press **ESC**.
4. Type **L0 6932** <enter>.
5. Type **R** <enter>.

6. Press **ESC** to display the **MSUP Command input screen**.

Fig. 370: MSUP Command Input screen (example)

```

**Command*****Syntax*****
R          Run Process Display.
B          Enter iButton menu.
T hh:mm:ss Set clock time.
D dd-mm-yy Set date.
xSN xxxxxxxx (A)larmbox, (M)agnet or (S)upervisory Serial Number.
xPN xxxxxxxx (A)larmbox, (M)agnet or (S)upervisory Part Number.
xRN xxx     (A)larmbox, (M)agnet or (S)upervisory Revision Number.
Hx nnn     Probe (1) or (2) ADC reading 100% He level.
Zx nnn     Probe (1) or (2) ADC reading 0% He level.
HT         Start Helium level test.
COMP ON/OFF Turn Compressor ON / OFF.
EIS ON/OFF  Turn EIS ON / OFF.
PH ON/OFF/NM Turn Pressure Heater ON / OFF / Normal Mode.
LEDTEST     Start LED Test Sequence.
VA         View Log Files.
VQ         View Quench Log File.
VH         View Calibration data.
*****
Command:

```

7. Type **B<enter>** this displays the **iButton menu**.

Fig. 371: MSUP 2306 iButton menu (example)

```

**Command*****Values*****
U          Back to up level.
A [TEXT]   Magnet type          OR105-01
           Magnet P/N, S/N, Rev  1, 26096631, 1
           Alarmbox P/N, S/N, Rev 1, 1, 1
           Helium calibration points Probe0:358-1
                                   Probel:358-1
B [f.ff]   Magnet heater resistance 20.5 Ohms
C [0/1]    Resistor/Ceramic         1
D [f:f:f]  4.2K:R77:RT              3160.0:1260.0:961.8
E [B]      Magnet indicator status  00000000000000000011
F [f.ff]   Magnet operating current 460.00A
G [n:n]    He Level warn and reset  50:52
H hh:mm:ss Supervisory test time    02:00:00
I [n]      Cold head sample day      1
           Z2 shim sensitivity       19000
J [n:n]    Link alarm and warning     94K: 77K
K [n:n]    Bore alarm and warning     102K: 84K
           Diode calibration reference 1
           Ramping Turret Temp Limit 550K
S          Test iButton
R          Refresh display menu
*****
Command:

```

8. **Capture the screenshot.**(→ Capturing a screenshot / Page 303)

9. Type **U** <enter>.
10. Type **R** <enter>.
11. **Email** the saved files to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

Capturing a screenshot

- Capture a **screenshot** of the **MSUP Terminal Emulation Screen**:
 - a) Press **Ctrl:Alt:Prt Sc**.
 - b) Go to: **Start:Programs:Accessories**:and open Paint.
 - c) Press **Ctrl:V** to paste screenshot.
 - d) **Name** and save as a **.jpg** or **.bmp**.
Name the file DD (**Day**), MM (**Month**), YYYY (**Year**), TTTT (**Time:24hour format**).
i.e. iButtonDDMMYYYYTTTT: MSUPDDMMYYYYTTTT.
- E-mail the saved file to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>).

7.1.1.4 Restoring system calibration parameters

Downloading MSUP 2306 data to the MRC

The **MRC (MRAWP) Host** must be updated with the **system parameters**.

Check the **serial/revision/part numbers** are correct for the **Alarm box, MSUP 2306, and Magnet**.

Access the Service software at the MRC (MRAWP) Host.

- Select menu item **"Magnet & Cooling."**

- Select **"Magnet Setup"**.

Fig. 372: MSUP 2306 - Entering system calibration parameters

Siemens Med Service Software - Windows Internet Explorer

Magnet & Cooling | Terminal | Event Log | Reports | Help

HOME

Magnet & Cooling

- Actions
- Cooling Status
- Magnet Status
- Compressor Status
- Cooling Setup
- Magnet Setup**

Magnet & Cooling - Magnet Setup

Magnet

Helium Level Probe 1	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe 2	Min. (0%)	85.5	Ohm
	Max. (100%)	0.2	Ohm
Helium Level Probe Selection		☒ Probe 1 ☐ Probe 2 ☐ Both Probes	
Helium Level Warning	Min.	50	%
Shield Temperature Warning 50K Link	Max.	77.0	K
Shield Temperature Warning 50K Bore	Max.	84.0	K
CCR Calibration	Room Temperature	980.0	Ohm
	Nitrogen Temperature	1259.0	Ohm
	Helium Temperature	2772.0	Ohm
Pressure Heater Resistance		20.5	Ohm
Time Setting	Daily at	02:00	O'Clock
MSUP	Serial Number	1599	
	Revision Number	7	
	Part Number	10523587	
Alarmbox	Serial Number	1508	
	Revision Number	3	
	Part Number	10523563	
Magnet	Magnet Type	OR105-01	
	Serial Number	21259456	
	Revision Number	43	
	Part Number	7758167	
Magnet Current		509.87	A

MSUP Data

Downloads the magnet setup/calibration data to the host

Uploads the magnet setup/calibration data from the host to the MSUP

☐ Save values as default

Ready | Save | Default | Update

- (1) Download button (MSUP data stored on MRC)
- (2) Save button - (transfer calibration values to MSUP)
- (3) Update button (refreshes on-screen data)

- **Compare** the values with the **Magnet Acceptance Document**. (→ Fig. 373 Page 306)
- **Correct** MSUP parameters as necessary. Compare **acceptance data** to the **serial data** captured **from the MSUP**.

- Click **"UPDATE"**
 - » This **refreshes** the on-screen values.(→ 3/Fig. 372 Page 304)
- Update the **serial/revision/part** numbers (if required).
- Click **"SAVE"** to transfer calibration values **to the MSUP**. (→ 2/Fig. 372 Page 304)
- Click **"DOWNLOAD"**
 - » This takes the **MSUP data** and stores it in a file on the **MRC**.
(→ 1/Fig. 372 Page 304)

Fig. 373: OR105 Magnet Acceptance Document

SIEMENS		Magnet Acceptance Documentation Installation data MR Magnet Technology Quality Regulations	
Cryostat number	21326022	SMT reference	21330107 12.11
System description	1.494T Actively Shielded Superconducting Magnet System OR105		
Measured magnet centre (Offset)	+1 (P) mm		
If magnetic offset is outside the spec (± 5mm), the magnet will be marked (± 5mm).			
 This system requires 0 Z2 rings and 2.92 kg of iron to shim. This is a theoretical calculation that may change due to site conditions.			
Operating parameters			
Description	Specification value	Measured value	
Operating current @ 1.494T	> 500 Amps	510.01 Amps	
Field/Current Ratio	N/A	12.48 kHz/0.1 Amp	
Magnet type:	OR105-01		
TDC calibration values		Carbon resistor values	
Room temperature (23 °C)	1009 Ohms	Calibration values	
Liquid nitrogen temperature (77K)	1309 Ohms	1009 Ohms	
Liquid helium temperature (4.2K)	2994 Ohms	1309 Ohms	
		2994 Ohms	
Position	Room temperature	Helium temperature	
TDC d - e	1034 Ohms	2837 Ohms	
Top a - e	980 Ohms	3014 Ohms	
Middle b - e	1110 Ohms	2681 Ohms	
Bottom c - e	1006 Ohms	2930 Ohms	
Helium level probe resistance (when using a 250mA supply)			
	Probe 1	Probe 2	
@ 0 resistance	85.36 Ohms	85.36 Ohms	
0% calibration	774	774	
100% resistance	0.22 Ohms	0.22 Ohms	
100% calibration	2	2	
Diode table no.	-1		
Pressure heater resistance	20.5 Ohms		
Final temperature sensor recon values			
Shield bore	Shield Link	Coldhead	Recon margin
47K	47K	42K	658mW
Checked by	Signed <u>ANOTHER</u>		
V Turner			
	Date 23/12/2011		
	Page 1 of 1		
	25-AD EPS 000 01		

7.1.2 Magnet Supervision (MSUP 2657B)

The MSUP is a **Field Replaceable Unit**. The fuses are the only elements which can be replaced if required.

This procedure can be used to fault find the MSUP. An understanding of Tim systems is required to use the following information.

7.1.2.1 MSUP fuses

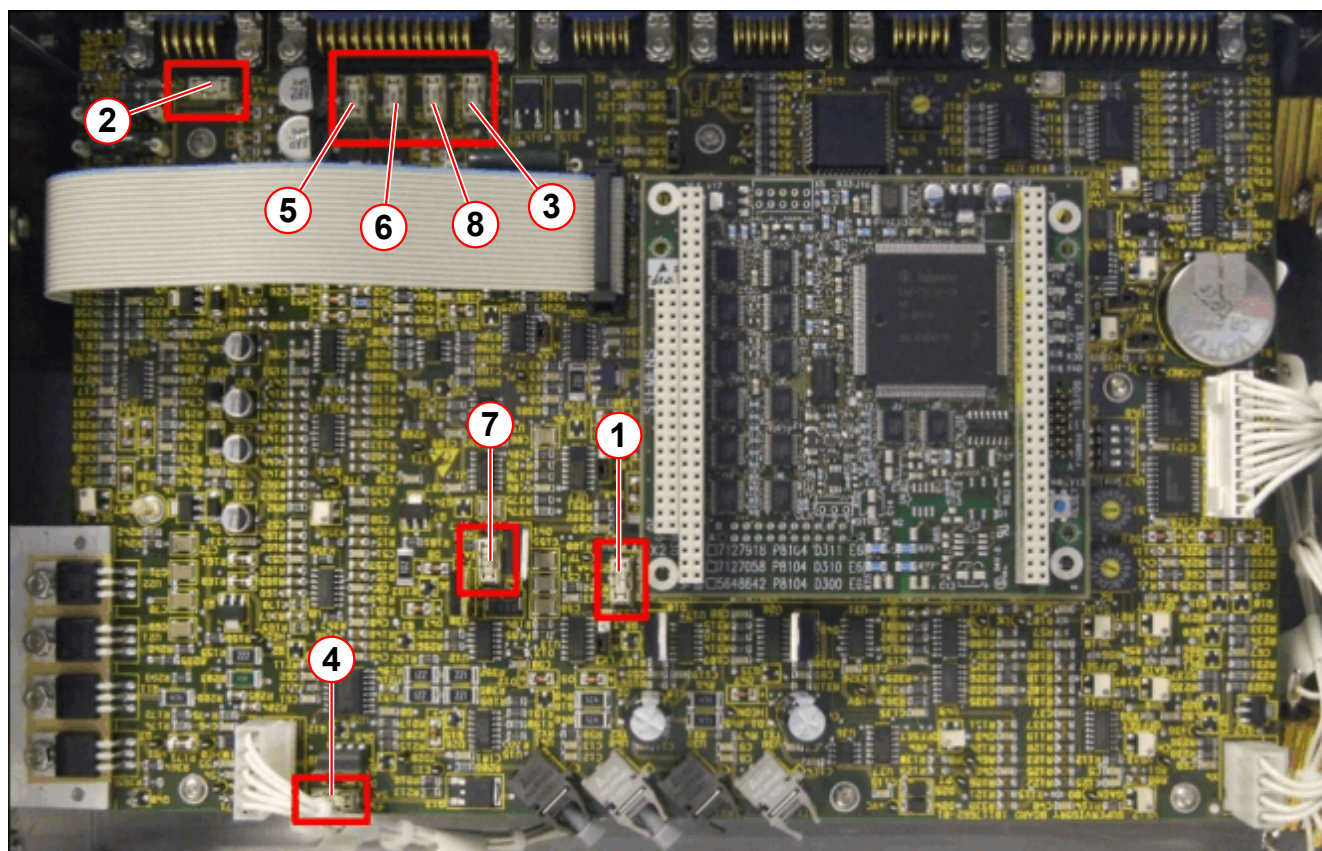
For location of these fuses, refer to (→ Fig. 374 Page 308).

Fuse	Rating	Title
F1	125 mA F/250 mA F*	EIS control
F2	4 A T	+36 V DC input supply
F3	500 mA T	He probe
F4	5 A T	Battery supply
F5	3 A F	Quench heater 1
F6	3 A F	Quench heater 2
F7	3 A F	Pressure heater control
F8	375 mA T/1A F*	Switch heater

*Higher value fitted after Rev 14.

7.1.2.2 Fuses

Fig. 374: MSUP 2657B - Fuses



- (1) Fuse F1, 125 mA, EIS Control (250 mAF for MSUP with Rev. = 14)
- (2) Fuse F2, 4 AT, +36 V MSUP supply
- (3) Fuse F3, 500 mA, LHe level probe
- (4) Fuse F4, 5 AT, Battery Supply
- (5) Fuse F5, 3 AF, Quench Heater 1
- (6) Fuse F5, 3 AF, Quench Heater 2
- (7) Fuse F7, 3 AF, Pressure Heater Control
- (8) Fuse F8, 375 mAF, Switch Heater (1 AF for MSUP with Rev. = 14)

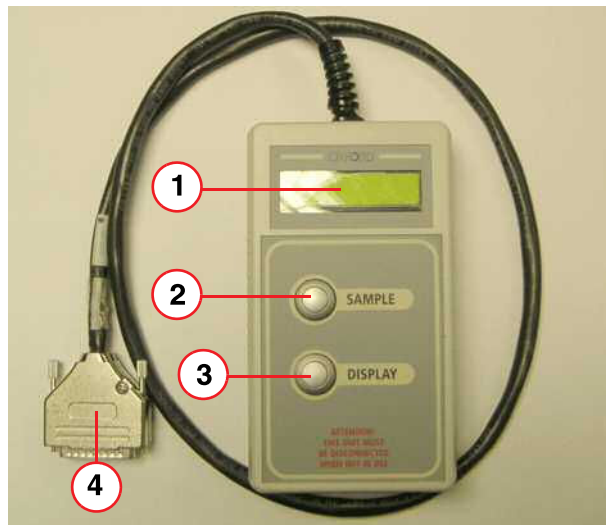
7.1.2.3 Hand-held MSUP Display-Unit

The only operations that can be performed independently of the controlling host computer are through the **hand-held MSUP Display-Unit**.

Required tools

Hand-held MSUP Display-Unit.

Fig. 375: MSUP- Display Unit



- (1) LCD Display
- (2) Sample button
- (3) Display scroll button
- (4) 25 way 'D' connector

Performing the procedure

- Connect the display unit to **X9** on the MSUP via the 25 way 'D' connector (4)
- Ensure that power has been applied to the MSUP.
- Press **SAMPLE** button (2).
 - The unit will carry out a helium level and battery test.
 - The helium fill level is shown in the display (1).
- Press **DISPLAY** (3), and **SAMPLE** (2) simultaneously to unlock the scroll function.

- Scroll through the parameters by pressing the **DISPLAY** button, one parameter for each press.

The parameters in order of presentation are:

1. He level (press sample to recheck level)
2. Battery volts
3. 50K link, bore (K, K)
4. Carbon resistor 1, 2 (ohm, ohm)
5. Carbon resistor 3, 4¹ (ohm, ohm)
6. Turret 1, 2 (K, K)
7. Switch heater (Ohm, status)
8. Quench heaters 1, 2 (Ohm)
9. Cold head (K)
10. ERDU buttons (status)
11. Magnet pressure (psiA)
12. Pressure heater (Off/On, %) (press Sample to switch on, 50 min time-out)
13. Pressure heater (average power over last hour of previous measurements)
14. EIS (Off, Sample to switch On, 130 min time-out)
15. Self-test
16. Compressor status (OFF/ON,P,T)
17. LVQD (Waiting trigger/quench location) Time & Date
 - Time and date

1. Carbon resistor 4 is shown in Kelvin for CCR type magnets.

- Magnet type
- Part no and software version



Disconnect the display unit from the MSUP after test is finished.

7.1.2.4 MSUP download instructions



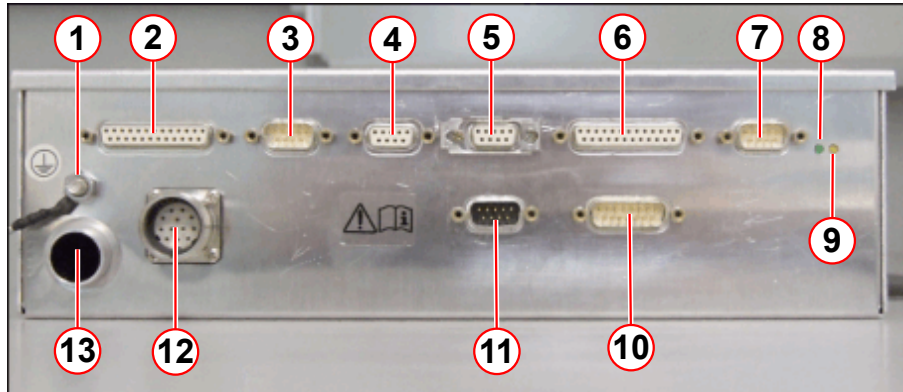
Ensure a safe distance from the magnet for troubleshooting and service actions which require the use of a laptop with extended cabling.

Instruction on how to download the **log file** of the MSUP

1. Turn on the laptop and open the hyper terminal application. Set to the following settings. Select the appropriate Com Port.
 - » Set bits per second to 9600
 - Set data bits to 8
 - Set parity to none
 - Set stop bits to 1
 - Set flow control to none
 - In the file, properties, settings tab, set Emulation to VT100.

2. Connect the laptop via an **RS232 cable** to connector **X5** on the MSUP (pos 3).

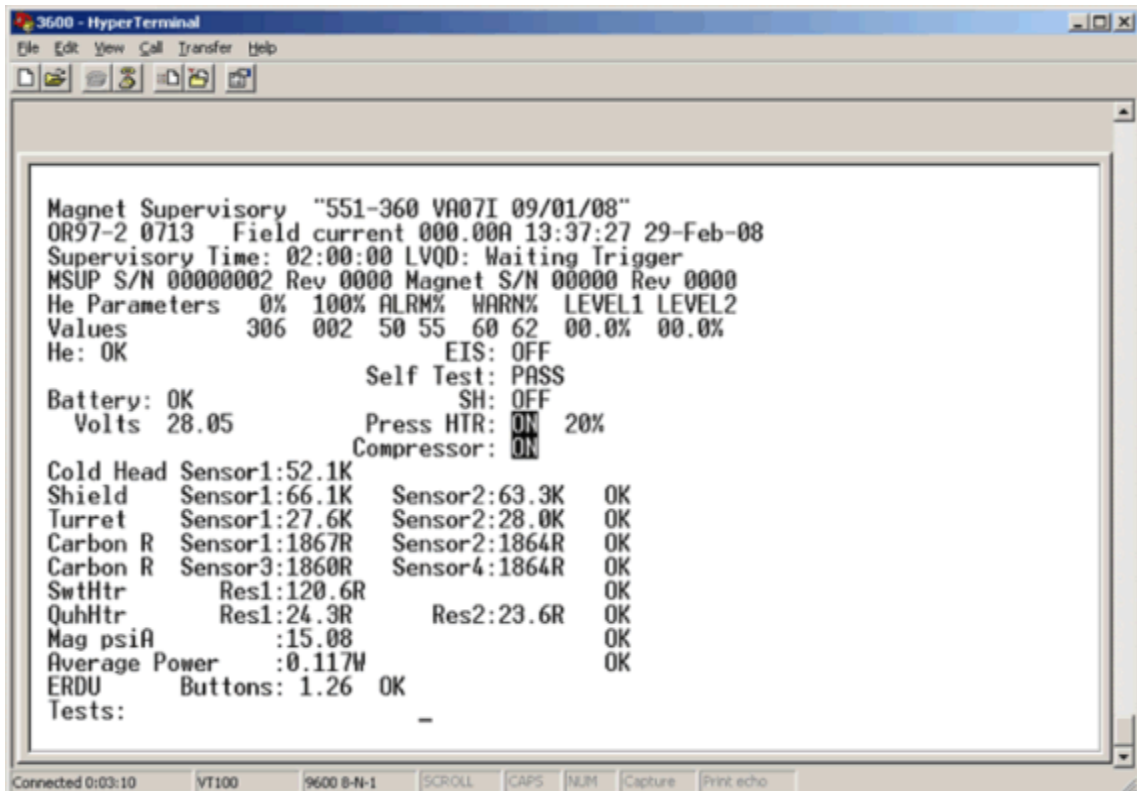
Fig. 376: MSUP 2657B - Indicators and input/output connections



- (1) Earth stud (6mm)
- (2) X9 DISPLAY
- (3) X5 RS232
- (4) X6 ERDU
- (5) X2 CH/DIAG/PRESS
- (6) X1 TURRET
- (7) X3 POWER/SL/UPS
- (8) Power ON LED (green)
- (9) Pressure heater LED (yellow)
- (10) X14 LVQD
- (11) X12 BATTERY
- (12) X7 Magnet ID
- (13) CAN Fiber optics wave guide

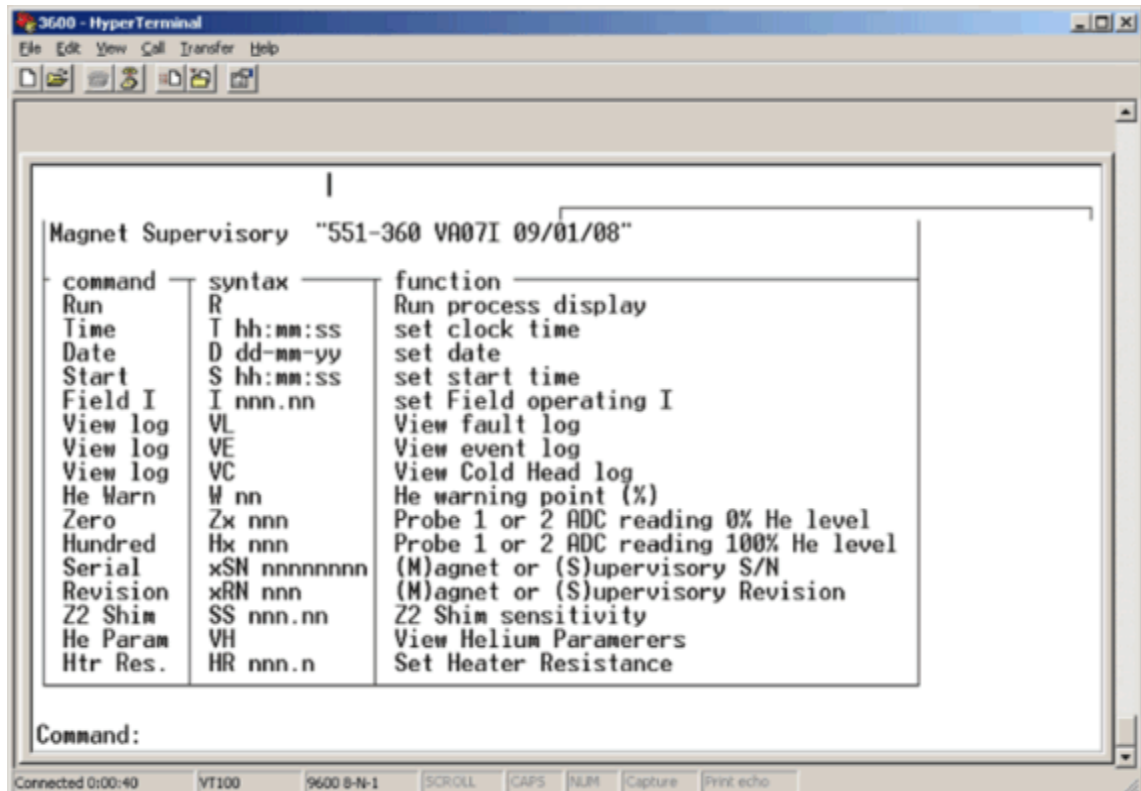
3. The **MSUP Process Display** is displayed.

Fig. 377: MSUP 2657B - Process Display screen (typical)



4. Press **Esc.** to display the **MSUP Command Input** screen.

Fig. 378: MSUP 2657B - Command Input screen (typical)



5. Click **"Transfer"**
6. Click **"Capture text"**
7. Click **"Browse"** and select a directory and a name for the file to be stored. Name should contain the **magnet serial number**.
e.g. C:\Temp\MSN 12345678log.txt
8. Click **"Start"**
9. Type the following
 - » VL <enter>
 - VE <enter>
 - VC <enter>
 - VH <enter>
 - VQ <enter>
 - VD <enter>
 - VP <enter>
 - VX <enter>
10. Click **"Transfer"**, **"capture text"**, **"Stop"** after downloaded data has been transferred.
11. Capture a **screenshot** of the **MSUP terminal emulation** screen:
 - a) Press **Ctrl:Alt:Prt Sc**
 - b) Go to: **Start:Programs:Accessories:**and open Paint

- c) Press **Ctrl:V** to paste screenshot
 - d) Name and save as a .jpg or .bmp.
12. E-mail the saved file and screenshot to (→ Magnet HSC, <mailto:magnet.wscshsc.healthcare@siemens.com>)

Tab. 30 MSUP 2657B Input Codes

Description	Command	Operation/Example
Time	T hh:mm:ss<R>	Set time in real time clock 1 hour 30mins 30secs - enter as T 13:30:30<R>
Date	D dd-mm-yy<R>	Set date in real time clock 1st Jan 2008 - enter as D 01-01-08<R>
Start test time	S hh:mm:ss<R>	Set supervisory test time 2 hour 45mins 45secs - enter as T 2:45:45<R>
Magnet operating current	I nnn.nn<R>	Set Mag Operating Current Operating current 685.02 - enter as I 685.02<R>
Helium warn point	W nn<R>	Set Helium Warning point 50% enter as W 50<R>
0% Calibration (probe 1)	Z1 nnnn<R>	Set He Probe 1 0% Probe1: 88.8 ohms -enter as Z1 805<R>
0% Calibration (probe 2)	Z2 nnnn<R>	Set He Probe 2 0% Probe2: 88.8 ohms - enter as Z2 805<R>
100% Calibration (probe 1)	H1 nnnn<R>	He Probe 1 100% Probe1: 1.8 ohms - enter as H1 16<R>
100% Calibration (probe 2)	H2 nnnn<R>	He Probe 2 100% Probe2: 1.8 ohms - enter as H2 16<R>
Magnet Serial No.	MSN nnnnnnnn<R>	Adjust Magnet Serial No Enter as MSN12345678<R>. Password required
Magnet Revision No.	MRN nnn<R>	Adjust Mag Rev No Magnet Rev 4 - enter as MRN4<R> Password required
Supervisory Serial No.	SSN nnnnnnnn<R>	Adjust Supervisory Serial No. Supervisory Serial No S0014 - enter as SSN S0014<R>
Supervisory Revision No.	SRN nnn<R>	Adjust Supervisory Revision No Supervisory Revision No 14 - enter as SRN14<R> Password required
Pressure heater resistance	HR nnn.n<R>	Adjust Pressure Heater Res 19.4 ohms - enter as HR 19.4<R> Password required
Shield bore warning	L4 2947<R> SBW nnn<R>	Shield Bore Warning Shield Bore warning of 77K - enter as SBW 77<R>

Description	Command	Operation/Example
CCR Calibration	SR <R> XXX.X <R> YYYY.Y <R> ZZZZ.Z <R>	Adjust CCR calibration to RT (XXX.X), 77K (YYYY.Y), 4.2K (ZZZZ.Z) Password required Refer to Replacement of Parts for Calibration of CCRs .
Run process image	R<R>	Refresh the screen
View status log	VL<R>	View status log
View error log	VE<R>	View error log
View month log	VM<R>	View month log
View cold head log	VC<R>	View cold head log
View quench log	VQ<R>	View quench log
View LVQD log	VD<R>	View LVQD log
View helium log	VH<R>	View the helium calibrations
View helium pulse log	VP<R>	View the helium pulse log
Notes: ■ <R> = 'Enter' or 'Return'. ■ Where a password is required, type L 9912<R> before the command is input. ■ CCR Calibration - The MSUP will prompt the user to input values after the command SR has been used. - Refer to the Magnet Acceptance Document for TDC CCR4 Calibration Values.		

7.2 Refrigerator

7.2.1 Helium gas charging CSW-71



Make sure the refrigerator is powered off and compressor and cold head temperatures are almost at room temperature before adjusting the helium gas pressure.

If the compressor has been running, switch off and leave for a minimum of 30 minutes before checking static pressure.



CAUTION

High pressure helium gas.

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves and eye protection must be worn.

Required tools

Charge and vent adapter (gas charging tool)(part number **08396660**)

Bleed adapter assembly (part number **10125874**)

High pressure regulator for helium gas bottle.

30bar Helium pressure hose (charging line).

Helium gas bottle, purity of 99.999% .



Use only helium of purity 99.999% or better.

3/4" (19mm) Open-end wrench/spanner.

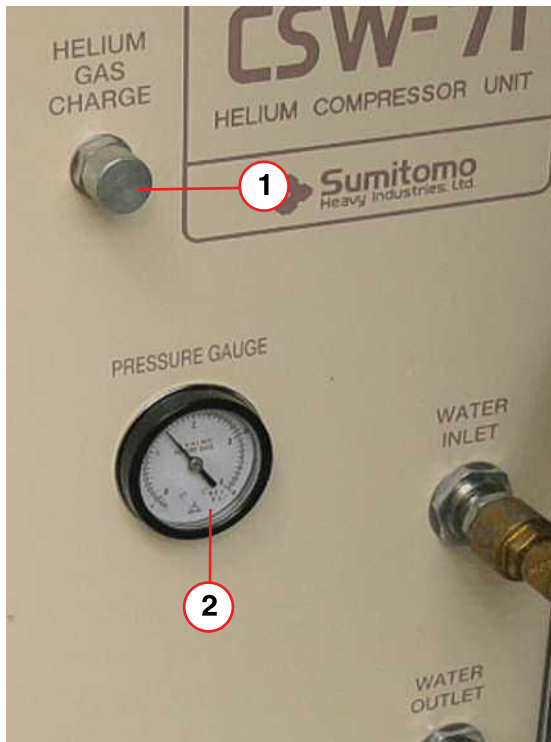
5/8" (16mm) Open-end wrench/spanner.

Performing the procedure

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm

- Wait for at least **30 minutes** to allow the static pressure to stabilize.

Fig. 379: Compressor He Gas charge & pressure gauge



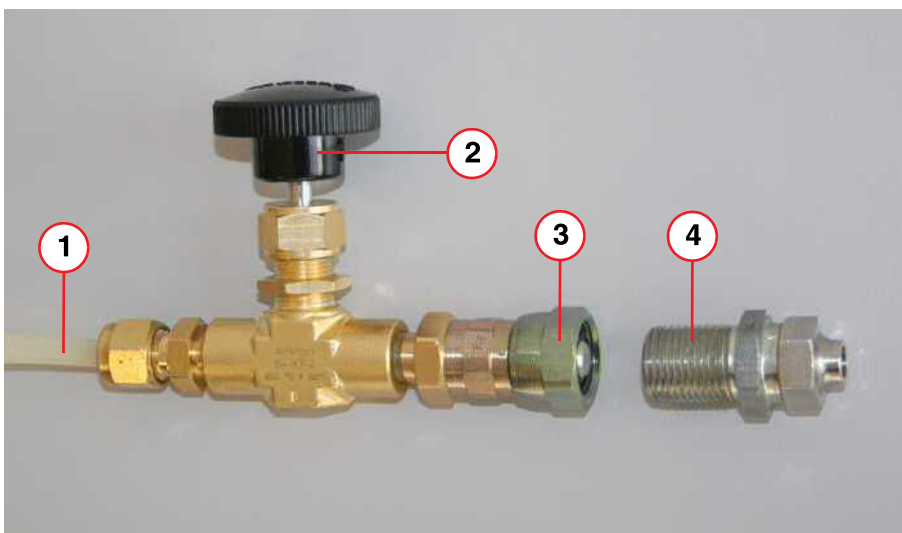
- Remove the **protective cap (1)** from the **helium gas charge coupling** on the compressor unit.
- Check the **rubber aeroquip gasket** for dirt or dust and ensure it is correctly fitted. Replace if required (part no **03861494**)

NOTE: Incorrect fitment may cause a helium gas leak.

- (1) Helium gas charge
(2) Helium pressure gauge

- Connect a **high pressure regulator** and **charging line** to the helium **gas bottle** (purity 99.999%).

Fig. 380: He compressor - charge and vent adaptor (gas charging tool)



- (1) Charging line (30bar)
(2) Charge and vent adaptor valve
(3) Aeroquip coupling (female)
(4) Bleed adaptor assembly

- **Open** the valve on the **helium gas bottle**.
 - Ensure the **regulator valve** is closed.
 - Ensure that **30bar** of helium gas is available.
- Screw the **charging line** (→ 1/ Fig. 380 Page 318) into the **gas charging tool**.
- Screw the **bleed adaptor assembly** (→ 4/ Fig. 380 Page 318) into the **gas charging tool**.
- **Purge the charging line** of any air contaminants by setting the high pressure regulator to 10psi for 30 seconds and **opening** the valve on the **gas charging tool**.
- Unscrew and remove the **bleed adaptor assembly**.
- **Close** the **gas charging tool valve** (→ 2/ Fig. 380 Page 318)
- Set the **gas pressure regulator** to:
 - » **16.5 - 17 bar or 239 - 246 psig**
- Using two wrenches, connect the **gas charging tool** to the **helium gas charge coupling** on the compressor unit.
- Slowly **open** the gas charging tool valve.

Fig. 381: Helium Compressor



(1) Gas charging tool with charging line

- Wait 60 seconds for gas pressure to **stabilize** on the compressor gauge.
- If the compressor pressure still indicates **lower** than the required:-
 - Adjust the gas pressure regulator until the compressor gauge reads **16.0 - 16.5 bar**.
 - Wait 60 seconds for gas pressure to **stabilize** on the compressor gauge.
- When pressure is achieved, **close** the **gas charge tool valve**.



It is possible to overpressurize the compressor. To adjust pressure, refer to (→ Adjusting the equalization pressure / Page 320).

- **Disconnect** the **gas charging tool** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Close** the valve on the **helium gas bottle**.
- Screw the **bleed adaptor assembly** into the end of the **charging line**.
- Slowly open the **gas charging tool valve** to release pressure from the **charging line**.
 - » The regulator gauge should read **zero**.
- **Unscrew** and remove the **bleed adaptor assembly**.
- **Disconnect** the high pressure **regulator** from the **gas bottle**.
- Store the **charge line** to keep it clean.
- Re-start the compressor.

7.2.1.1 Adjusting the equalization pressure

If the compressor has been **overpressurized**, the excess helium gas must be vented to atmosphere.



The compressor should not be operated above the specified pressure. Overpressure will cause the compressor to run in continuous bypass mode.

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red **COMPRESSOR** LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- Wait for at least **30 minutes** to allow the static pressure to stabilize.
- **Remove** the **protective cap** from the **helium gas charge coupling** on the compressor unit (→ 1/Fig. 379 Page 318).
- Make sure the **gas charge tool valve** is **closed** (→ 2/Fig. 380 Page 318).
- Using two wrenches, connect the **gas charging tool** to the **helium gas charge coupling** on the compressor unit.
- **Slowly open** the **gas charging tool valve**.
- **Vent** the **helium gas** until the required **equalization pressure** is achieved.
- **Close** the **gas charging tool valve**.
- **Disconnect** the **gas charging tool** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- Re-start the compressor.

7.2.2 Helium gas charging F70



Make sure the refrigerator is powered off and compressor and cold head temperatures are almost at room temperature before adjusting the helium gas pressure.



CAUTION

High-pressure helium gas

Failure to observe the following safety warning may result in personal injury.

- » Protective clothing, gloves, and eye protection must be worn.

Required tools

- **Charge & Vent tool Kit** (part number **10125858**)
 - Adaptor valve (gas charging tool) (part number **10125873**)
 - Bleed adapter assembly (part number **10125874**)
- High pressure regulator for helium gas bottle
- 30bar Helium pressure hose (charging line)
- Helium gas bottle, purity of 99.999%



Use only helium of purity 99.999% or better.

3/4" (19 mm) Open-end wrench/spanner.

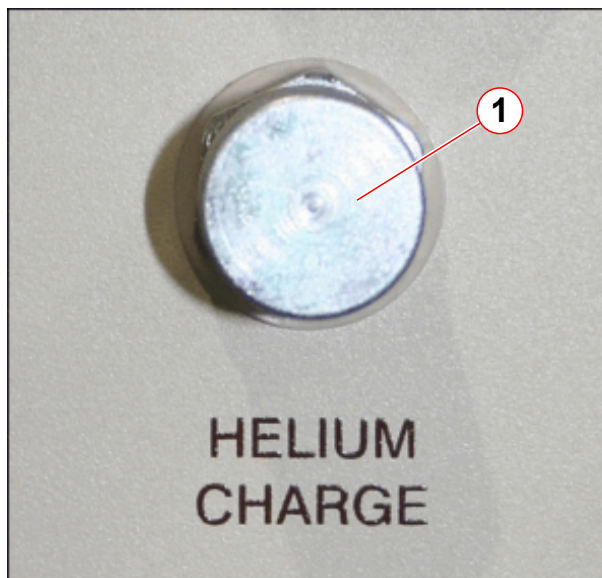
5/8" (16 mm) Open-end wrench/spanner.

Performing the procedure

- Turn **off** the compressor at the main power switch.
 - » The Alarm Box will display a warning LED, and an **audible** alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm

- Wait for at least **30 minutes** to allow the static pressure to stabilize.

Fig. 382: F70 Compressor - Helium charge port



(1) Protective cap

- Remove the **protective cap (1)** from the **helium gas charge coupling** on the compressor unit.

Fig. 383: F70 Compressor - Helium Charge fill port gas-ket seal



(1) Aeroquip gasket seal

- Check the **rubber aeroquip gasket** for dirt or dust and ensure it is correctly fitted. Replace if required (part number **03861494**).

NOTE: Incorrect fitment may cause a helium gas leak.

- Connect a **high pressure regulator** and **charging line** to the helium **gas bottle** (purity 99.999%).

Fig. 384: He compressor - charge and vent adaptor (gas charging tool)



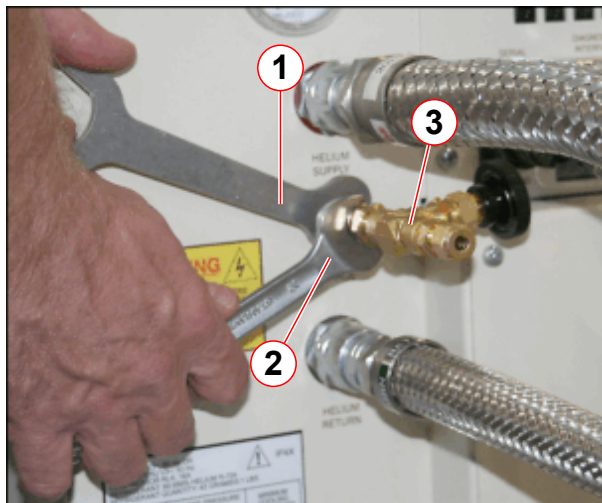
- (1) Charging line (30bar)
- (2) Charge and vent adaptor valve
- (3) Aeroquip coupling (female)
- (4) Bleed adaptor assembly

- **Open** the valve on the **helium gas bottle**.
 - Ensure the **regulator valve** is closed.
 - Ensure that **30bar** of helium gas is available.
- Screw the **charging line** (→ 1/Fig. 384 Page 323) into the **adaptor valve**.
- Screw the **bleed adaptor assembly** (→ 4/Fig. 384 Page 323) into the **adaptor valve**.
- **Purge the charging line** of any air contaminants by setting the **high pressure regulator** to 10 psi for 30 seconds and **opening** the valve on the **adaptor valve**.
- Unscrew and remove the **bleed adapter assembly**.
- **Close** the **adaptor valve**.
- Set the **gas pressure regulator** as required - see table below:-

Tab. 31 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4 K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal) 196 - 203 psi (G) 1350 - 1400 kPa	21 - 22 bar 304 - 319 psi (G) 2100 - 2200 kPa

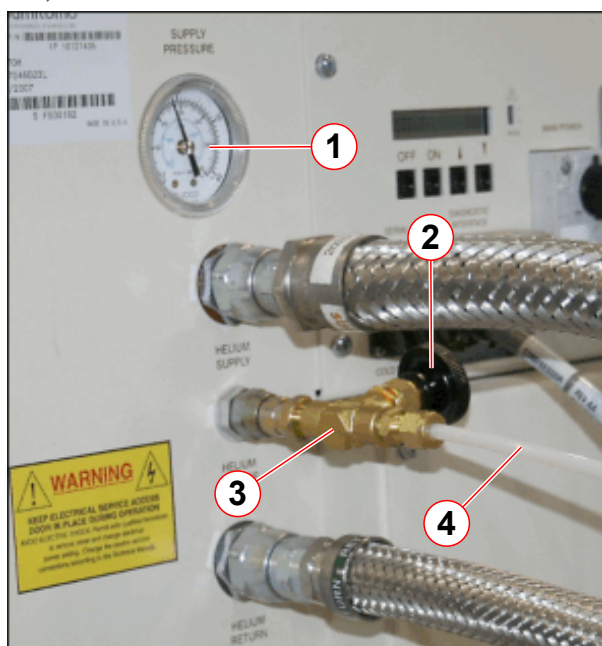
Fig. 385: F70 Compressor - Fit/remove charge & vent adaptor (gas charging tool)



- (1) Turn 5/8" (16mm) spanner/wrench
- (2) Hold 3/4" (19mm) spanner/wrench
- (3) Charge & vent adaptor (gas charging tool)

- Using two wrenches, connect the **adaptor valve** to the **helium gas charge coupling** on the compressor unit.

Fig. 386: F70 - Charge & vent adaptor (gas charging tool)



- (1) Pressure gauge
- (2) Valve
- (3) Charge and vent adaptor (gas charging tool)
- (4) Charging line

- **Slowly open** the adaptor valve 2.

- Wait **60 seconds** for gas pressure to **stabilize** on the compressor gauge.
- If the compressor pressure still indicates **lower** than the required:-
 - Slowly open the **high pressure regulator** until the required pressure has been achieved.
 - Wait 60 seconds for gas pressure to stabilize on the compressor gauge.

- When pressure is achieved, **close** the **adaptor valve**. (→ 2/Fig. 386 Page 324)



It is possible to overpressurize the compressor. To adjust pressure, refer to Adjusting the equalization pressure.

- **Disconnect** the **adaptor valve** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
- Replace the **protective cap** on the **helium gas charge coupling**.
- **Close** the valve on the **helium gas bottle**.
- Screw the **bleed adaptor assembly** into the end of the **charging line**.
- **Slowly** open the **adaptor valve** to release pressure from the **charging line**.
 - » The regulator gauge should read **zero**.
- **Unscrew** and remove the **bleed adaptor assembly**.
- **Disconnect** the high pressure **regulator** from the **gas bottle**.
- Store the **charge line** to keep it clean.
- Restart the compressor.

7.2.2.1 Adjusting the equalization pressure

If the compressor has been **over pressurized**, the excess helium gas must be vented to atmosphere.

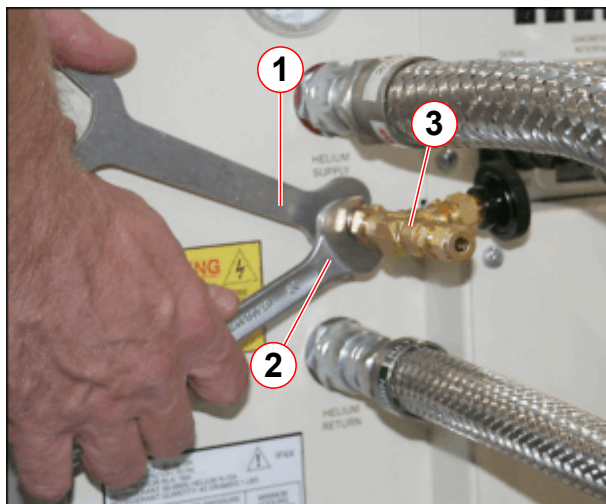


The compressor should not be operated above the specified pressure. Overpressure will cause the compressor to run in continuous bypass mode.

- Turn **off** the compressor at the **main power switch**.
 - » The Alarm Box will display a red warning LED, and an audible alarm will sound.
 - » Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- Wait for at least **30 minutes** to allow the static pressure to stabilize.
- **Remove** the **protective cap** from the **helium gas charge coupling** on the compressor unit.

- Make sure the **gas charge tool valve** is closed.

Fig. 387: F70 Compressor - Fit/remove charge & vent adaptor (gas charging tool)



- Using two wrenches, connect the **adaptor valve** to the **helium gas charge coupling** on the compressor unit.
- (1) Turn 5/8" (16mm) spanner/wrench
(2) Hold 3/4" (19mm) spanner/wrench
(3) Charge & vent adaptor (gas charging tool)
- **Slowly open the adaptor valve.**
 - **Vent the helium gas** until the required **equalization pressure** is achieved.
 - **Close the adaptor valve.**
 - **Disconnect the adaptor valve** from the **compressor**.
 - Take care not to dislodge the aeroquip gasket seal.
 - Replace the **protective cap** on the **helium gas charge coupling**.
 - **Restart** the compressor.

- » The common print number (M6-020.841.01) from the Replacement of Parts is divided into separate Component print numbers.

Magnet System	
Chapter Refrigerator	Lifting the compressor

Version 02

- Magnet System
 - Burst disks reused from SMT document
 - Helium compressor reused from SMT document
 - MSUP battery reused from SMT document

Version 03

- Magnet Supervisory System
 - MSUP 2657B
- Magnet
 - Update of pressure switch transducer assembly.
- Refrigerator
 - Update of adsorber exchange.

Version 04

- Magnet
 - Replaced F2 with F3.
- Refrigerator
 - CSW71: Corrected dynamic pressure to 19.0 - 22 bar.

Version 05

- Magnet
 - New burst disc rating.
- Refrigerator
 - Update of replacing the 4K cold head.
- Alarmbox
 - Replacement of base plate.

Version 06

- FRU Overview
 - Hint: "Protective earth connection test" was added.
 - Editorial changes.
- Refrigerator
 - Update of replacing the compressor F70.
 - Chapter "Replacing the adsorber of compressor F70" deleted.

Version 07

- FRU Overview
 - Update of FRU overview
- Magnet Supervisory Unit
 - Update with the MSUP for Verio Dot Ex-factory
- Alarm box
 - Update with the Alarm box for Verio dot Ex-factory

Version 08

- Magnet supervisory
 - "the initial replacement in this case occurs in two years" removed

Version 09

- Magnet Supervisory Unit
 - Update MSUP 2306 for ESPREE (→ MSUP for Avanto, Espree (VB19) / Avanto Dot Ex-factory / Page 235)
- Alarm box
 - Update Alarm box for ESPREE (→ Alarm Box for Avanto, Espree (VB19)/ Avanto Dot Ex-factory / Page 280)

Version 10

- Alarm Box
 - Erdu test load, material number changed in 3861197. (→ Tools / Page 276)

Version 11

- Refrigerator
 - Chapter of removing and fitting the Cold Head added. (→ Cold Head Replacement / Page 74)

Version 12

- Refrigerator
 - Chapter Cold Head Replacement updated. (→ Cold Head Replacement / Page 74) (Charm_ MR_00448144).
- MSUP
 - Chapter "Replacing the ERDU battery box" updated.

Version 13

- Magnet
 - Chapter Replacing the OVC burst disk updated (Charm_ MR_00453494).
- Refrigerator
 - Chapter (→ Replacing the F70 Compressor with Tooling kit / Page 145) Compressor tooling kit added (Charm_ MR_00451966).
 - Chapter (→ Cold Head Replacement / Page 74) New cold head shield added (Charm_ MR_00454101).
 - Chapter (→ Cold Head Replacement / Page 74) Temperature Sensor updated (Charm_ MR_00450528).

- MSUP
 - Chapter (→ Replacing the ERDU battery box / Page 273) new note: Do not short the pins on the battery connector (Charm_MR_00438066).

Version 14

- Chapter (→ Procedures / Page 298) added.
- Links updated.

Version 15

- Chapter (→ Magnet / Page 12)
 - Procedure (→ Replacing the OVC Burst Disk / Page 36) updated (Charm_MR_00466629).

There are no Hazard IDs in this document.

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Print No.: M6-020.841.17.15.02 | Replaces: M6-020.841.17.14.02
Doc. Gen. Date: 06.16 | Language: English
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